Node: 1. Node 1:
Condensate Feed from Condenensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column
(Pumps PM-111A/B, Filter FI-101A/B, Heat Exchanger EX-101/102/103/104/105)

Drawings / References:

40-PI-9004 ; 40-PI-9004 Redline; 40-PI-9005; 40-PI-9005 Redline; 40-PI-9006; 40-PI-9007

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
1.1. Low/No Flow	1.1.1. 8" manual valve downstream of Condensate Feed Pumps PM-111A/B/C inadvertently closed	1.1.1.1. Blocked discharge from Condensate Feed Pumps PM-111A/B/C; Pump deadhead; Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		1.1.1.2. Loss of feed flow through the downstream exchanges; See high temperature deviation, this node							
	1.1.2. Flow control valve FCV-12425 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	1.1.2.1. Blocked discharge from Condensate Feed Pumps PM-111A/B/C; Pump deadhead; Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		1.1.2.2. Loss of feed flow through the downstream exchanges; See high temperature deviation, this node							
	1.1.3. 8" manual valve downstream of EX-102 inadvertently closed	1.1.3.1. Blocked in flow from EX-102; Potential thermal expansion and release with potential ignition and personnel impact	452. 8" valve on feed outlet from EX-102/103 is CSO 453. PSV-10010 (set @ 305 psig) on outlet of EX-102 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
1.2. High Flow	1.2.1. Flow control valve FCV-12425 fails open or bypass valve inadvertently opened	1.2.1.1. Increased feed to Naphtha Stabilizer Column T-101, reduced temperature in T-101, potential for off-spec product, potential financial impact	578. LAH-10525 high level alarm on Stabilizer T-101 will prompt operator response	F	2	3	2	61. No recommendations required.	
1.3. Reverse/Misdirected Flow	1.3.1. Tube leak/rupture on Heat Exchanger EX-101	1.3.1.1. Shell and Tube MAWP both listed as 305 psig @ 450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	

Node: 1. Node 1:
Condensate Feed from Condenensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column (Pumps PM-111A/B, Filter FI-101A/B, Heat Exchanger EX-101/102/103/104/105)

Drawings / References:

40-PI-9004 ; 40-PI-9004 Redline; 40-PI-9005; 40-PI-9005 Redline; 40-PI-9006; 40-PI-9007

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		no concern of tube rupture; Loss of feed to product, product quality issue; Negligible downtime							
	1.3.2. Tube leak/rupture on Heat Exchanger EX-102	1.3.2.1. Shell and Tube MAWP both listed as 305 psig @ 450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Negligible downtime	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	1.3.3. Tube leak/rupture on Heat Exchanger EX-103	1.3.3.1. Shell and Tube MAWP both listed as 305 psig @ 460/450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Inability to heat up feed enough for the unit; Potential downtime greater than 5 days	449. Spare bundles available 450. EX-103/104 tubes inspected once per year	F	2	2	1	61. No recommendations required.	
	1.3.4. Tube leak/rupture on Heat Exchanger EX-104	1.3.4.1. Shell and Tube MAWP both listed as 311 psig @ 650 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Inability to heat up feed enough for the unit; Potential downtime greater than 5 days	449. Spare bundles available 450. EX-103/104 tubes inspected once per year	F	2	2	1	61. No recommendations required.	
	1.3.5. Tube leak/rupture on Heat Exchanger EX-105	1.3.5.1. Potential to lose atmospheric gas oil (AGO) into feed, inefficient operation; Operability issue only, no hazardous consequences identified							
	1.3.6. Bypass open on Heat Exchanger EX-103 or Heat Exchanger EX-104	1.3.6.1. Loss of feed flow through the tube side of EX-103; Flow through shell side coming in at ~420 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.3.6.2. Loss of feed flow through the tube side of EX-104; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.3.6.3. Lower temperature feed to Naphtha Stabilizer Column T-101, operability issue							
		1.3.6.4. Lower temperature feed to desalter, reduced desalter efficiency and some additional water carryover to process, reduced process efficiency							
	1.3.7. Bypass open on Bypass open on Heat Exchanger EX-101	1.3.7.1. Loss of feed flow through the tube side of EX-101; Flow through the shell side coming in at ~94 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
	1.3.8. Manual valve open on hydrocarbon drain	1.3.8.1. Potential to send feed to Hydrocarbon Sump TK-101; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 1. Node 1:
Condensate Feed from Condenensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column
(Pumps PM-111A/B, Filter FI-101A/B, Heat Exchanger EX-101/102/103/104/105)

Drawings / References:

40-PI-9004 ; 40-PI-9004 Redline; 40-PI-9005; 40-PI-9005 Redline; 40-PI-9006; 40-PI-9007

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Sumps TK-101/201 will prompt operator response to cordon off the area						
1.4. High Pressure	1.4.1. EX-102 Exchanger Isolation	1.4.1.1. Potential thermal expansion and release with potential ignition and personnel impact	452. 8" valve on feed outlet from EX-102/103 is CSO 453. PSV-10010 (set @ 305 psig) on outlet of EX-102 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
	1.4.2. EX-103 Exchanger Isolation	1.4.2.1. Potential thermal expansion and release with potential ignition and personnel impact	452. 8" valve on feed outlet from EX-102/103 is CSO 454. PSV-10020 (set @ 305 psig) on outlet of EX-103 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
	1.4.3. Exchanger plugging - Heat Exchanger EX-101	1.4.3.1. Gradual restricted flow, see low/no flow consequences in this node							
1.5. Low Pressure	1.5.1. No additional causes identified								
1.6. High Level	1.6.1. Not applicable								
1.7. Low Level	1.7.1. Not applicable								
1.8. High Interface Level	1.8.1. Not applicable								
1.9. Low Interface Level	1.9.1. Not applicable								
1.10. High Temperature	1.10.1. Low feed flow from upstream Condensate Feed Pumps PM-111A/B/C	1.10.1.1. Loss of feed to Naphtha Stabilizer Column T-101; Potential to build up temperature in T-101; Potential overpressure of T-101 with loss of containment and potential personnel impact	674. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for blocked flow 466. PDAH-10235 high differential pressure alarm on Stabilizer T-101 will prompt operator response 456. LAL-10525 low level alarm on Stabilizer T-101 will	P	4	1	2	61. No recommendations required.	

Node: 1. Node 1:
Condensate Feed from Condenensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column
(Pumps PM-111A/B, Filter FI-101A/B, Heat Exchanger EX-101/102/103/104/105)

Drawings / References:

40-PI-9004 ; 40-PI-9004 Redline; 40-PI-9005; 40-PI-9005 Redline; 40-PI-9006; 40-PI-9007

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
		1.10.1.2. Loss of feed to Combi Tower T-102; Increased vapor traffic; Potential operational upset							
		1.10.1.3. Loss of feed flow through the tube side of EX-101; Flow through the shell side coming in at ~94 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.10.1.4. Loss of feed flow through the tube side of EX-102; Flow through shell side coming in at ~200 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.10.1.5. Loss of feed flow through the tube side of EX-103; Flow through shell side coming in at ~420 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.10.1.6. Loss of feed flow through the tube side of EX-104; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		1.10.1.7. Loss of feed flow through the tube side of EX-105; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
1.11. Low Temperature	1.11.1. Low ambient temperature	1.11.1.1. No history of issues with feed lines							
	1.11.2. Exchanger fouling	1.11.2.1. Low feed temperature into Naphtha Stabilizer Column T-101, operability issue only, no hazardous consequences identified							
1.12. Contamination/ Composition	1.12.1. No additional causes identified								
1.13. Corrosion	1.13.1. No additional causes identified								
1.14. Additional Phase	1.14.1. Not applicable								
1.15. Rupture/Loss of Containment	1.15.1. Flange/gasket leaks	1.15.1.1. Release of material, potential ignition if not observed in incipient stages	3. Leak detection and repair program LDAR program 455. Preventative Maintenance (PM) for flanges/gaskets	P	3	2	2	61. No recommendations required.	
1.16. Loss of Utilities	1.16.1. Instrument air failure	1.16.1.1. Valves revert to fail-safe position							

Node: **1. Node 1:**
Condensate Feed from Condenensate Feed Pumps to Condensate Feed Filter/Coalescer through Preheaters to Naphtha Stabilizer Column
(Pumps PM-111A/B, Filter FI-101A/B, Heat Exchanger EX-101/102/103/104/105)

Drawings / References:

40-PI-9004 ; 40-PI-9004 Redline; 40-PI-9005; 40-PI-9005 Redline; 40-PI-9006; 40-PI-9007

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
1.17. Start-up/Shutdown hazards/ Maintenance	1.17.1. Failure to purge EX-101/2/3/4 tubes with nitrogen after a shutdown	1.17.1.1. Potential oxygen ingress with potential explosive atmosphere; Potential personnel impact	458. Start-up procedure includes nitrogen purge 459. Oxygen check after nitrogen purging	P	4	1	2	61. No recommendations required.	
1.18. General	1.18.1. No additional causes identified								

Node: **2. Node 2: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-101, Heat Exchanger EX 106, 107)**

Drawings / References:

40-PI-9007; 40-PI-9010; 40-PI-9010 Redline; 40-PI-9014

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
2.1. Low/No Flow	2.1.1. Level control valve LCV-10525 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	2.1.1.1. High level in Naphtha Stabilizer Column T-101; Potential carryover to Recycle Gas Compressors; Potential carryover from T-101 Condenser and downstream into Accumulator and Spheres; Potential off-spec product resulting in need to send product back and re-process; Potential roof failure and release of vapors with potential permit exceedance; Potential financial impact	762. PDI-10235 Differential Pressure Alarm will prompt operator response on high and high-high	F	4	3	4	1. Perform an engineering evaluation to provide additional safeguards for the scenario of high level in Naphtha Stabilizer Column T-101/201 resulting in potential carryover to spheres as well as loss of reboiler duty and tray damage, leading to financial impact.	
		2.1.1.2. Loss of reboiler duty; Potential tray damage in T-101 with potential financial impact and downtime	762. PDI-10235 Differential Pressure Alarm will prompt operator response on high and high-high	F	3	3	3	1. Perform an engineering evaluation to provide additional safeguards for the scenario of high level in Naphtha Stabilizer Column T-101/201 resulting in potential carryover to spheres as well as loss of reboiler duty and tray damage, leading to financial impact.	
		2.1.1.3. Loss of feed to Combi Tower T-102; Increased vapor traffic; Potential operational upset							
2.2. High Flow	2.2.1. Level control valve LCV-10525 fails open or bypass valve inadvertently opened	2.2.1.1. Loss of level of Naphtha Stabilizer Column T-101, potential blow by of vapors at 90 psig to Combi Tower T-102 rated at 50 psig; Potential overpressure between 1.5-1.8x MAWP of T-102 - potential release through gaskets; Potential ignition and personnel impact Considered a low initial likelihood due to the need to overcome a significant liquid head pressure to result in a gas blowby case	4. PAH-11271 High pressure alarm on Combi Tower T-102 will prompt operator response	P	3	2	2	61. No recommendations required.	
2.3. Reverse/Misdirected Flow	2.3.1. Tube leak/rupture in Heat exchanger EX-106	2.3.1.1. Potential leak of hot oil into the Combi Tower feed; Loss of hot oil; Operability issue only, no hazardous consequences identified							
	2.3.2. Manual valve open on hydrocarbon drain	2.3.2.1. Potential to send feed to Hydrocarbon Sump TK-101; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 2. Node 2: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-101, Heat Exchanger EX 106, 107)

Drawings / References:

40-PI-9007; 40-PI-9010; 40-PI-9010 Redline; 40-PI-9014

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
2.4. High Pressure	2.4.1. No additional causes identified								
2.5. Low Pressure	2.5.1. No additional causes identified								
2.6. High Level	2.6.1. No additional causes identified								
2.7. Low Level	2.7.1. No additional causes identified								
2.8. High Interface Level	2.8.1. Not applicable								
2.9. Low Interface Level	2.9.1. Not applicable								
2.10. High Temperature	2.10.1. TIC-10265 malfunctions	2.10.1.1. Increased hot oil flow through EX-106; Increased temperature in T-101; Potential to build up temperature in T-101; Potential increased vapor traffic; Potential overpressure of T-101; Potential to damage trays in T-101 with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 461. TAH-10310 high temperature alarm on overheads from Stabilizer T-101 will prompt operator response	F	3	2	2	61. No recommendations required.	
	2.10.2. External fire around T-101 Naphtha Stabilizer Column	2.10.2.1. The vessel is above surface that is sloped away towards the hydrocarbon drain header and the skirt is fire-proofed, therefore accumulation of flammable liquids and external fire is not considered credible							
2.11. Low Temperature	2.11.1. TIC-10265 malfunctions	2.11.1.1. Decreased hot oil flow through EX-106; Decreased temperature in T-101; Potential off-spec product; Operability issue only, no hazardous consequences identified							
2.12. Contamination/ Composition	2.12.1. No additional causes identified								
2.13. Corrosion	2.13.1. Corrosion of T-101 Stabilizer Column Overheads	2.13.1.1. Potential loss of containment from T-101 with potential ignition and personnel impact	462. T-101/201 inspections 2x/year 463. T-101/201 AUT scan every 6 months	P	3	2	2	61. No recommendations required.	

Node: 2. Node 2: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-101, Heat Exchanger EX 106, 107)

Drawings / References:

40-PI-9007; 40-PI-9010; 40-PI-9010 Redline; 40-PI-9014

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
2.14. Additional Phase	2.14.1. Not applicable								
2.15. Rupture/Loss of Containment	2.15.1. Flange/gasket leaks	2.15.1.1. Release of material, potential ignition if not observed in incipient stages	3. Leak detection and repair program LDAR program 455. Preventative Maintenance (PM) for flanges/gaskets	P	3	2	2	61. No recommendations required.	
2.16. Loss of Utilities	2.16.1. Loss of instrument air	2.16.1.1. Actuated valves will go to fail-state position							
2.17. Start-up/Shutdown hazards/ Maintenance	2.17.1. Tube leak/rupture on Heat exchanger EX-107A/B (hot oil pumps shutdown)	2.17.1.1. Potential to contaminate hot oil with product; Potential coking of Hot Oil Heater F-101 tubes over time with potential tube failure; Potential financial impact	464. Monthly hot oil sampling 5. LAH-11965 High Level Alarm on Hot Oil Surge Tank pressure vessel PV-103 will prompt operator response	F	3	2	2	61. No recommendations required.	
2.18. General	2.18.1. No additional causes identified								

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
3.1. Low/No Flow	3.1.1. Naphtha Stabilizer Overhead Pump PM-101A/B trips	3.1.1.1. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-101, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	6. FAL-10400 Low flow alarm on reflux flow to Stabilizer T-101 will prompt operator response 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 8. Standby pump, remote startup from control room 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-	F	5	1	2	61. No recommendations required.	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response						
		3.1.1.2. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-101; Increased temperature in T-101; Potential to build up temperature in T-101; Potential increased vapor traffic; Potential overpressure of T-101; Potential to damage trays in T-101 with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 461. TAH-10310 high temperature alarm on overheads from Stabilizer T-101 will prompt operator response	F	3	2	2	61. No recommendations required.	
	3.1.2. Manual valve on suction line to Naphtha Stabilizer Overhead Pump PM-101A/B inadvertently closed	3.1.2.1. Loss of suction to Naphtha Stabilizer Overhead Pump PM-101A/B; Potential to cavitate pumps with potential seal damage with potential loss of containment, potential ignition with personnel impact	465. Dual seals on pumps PM-101A/B 602. Seal failure alarm SFA-10415/10440 will shutdown pumps PM-101A/B 627. Lower Explosive Limit LEL alarms AI-10416/10441/20416/20441 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		3.1.2.2. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-101, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	6. FAL-10400 Low flow alarm on reflux flow to Stabilizer T-101 will prompt operator response 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber	F	5	1	2	61. No recommendations required.	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pressure vessel PV-109 will prompt operator response						
			10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response						
		3.1.2.3. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-101; Increased temperature in T-101; Potential to build up temperature in T-101; Potential increased vapor traffic; Potential overpressure of T-101; Potential to damage trays in T-101 with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response	F	3	2	2	61. No recommendations required.	
			461. TAH-10310 high temperature alarm on overheads from Stabilizer T-101 will prompt operator response						
	3.1.3. Manual valve on discharge line from Naphtha Stabilizer Overhead Pump PM-101A/B inadvertently closed	3.1.3.1. Blocked discharge from pumps with potential to deadhead pumps; Pump max discharge pressure is 55 psig and downstream piping is rated for up to 285 psig, therefore no overpressure concern; Potential seal damage with potential loss of containment, potential ignition with personnel impact	465. Dual seals on pumps PM-101A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			602. Seal failure alarm SFA-10415/10440 will shutdown pumps PM-101A/B						
			627. Lower Explosive Limit LEL alarms AI-10416/10441/20416/20441 will prompt operators to cordon off the area						
		3.1.3.2. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-101, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	6. FAL-10400 Low flow alarm on reflux flow to Stabilizer T-101 will prompt operator response	F	5	1	2	61. No recommendations required.	
			7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response						
			9. LAH-13540 High Level Alarm on Recycle Compressor						

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response						
		3.1.3.3. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-101; Increased temperature in T-101; Potential to build up temperature in T-101; Potential increased vapor traffic; Potential overpressure of T-101; Potential to damage trays in T-101 with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 461. TAH-10310 high temperature alarm on overheads from Stabilizer T-101 will prompt operator response	F	3	2	2	61. No recommendations required.	
	3.1.4. FCV-10400 malfunctions closed or upstream/downstream valve(s) inadvertently closed	3.1.4.1. No reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-101; Increased temperature in T-101; Potential to build up temperature in T-101; Potential increased vapor traffic; Potential overpressure of T-101; Potential to damage trays in T-101 with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling	F	3	1	1	61. No recommendations required.	
	3.1.5. LCV-10350 malfunctions closed or upstream/downstream valve(s) inadvertently closed	3.1.5.1. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-101, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	6. FAL-10400 Low flow alarm on reflux flow to Stabilizer T-101 will prompt operator response 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor	F	5	1	2	61. No recommendations required.	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Discharge Scrubber pressure vessel PV-109 will prompt operator response						
3.2. High Flow	3.2.1. LCV-10350 on Y-Grade product line fails open or bypass inadvertently opened	3.2.1.1. Loss of level in Naphtha Stabilizer Accumulator pressure vessel PV-101, potential to run Naphtha Stabilizer Overhead Pump PM-101A/B dry, pump cavitation and damage, potential release with personnel impact	11. Level alarm low LALL-10365 on Naphtha Stabilizer Accumulator pressure vessel PV-101 trips Naphtha Stabilizer Overhead Pumps PM-101A/B 465. Dual seals on pumps PM-101A/B 602. Seal failure alarm SFA-10415/10440 will shutdown pumps PM-101A/B	P	4	1	2	61. No recommendations required.	
	3.2.2. FCV-10400 fails open or bypass inadvertently opened on reflux line	3.2.2.1. Increased flow of reflux to Naphtha Stabilizer Column T-101; Increased liquid traffic; Potential to flood T-101, increasing deltaP, restricting vapor traffic; Potential to damage the T-101 trays with financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 466. PDAH-10235 high differential pressure alarm on Stabilizer T-101 will prompt operator response	F	2	2	1	61. No recommendations required.	
		3.2.2.2. Increased flow of reflux to Naphtha Stabilizer Column T-101; Increased liquid traffic; Potential increased pressure to the Combi Tower T-102 and increased gas volume to the RGC; Potential increased backpressure on the RGC resulting in RGC shutdown leading to flaring and potential environmental impact (unplanned flaring event)	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 466. PDAH-10235 high differential pressure alarm on Stabilizer T-101 will prompt operator response	F	1	2	1	3. Evaluate the engineering studies on flaring emissions for the Splitters.	
3.3. Reverse/Misdirected Flow	3.3.1. Inadvertent opening of Y-Grade rerun valve	3.3.1.1. Excessive Y-Grade to Feed Tank (Tk-200-1/2/3 & TK-100-14); Potential floating roof damage and release of light ends, ignition, personnel impact.	467. Lower explosive limit LEL alarm AT-9101,9102,9103,9104; AT-9301,9302,9393,9304 at Feed Tanks and operator response to cordon off area 468. Administrative procedures give	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			direction on how to send Y-Grade to Re-Run						
	3.3.2. Manual valve open on hydrocarbon drain	3.3.2.1. Potential to send feed to Hydrocarbon Sump TK-101; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
3.4. High Pressure	3.4.1. Naphtha Stabilizer Condenser AF-101-1/2 fin Fan motor mechanical failure/ throws a blade	3.4.1.1. Loss of individual Naphtha Stabilizer Condenser AF-101-1/2 fin Fan, slow pressurization of Naphtha Stabilizer Column T-101, potential overpressure and loss of containment, ignition and personnel impact	12. Pressure safety valve PSV-10195/10215/10225 set at 123 psig on Stabilizer T-101 (sized for electric power failure leading to loss of cooling) 469. XI-10315A/B Condenser Run Alarm will prompt operator response 470. VAH-10316A/B and VAH-10331A/B high vibration alarm will prompt operator response	P	4	1	2	61. No recommendations required.	
	3.4.2. PCV-10395B malfunctions open or bypass valve inadvertently open	3.4.2.1. Inadvertently sending hot vapors to Accumulator PV-101; Potential slow pressurization of Naphtha Stabilizer Column T-101, potential overpressure and loss of containment, ignition and personnel impact	12. Pressure safety valve PSV-10195/10215/10225 set at 123 psig on Stabilizer T-101 (sized for electric power failure leading to loss of cooling)	P	4	2	3	4. Provide an additional safeguard for the scenario of PCV-10395B/20395B malfunctioning open sending hot vapors to Accumulator PV-101/201 resulting in potential overpressure of Stabilizer Column T-101/201.	
3.5. Low Pressure	3.5.1. No additional causes identified								
3.6. High Level	3.6.1. LCV-10380 fails closed or manual valve(s) inadvertently closed	3.6.1.1. Water carryover into the reflux and Y-grade storage, loss of fractionation, operability issues							
3.7. Low Level	3.7.1. LCV-10380 malfunctions open or bypass inadvertently opened		663. Lower explosive limit alarms AI-	P	4	3	4	5. Provide additional safeguards for the scenario of the PV-101/201 Accumulator	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		3.7.1.1. Excessive hydrocarbon in the waste water system (Tank 5), Potential floating roof damage and release of light ends, ignition, personnel impact.	9110A/B/C at Tank TK-5 will prompt operators to cordon off the area					Water Boot LCV malfunctioning open leading to excessive hydrocarbons in Tank 5 and floating roof damage with release and ignition. 60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
3.8. High Interface Level	3.8.1. Not applicable								
3.9. Low Interface Level	3.9.1. Not applicable								
3.10. High Temperature	3.10.1. External fire around PV-101 Naphtha Stabilizer Accumulator	3.10.1.1. The vessel is located on a grated surface and is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
3.11. Low Temperature	3.11.1. No additional causes identified								
3.12. Contamination/ Composition	3.12.1. Elevated chlorides	3.12.1.1. Excess corrosion with potential loss of containment and potential ignition	13. Relief valves and rupture disks designed for chlorides 471. Mechanical integrity program 14. Corrosion inhibitor injection	P	4	2	3	6. Perform engineering evaluation to ensure corrosion monitoring program is adequate and properly implemented throughout the Naphtha Stabilizer overhead system to monitor for potential corrosion due to elevated chlorides.	
		3.12.1.2. Deposits causing flow restrictions, see consequences related to low/no flow in this node							
3.13. Corrosion	3.13.1. No additional causes identified								
3.14. Additional Phase	3.14.1. Not applicable								
3.15. Rupture/Loss of Containment	3.15.1. No additional causes identified								
3.16. Loss of Utilities	3.16.1. Power loss due to lightning storm results in loss of control and off-spec product	3.16.1.1. Off-spec product - unable to divert Y-Grade and heavier material from storage to rerun during lightning storm (lowers ability to safely access valves during storm/lightning alert and heavier product needed to balance Y-Grade vapor pressure). Potential financial impact	472. Lightning arrestors 473. Lightning dissipaters	F	2	3	2	2. Evaluate the need to automate the valves on product lines to the rerun line. This will allow for faster and safer switchover to and from rerun line from control room and eliminate the need for operations to manually manipulate valves during lightning storms and severe weather events.	
		3.16.1.2. Off-spec product requires opening manual valves to divert material to rerun line with potential personnel impact during lightning in the area	473. Lightning dissipaters 472. Lightning arrestors	P	4	3	4	2. Evaluate the need to automate the valves on product lines to the rerun line. This will allow for faster and safer switchover to and from rerun line from	

Node: 3. Node 3: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-101, PV-101, PM-101A/B)

Drawings / References:

40-PI-9007; 40-PI-9008; 40-PI-9009; 40-PI-9009 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								control room and eliminate the need for operations to manually manipulate valves during lightning storms and severe weather events.	
	3.16.2. Localized loss of power	3.16.2.1. See PM-101A/B pump trip scenario in this node							
		3.16.2.2. See AF-101 fan failure scenario in this node							
	3.16.3. Loss of instrument air	3.16.3.1. Valves revert to fail-safe position							
3.17. Start-up/Shutdown hazards/ Maintenance	3.17.1. Cooling during Naphtha Stabilizer Column T-101 shutdown	3.17.1.1. Potential for vacuum as Naphtha Stabilizer Column T-101 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
	3.17.2. Naphtha Stabilizer Condenser AF-101-1/2 fin fan access for maintenance or cleaning	3.17.2.1. Currently no means for personnel to tie off for fall protection, potential fall hazard, personnel impact	448. No specific safeguards identified	P	3	3	3	8. Evaluate the need to provide means for personnel to access and tie-off for fall protection during maintenance and cleaning activities for the Naphtha Stabilizer Condenser AF-101/201-1/2 fin fans.	
3.18. General	3.18.1. No additional causes identified								

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
4.1. Low/No Flow	4.1.1. Combi Tower Bottoms Product Pump PM-104A/B trips	4.1.1.1. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	15. Second Combi Tower Bottoms Product Pump PM-104A/B available for remote startup 16. LAH-11740 High Level Alarm on Combi Tower T-102 will	F	1	1	1	61. No recommendations required.	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response						
	4.1.2. Manual valve on suction to Combi Tower Bottoms Product Pump PM-104A/B inadvertently closed	4.1.2.1. Loss of suction to PM-104A/B; Potential to cavitate pump resulting in pump damage; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	4.1.3. Manual valve on discharge from Combi Tower Bottoms Product Pump PM-104A/B inadvertently closed	4.1.3.1. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		4.1.3.2. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response	F	1	1	1	61. No recommendations required.	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response						
	4.1.4. 6" manual valve from EX-105 inadvertently closed	4.1.4.1. Product flow out from the shell side of EX-105 blocked; No overpressure concerns, as the max pressure from the upstream pumps PM-104A/B is 104 psig, and the shell side is MAWP 305 psig; High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		4.1.4.2. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		4.1.5.1. Inadvertently sending all flow through EX-112A/B; Potential for higher temperature wash water to desalter; Potential wash water							

Node: **4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)**

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	4.1.5. TCV-18320B malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	steaming; Operability issue only, no hazardous consequences identified							
		4.1.5.2. Potential to back-up flow to Combi Tower; High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
	4.1.6. TCV-18320A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	4.1.6.1. No longer heating up the wash water for the Desalter using EX-112A/B; Potential to send colder temperature wash water to the Desalter; Potential cold Desalter temperature; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	4.1.7. 6" manual valve on inlet to AF-103 Bottoms Product Cooler inadvertently closed	4.1.7.1. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		4.1.7.2. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	4.1.8. 6" manual valve on outlet from AF-103 Bottoms Product Cooler inadvertently closed	4.1.8.1. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		4.1.8.2. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	4.1.9. TCV-11240A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	4.1.9.1. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 507. LAHH-13505 high-high level alarm	F	1	1	1	61. No recommendations required.	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response						
		4.1.9.2. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	4.1.10. LCV-11740 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	4.1.10.1. High level in Combi Tower T-102; Potential to overfill T-102; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response	F	1	2	1	61. No recommendations required.	
		4.1.10.2. Blocked discharge from pumps PM-104A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/2	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			1481 will prompt operators to cordon off the area						
	4.1.11. FCV-15010 fails closed or upstream/downstream manual valve(s) inadvertently closed	4.1.11.1. No consequences, this valve is normally kept closed							
4.2. High Flow	4.2.1. TCV-18320A malfunctions open or bypass inadvertently open	4.2.1.1. Potential for higher temperature wash water to desalter; Operability issue only, no hazardous consequences identified							
	4.2.2. TCV-11240B malfunctions open or bypass inadvertently open	4.2.2.1. Inadvertently bypass flow around AF-103 Bottoms Product Cooler; Potential to send higher temperature product downstream; If downstream HX-1/2 exchangers are offline, potential to send higher temperature (up to ~ 380 degF) product to the storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	448. No specific safeguards identified	F	3	2	2	61. No recommendations required.	1. Considered a low initial likelihood, as downstream HX-1/2 exchangers are rarely offline.
	4.2.3. LCV-11740 fails open or bypass inadvertently open	4.2.3.1. High flow results in low level in Combi Tower T-102; Potential loss of suction to Bottoms Pumps PM-104A/B; Potential to cavitate pump resulting in pump damage; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	474. Dual seals on PM-104A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 603. LAL-11740 low level alarm on T-102 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/2 1481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	4.2.4. FCV-15010 malfunctions open or bypass inadvertently opened	4.2.4.1. Potential for colder temperature from EX-107; Operability issue only, no hazardous consequences identified							
4.3. Reverse/Misdirected Flow	4.3.1. 6" manual valve on bypass around EX-105 shell side inadvertently opened	4.3.1.1. Inadvertently partially bypassing product flow around EX-105; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; No damage expected to Storage Tank or environmental impacts expected; Operability issue only, no hazardous consequences identified							
		4.3.1.2. Potential reduced ability to heat the feed in EX-105; Potential for low feed temperature into Naphtha Stabilizer Column T-101; Potential reduced rates; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
	4.3.2. 2" manual valve on AF-103 flush to EX-107 inadvertently opened	4.3.2.1. Potential for colder temperature from EX-107; Operability issue only, no hazardous consequences identified							
	4.3.3. 2" manual valve on diesel flush line to AF-103 inadvertently opened	4.3.3.1. Sending diesel to AF-103 inadvertently would require multiple 2" manual valves to be opened, which is not credible							

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
4.4. High Pressure	4.4.1. Bottoms Product Cooler AF-103-1/2 fin fan motor mechanical failure/throws a blade	4.4.1.1. Loss of individual AF-103-1/2 fin fan; Potential to send higher temperature product downstream; If downstream HX-1/2 exchangers are offline, potential to send higher temperature (up to ~ 380 degF) product to the storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	448. No specific safeguards identified	F	3	2	2	61. No recommendations required.	1. Considered a low initial likelihood, as downstream HX-1/2 exchangers are rarely offline.
4.5. Low Pressure	4.5.1. Too many fin fans in AF-103 running in winter	4.5.1.1. Potential to excessively cool the bottoms with potential to result in the lines waxing up and plugging; Potential restricted flow; See no/low flow consequences in this node							
4.6. High Level	4.6.1. No additional causes identified								
4.7. Low Level	4.7.1. No additional causes identified								
4.8. High Interface Level	4.8.1. Not applicable								
4.9. Low Interface Level	4.9.1. Not applicable								
4.10. High Temperature	4.10.1. External fire around T-102 Combi Tower	4.10.1.1. The vessel is above surface that is sloped away towards the hydrocarbon drain header and the skirt is fire-proofed, therefore accumulation of flammable liquids and external fire is not considered credible							
4.11. Low Temperature	4.11.1. Ambient conditions below freezing	4.11.1.1. No known issues identified							
4.12. Contamination/ Composition	4.12.1. No additional causes identified								
4.13. Corrosion	4.13.1. Salt build-up	4.13.1.1. Deposits causing flow restrictions, see consequences related to low/no flow in this node							
4.14. Additional Phase	4.14.1. Not applicable								
4.15. Rupture/Loss of Containment	4.15.1. No additional causes identified								
4.16. Loss of Utilities	4.16.1. Localized loss of power	4.16.1.1. See PM-104A/B pump trip scenario in this node							
		4.16.1.2. See AF-103 fan failure scenario in this node							
	4.16.2. Loss of instrument air	4.16.2.1. Valves revert to fail-safe position							
4.17. Start-up/Shutdown hazards/ Maintenance	4.17.1. LT-11740 maintenance	4.17.1.1. Personnel exposure to LT-11740 during maintenance activities (radiation)	476. Radiation Safety Plan	P	1	3	1	61. No recommendations required.	
	4.17.2. Cooling during Combi Tower T-102 shutdown	4.17.2.1. Potential for vacuum as Combi Tower T-103 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha	

Node: 4. Node 4: Combi Tower, Combi Tower Overheads Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Wash Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-102, PM-104A/B, AF-103, Heat Exchanger EX-105, Heat Exchanger EX-112A/B)

Drawings / References:

40-PI-9006; 40-PI-9007; 40-PI-9014; 40-PI-9025; 40-PI-9026; 40-PI-9059A; 40-PI-9005; 40-PI-9005 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
4.18. General	4.18.1. No additional causes identified								

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
5.1. Low/No Flow	5.1.1. Combi Tower Overhead Pumps PM-103A/B trip	5.1.1.1. Inability to send liquids out of PV-102 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	478. Second Combi Tower Overhead Pump PM-103A/B available for remote startup 479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response	F	1	2	1	61. No recommendations required.	
		5.1.1.2. Loss of reflux flow resulting in loss of cooling in Combi Tower T-102, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling 478. Second Combi Tower Overhead Pump PM-103A/B available for remote startup 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will	P	4	1	2	61. No recommendations required.	

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
	5.1.2. 12" manual valve on inlet to Combi Tower Overhead Pumps PM-103A/B inadvertently closed	5.1.2.1. Loss of suction to Combi Tower Overhead Pumps PM-103A/B; Potential to cavitate pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	481. Dual seals on PM-103A/B 482. Seal failure alarm SF-11360/11385 will shutdown pumps	P	4	1	2	61. No recommendations required.	
		5.1.2.2. Inability to send liquids out of PV-102 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	1	1	61. No recommendations required.	
		5.1.2.3. Loss of reflux flow resulting in loss of cooling in Combi Tower T-102, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response	P	4	1	2	61. No recommendations required.	
	5.1.3. 10" manual valve on outlet from Combi Tower Overhead Pumps PM-103A/B inadvertently closed	5.1.3.1. Blocked discharge from Combi Tower Overhead Pumps PM-103A/B; Potential to deadhead pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	481. Dual seals on PM-103A/B 482. Seal failure alarm SF-11360/11385 will shutdown pumps	P	4	1	2	61. No recommendations required.	
		5.1.3.2. Inability to send liquids out of PV-102 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	1	1	61. No recommendations required.	
		5.1.3.3. Loss of reflux flow resulting in loss of cooling in Combi Tower T-102, eventual increase in pressure, and potential to	480. PSV-11690/11710/11720	P	4	1	2	61. No recommendations required.	

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		overpressure column, loss of containment, potential ignition and personnel impact	(set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling						
			27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response						
	5.1.4. FCV-11345 reflux control valve fails closed or upstream/downstream manual valve(s) inadvertently closed	5.1.4.1. Loss of reflux flow resulting in loss of cooling in Combi Tower T-102, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling	P	4	1	2	61. No recommendations required.	
			479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response						
	5.1.5. 6" manual valve on shell side outlet from EX-101 inadvertently closed	5.1.5.1. Potential to build up liquid level in PV-102 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response	F	1	1	1	61. No recommendations required.	
			507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B						
		5.1.5.2. Loss of light naphtha product to storage; Potential financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response	F	1	2	1	61. No recommendations required.	
	5.1.6. LCV-11320 fails closed or upstream/downstream manual valve(s) inadvertently closed	5.1.6.1. Potential to build up liquid level in PV-102 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	3	1	61. No recommendations required.	
		5.1.6.2. Loss of light naphtha product to storage; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
5.2. High Flow	5.2.1. FCV-11345 reflux valve fails open or bypass valve open	5.2.1.1. Excess reflux sent to Combi Tower T-102, potential increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig	P	4	1	2	61. No recommendations required.	

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			respectively) is sized for loss of cooling 483. LAL-11320 low level alarm on PV-102 will shutdown pumps PM103A/B						
	5.2.2. Level control valve LCV-11320 light naphtha to storage fails open or bypass opened	5.2.2.1. Potential loss of level in Accumulator PV-102; Loss of suction to Combi Tower Overhead Pumps PM-103A/B; Potential to cavitate pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	481. Dual seals on PM-103A/B 482. Seal failure alarm SF-11360/11385 will shutdown pumps 483. LAL-11320 low level alarm on PV-102 will shutdown pumps PM103A/B	P	4	1	2	61. No recommendations required.	
		5.2.2.2. Potential loss of level in Accumulator PV-102; Loss of reflux to Combi Tower T-102; Loss of reflux flow resulting in loss of cooling in Combi Tower T-102, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response	P	4	1	2	61. No recommendations required.	
5.3. Reverse/Misdirected Flow	5.3.1. PCV-11315B malfunctions open or bypass inadvertently open	5.3.1.1. Inadvertently route some of the flow around the AF-102 Condensers (most of the flow would still go through the Condensers); Potential to send hot vapors to Accumulator PV-102 (would not exceed PV-102 design temperature); RCG would shutdown on high suction pressure, and the hot vapors would be unable to flow to the Hot Oil Furnace F-101; Potential to back-up pressure to the Combi Tower T-102; Potential overpressure of T-102 with loss of containment and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 457. PAHH-11715 (2oo3) high high pressure interlock on T-102 will trip heater F-101A/B, Ammonia Skid, and Hot Oil Pumps	P	4	1	2	61. No recommendations required.	
	5.3.2. 6" manual bypass around EX-101 opened	5.3.2.1. This valve is regularly open; No adverse consequences identified							
5.4. High Pressure	5.4.1. No additional causes identified								
5.5. Low Pressure	5.5.1. No additional causes identified								

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
5.6. High Level	5.6.1. Failure to drain water boot for PV-102 Accumulator	5.6.1.1. Potential water carryover into the reflux and Light Naphtha storage; Potential reduced reflux efficiency; Potential increased moisture in system; Potential corrosion; Potential loss of fractionation; Reduced efficiency; Operability issue only, no hazardous consequences identified							
5.7. Low Level	5.7.1. Improper drainage of water boot for PV-102 Accumulator	5.7.1.1. Potential to send light naphtha to Hydrocarbon Sump TK-101; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
5.8. High Interface Level	5.8.1. Not applicable								
5.9. Low Interface Level	5.9.1. Not applicable								
5.10. High Temperature	5.10.1. AF-102 Combi Tower Condenser fin fan trip/failure	5.10.1.1. A loss of a fin fan for AF-102 would not result in a significant high temperature consequence downstream; Operability issue only, no hazardous consequences identified							
	5.10.2. External fire around PV-102 Accumulator	5.10.2.1. The vessel is located on a grated surface and is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
5.11. Low Temperature	5.11.1. No additional causes identified								
5.12. Contamination/ Composition	5.12.1. No additional causes identified								
5.13. Corrosion	5.13.1. No additional causes identified								
5.14. Additional Phase	5.14.1. Not applicable								
5.15. Rupture/Loss of Containment	5.15.1. No additional causes identified								
5.16. Loss of Utilities	5.16.1. Power failure	5.16.1.1. See PM-103A/B pump trip scenario in this node							
	5.16.2. Loss of instrument air	5.16.2.1. Valves revert to fail-safe position							

Node: 5. Node 5: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-102, PV-102, PM-103 A/B, Heat Exchanger EX-101) to tanks 24/25

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9015; 40-PI-9016; 40-PI-9017; 40-PI-9018; 40-PI-9022 ; 40-PI-9022 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
5.17. Start-up/Shutdown hazards/ Maintenance	5.17.1. H2S in the LVN	5.17.1.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 109. H2S Scavenger requires good mixing to reduce the H2S present in the LVN tank 606. AI-20499/21699 (Splitter 2) Fixed H2S monitors will alert in the control room 607. AI-10499/11699 (Splitter 1) Fixed H2S monitors will alert in the control room	P	2	2	1	61. No recommendations required.	
		5.17.1.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
5.18. General	5.18.1. No additional causes identified								

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
6.1. Low/No Flow	6.1.1. Heavy Naphtha Stripper Product Pump PM-105B trips	6.1.1.1. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response	F	1	3	1	61. No recommendations required.	
		6.1.1.2. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. Confirmed that LELs AI-11461/11481/21461/21481 are located within 15 feet of T-102 and T-202.

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			operators to cordon off the area						
	6.1.2. 6" manual valve on suction to Heavy Naphtha Stripper Product Pump PM-105B inadvertently closed	6.1.2.1. Blocked suction to PM-105B; Potential pump cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.2.2. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response	F	1	2	1	61. No recommendations required.	
		6.1.2.3. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	6.1.3. 6" manual valve on discharge from Heavy Naphtha Stripper Product Pump PM-105B inadvertently closed	6.1.3.1. Blocked discharge from PM-105B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.3.2. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-	F	1	2	1	61. No recommendations required.	

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			103 will prompt operator response						
		6.1.3.3. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	6.1.4. 6" manual valve on EX-102 shell side outlet inadvertently closed	6.1.4.1. Blocked discharge from PM-105B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.4.2. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response	F	1	2	1	61. No recommendations required.	
		6.1.4.3. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operators to cordon off area						
	6.1.5. FCV-11191 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	6.1.5.1. Blocked discharge from PM-105B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	2	3	9. Provide an additional safeguard for the scenario of FCV-11191/21191 malfunctioning closed resulting in either: 1) Pump PM-105B/205B deadhead, with seal failure and loss of containment or 2) Overfill of Naphtha Stripper T-103/203 liquids back to Combi Tower T-102/202 with possible loss of containment. (Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop).	
			485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.5.2. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	2	3	9. Provide an additional safeguard for the scenario of FCV-11191/21191 malfunctioning closed resulting in either: 1) Pump PM-105B/205B deadhead, with seal failure and loss of containment or 2) Overfill of Naphtha Stripper T-103/203 liquids back to Combi Tower T-102/202 with possible loss of containment. (Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop).	
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.5.3. High level in Heavy Naphtha Stripper T-103, potential to overfill T-103 back to Combi Tower T-102, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	6.1.6. 6" manual valve on product to re-run line inadvertently closed	6.1.6.1. This valve is normally closed; No adverse consequences identified							
	6.1.7. LCV-11595 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	6.1.7.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			18. FAL-11191 Low Flow Alarm downstream of AF-105 Heavy Naphtha Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		6.1.7.2. Potential Combi Tower T-102 flooding; leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	18. FAL-11191 Low Flow Alarm downstream of AF-105 Heavy Naphtha Cooler will prompt operator response 487. LAL-11595 low level alarm on T-103 Stripper will prompt operator response	F	1	2	1	61. No recommendations required.	1. LAH/LAL-11595 is not tied to LCV-11595.
	6.1.8. T-102 Combi Tower tray pluggage	6.1.8.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 18. FAL-11191 Low Flow Alarm downstream of AF-105 Heavy Naphtha Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		6.1.8.2. Potential Combi Tower T-102 flooding; leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	F	1	3	1	61. No recommendations required.	
6.2. High Flow	6.2.1. 6" manual valve on bypass around EX-102 shell side inadvertently opened	6.2.1.1. Reduced preheat for feed flow (at most a 10 degF difference); Operability issue only, no hazardous consequences identified							

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	6.2.2. 6" manual valve on product to re-run line inadvertently open	6.2.2.1. Potential reduced flow of heavy naphtha product to storage tanks; Negligible financial loss							
	6.2.3. LCV-11595 fails open or bypass inadvertently open	6.2.3.1. Potential to draw kerosene and diesel into the heavy naphtha draw off line; Potential to dry out the tray; Potential off-spec product/quality issues; Potential financial impact	488. TAH-11617 high temperature alarm on EX-108 reboiler	F	1	3	1	61. No recommendations required.	
	6.2.4. FCV-11191 fails open or bypass valve inadvertently open	6.2.4.1. Potential to lose level in Stripper T-103; Potential pump PM-105B cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	2	3	10. Provide an additional safeguard for the scenario of FCV-11191/21191 failing open resulting in loss of level in Stripper T-103/203 and pump PM-105B/205B cavitation, with seal failure and possible loss of containment. (Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop).	1. LAH/LAL-11595/21595 on Stripper T-103/203 is tied to the FIC-11191 control loop.
6.3. Reverse/Misdirected Flow	6.3.1. No additional causes identified								
6.4. High Pressure	6.4.1. External fire around T-103 Heavy Naphtha Stripper	6.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
6.5. Low Pressure	6.5.1. No additional causes identified								
6.6. High Level	6.6.1. No additional causes identified								
6.7. Low Level	6.7.1. No additional causes identified								
6.8. High Interface Level	6.8.1. Not applicable								
6.9. Low Interface Level	6.9.1. Not applicable								
6.10. High Temperature	6.10.1. Heavy Naphtha Product Cooler AF-105 fan trips	6.10.1.1. Higher temperature heavy naphtha to storage, minimal consequences at storage tanks - not exceeding storage tank design temperature; Negligible increased venting; Operability issue only, no hazardous consequences identified							
6.11. Low Temperature	6.11.1. No additional causes identified								
6.12. Contamination/Composition	6.12.1. No additional causes identified								
6.13. Corrosion	6.13.1. No additional causes identified								

Node: 6. Node 6: Heavy Naphtha Side Draw From T-102 to Heavy Naphtha Stripper (T-103), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-102 (T-103, Heat Exchanger EX-108, PM-105A/B, Heat Exchanger EX-102, AF-105)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9019; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9023

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
6.14. Additional Phase	6.14.1. Not applicable								
6.15. Rupture/Loss of Containment	6.15.1. No additional causes identified								
6.16. Loss of Utilities	6.16.1. Loss of power	6.16.1.1. See AF-105 fan failure scenario, this node							
		6.16.1.2. See See PM-105B pump trip scenario, this node							
	6.16.2. Loss of instrument air	6.16.2.1. Valves revert to fail-safe position							
6.17. Start-up/Shutdown hazards/ Maintenance	6.17.1. Cooling during Heavy Naphtha Stripper T-103 shutdown	6.17.1.1. Potential for vacuum as Heavy Naphtha Stripper T-103 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
6.18. General	6.18.1. No additional causes identified								

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
7.1. Low/No Flow	7.1.1. Level control valve LCV-11660 fails closed	7.1.1.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 489. LAL-11660 low level alarm on T-104 Kero Stripper will prompt operator response 490. FAL-11221 low flow alarm downstream	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. LAH/LAL-11660 on Kero Stripper is not tied to the LCV-11660 control loop.

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			of AF-104 will shutdown pumps PM-106A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		7.1.1.2. Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	489. LAL-11660 low level alarm on T-104 Kero Stripper will prompt operator response	F	1	3	1	61. No recommendations required.	
	7.1.2. T-102 Combi Tower Tray Pluggage	7.1.2.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 489. LAL-11660 low level alarm on T-104 Kero Stripper will prompt operator response 490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		7.1.2.2. Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B	F	1	3	1	61. No recommendations required.	
	7.1.3. Kero Stripper Product Pump PM-106A/B trips or loss of power	7.1.3.1. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 493. LAH-11660 high level alarm on Kero	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)**

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Stripper T-104 will prompt operator response 21. Standby pump available - Kero Stripper Product Pump PM-106A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		7.1.3.2. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 21. Standby pump available - Kero Stripper Product Pump PM-106A/B	F	1	2	1	61. No recommendations required.	
	7.1.4. Manual valve closed inadvertently in Kero Stripper Product Pump PM-106A/B suction or basket strainers on suction lines plugged	7.1.4.1. Kero Stripper Product Pump PM-106A/B cavitation, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		7.1.4.2. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		7.1.4.3. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
	7.1.5. Manual valve closed inadvertently in Kero Stripper Product Pump PM-106A/B discharge	7.1.5.1. Kero Stripper Product Pump PM-106A/B deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		7.1.5.2. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
		7.1.5.3. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	7.1.6. 8" manual valve on EX-103 shell side outlet inadvertently closed	7.1.6.1. Blocked discharge from Kero Stripper Product Pump PM-106A/B; Potential pump deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		7.1.6.2. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
		7.1.6.3. High level in Kero Stripper T-104; Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	7.1.7. 8" manual valve upstream/downstream of AF-104 Kero Product Cooler inadvertently closed	7.1.7.1. Inadvertently route all flow through either AF-104-1 or AF-104-2; Potential reduced cooling ability, sending higher temperature product to storage tanks; Not enough to exceed design temp of storage tanks (200 degF); Potential reduced flow and potential increased level in Kero Stripper T-104; Potential high level in	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Stripper T-104; Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		7.1.7.2. Inadvertently route all flow through either AF-104-1 or AF-104-2; Potential reduced cooling ability, sending higher temperature product to storage tanks; Not enough to exceed design temp of storage tanks (200 degF); Potential reduced flow and potential increased level in Kero Stripper T-104; Potential high level in Stripper T-104; Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
	7.1.8. FCV-11221 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed or 8" manual valve on kero product to rerun inadvertently closed	7.1.8.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area 16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		7.1.8.2. Blocked discharge from Kero Stripper Product Pump PM-106A/B; Potential pump deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/2	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			1461 will alarm and prompt operators to cordon off area						
		7.1.8.3. Potential Combi Tower T-102 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
7.2. High Flow	7.2.1. LCV-11660 fails open or bypass valve inadvertently open	7.2.1.1. Potential to draw diesel into the kero draw off line; Potential to dry out the tray; Potential off-spec product/quality issues; Potential financial impact	494. TAH-11656 high temperature alarm from EX-109 Reboiler will prompt operator response	F	1	3	1	61. No recommendations required.	
		7.2.1.2. Potential to flood the Kero Stripper T-104; Potential to send out liquids in the vapor return line back to the Combi Tower T-102; Potential damage to vapor return piping; Potential release with ignition and personnel impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. LAH/LAL-11660 on Kero Stripper is not tied to the LCV-11660 control loop.
			494. TAH-11656 high temperature alarm from EX-109 Reboiler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	7.2.2. 8" manual valve on bypass around EX-103 inadvertently opened	7.2.2.1. Potential reduced ability to heat the feed in EX-103; Potential for low feed temperature into Naphtha Stabilizer Column T-101; Potential to swing T-101; Potential offspec product/quality; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		7.2.2.2. Inadvertently partially bypassing product flow around EX-103; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; No damage expected to Storage Tank or environmental impacts expected; Operability issue only, no hazardous consequences identified							
	7.2.3. FCV-11221 fails open or bypass valve inadvertently open	7.2.3.1. Loss of level in Kero Stripper T-104, potential to vapor lock the stripper, potential to cavitate Kero Stripper Product Pump PM-106A/B; Potential seal failure with loss of containment, ignition, and personnel impact	491. Dual seals on PM-106A/B and PM-206A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B						
			484. Lower explosive limit LEL-						

Node: 7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			11481/11461/21481/21461 will alarm and prompt operators to cordon off area						
	7.2.4. Both Kero Product Pumps PM-106A/B running at once	7.2.4.1. Potential loss of level in Kero Stripper T-104, potential to vapor lock the stripper, potential to cavitate Kero Stripper Product Pump PM-106A/B; Potential seal failure with loss of containment, ignition, and personnel impact	489. LAL-11660 low level alarm on T-104 Kero Stripper will prompt operator response 491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
7.3. Reverse/Misdirected Flow	7.3.1. Tube leak/rupture on kero Stripper Reboiler EX-109	7.3.1.1. Potential to contaminate kero product with hot oil; Potential loss of hot oil resulting in potential need to shutdown and financial impact	20. LAL-11965 Low level alarm on Hot Oil Surge Tank pressure vessel PV 103 will prompt operator response 495. LALL-11975 low-low level alarm on Hot Oil Surge Tank PV-103 will prompt operator response	F	3	2	2	61. No recommendations required.	
	7.3.2. Inadvertent opening of spec valve to rerun	7.3.2.1. Inefficient operation - sending product to rerun; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
7.4. High Pressure	7.4.1. External fire around T-104 Kero Stripper	7.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
7.5. Low Pressure	7.5.1. No additional causes identified								
7.6. High Level	7.6.1. No additional causes identified								
7.7. Low Level	7.7.1. No additional causes identified								

Node: **7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)**

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
7.8. High Interface Level	7.8.1. Not applicable								
7.9. Low Interface Level	7.9.1. Not applicable								
7.10. High Temperature	7.10.1. Kero Product Cooler AF-104 fan trips	7.10.1.1. Potential to send higher temperature Kerosene product to downstream storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	496. XI-11215A/B and XI-11235A/B Kero Product Cooler AF-104 fan run indication will alarm and prompt operator response 497. TAH-11232A/B high temperature alarm will prompt operator response	F	3	2	2	61. No recommendations required.	
7.11. Low Temperature	7.11.1. No additional causes identified								
7.12. Contamination/ Composition	7.12.1. No additional causes identified								
7.13. Corrosion	7.13.1. No additional causes identified								
7.14. Additional Phase	7.14.1. Not applicable								
7.15. Rupture/Loss of Containment	7.15.1. No additional causes identified								
7.16. Loss of Utilities	7.16.1. Loss of power	7.16.1.1. See PM-106A/B pump trip scenario, this node							
		7.16.1.2. See AF-104 fan failure scenario, this node							
	7.16.2. Loss of instrument air	7.16.2.1. Valves revert to fail-safe position							
7.17. Start-up/Shutdown hazards/ Maintenance	7.17.1. Cooling during Kero Stripper T-104 shutdown	7.17.1.1. Potential for vacuum as Kero Stripper T-104 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid	

Node: **7. Node 7: Kero Side draw from T-102 to Kero Stripper (T-104), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-103 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-102 (T-104, Heat Exchanger EX-109, PM-106A/B, Heat Exchanger EX-103, AF-104)**

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9014; 40-PI-9020; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9024

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								pulling vacuum and update procedures if necessary.	
7.18. General	7.18.1. No additional causes identified								

Node: **8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)**

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
8.1. Low/No Flow	8.1.1. Level control valve LCV-15090 fails closed or T-102 tower tray plugs	8.1.1.1. Potential to drop diesel into the bottom of the Combi Tower T-102, potential to send out diesel with the AGO; Potential loss of level in Diesel Stripper T-105; Potential to cavitate pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 500. LAL-15090 low level alarm on T-105 Diesel Stripper will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	1. LAH/LAL-15090 on T-105/205 Diesel Stripper is not tied to LCV-15090/25090
		8.1.1.2. Potential to drop diesel into the bottom of the Combi Tower T-102, potential to send out diesel with the AGO; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response 500. LAL-15090 low level alarm on T-105 Diesel Stripper will prompt operator response 16. LAH-11740 High Level Alarm on Combi Tower T-102 will	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. Confirmed that LELs AI-11461/11481/21461/21481 are located within 15 feet of T-102 and T-202.

Node: **8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)**

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	8.1.2. Diesel Stripper Product Pumps PM-107A/B trips	8.1.2.1. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response 16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	8.1.3. Manual valve closed on suction to Diesel Stripper Product Pumps PM-107A/B or basket strainers plugged on suction piping	8.1.3.1. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response 16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			11461/11481/21461/21481 will prompt operators to cordon off the area						
		8.1.3.2. Potential to cavitate pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response	P	4	1	2	61. No recommendations required.	
	8.1.4. Manual valve closed on discharge from Diesel Stripper Product Pumps PM-107A/B	8.1.4.1. Potential to deadhead pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	
		8.1.4.2. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response 16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			operators to cordon off the area						
	8.1.5. 6" manual valve on EX-104 shell side outlet inadvertently closed	8.1.5.1. Potential to deadhead pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B	P	4	1	2	61. No recommendations required.	
			499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B						
			502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response						
			501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response						
		8.1.5.2. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response						
			501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	8.1.6. 6" manual valve(s) upstream/downstream AF-106 Diesel Product Cooler inadvertently closed	8.1.6.1. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			16. LAH-11740 High Level Alarm on Combi						

Node: **8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)**

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Tower T-102 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		8.1.6.2. Potential to deadhead pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 502. LAH-15090 high level alarm on Diesel Stripper T-105 will prompt operator response 501. FAL-16065 low flow alarm downstream of AF-106 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	
	8.1.7. FCV-16065 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	8.1.7.1. Potential to deadhead pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		8.1.7.2. High level in Diesel Stripper T-105; Inability to draw off diesel from Combi Tower T-102; Potential to ultimately flood level in T-102; Potential to send liquids in the vapor return lines; Potential	16. LAH-11740 High Level Alarm on Combi Tower T-102 will	P	4	2	3	11. Provide an additional safeguard for the scenario of FCV-16065/26065 malfunctioning closed resulting in overfill	

Node: 8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		damage to vapor return piping; Potential release with ignition and personnel impact	prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area					of Diesel Stripper T-105/205 and flooding of Combi Tower T-102/202 with potential loss of containment. (Note - LIC-15090/25090 on Stripper T-105/205 is tied to the FCV-16065/26065 control loop).	
8.2. High Flow	8.2.1. 6" manual bypass around EX-104 inadvertently open	8.2.1.1. Potential reduced ability to heat the feed in EX-104; Potential for low feed temperature into Naphtha Stabilizer Column T-101; Potential to swing T-101; Potential offspec product/quality; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		8.2.1.2. Inadvertently partially bypassing product flow around EX-104; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	503. TAH-16055 high temperature alarm downstream AF-106 Diesel Product Cooler will prompt operator response	F	2	2	1	61. No recommendations required.	
	8.2.2. LCV-15090 fails open or bypass inadvertently open	8.2.2.1. Potential to dry out Combi Tower T-102 tray, leading to potential tray damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	8.2.3. FCV-16065 fails open or bypass inadvertently open	8.2.3.1. Potential loss of level in Diesel Stripper T-105; Potential to cavitate pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	8.2.4. Both PM-107A/B Diesel Stripper Product Pumps running at once	8.2.4.1. Potential loss of level in Diesel Stripper T-105; Potential to cavitate pumps PM-107A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 500. LAL-15090 low level alarm on T-105 Diesel Stripper will prompt operator response	P	4	1	2	61. No recommendations required.	
8.3. Reverse/Misdirected Flow	8.3.1. Tube leak/rupture on Diesel Stripper Reboiler EX-110	8.3.1.1. Potential to contaminate diesel product with hot oil; Potential loss of hot oil resulting in potential need to shutdown and financial impact	20. LAL-11965 Low level alarm on Hot Oil Surge Tank pressure	F	3	2	2	61. No recommendations required.	

Node: 8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			vessel PV 103 will prompt operator response 495. LALL-11975 low-low level alarm on Hot Oil Surge Tank PV-103 will prompt operator response						
	8.3.2. Inadvertent opening of spec valve to rerun	8.3.2.1. Inefficient operation - sending product to rerun; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
8.4. High Pressure	8.4.1. External fire around T-105 Diesel Stripper	8.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
8.5. Low Pressure	8.5.1. No additional causes identified								
8.6. High Level	8.6.1. No additional causes identified								
8.7. Low Level	8.7.1. No additional causes identified								
8.8. High Interface Level	8.8.1. Not applicable								
8.9. Low Interface Level	8.9.1. Not applicable								
8.10. High Temperature	8.10.1. Diesel Product Cooler AF-106 fan trips	8.10.1.1. Potential to send higher temperature Kerosene product to downstream storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	504. XI-16040/16050A Diesel Product Cooler AF-106 fan run indication will alarm and prompt operator response 503. TAH-16055 high temperature alarm downstream AF-106 Diesel Product Cooler will prompt operator response	F	3	2	2	61. No recommendations required.	
8.11. Low Temperature	8.11.1. No additional causes identified								
8.12. Contamination/ Composition	8.12.1. No additional causes identified								
8.13. Corrosion	8.13.1. No additional causes identified								
8.14. Additional Phase	8.14.1. Not applicable								

Node: 8. Node 8: Diesel Stripper (T-105, Heat Exchanger EX-110, AF-106, PM-107A/B)

Drawings / References:

40-PI-9006; 40-PI-9014; 40-PI-9021; 40-PI-9022 ; 40-PI-9022 Redline; 40-PI-9025; 40-PI-9026; 40-PI-9049; 40-PI-9050

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
8.15. Rupture/Loss of Containment	8.15.1. No additional causes identified								
8.16. Loss of Utilities	8.16.1. Loss of power	8.16.1.1. See AF-106 fan failure scenario in this node							
		8.16.1.2. See PM-107A/B pump trip scenario in this node							
8.17. Start-up/Shutdown hazards/ Maintenance	8.17.1. Cooling during Diesel Stripper T-105 shutdown	8.17.1.1. Potential for vacuum as Diesel Stripper T-105 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
8.18. General	8.18.1. No additional causes identified								

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
9.1. Low/No Flow	9.1.1. PCV-11315A fails closed or upstream/downstream manual valve(s) inadvertently closed	9.1.1.1. Loss of suction to Recycle Gas Compressor; Potential compressor rod load damage with potential financial impact	505. PIC-13565/13611 low suction pressure shutdown will shutdown CM-101A/CM-101B compressors	F	1	3	1	61. No recommendations required.	
		9.1.1.2. Loss of vapor flow out of Combi Tower Accumulator pressure vessel PV-102, increased pressure in Combi Tower T-102, overpressure Combi Tower, loss of containment, potential ignition and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 4. PAH-11271 High pressure alarm on Combi Tower T-102	P	4	1	2	61. No recommendations required.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			will prompt operator response 506. PAHH-11715 (2oo3) high high pressure alarm on Combi Tower T-102 will prompt operator response						
	9.1.2. PCV-10395A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	9.1.2.1. Potential to overpressure PV-101 Naphtha Stabilizer Accumulator with loss of containment, potential ignition and personnel impact	509. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for blocked vapor outlet	P	4	1	2	61. No recommendations required.	
	9.1.3. Suction Scrubber pressure vessel PV-107 demister pad saturated	9.1.3.1. Slow increase in pressure on Combi Tower T-102; Potential carryover of water to RGC Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	2	1	61. No recommendations required.	
	9.1.4. 10" manual valve on suction to Recycle Compressor CM-101A/B inadvertently closed	9.1.4.1. Potential increase in pressure in Combi Tower T-102, and potential to overpressure column, loss of containment, potential ignition and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response 4. PAH-11271 High pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
	9.1.5. XV-13576/13627 on discharge from Recycle Compressor CM-101A/B fails closed		520. PSV-13574/13622 (set @	P	4	1	2	61. No recommendations required.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		9.1.5.1. Blocked CM-101A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	125 psig) is sized for blocked flow 521. PAHH-13584/13626 high-high pressure alarm on CM-101A/B discharge will shutdown compressor						
	9.1.6. Recycle Compressor CM-101A or B trips	9.1.6.1. Potential increase in pressure in Combi Tower T-102, and potential to overpressure column, loss of containment, potential ignition and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response 4. PAH-11271 High pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
	9.1.7. Pressure control valve PCV-13685 fails closed	9.1.7.1. Blocked CM-101A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	520. PSV-13574/13622 (set @ 125 psig) is sized for blocked flow 521. PAHH-13584/13626 high-high pressure alarm on CM-101A/B discharge will shutdown compressor	P	4	1	2	61. No recommendations required.	
	9.1.8. FCV-13462 fails closed or upstream/downstream manual valve(s) inadvertently closed	9.1.8.1. Blocked CM-101A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	520. PSV-13574/13622 (set @ 125 psig) is sized for blocked flow 521. PAHH-13584/13626 high-high pressure alarm on CM-101A/B discharge will shutdown compressor	P	4	1	2	61. No recommendations required.	
9.2. High Flow	9.2.1. PCV-11315A malfunctions open or bypass inadvertently open	9.2.1.1. Reduced pressure in Combi Tower T-102, increased level in Combi Tower Accumulator pressure vessel PV-102; Potential to overfill PV-102 and carryover to Recycle Gas Compressor Suction	479. LAH-11320 high level alarm on PV-102 Accumulator will	F	1	2	1	61. No recommendations required.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Scrubber PV-107; Potential to overfill PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B						
	9.2.2. PCV-10395A fails open or bypass valve open	9.2.2.1. T-101 Stabilizer pressure would drop; Potential liquid carryover to the PV-101 Accumulator; Potential overfill PV-101 and carryover to Recycle Compressor Discharge Scrubber PV-109; Potential to overfill PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response	F	5	1	2	61. No recommendations required.	
	9.2.3. 10" manual valve on suction to offline CM-101A (or CM-101B) left open while the other compressor is running	9.2.3.1. Reduced flow of suction to online CM-101A/B; Operability issue only, no hazardous consequences identified							
	9.2.4. Pressure control valve PCV-13685 fails open (compressor discharge)	9.2.4.1. Reduced residence time in PV-108 Lube Oil Separator Vessel; Potential inability to separate out off-gas; Potential to damage the CM-101A/B compressor casing, slide valve, and bearings; Potential to send contaminated lube oil to Heater; Potential asset damage; Potential environmental impact	522. LSLL-13668 low-low level switch on PV-108 Lube Oil Separator will shutdown CM-101A/B compressor	F	2	3	2	61. No recommendations required.	
	9.2.5. FCV-13462 malfunctions open or bypass valve inadvertently open	9.2.5.1. Potential low backpressure on the Discharge Scrubber PV-109; Potential to carryover liquids to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	518. LAL-13540 low level alarm on PV-109 will prompt operator response 524. PAL-13663 low pressure alarm on PV-109 outlet will prompt operator response	F	5	2	4	14. Provide an additional safeguard for the scenario of FCV-13462/23462 malfunctioning open on the outlet of PV-109/PV-209 Discharge Scrubber causing low backpressure on PV-109/PV-209 resulting in a potential liquid carryover to the Hot Oil Furnace F-101/F-201 fuel inlet causing potential explosion with personnel impace and significant downtime.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	9.2.6. Pressure control valve PCV-13460 fails open to flare	9.2.6.1. Unintended flaring event; Potential environmental impact	448. No specific safeguards identified	E	1	4	2	61. No recommendations required.	
9.3. Reverse/Misdirected Flow	9.3.1. BDV-13575/13619 malfunctions open when not needed	9.3.1.1. Inadvertently sending compressor suction gas to vent header; Potential unplanned flaring event with potential environmental impact	448. No specific safeguards identified	E	1	4	2	61. No recommendations required.	
9.4. High Pressure	9.4.1. No additional causes identified								
9.5. Low Pressure	9.5.1. No additional causes identified								
9.6. High Level	9.6.1. Suction scrubber pump PM-113A/B failure	9.6.1.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-107; Potential to overfill PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	2	1	61. No recommendations required.	
	9.6.2. Manual valve(s) on suction to Suction scrubber pump PM-113A/B inadvertently closed	9.6.2.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-107; Potential to overfill PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	2	1	61. No recommendations required.	
		9.6.2.2. Loss of suction to pumps PM-113A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	510. LAL-13507 low level alarm on Suction Scrubber P-107 will prompt operator response 511. LALL-13507 low-low alarm on Suction Scrubber P-107 will shutdown pumps PM-113A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	9.6.3. Manual valve(s) on discharge from Suction scrubber pump PM-113A/B inadvertently closed	9.6.3.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-107; Potential to overfill PV-107 and send liquids to the RGC; Potential to damage compressor leading to financial impact	479. LAH-11320 high level alarm on PV-102 Accumulator will prompt operator response 507. LAHH-13505 high-high level alarm on Suction Scrubber PV-107 will shutdown compressors CM-101A/B	F	1	2	1	61. No recommendations required.	
		9.6.3.2. Loss of discharge from pumps PM-113A/B with potential deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	510. LAL-13507 low level alarm on Suction Scrubber P-107 will prompt operator response 511. LALL-13507 low-low alarm on Suction Scrubber P-107 will shutdown pumps PM-113A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	9.6.4. Discharge scrubber pump PM-117A/B failure	9.6.4.1. Potential high level in PV-109 Discharge Scrubber; Potential to overfill PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 517. Standby pump PM-117A/B available for remote start-up	F	5	1	2	61. No recommendations required.	
	9.6.5. Manual valve on suction to Discharge Scrubber Pump PM-117A/B inadvertently closed	9.6.5.1. Potential to overfill PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber	F	5	1	2	61. No recommendations required.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response						
		9.6.5.2. Loss of suction to pumps PM-117A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	9.6.6. Manual valve on discharge from Discharge Scrubber Pump PM-117A/B inadvertently closed	9.6.6.1. Blocked discharge from pumps PM-117A/B with potential deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area 9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		9.6.6.2. Potential to overfill PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response	F	5	1	2	61. No recommendations required.	
9.7. Low Level	9.7.1. Suction scrubber pump PM-113A/B fails to stop	9.7.1.1. Potential to pump out all level in Suction Scrubber PV-107; Potential loss of suction to pumps PM-113A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	510. LAL-13507 low level alarm on Suction Scrubber P-107 will prompt operator response 511. LALL-13507 low-low alarm on Suction Scrubber P-107 will shutdown pumps PM-113A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	9.7.2. Discharge scrubber pump PM-117A/B fails to stop	9.7.2.1. Potential to pump out all level in Discharge Scrubber PV-109; Potential loss of suction to pumps PM-117A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	518. LAL-13540 low level alarm on PV-109 will prompt operator response 519. LALL-13540 low-low level alarm on PV-109 will shut off PM-117A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
9.8. High Interface Level	9.8.1. Not applicable								
9.9. Low Interface Level	9.9.1. Not applicable								

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
9.10. High Temperature	9.10.1. Electric Gas Preheater HTR-101 stuck in "on" position	9.10.1.1. High suction temperature, increased load on Recycle Compressor CM-102A/B, high discharge temperature from compressor, potential overheating of compressor lube oil and compressor damage (bearings, etc.)	22. TAH-13510 High Temperature Alarm on Heater H-101 will prompt operator response 23. TAHH-13510 High-High Temperature Alarm on Electric Gas Preheater HTR-101 will prompt operator response 514. TAH-13625/13583 high temperature alarm on CM-101A/B Compressor discharge will prompt operator response 515. TAHH-13625/13583 high temperature alarm on CM-101A/B Compressor discharge will prompt operator response	F	1	1	1	61. No recommendations required.	
	9.10.2. TCV-13515 receives incorrect signal (opens inadvertently)	9.10.2.1. Inadvertently sending compressor discharge flow around Gas-to-Gas Exchanger EX-111; Inability to cool discharge gas with EX-111; Potential to send higher temperature discharge gas to downstream AF-108 (design temp 350 degF); Not expected to exceed design temp of AF-108; Operability issue only, no hazardous consequences identified							
		9.10.2.2. Inadvertently sending compressor discharge flow around Gas-to-Gas Exchanger EX-111; Inability to heat suction gas with EX-111; Potential to send lower temperature gas to compressor; Potential liquids in compressor resulting in potential compressor damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	9.10.3. AF-108 Gas Aftercooler Fan Failure	9.10.3.1. Reduced ability to cool CM-101A/B compressor discharge gas; Potential higher BTUs; Potential increased offgas to Heater; Potential increased emissions with potential environmental impact	516. CM-101A/B run permissive requires at least one AF-108 fan running to remain online	F	1	3	1	61. No recommendations required.	
9.11. Low Temperature	9.11.1. Electric Gas Preheater HTR-101 trips when needed (winter)	9.11.1.1. Lower Recycle Compressor CM-102A/B suction temperature, potential for liquids to reach compressor, compressor damage (no loss of containment)	24. Gas to Gas Heat Exchanger EX-111 downstream of Electric Gas Preheater HTR-101	F	1	2	1	61. No recommendations required.	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			513. TAL-13564/13610 low temperature alarm on suction to Recycle Compressor CM-101A/B will prompt operator response						
	9.11.2. TCV-13515 receives incorrect signal (closes inadvertently)	9.11.2.1. Inadvertently sending compressor discharge flow through Gas-to-Gas Exchanger EX-111, further cooling the discharge gas when not needed; Potential liquid carryover PV-109 Discharge Scrubber; Potential to overfill PV-109 and send liquid to Hot Oil Furnace F-101 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	9. LAH-13540 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response 10. LAH-13538 High Level Alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-109 will prompt operator response	F	5	2	4	12. Provide an additional safeguard for the scenario of inadvertently sending CM-101A/B / CM-201A/B compressor discharge gas through the EX-111/211 Gas-to-Gas Exchanger, further cooling the gas and leading to potential PV-109/209 Discharge Scrubber liquid carryover to the Hot Oil Furnace F-101/201 fuel inlet causing potential explosion with personnel impace and significant downtime.	
9.12. Contamination/ Composition	9.12.1. Product infiltration into compressor oil	9.12.1.1. Contamination of Recycle Compressor CM-102A/B oil; Potential to damage the CM-101A/B compressor casing, slide valve, and bearings; Potential to send contaminated lube oil to Heater; Potential asset damage; Potential environmental impact	25. Coalescing Filters in PV-108/208	F	2	3	2	61. No recommendations required.	
9.13. Corrosion	9.13.1. No specific causes identified								
9.14. Additional Phase	9.14.1. Not applicable								
9.15. Rupture/Loss of Containment	9.15.1. Tube leak/rupture on EX-111 Gas-to-Gas Exchanger	9.15.1.1. Potential to overpressure and blow compressor seals; Potential to contaminate compressor lube oil; Potential compressor damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	9.15.2. Recycle Compressor CM-101A/B mechanical seal leaks	9.15.2.1. Loss of barrier fluid; Potential loss of containment of barrier fluid; Potential release of lube oil with entrained gas; Potential environmental/financial impact	522. LSLL-13668 low-low level switch on PV-108 Lube Oil Separator will shutdown CM-101A/B compressor	F	1	3	1	61. No recommendations required.	
9.16. Loss of Utilities	9.16.1. Loss of power	9.16.1.1. Compressor CM-101A/B shuts down upon loss of power; Potential flaring event with potential environmental impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	9.16.2. Loss of instrument air	9.16.2.1. Valves revert to fail-safe position							
9.17. Start-up/Shutdown hazards/ Maintenance	9.17.1. Recycle Compressor CM-101A/B accessibility issues (congested area)	9.17.1.1. Potential personnel impacts while performing maintenance activities	523. Platforms built with access to CM-101A/B valves	P	1	4	2	61. No recommendations required.	
9.18. General	9.18.1. Compressor CM-101A/B designed for clean-gas service	9.18.1.1. Potential compressor wear and tear due to dirty gas service, requiring excessive maintenance activities; Potential for	448. No specific safeguards identified	F	2	5	4	13. Perform an engineering study to determine if a different type of Recycle Gas Compressor should replace the	

Node: 9. Node 9: From PV-102 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-107, AF-108, PV-109, PM-117, PM-113)

Drawings / References:

40-PI-9005; 40-PI-9005 Redline; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9013; 40-PI-9018; 40-PI-9030 Redline; 40-PI-9030 ; E000318-01-101 SHT 1 of 5; E000318-01-101 SHT 2 of 5; E000318-01-101 SHT 3 of 5; E000318-01-101 SHT 4 of 5; E000318-01-101 SHT 5 of 5; E000318-02-102 SHT 2; E000318-02-102-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		environmental impacts and financial impacts due to increased flaring events						existing CM-101A/B / CM-201A/B compressors, as existing compressors are designed for clean-gas service only. Financial risk only.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
10.1. Low/No Flow	10.1.1. One hot oil Hot Oil Circulation Pump PM-102A/B/C/D trips	10.1.1.1. Reduced flow of hot oil to the heat exchangers; Decreased heat transfer, operability issue only, no hazardous consequences identified							
	10.1.2. Manual valve closed in Hot Oil Circulation Pump PM-102A/B/C/D suction	10.1.2.1. Loss of suction to Hot Oil Circulation Pump PM-102A/B/C/D; Potential pump cavitation with seal failure and loss of containment, potential ignition and personnel impact	26. VAH-10485/10505/10525/10545 high vibration alarm on Hot Oil Circulation Pump PM-102A/B/C/D will prompt operator response 525. Dual seals on PM-102A/B/C/D	P	4	1	2	61. No recommendations required.	
	10.1.3. Manual valve closed in Hot Oil Circulation Pump PM-102A/B/C/D discharge	10.1.3.1. Blocked discharge from Hot Oil Circulation Pump PM-102A/B/C/D; Potential pump deadhead with seal failure and loss of containment, potential ignition and personnel impact	26. VAH-10485/10505/10525/10545 high vibration alarm on Hot Oil Circulation Pump PM-102A/B/C/D will prompt operator response 525. Dual seals on PM-102A/B/C/D	P	4	1	2	61. No recommendations required.	
	10.1.4. Multiple Hot Oil Circulation Pumps PM-102A/B/C/D trip	10.1.4.1. Reduced hot oil flow, potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	28. All Hot Oil Circulation Pumps PM-102A/B/C/D down - trips Hot Oil Heater F-101 526. PALL-10625 low-low pressure alarm on Hot Oil Heater F-101 discharge will trip F-101 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will	P	4	1	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D						
10.2. High Flow	10.2.1. All Hot Oil Circulation Pumps PM-102A/B/C/D online at once	10.2.1.1. No adverse consequences identified							
10.3. Reverse/Misdirected Flow	10.3.1. No specific causes identified								
10.4. High Pressure	10.4.1. Pressure control valve PCV-11940 set too high	10.4.1.1. Potential to pop the rupture disk, resulting in PSV-11945 release and unintended flaring event, potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.4.2. Hot oil blocked in at Hot Oil Heater F-101	10.4.2.1. Potential to overpressure hot oil piping, loss of containment, release, personnel impact	30. Pressure safety valve PSV-10630 set @ 300 psig, sized for hydraulic expansion 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D	P	4	1	2	61. No recommendations required.	
10.5. Low Pressure	10.5.1. Pressure control valve PCV-11940 set too low	10.5.1.1. Insufficient suction head for Hot Oil Circulation Pump PM-102A/B/C/D, potential for insufficient pump suction head, pump cavitation and potential pump damage	26. VAH-10485/10505/10525/10545 high vibration alarm on Hot Oil Circulation Pump PM-102A/B/C/D will prompt operator response 525. Dual seals on PM-102A/B/C/D	P	4	1	2	61. No recommendations required.	
10.6. High Level	10.6.1. Thermal expansion of heat transfer fluid on startup	10.6.1.1. Potential to overfill Hot Oil Surge Tank pressure vessel PV-103, requires manual removal of excess heat transfer fluid after cooling it down and removing via vac truck; Potential downtime required to cool down the heat transfer fluid prior to removal (~1 to 5 days of downtime)	31. Startup procedures to account for thermal expansion	F	2	3	2	61. No recommendations required.	
10.7. Low Level	10.7.1. Thermal expansion of heat transfer fluid on startup requires low level in Hot Oil Surge Tank pressure vessel PV-103 at startup	10.7.1.1. Hot Oil Circulation Pump PM-102A/B/C/D cavitation at startup/shutdown due to insufficient suction head, potential seal failure and loss of containment, potential ignition and personnel impact	31. Startup procedures to account for thermal expansion 26. VAH-10485/10505/10525/10545 high vibration alarm on Hot Oil Circulation Pump PM-	P	4	1	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			102A/B/C/D will prompt operator response						
			525. Dual seals on PM-102A/B/C/D						
10.8. High Interface Level	10.8.1. Not applicable								
10.9. Low Interface Level	10.9.1. Not applicable								
10.10. High Temperature	10.10.1. Reduced heat transfer fluid flow	10.10.1.1. Reduced hot oil flow, potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	526. PALL-10625 low-low pressure alarm on Hot Oil Heater F-101 discharge will trip F-101	P	4	1	2	61. No recommendations required.	
			27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response						
			29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D						
	10.10.2. FCV-11681 on hot oil from Kero Stripper Reboiler EX-109 fails open or bypass inadvertently open	10.10.2.1. Potential reduced hot oil flow to other heat exchangers; Potential off-spec product with financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.10.3. FCV-10521 fails open or bypass inadvertently open	10.10.3.1. Increased temperature in Naphtha Stabilizer Column T-101, push heavier components overhead and contaminate Y-Grade product, product quality issues, potential to exceed Maximum Allowable Working Temperature (MAWT) of T-101 of 375F, mechanical integrity issues, potential release, potential ignition and personnel impact	461. TAH-10310 high temperature alarm on overheads from Stabilizer T-101 will prompt operator response	P	4	1	2	61. No recommendations required.	
			460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response						
			527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling						
	10.10.4. FCV-10475 fails open or bypass inadvertently open	10.10.4.1. Increase in Combi Tower T-102 temperature and pressure, potential column overpressure, loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50	P	4	1	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			psig, and 51.25 psig respectively) is sized for loss of cooling						
			4. PAH-11271 High pressure alarm on Combi Tower T-102 will prompt operator response						
			506. PAHH-11715 (2oo3) high high pressure alarm on Combi Tower T-102 will prompt operator response						
	10.10.5. FCV-10522 fails open or bypass inadvertently open	10.10.5.1. Inadvertent bypass of hot oil flow around EX-107 Combi Tower Feed Preheater; Potential reduced heat to Combi Tower T-102 feed; Potential off-spec product; Potential trip to Heater; Potential flaring; Potential downtime; Potential environmental and environmental impact	29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D	F	2	3	2	61. No recommendations required.	
	10.10.6. FCV-16030 on hot oil from Diesel Stripper Reboiler EX-110 fails open or bypass inadvertently open	10.10.6.1. Potential reduced hot oil flow to other heat exchangers; Potential off-spec product with financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
10.11. Low Temperature	10.11.1. FCV-11681 malfunctions closed or manual valve(s) inadvertently closed	10.11.1.1. Loss of hot oil circulation through EX-109 Kero Stripper Reboiler; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.11.2. FCV-10521 fails closed or manual valve(s) inadvertently closed	10.11.2.1. Loss of hot oil circulation through Naphtha Stabilizer Reboiler EX-106; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.11.3. FCV-10475 fails closed or manual valve(s) inadvertently closed	10.11.3.1. Loss of heat to Combi Tower T-102 feed; Potential off-spec product; Potential trip to Heater; Potential flaring; Potential downtime; Potential environmental and environmental impact	29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D	F	2	3	2	61. No recommendations required.	
	10.11.4. FCV-16030 fails closed or manual valve(s) inadvertently closed	10.11.4.1. Loss of hot oil circulation through Diesel Stripper Reboiler EX-110; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
10.12. Contamination/ Composition	10.12.1. No additional causes identified								
10.13. Corrosion	10.13.1. No specific causes identified								
10.14. Additional Phase	10.14.1. Not applicable								
10.15. Rupture/Loss of Containment	10.15.1. No additional causes identified								

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
10.16. Loss of Utilities	10.16.1. Loss of instrument air	10.16.1.1. Valves revert to fail-safe position							
	10.16.2. Loss of power	10.16.2.1. See Hot Oil Heater F-101 trip scenarios, this node							
10.17. Start-up/Shutdown hazards/ Maintenance	10.17.1. Hot Oil Circulation Pump PM-102A/B/C/D aging and hot oil quality	10.17.1.1. High frequency of Hot Oil Circulation Pump PM-102A/B/C/D seal failures (currently weekly). Pump temperatures require significant time for cooling before maintenance, requiring downtime	448. No specific safeguards identified	F	2	5	4	15. Implement the following hot oil system improvements to minimize the frequency of hot oil system maintenance: 1) Additional spare hot oil pumps; 2) Hot oil filtration; 3) Hot Oil Day Tank; 4) Nitrogen blanket on Hot Oil Day Tank and Hot Oil Surge Drums; 5) Hot Oil Seal Coolers; 6) Upgrade heat transfer fluid.	
	10.17.2. H2S in the CEMS building	10.17.2.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 610. AT-10581/20581 located inside CEMS 1/2 buildings will alert in the control room	P	2	2	1	61. No recommendations required.	
		10.17.2.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
10.18. General	10.18.1. **What if vessel temperature control loop fails open or closed?**	10.18.1.1. Fails open - increased heating of hot oil; potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 526. PALL-10625 low-low pressure alarm on Hot Oil Heater F-101 discharge will trip F-101 27. FALL-11350 low-low flow alarm on Hot Oil Heater F-101 will prompt operator response 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D	P	4	1	2	61. No recommendations required.	
		10.18.1.2. Fails closed - Hot Oil Heater F-101 shuts down on low pressure; Unit would go to re-run mode; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	10.18.2. **What if fuel supply is cut off?**	10.18.2.1. Loss of firing; Hot Oil Heater F-101 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.18.3. **What if flame fails?**	10.18.3.1. Potential for unburned fuel in Hot Oil Heater F-101 firebox, ignition and heat damage, personnel impact	33. 20 minute purge cycle prior to startup 32. Hot Oil Heater F-101/201 cells must be balanced, loss of a burner in one cell requires cutting burner on the other (BMS)	P	4	1	2	61. No recommendations required.	
	10.18.4. **What if air damper fails closed?**	10.18.4.1. Mechanical failure (all dampers in manual) - pressure swings on ID fan, air imbalance, potential for high Hot Oil Heater F-101 duct temperatures; Potential heater damage and downtime	35. Pressure alarm high-high PAH-14137 at common duct alarm will prompt operator response 34. Pressure alarm high-high PAHH-14137 at common duct trips Hot Oil Heater F-101 36. Pressure alarm high-high PAHH-14102/14122 on individual cells trips Hot Oil Heater F-101	F	3	1	1	61. No recommendations required.	
	10.18.5. **What if air damper fails open?**	10.18.5.1. Positive pressure and Hot Oil Heater F-101 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.18.6. **What if blower or motor fails?**	10.18.6.1. Potential for over temperature of Hot Oil Heater F-101/ducts, heater damage	37. FD Fan BM-103 or ID Fan BM-104 shutdown trips Hot Oil Heater F-101 38. Temperature alarm high TAH-14105/14125 - high duct temperature alarm will prompt operator response 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D	F	3	1	1	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	10.18.7. **What if fuel supply pressure increases?**	10.18.7.1. Increased heating of hot oil; potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	
	10.18.8. **What if water is entrained in fuel supply?**	10.18.8.1. Loss of flame at one or more burners; Potential for unburned fuel in Hot Oil Heater F-101 firebox, ignition and heat damage, personnel impact	33. 20 minute purge cycle prior to startup 32. Hot Oil Heater F-101/201 cells must be balanced, loss of a burner in one cell requires cutting burner on the other (BMS)	P	4	1	2	61. No recommendations required.	
	10.18.9. **What if fuel supply regulator fails open?**	10.18.9.1. Increased heating of hot oil; potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	530. FCV-14100/14120 fuel supply control valve will further control the fuel gas supply pressure 528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	
	10.18.10. **What if fuel supply regulator fails closed?**	10.18.10.1. Loss of firing; Hot Oil Heater F-101 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	10.18.11. **What if fuel contains excessive solids?**	10.18.11.1. Potential for burner pluggage, loss of flame at one or more burners; Hot Oil Heater F-101 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	10.18.12. **What if burner tube skin temperature increases? (Caused by flow restriction in tube)**	10.18.12.1. Potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	
	10.18.13. **What if burner tube skin temperature decreases?**	10.18.13.1. Not considered a stand-alone credible scenario							
	10.18.14. **What if burner tube ruptures?**	10.18.14.1. Potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	
	10.18.15. **What if solids or coke builds up on tube external surfaces?**	10.18.15.1. Potential to damage heater tubes; Potential reduced heating efficiency; Operability issue only, no hazardous consequences identified							
	10.18.16. **What if the BTU value of the gas changes?**	10.18.16.1. Potential increased gas usage on the low end; Potential reduced heating efficiency; Operability issue only, no hazardous consequences identified							
	10.18.17. **What if solids build up on tube internal surfaces?**	10.18.17.1. Flow restriction in tubes; Potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	

Node: 10. Node 10: Hot Oil Heater, F-101; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-103, PM-102), including natural gas and combustion air

Drawings / References:

40-PI-9010; 40-PI-9010 Redline; 40-PI-9011 ; 40-PI-9011 Redline 1; 40-PI-9011 Redline 2; 40-PI-9012; 40-PI-9019; 40-PI-9020; 40-PI-9049; 13040-D-100-1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	10.18.18. **What if the purge is inadequate? (BMS Sequence Failure)**	10.18.18.1. Potential for burner explosion on restart if un reacted fuel is present	39. Timer failure results in burner management system (BMS) trip of the Hot Oil Furnace F-101/201 531. AIC-14103/14123 low oxygen analyzer alarm will prompt operator response	P	4	1	2	61. No recommendations required.	
	10.18.19. **What if hot oil flow through the tubes stops?**	10.18.19.1. Flow restriction in tubes; Potential for overheating of Hot Oil Heater F-101 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	528. PAHH-14009/14029 high-high pressure alarm will trip Hot Oil Heater F-101 29. High flue gas duct temperature - trips Hot Oil Heater F-101 TI-14104A/B/C/D and TI-14124A/B/C/D 529. TAHH-10615 high high temperature alarm will trip Hot Oil Heater F-101	P	4	1	2	61. No recommendations required.	

Node: 11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack

Drawings / References:

40-PI-9038; 40-PI-9039; 40-PI-9046; 40-PI-9051; 40-PI-9052; 40-PI-9054

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
11.1. Low/No Flow	11.1.1. Multi-Stage Vertical Turbine Pumps PM-416A/B/C trip	11.1.1.1. Inability to send product to truck loading skid; Operability issue only, no hazardous consequences identified							
	11.1.2. Manual valve closed in suction of Multi-Stage Vertical Turbine Pumps PM-416A/B/C	11.1.2.1. Block suction to Multi-Stage Vertical Turbine Pumps PM-416A/B/C, cavitation, seal leak, potential ignition, personnel impact	532. Dual seals on PM-416A/B/C pumps 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 534. Seal failure alarm SFA-40023A/B/C will shutdown pumps PM-416A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	11.1.3. Manual valve closed in discharge of Multi-Stage Vertical Turbine Pumps PM-416A/B/C or FV-40030A/B/C malfunctions closed when not loading or upstream/downstream manual valve(s) inadvertently closed	11.1.3.1. Block discharge from Multi-Stage Vertical Turbine Pumps PM-416A/B/C, deadhead, seal leak, potential ignition, personnel impact	532. Dual seals on PM-416A/B/C pumps 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack

Drawings / References:

40-PI-9038; 40-PI-9039; 40-PI-9046; 40-PI-9051; 40-PI-9052; 40-PI-9054

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			534. Seal failure alarm SFA-40023A/B/C will shutdown pumps PM-416A/B/C						
	11.1.4. SDV-40017/40117 fails closed or downstream manual valve inadvertently closed	11.1.4.1. Block suction to Multi-Stage Vertical Turbine Pumps PM-416A/B/C, cavitation, seal leak, potential ignition, personnel impact	532. Dual seals on PM-416A/B/C pumps 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 534. Seal failure alarm SFA-40023A/B/C will shutdown pumps PM-416A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	11.1.5. FV-40030A/B/C malfunctions closed during truck loading	11.1.5.1. Block discharge from Multi-Stage Vertical Turbine Pumps PM-416A/B/C, deadhead, seal leak, potential ignition, personnel impact	532. Dual seals on PM-416A/B/C pumps 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 534. Seal failure alarm SFA-40023A/B/C will shutdown pumps PM-416A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
11.2. High Flow	11.2.1. More than one Multi-Stage Vertical Turbine Pumps PM-416A/B/C at once	11.2.1.1. More than one Multi-Stage Vertical Turbine Pumps PM-416A/B/C operating, overfill of truck, deadheading PM-416A/B/C pumps; seal leak, potential ignition, personnel impact	532. Dual seals on PM-416A/B/C pumps 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 534. Seal failure alarm SFA-40023A/B/C will shutdown pumps PM-416A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
11.3. Reverse/Misdirected Flow	11.3.1. Shutoff valve XV-40016/40116 fails open due to debris or left open (Water draw off pot Interface level control, human intervention to actuate valve)	11.3.1.1. Potential to send Y Grade to Process Water Tank TK 5-201. Potential release to atmosphere from vapor space above floating roof, ignition, personnel impact.	40. Constant operator monitoring 535. Lower explosive limit LEL AT-9301/2/3/4 will prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	11.3.2. Lining up wrong Y-Grade Storage Sphere pressure vessel PV-410/411 for unit production	11.3.2.1. Potential overfill of Y-Grade Storage Sphere pressure vessel PV-410/411 to flare, potential to send liquids out of the flare. Potential personnel impacts. Potential environmental impact	536. LAH-40007/40107 high level alarm on PV-410/411 will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack

Drawings / References:

40-PI-9038; 40-PI-9039; 40-PI-9046; 40-PI-9051; 40-PI-9052; 40-PI-9054

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			537. LAH-40004/40104 high level shutdown on PV-410/411 will close inbound valves SDV-40001/40101						
		11.3.2.2. Potential to send product to certified tank (instead of rundown tank); Potential operations delay (requiring re-testing), no hazardous consequences identified							
	11.3.3. Lining up correctly to truck rack, with wrong connection to return from truck rack. (Incorrect valve lineup to sphere)	11.3.3.1. Potential overfill of Y-Grade Storage Sphere pressure vessel PV-410/411 to flare, potential to send liquids out of the flare. Potential personnel impacts. Potential environmental impact	536. LAH-40007/40107 high level alarm on PV-410/411 will prompt operator response 537. LAH-40004/40104 high level shutdown on PV-410/411 will close inbound valves SDV-40001/40101	P	4	1	2	61. No recommendations required.	
11.4. High Pressure	11.4.1. External fire around PV-410/411 Storage Spheres	11.4.1.1. Potential overpressure of vessel with loss of containment, ignition, and personnel impact	538. PSV-40010/40012 and PSV-40110/40112 (set @ 120 psig) is sized for external fire 41. Deluge system	P	4	1	2	61. No recommendations required.	
	11.4.2. Blocked in piping on LR-501/502 Truck Loading Skid	11.4.2.1. Potential thermal expansion and release with potential ignition and personnel impact	448. No specific safeguards identified	P	4	3	4	16. Perform engineering evaluation to ensure LR-501/502 Truck Loading Skid piping has adequate thermal expansion relief valves. Develop plan to install additional relief valve if required and provide adequate access for maintenance.	
11.5. Low Pressure	11.5.1. No additional causes identified								
11.6. High Level	11.6.1. No additional causes identified								
11.7. Low Level	11.7.1. No additional causes identified								
11.8. High Interface Level	11.8.1. Not applicable								
11.9. Low Interface Level	11.9.1. Not applicable								
11.10. High Temperature	11.10.1. No additional causes identified								

Node: 11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack
Drawings / References:
40-PI-9038; 40-PI-9039; 40-PI-9046; 40-PI-9051; 40-PI-9052; 40-PI-9054

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
11.11. Low Temperature	11.11.1. Loss of heat tracing on water line drain	11.11.1.1. Potential to freeze water at water draw off pot, pipe failure upon thaw, release, ignition, personnel impact	533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 539. Lower Explosive Limit LEL AI-40020/40022 and AI-40120/40122 alarm will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
11.12. Contamination/ Composition	11.12.1. Solid contamination upstream of PM-416A/B (only PM-416C has inline strainer)	11.12.1.1. Potential off-spec product; Potential pump damage from solid contamination and financial impact Operations normally only run PM-416C due to it being the only pump with an inline strainer	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
11.13. Corrosion	11.13.1. No specific causes identified								
11.14. Additional Phase	11.14.1. Not applicable								
11.15. Rupture/Loss of Containment	11.15.1. No additional causes identified								
11.16. Loss of Utilities	11.16.1. Loss of power	11.16.1.1. See pump PM-416A/B/C trip scenario, this node							
11.17. Start-up/Shutdown hazards/ Maintenance	11.17.1. Ladder hand rails on Storage Spheres at improper height (per OSHA standard)	11.17.1.1. Ladder hand rails on Storage Spheres at improper height (per OSHA standard); Potential fall hazard to personnel	448. No specific safeguards identified	P	3	3	3	17. Evaluate the need to re-design the ladder hand rails on Storage Spheres PV-410/411 to comply with OSHA standards.	
	11.17.2. H2S in the y-grade	11.17.2.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 611. H2S Scavenger to reduce H2S in the storage sphere	P	2	2	1	61. No recommendations required.	
		11.17.2.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
11.18. General	11.18.1. Driver pulls away while loading or hose rupture	11.18.1.1. Potential release from hose/piping, ignition, personnel impact	540. Breakaway fittings on hose connections 42. Ground system trips system on loss of connectivity 43. Drivers not allowed in trucks during unloading	P	4	1	2	61. No recommendations required.	

Node: 11. Node 11: Butane/Y Grade Storage Sphere, pumps and truck rack

Drawings / References:

40-PI-9038; 40-PI-9039; 40-PI-9046; 40-PI-9051; 40-PI-9052; 40-PI-9054

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			44. Wheel chocks required during unloading						
	11.18.2. Driver pulls away after loading without disconnecting hoses or grounding cables	11.18.2.1. Potential damage to hose fittings with potential financial impact	540. Breakaway fittings on hose connections 45. Driver training 44. Wheel chocks required during unloading	F	1	1	1	61. No recommendations required.	
	11.18.3. Driver impacts the loading rack	11.18.3.1. Potential loss of containment from loading rack, potential ignition and personnel impact	541. Racks protected by bollards 45. Driver training	P	4	1	2	61. No recommendations required.	
	11.18.4. Meter is not calibrated properly	11.18.4.1. Potential SOX exceptions with potential financial impact	46. Meters calibrated annually by third party	F	1	3	1	61. No recommendations required.	
	11.18.5. Trailer is not properly grounded before loading	11.18.5.1. Potential ignition inside trailer, personnel impact	42. Ground system trips system on loss of connectivity 45. Driver training	P	4	1	2	61. No recommendations required.	1. Low likelihood of oxygen inside the trailer.

Node: 12. Node 12: Flare, Flare Knockout Drum, Flare Knockout Drum Pump (PV-305, PM-312) and Flare, Flow switch FL-301

Drawings / References:

40-PI-9027; 40-PI-9043; 40-PI-9231; YA-1770-S1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
12.1. Low/No Flow	12.1.1. 3" manual valve on suction to PM-312A/B pumps inadvertently closed	12.1.1.1. Loss of suction to PM-312A/B pumps; Potential to cavitate pumps with seal failure, potential ignition and personnel impact	533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 551. FALL-30060 low-low flow alarm will shutdown pumps PM-312A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	12.1.2. Manual valve(s) downstream of PM-312A/B pumps inadvertently closed	12.1.2.1. Blocked discharge from PM-312A/B pumps; Potential to deadhead pumps with seal failure, potential ignition and personnel impact	533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 551. FALL-30060 low-low flow alarm will shutdown pumps PM-312A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	12.1.3. PCV-30235 pressure regulator fails fully closed or upstream manual valve(s) inadvertently closed	12.1.3.1. Loss of fuel gas supply to the flare pilot; Pilot flame extinguished with incomplete combustion and potential environmental/financial impact	552. TAL-30180/30185/30190 low temperature alarm on the pilots will prompt operator response	F	1	3	1	61. No recommendations required.	

Node: 12. Node 12: Flare, Flare Knockout Drum, Flare Knockout Drum Pump (PV-305, PM-312) and Flare, Flow switch FL-301

Drawings / References:

40-PI-9027; 40-PI-9043; 40-PI-9231; YA-1770-S1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	12.1.4. PCV-30220 pressure regulator fails fully closed or upstream manual valve(s) inadvertently closed or SV-30225 fails closed or XV-30205 fails closed	12.1.4.1. Loss of air supply to the flare pilots; Pilot flame extinguished with incomplete combustion and potential environmental/financial impact	552. TAL-30180/30185/30190 low temperature alarm on the pilots will prompt operator response	F	1	3	1	61. No recommendations required.	
	12.1.5. SV-30240 fails closed or manual valve(s) inadvertently closed	12.1.5.1. Inability to route instrument air supply to the flare pilot fuel gas supply line; Potential incorrect ratio of instrument air and fuel gas to the pilots; Potential loss of pilots, potential incomplete combustion and potential environmental/financial impact	552. TAL-30180/30185/30190 low temperature alarm on the pilots will prompt operator response	F	1	3	1	61. No recommendations required.	
	12.1.6. Loss of fuel gas supply to the flare	12.1.6.1. Loss of sweep gas, lowered BTU content of fuel; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	12.1.7. BM-301-A/B flare blower failure	12.1.7.1. Power outage or mechanical trip; Potential incomplete combustion; Potential environmental/financial impact	47. Spare blower motor in stock 48. Secondary blower available - start from control room	F	1	2	1	61. No recommendations required.	
12.2. High Flow	12.2.1. Both BM-301-A/B flare blowers online	12.2.1.1. Excessive combustion air; Potential to snuff out the flames; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
12.3. Reverse/Misdirected Flow	12.3.1. No specific causes identified								
12.4. High Pressure	12.4.1. PCV-30235 pressure regulator fails fully open	12.4.1.1. Fuel gas is coming into this line at 15 psig; PCV-30235 regulator is a protection from the upstream pressure regulation; This regulator failing fully open would not result in any adverse consequences							
	12.4.2. PCV-30220 pressure regulator fails fully open	12.4.2.1. Inability to reduce instrument air supply pressure to 15 psig; Potential to send up to 100 psig instrument air supply to flare pilots; Potential to blow out pilots; Pilot flame extinguished with incomplete combustion and potential environmental/financial impact	552. TAL-30180/30185/30190 low temperature alarm on the pilots will prompt operator response	F	1	3	1	61. No recommendations required.	
12.5. Low Pressure	12.5.1. No additional causes identified								
12.6. High Level	12.6.1. Failure to pump out level in PV-305 Flare KO Drum	12.6.1.1. High level in PV-305; Potential liquid carryover to flare; Potential liquid out of the flare with potential fire and area personnel impact	548. LAHH-30045 high-high level alarm on PV-305 will auto start pumps PM-312A/B 549. LAHH-30040 high-high level alarm on PV-305 will initiate unit shutdown	P	3	1	1	61. No recommendations required.	
12.7. Low Level	12.7.1. PM-312A/B pumps fail to trip off		550. LAL-30040 low level alarm on PV-305	P	4	1	2		

Node: 12. Node 12: Flare, Flare Knockout Drum, Flare Knockout Drum Pump (PV-305, PM-312) and Flare, Flow switch FL-301

Drawings / References:

40-PI-9027; 40-PI-9043; 40-PI-9231; YA-1770-S1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		12.7.1.1. Potential to drain all level from PV-305 Flare KO Drum; Potential to lose suction to PM-312A/B pumps; Potential to cavitate pumps with seal failure, potential ignition and personnel impact	will prompt operator response 533. Lower Explosive Limit LEL-40027 will prompt operators to cordon off the area 551. FALL-30060 low-low flow alarm will shutdown pumps PM-312A/B					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
12.8. High Interface Level	12.8.1. Not applicable								
12.9. Low Interface Level	12.9.1. Not applicable								
12.10. High Temperature	12.10.1. No specific causes identified								
12.11. Low Temperature	12.11.1. Low ambient temperatures	12.11.1.1. Potential increased liquids removed in the PV-305 Flare KO Drum; No stand-alone hazardous consequences identified							
12.12. Contamination/ Composition	12.12.1. Solids/coke build-up on stacks or nozzles	12.12.1.1. Potential incomplete combustion and potential environmental/financial impact	554. API flare inspection criteria	E	1	2	1	61. No recommendations required.	
12.13. Corrosion	12.13.1. No specific causes identified								
12.14. Additional Phase	12.14.1. Not applicable								
12.15. Rupture/Loss of Containment	12.15.1. No additional causes identified								
12.16. Loss of Utilities	12.16.1. Loss of power	12.16.1.1. See flare blower failure scenario in this node							
12.17. Start-up/Shutdown hazards/ Maintenance	12.17.1. Excessive radiant heat levels from flare	12.17.1.1. Potential for area grass fire and potential personnel impact	553. Flare exclusion zone marked in the area	P	3	1	1	61. No recommendations required.	
	12.17.2. H2S in the calibration gas	12.17.2.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 609. AT-30400A/B/C Fixed H2S Monitors located at the flare will alert in the control room	P	2	2	1	61. No recommendations required.	
		12.17.2.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							

Node: **12. Node 12: Flare, Flare Knockout Drum, Flare Knockout Drum Pump (PV-305, PM-312) and Flare, Flow switch FL-301**

Drawings / References:

40-PI-9027; 40-PI-9043; 40-PI-9231; YA-1770-S1

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
12.18. General	12.18.1. No additional causes identified								

Node: **13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)**

Drawings / References:

40-PI-9012; 40-PI-9013; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
13.1. Low/No Flow (Ammonia)	13.1.1. Ammonia Forwarding Pump PM-119A/B trips	13.1.1.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	50. Redundant Ammonia Forwarding Pumps PM-119A/B 555. PIT-14815 low pump discharge pressure alarm will prompt operator response 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
	13.1.2. Manual valve(s) on suction to Ammonia Forwarding Pump PM-119A/B inadvertently closed or upstream strainer plugged	13.1.2.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	555. PIT-14815 low pump discharge pressure alarm will prompt operator response 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	
		13.1.2.2. NH3 Forwarding Pump PM-119A/B cavitation, potential seal damage, release of aqueous ammonia, personnel impact	555. PIT-14815 low pump discharge pressure alarm will prompt operator response 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	13.1.3. Manual valve(s) on discharge from Ammonia Forwarding Pump PM-119A/B inadvertently closed	13.1.3.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	555. PIT-14815 low pump discharge pressure alarm will prompt operator response 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	

Node: 13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)

Drawings / References:

40-PI-9012; 40-PI-9013; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		13.1.3.2. NH3 Forwarding Pump PM-119A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	557. PSV-14812A/B (set @ 95 psig) is sized for blocked discharge 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	13.1.4. Closure of vapor return line on tank filling	13.1.4.1. Potential overpressure of Ammonia Storage Tank PV-106 with release of aqueous ammonia to surrounding area and potential personnel impact	448. No specific safeguards identified	P	4	3	4	18. Perform engineering evaluation to verify that PSV-14806 on Ammonia Storage Tank PV-106 is sized for loss of vapor return during truck filling resulting in potential overpressure of PV-106. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
	13.1.5. Failure of expansion joint EJ1500	13.1.5.1. Release of aqueous ammonia to surrounding area and potential personnel impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	P	3	2	2	61. No recommendations required.	
	13.1.6. Failure of expansion joint EJ-1610	13.1.6.1. Release of aqueous ammonia to surrounding area and potential personnel impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	P	3	2	2	61. No recommendations required.	
	13.1.7. FCV-14821 fails closed or upstream/downstream manual valve(s) inadvertently closed or upstream filters FIL-1500A/B plugged	13.1.7.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	555. PIT-14815 low pump discharge pressure alarm will prompt operator response 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	
		13.1.7.2. NH3 Forwarding Pump PM-119A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	557. PSV-14812A/B (set @ 95 psig) is sized for blocked discharge 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	13.1.8. NH3 injection nozzles plugged at furnace SCR	13.1.8.1. Potential inadequate ammonia injection; Potential inability to control air emissions with potential environmental/financial impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	3	1	61. No recommendations required.	

Node: 13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)

Drawings / References:

40-PI-9012; 40-PI-9013; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		13.1.8.2. NH3 Forwarding Pump PM-119A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	557. PSV-14812A/B (set @ 95 psig) is sized for blocked discharge 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
13.2. Low/No Flow (Air)	13.2.1. Air blower failure BM-105A/B	13.2.1.1. Inability to send dilution air, inability to disperse ammonia properly; Potential reduced ability to control air emissions with potential environmental/financial impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response 556. FIT-14820 low flow alarm on pump discharge will prompt operator response 558. Spare BM-105A/B available for start-up	E	1	1	1	61. No recommendations required.	
13.3. High Flow	13.3.1. NOx analyzer out of calibration, flow control valve FCV-14821 fails open or flow element (FE) reading low, or flow control valve FCV bypass open or leaking through	13.3.1.1. High level of ammonia injection; Potential high NH3 concentration out of the stack with potential environmental/financial impact	51. NOx analyzers calibrated daily	E	1	4	2	61. No recommendations required.	
	13.3.2. Ammonia Forwarding Pump PM-119A/B fail to turn off	13.3.2.1. Potential to pull vacuum on PV-106 Ammonia Storage Tank; Potential release of aqueous ammonia with potential personnel impact	559. LIT-14805 low level interlock on PV-106 will trip pumps PM-119A/B and PM-120A/B	P	3	3	3	19. Perform engineering evaluation to verify that PSV-14807 on PV-106 Ammonia Storage Tank is sized for vacuum. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
13.4. Reverse/Misdirected Flow	13.4.1. No specific causes identified								
13.5. High Pressure	13.5.1. PCV-14813 pressure regulator fails fully open	13.5.1.1. Inability to reduce PM-119A/B pump discharge pressure to 80 psig; Potential to send up to 95 psig ammonia back to PV-106 Ammonia Storage Tank; Not expected to result in an overpressure scenario; No adverse consequences identified							
13.6. Low Pressure	13.6.1. PCV-14813 pressure regulator fails fully closed	13.6.1.1. NH3 Forwarding Pump PM-119A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	557. PSV-14812A/B (set @ 95 psig) is sized for blocked discharge 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
13.7. High Level	13.7.1. Overfilling Ammonia Storage Tank PV-106		560. Filling PV-106 Ammonia Storage	P	4	1	2	61. No recommendations required.	

Node: 13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)

Drawings / References:

40-PI-9012; 40-PI-9013; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		13.7.1.1. Potential overpressure of Ammonia Storage Tank PV-106 with release of aqueous ammonia to surrounding area and potential personnel impact	Tank is a continuously monitored operation 561. LIT-14805 high level alarm will prompt operator response						
13.8. Low Level	13.8.1. Failure to re-fill Ammonia Storage Tank PV-106	13.8.1.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	555. PIT-14815 low pump discharge pressure alarm will prompt operator response 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response 556. FIT-14820 low flow alarm on pump discharge will prompt operator response	E	1	1	1	61. No recommendations required.	
13.9. High Interface Level	13.9.1. Not applicable								
13.10. Low Interface Level	13.10.1. Not applicable								
13.11. High Temperature	13.11.1. Air Heater HTR-1640 control failure	13.11.1.1. Excess hot air supply, potential catalyst damage with potential environmental/financial impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response 562. Temperature indicator TI-14827 high NH3/air temperature indication will prompt operator response	E	1	2	1	61. No recommendations required.	
	13.11.2. External fire around PV-106 Ammonia Storage Tank	13.11.2.1. Potential thermal expansion and release with potential personnel impact	448. No specific safeguards identified	P	3	3	3	20. Perform engineering evaluation to determine if Ammonia Storage Tank PV-106 requires overpressure protection for external fire scenario. If so, determine if PSV-14806 on PV-106 is sized appropriately and update the PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
13.12. Low Temperature	13.12.1. Heater unit failure HTR-1640	13.12.1.1. Reduced vaporization of NH3 prior to heater injection, potential impaired distribution of NH3 into Hot Oil Furnace H-101/201; Potential environmental/financial impact	49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	

Node: 13. Node 13: Aqueous Ammonia Heater System (PV-106, PM-119A/B, BM-105A/B)

Drawings / References:

40-PI-9012; 40-PI-9013; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			52. Temperature indicator TI-14827 low NH3/air temperature indication will prompt operator response						
13.13. Contamination/ Composition	13.13.1. Out of spec ammonia - high or low aqueous NH3 concentration	13.13.1.1. Higher ammonia with potential environmental/financial impact	563. Ammonia spec is checked upon arrival 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
		13.13.1.2. Higher NOX with potential environmental/financial impact	563. Ammonia spec is checked upon arrival 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
		13.13.1.3. Potential mineralization on vaporizer; Potential pluggage of vaporizer with potential reduced/loss of flow of ammonia; Potential environmental/financial impact	563. Ammonia spec is checked upon arrival 49. Analyzer indicator AI-10575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
13.14. Corrosion	13.14.1. No specific causes identified								
13.15. Additional Phase	13.15.1. Not applicable								
13.16. Rupture/Loss of Containment	13.16.1. No additional causes identified								
13.17. Loss of Utilities	13.17.1. Loss of instrument air	13.17.1.1. Loss of programmable logic controller (PLC) cabinet purge; Potential to overheat the PLC with potential environmental/impact	53. Pressure switch low PSL-14824 on instrument air line to system (including cabinet purge) will prompt operator response	E	1	3	1	61. No recommendations required.	
	13.17.2. Loss of power	13.17.2.1. See pump trip scenario, this node							
		13.17.2.2. See blower trip scenario, this node							
13.18. Start-up/Shutdown hazards/ Maintenance	13.18.1. Driver pulls away while loading or hose rupture	13.18.1.1. Potential release from hose/piping, personnel impact	564. Wheel chocks required during loading	P	3	2	2	61. No recommendations required.	
13.19. General	13.19.1. No additional causes identified								

Node: **14. Node 14: Natural Gas/Fuel Gas Header**

Drawings / References:

40-PI-9009; 40-PI-9009 Redline; 40-PI-9014; 40-PI-9030 Redline; 40-PI-9030 ; 40-PI-9043; 40-PI-9048; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
14.1. Low/No Flow	14.1.1. PCV-10355 fails closed or upstream/downstream manual valve(s) inadvertently closed	14.1.1.1. Loss of make-up natural gas to PV-101 Naphtha Stabilizer Accumulator; Inability to send naphtha to Combi Tower T-102; Potential higher temperature in T-102; Potential higher temperature to the heater F-101; Potential increased vapor traffic from T-101; Potential off-spec product with potential financial impact	466. PDAH-10235 high differential pressure alarm on Stabilizer T-101 will prompt operator response	F	1	3	1	61. No recommendations required.	
	14.1.2. FCV-11760 fails closed or upstream/downstream manual valve(s) inadvertently closed	14.1.2.1. Loss of stripping gas to T-102 Combi Tower; Potential off-spec AGO product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	14.1.3. FCV-13480 fails closed or upstream/downstream manual valve(s) inadvertently closed	14.1.3.1. Loss of stripping gas to T-102 Combi Tower; Potential off-spec AGO product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	14.1.4. FCV-15070 fails closed or upstream/downstream manual valve(s) inadvertently closed	14.1.4.1. Loss of sweep gas to flare header; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	14.1.5. PCV-15080 fails fully closed	14.1.5.1. Loss of sweep gas to flare header; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	14.1.6. Manual valve closed in main header or supply line to users	14.1.6.1. Unit/plant shutdown - see individual gas users for specific scenarios							
14.2. High Flow	14.2.1. PCV-10355 malfunctions fully open or bypass valve inadvertently open	14.2.1.1. Potential to overpressure T-101 Stabilizer with potential loss of containment, ignition and personnel impact	509. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for blocked vapor outlet 508. PCV-10395C on vapor overheads from PV-101 will open to flare on high overhead pressure	P	4	1	2	61. No recommendations required.	
	14.2.2. FCV-11760 malfunctions open or bypass inadvertently opened	14.2.2.1. Excess stripping gas at 65 psig, increase in Combi Tower T-102 pressure; Potential to overpressure T-102 with loss of containment, potential ignition and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 565. PAH-11715A/B/C high pressure alarm will prompt operator response 4. PAH-11271 High pressure alarm on Combi Tower T-102	P	4	1	2	61. No recommendations required.	

Node: **14. Node 14: Natural Gas/Fuel Gas Header**

Drawings / References:

40-PI-9009; 40-PI-9009 Redline; 40-PI-9014; 40-PI-9030 Redline; 40-PI-9030 ; 40-PI-9043; 40-PI-9048; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			will prompt operator response						
	14.2.3. FCV-13480 malfunctions open or bypass inadvertently opened	14.2.3.1. Excess stripping gas, increase in Combi Tower T-102 pressure; Potential to overpressure T-102 with loss of containment, potential ignition and personnel impact	477. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-102 is sized for blocked vapor outlet 565. PAH-11715A/B/C high pressure alarm will prompt operator response 4. PAH-11271 High pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
	14.2.4. FCV-15070 malfunctions open or bypass inadvertently opened	14.2.4.1. Potential for excess gas flow to users; No high pressure or environmental concerns identified; Operability issue only, no hazardous consequences identified							
14.3. Reverse/Misdirected Flow	14.3.1. No specific causes identified								
14.4. High Pressure	14.4.1. PCV-15080 fails fully open	14.4.1.1. Inability to reduce natural gas to 40 psig; Potential larger flame at flare; Potential heater shutdown; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
14.5. Low Pressure	14.5.1. No additional causes identified								
14.6. High Level	14.6.1. No additional causes identified								
14.7. Low Level	14.7.1. No additional causes identified								
14.8. High Interface Level	14.8.1. Not applicable								
14.9. Low Interface Level	14.9.1. Not applicable								
14.10. High Temperature	14.10.1. No specific causes identified								
14.11. Low Temperature	14.11.1. No specific causes identified								
14.12. Contamination/ Composition	14.12.1. Contaminants in Natural Gas - pipe scale	14.12.1.1. Potential for blockage of heater burners, control valves, regulator; Potential financial/environmental impact	54. Filter FL-208 natural gas	F	1	3	1	61. No recommendations required.	

Node: 14. Node 14: Natural Gas/Fuel Gas Header

Drawings / References:

40-PI-9009; 40-PI-9009 Redline; 40-PI-9014; 40-PI-9030 Redline; 40-PI-9030 ; 40-PI-9043; 40-PI-9048; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			filter/separator on gas supply 55. Strainers on gas line to heaters						
14.13. Corrosion	14.13.1. Corrosion on natural gas piping	14.13.1.1. Potential long-term mechanical integrity issues, no acute hazards identified							
14.14. Additional Phase	14.14.1. Not applicable								
14.15. Rupture/Loss of Containment	14.15.1. No additional causes identified								
14.16. Loss of Utilities	14.16.1. Loss of instrument air	14.16.1.1. Valves revert to fail-safe position							
14.17. Start-up/Shutdown hazards/ Maintenance	14.17.1. No additional causes identified								
14.18. General	14.18.1. No additional causes identified								

Node: 15. Node 15: Hydrocarbon Drain

Drawings / References:

40-PI-9044

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
15.1. Low/No Flow	15.1.1. Hydrocarbon Sump Pump PM-118A/B trips	15.1.1.1. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 567. LAH-12501 high level in TK-101 Sump will kick on other pump PM-118A/B 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	15.1.2. Manual valve closed in Hydrocarbon Sump Pump PM-118A/B discharge	15.1.2.1. Hydrocarbon Sump Pump PM-118A/B deadheaded, potential pump damage; Potential seal failure with potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 15. Node 15: Hydrocarbon Drain

Drawings / References:

40-PI-9044

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
		15.1.2.2. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
15.2. High Flow	15.2.1. Drain valve left open	15.2.1.1. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
	15.2.2. High rate of flow from vacuum truck to sump	15.2.2.1. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						

Node: 15. Node 15: Hydrocarbon Drain

Drawings / References:

40-PI-9044

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
15.3. Reverse/Misdirected Flow	15.3.1. Reverse flow from Feed Tanks to sump	15.3.1.1. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	15.3.2. Send Y-Grade to sump (drain valve left open from pumps PM-101A/B)	15.3.2.1. Potential release through Hydrocarbon Sump TK-101 vent; Potential ignition and personnel impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area 56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
15.4. High Pressure	15.4.1. No additional causes identified								
15.5. Low Pressure	15.5.1. No additional causes identified								
15.6. High Level	15.6.1. No additional causes identified								
15.7. Low Level	15.7.1. No additional causes identified								
15.8. High Interface Level	15.8.1. Not applicable								
15.9. Low Interface Level	15.9.1. Not applicable								
15.10. High Temperature	15.10.1. Drain valve left open sending hot material to sump	15.10.1.1. Potential release through Hydrocarbon Sump TK-101 vent; Potential ignition and personnel impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 15. Node 15: Hydrocarbon Drain

Drawings / References:

40-PI-9044

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response						
15.11. Low Temperature	15.11.1. No specific causes identified								
15.12. Contamination/ Composition	15.12.1. Trash in sump (for example from vacuum trucks) can plug pump screen or damage pump	15.12.1.1. Hydrocarbon Sump Pump PM-118A/B deadheaded, potential pump damage; Potential seal failure with potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		15.12.1.2. Potential to overfill the Hydrocarbon Sump TK-101, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area 56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
15.13. Corrosion	15.13.1. No specific causes identified								
15.14. Additional Phase	15.14.1. Not applicable								
15.15. Rupture/Loss of Containment	15.15.1. No additional causes identified								
15.16. Loss of Utilities	15.16.1. Loss of power	15.16.1.1. See pump trip scenario, this node							
15.17. Start-up/Shutdown hazards/ Maintenance	15.17.1. Maintenance activities at sump	15.17.1.1. Potential for vapor release during maintenance activities, ignition of vapors from maintenance activity, personnel impact	57. Hot work permit required for work in	P	3	2	2		

Node: **15. Node 15: Hydrocarbon Drain**

Drawings / References:

40-PI-9044

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Class I, Div. 1 area with continuous monitoring 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
15.18. General	15.18.1. No additional causes identified								

Node: **16. Node 16: Wastewater Drain**

Drawings / References:

40-PI-9004 Redline; 40-PI-9004 ; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9045

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
16.1. Low/No Flow	16.1.1. Contact Water Sump Pump PM-115A/B trip	16.1.1.1. Potential to overfill Contact Water Sump TK-102; Potential release through TK-102 atmospheric vent; Potential to flood unit with potential environmental impact	568. LAH-12504 high level alarm on TK-102 will prompt operator response	E	2	3	2	61. No recommendations required.	
	16.1.2. Manual valve on discharge from Contact Water Sump Pump PM-115A/B inadvertently closed	16.1.2.1. Potential to overfill Contact Water Sump TK-102; Potential release through TK-102 atmospheric vent; Potential to flood unit with potential environmental impact	568. LAH-12504 high level alarm on TK-102 will prompt operator response	E	2	2	1	61. No recommendations required.	
		16.1.2.2. Potential to deadhead Contact Water Sump Pump PM-115A/B with potential pump damage and financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
16.2. High Flow	16.2.1. Drain valve left open or opened inadvertently	16.2.1.1. Potential to send hydrocarbons to Contact Water Sump TK-102, with vent to atmosphere, send to Gulf Coast Water Authority; Potential environmental/financial impact	569. LI-18500A high interface level alarm on TK 5-0 will prompt operator response	F	1	2	1	61. No recommendations required.	
16.3. Reverse/Misdirected Flow	16.3.1. Check valve failure	16.3.1.1. Flow of contact water into TK-102 from Tank 5; Potential to overfill TK-102; Potential release through TK-102 atmospheric vent; Potential to flood unit with potential environmental impact	570. LAH-12504 high level alarm on TK-102 will prompt operator response and will turn on both pumps PM-115A/B	E	2	3	2	61. No recommendations required.	
16.4. High Pressure	16.4.1. No additional causes identified								
16.5. Low Pressure	16.5.1. No additional causes identified								
16.6. High Level	16.6.1. No additional causes identified								
16.7. Low Level	16.7.1. No specific causes identified								

Node: **16. Node 16: Wastewater Drain**

Drawings / References:

40-PI-9004 Redline; 40-PI-9004 ; 40-PI-9009; 40-PI-9009 Redline; 40-PI-9045

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
16.8. High Interface Level	16.8.1. Not applicable								
16.9. Low Interface Level	16.9.1. Not applicable								
16.10. High Temperature	16.10.1. No specific causes identified								
16.11. Low Temperature	16.11.1. No specific causes identified								
16.12. Contamination/ Composition	16.12.1. Trash in sump (for example from vacuum trucks) can plug pump screen or damage pump	16.12.1.1. Potential loss of pumping capacity; Potential to overfill TK-102; Potential release through TK-102 atmospheric vent; Potential to flood unit with potential environmental impact	570. LAH-12504 high level alarm on TK-102 will prompt operator response and will turn on both pumps PM-115A/B	E	2	3	2	61. No recommendations required.	
16.13. Corrosion	16.13.1. No specific causes identified								
16.14. Additional Phase	16.14.1. Not applicable								
16.15. Rupture/Loss of Containment	16.15.1. No additional causes identified								
16.16. Loss of Utilities	16.16.1. Power interruption	16.16.1.1. See pump trip scenario, this node							
16.17. Start-up/Shutdown hazards/ Maintenance	16.17.1. Access to outfall #1 and #2 for sampling	16.17.1.1. Requires personnel to go through standing water; Potential snakes in area with potential personnel impact	448. No specific safeguards identified	P	2	3	2	61. No recommendations required.	
16.18. General	16.18.1. No additional causes identified								

Node: **17. Node 17: Air Compressor System**

Drawings / References:

40-PI-9041; 40-PI-9047; 1318317118 SHT 1; 1318317118 SHT 2; 1318317118 SHT 3; 1318317118 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
17.1. Low/No Flow	17.1.1. CM-102A/B/CM-202A/B Air Compressors trip	17.1.1.1. Cabinets using air purge for electrical classification unprotected; Potential flammable atmosphere, potential ignition and personnel impact	58. Standby Air Compressors CM-102A/B automatic startup - lead/lag system 59. Pressure alarm low PAL-40001 unit air header low air pressure alarm will	P	3	2	2	61. No recommendations required.	

Node: **17. Node 17: Air Compressor System**

Drawings / References:

40-PI-9041; 40-PI-9047; 1318317118 SHT 1; 1318317118 SHT 2; 1318317118 SHT 3; 1318317118 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
		17.1.1.2. Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	58. Standby Air Compressors CM-102A/B automatic startup - lead/lag system	F	1	3	1	61. No recommendations required.	
	17.1.2. Air filter plugged	17.1.2.1. Cabinets using air purge for electrical classification unprotected; Potential flammable atmosphere, potential ignition and personnel impact	59. Pressure alarm low PAL-40001 unit air header low air pressure alarm will prompt operator response	P	3	2	2	61. No recommendations required.	
			571. Air filters are on a scheduled preventative maintenance schedule						
		17.1.2.2. Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
17.2. High Flow	17.2.1. CM-102A/B/CM-202A/B Air Compressors both online	17.2.1.1. No adverse consequences identified							
17.3. Reverse/Misdirected Flow	17.3.1. No specific causes identified								
17.4. High Pressure	17.4.1. High air pressure from CM-102A/B/CM-202A/B Air Compressors due to line regulator failure.	17.4.1.1. Maximum Air Compressors CM-102A/B supply is lower than pressure rating of downstream systems (listed at 175 psig); No adverse consequences identified							
17.5. Low Pressure	17.5.1. No additional causes identified								
17.6. High Level	17.6.1. No additional causes identified								
17.7. Low Level	17.7.1. No specific causes identified								
17.8. High Interface Level	17.8.1. Not applicable								
17.9. Low Interface Level	17.9.1. Not applicable								
17.10. High Temperature	17.10.1. External fire around PV-111 Air Receiver	17.10.1.1. Potential thermal expansion and release with potential ignition and personnel impact	448. No specific safeguards identified	P	3	3	3	21. Perform engineering evaluation to determine if PV-111/211 Air Receiver requires overpressure protection for external fire scenario. If so, determine if PSV-13470 on PV-111 and PSV-23470 on PV-211 are sized appropriately and update the PSV sizing documentation.	

Node: 17. Node 17: Air Compressor System

Drawings / References:

40-PI-9041; 40-PI-9047; 1318317118 SHT 1; 1318317118 SHT 2; 1318317118 SHT 3; 1318317118 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								If evaluation determines PSV-13470 on PV-111 and PSV-23470 on PV-211 are not sized for this scenario, develop a plan to add adequate overpressure protection.	
17.11. Low Temperature	17.11.1. Low ambient temperature	17.11.1.1. Potential freezing air dryers; Potential loss of instrument air to downstream users; Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	448. No specific safeguards identified	F	2	5	4	22. Complete the existing project to replace the air compressor system with more reliable equipment, including heat tracing on equipment and building enclosure.	
17.12. Contamination/ Composition	17.12.1. Water in line - desiccant dryer not removing moisture from air or poor desiccant maintenance	17.12.1.1. Potential to accumulate water in pneumatic valves or lines that can freeze during winter conditions damaging valve actuator or loss of valve actuation, respectively; Potential financial impact	60. Lines with potential to see water are heat traced	F	1	3	1	61. No recommendations required.	
	17.12.2. Accumulation of water in Air Receiver pressure vessel PV-111, resulting in corrosion (receiver located upstream of instrument air dryer)	17.12.2.1. Failure of shell integrity on the Air Receiver pressure vessel PV-111; Potential loss of containment with potential financial impact	61. Air Receiver pressure vessel PV-111 drain trap (and manual valve cracked open)	F	1	3	1	61. No recommendations required.	
		17.12.2.2. Potential for rust particles in air headers; Potential build-up in instrument air equipment; Potential to block off instrument air tubing to control valve users; See valve fail-state scenarios in this PHA							
17.13. Corrosion	17.13.1. Potential corrosion due to wet air	17.13.1.1. Potential increased corrosion in air compressor system, ultimate potential loss of containment and financial impact	448. No specific safeguards identified	F	2	4	3	22. Complete the existing project to replace the air compressor system with more reliable equipment, including heat tracing on equipment and building enclosure.	
17.14. Additional Phase	17.14.1. Not applicable								
17.15. Rupture/Loss of Containment	17.15.1. No additional causes identified								
17.16. Loss of Utilities	17.16.1. Loss of power to facility	17.16.1.1. Loss of pressure causing operation valves fail to safe position							
	17.16.2. Power surge	17.16.2.1. Circuit breakers trip, fuses blow; potential to damage various powered equipment; Valves will go to fail-safe position							
17.17. Start-up/Shutdown hazards/ Maintenance	17.17.1. No additional causes identified								
17.18. General	17.18.1. No additional causes identified								

Node: 18. Node 18: Rerun Manifold Header

Drawings / References:
40-PI-9022

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
18.1. Low/No Flow	18.1.1. No specific causes identified								
18.2. High Flow	18.2.1. Both Splitter units are in startup or off spec at the same time.	18.2.1.1. Potential to exceed capacity of rerun line, but has not been seen on site to date. In this scenario, both splitters would be at reduced rates (not expected to result in backup of product in splitters); Operability issue only, no hazardous consequences identified							
18.3. Reverse/Misdirected Flow	18.3.1. Startup Process - Valve mis-alignment	18.3.1.1. Flow lined up to feed tank instead of 100-14; Operability issue only, no hazardous consequences identified							
18.4. High Pressure	18.4.1. No specific causes identified								
18.5. Low Pressure	18.5.1. No specific causes identified								
18.6. High Level	18.6.1. Not applicable								
18.7. Low Level	18.7.1. Not applicable								
18.8. High Interface Level	18.8.1. Not applicable								
18.9. Low Interface Level	18.9.1. Not applicable								
18.10. High Temperature	18.10.1. High temperature product to the feed tanks (if bypassing the product coolers)	18.10.1.1. Potential to exceed the design temperature of the feed tanks (200 degF); Potential IFR roof damage; Potential financial impact and environmental impact	572. TI-11232A/B, TI-11263A/B, and TI-16055/16060 high temperature alarms on product coolers will prompt operator response	F	3	2	2	61. No recommendations required.	
18.11. Low Temperature	18.11.1. No specific causes identified								
18.12. Contamination/Composition	18.12.1. No specific causes identified								
18.13. Corrosion	18.13.1. No specific causes identified								
18.14. Additional Phase	18.14.1. Not applicable								
18.15. Rupture/Loss of Containment	18.15.1. No specific causes identified								

Node: 18. Node 18: Rerun Manifold Header

Drawings / References:
40-PI-9022

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
18.16. Loss of Utilities	18.16.1. No specific causes identified								
18.17. Start-up/Shutdown hazards/ Maintenance	18.17.1. No additional causes identified								
18.18. General	18.18.1. No additional causes identified								

Node: 19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX- 201,202,203, 204, 205)

Drawings / References:
40-PI-9204 Redline; 40-PI-9204 ; 40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9206; 40-PI-9207

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
19.1. Low/No Flow	19.1.1. 8" manual valve downstream of Condensate Feed Pumps PM-111A/B/C inadvertently closed	19.1.1.1. Blocked discharge from Condensate Feed Pumps PM-111A/B/C; Pump deadhead; Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		19.1.1.2. Loss of feed flow through the downstream exchanges; See high temperature deviation, this node							
	19.1.2. Flow control valve FCV-22425 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	19.1.2.1. Blocked discharge from Condensate Feed Pumps PM-111A/B/C; Pump deadhead; Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		19.1.2.2. Loss of feed flow through the downstream exchanges; See high temperature deviation, this node							

Node: 19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX- 201,202,203, 204, 205)

Drawings / References:

40-PI-9204 Redline; 40-PI-9204 ; 40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9206; 40-PI-9207

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	19.1.3. 8" manual valve downstream of EX-202 inadvertently closed	19.1.3.1. Blocked in flow from EX-202; Potential thermal expansion and release with potential ignition and personnel impact	675. 8" valve on feed outlet from EX-202/203 is CSO 676. PSV-20010 (set @ 305 psig) on outlet of EX-202 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
19.2. High Flow	19.2.1. Flow control valve FCV-22425 fails open or bypass valve inadvertently opened	19.2.1.1. Increased feed to Naphtha Stabilizer Column T-201, reduced temperature in T-201, potential for off-spec product, potential financial impact	652. LAH-20525 high level alarm on Stabilizer T-201 will prompt operator response	F	2	3	2	61. No recommendations required.	
19.3. Reverse/Misdirected Flow	19.3.1. Tube leak/rupture on Heat Exchanger EX-201	19.3.1.1. Shell and Tube MAWP both listed as 305 psig @ 450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Negligible downtime	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	19.3.2. Tube leak/rupture on Heat Exchanger EX-202	19.3.2.1. Shell and Tube MAWP both listed as 305 psig @ 450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Negligible downtime	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	19.3.3. Tube leak/rupture on Heat Exchanger EX-203	19.3.3.1. Shell and Tube MAWP both listed as 305 psig @ 460/450 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Inability to heat up feed enough for the unit; Potential downtime greater than 5 days	449. Spare bundles available 677. EX-203/204 tubes inspected once per year	F	2	2	1	61. No recommendations required.	
	19.3.4. Tube leak/rupture on Heat Exchanger EX-204	19.3.4.1. Shell and Tube MAWP both listed as 311 psig @ 650 degF; Feed typically flows through the tubes at ~200-250 psig, therefore no concern of tube rupture; Loss of feed to product, product quality issue; Inability to heat up feed enough for the unit; Potential downtime greater than 5 days	449. Spare bundles available 677. EX-203/204 tubes inspected once per year	F	2	2	1	61. No recommendations required.	
	19.3.5. Tube leak/rupture on Heat Exchanger EX-205	19.3.5.1. Potential to lose atmospheric gas oil (AGO) into feed, inefficient operation; Operability issue only, no hazardous consequences identified							
	19.3.6. Bypass open on Heat Exchanger EX-203 or Heat Exchanger EX-204	19.3.6.1. Loss of feed flow through the tube side of EX-203; Flow through shell side coming in at ~420 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.3.6.2. Loss of feed flow through the tube side of EX-204; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.3.6.3. Lower temperature feed to Naphtha Stabilizer Column T-201, operability issue							
		19.3.6.4. Lower temperature feed to desalter, reduced desalter efficiency and some additional water carryover to process, reduced process efficiency							

Node: 19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX- 201,202,203, 204, 205)

Drawings / References:

40-PI-9204 Redline; 40-PI-9204 ; 40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9206; 40-PI-9207

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	19.3.7. Bypass open on Bypass open on Heat Exchanger EX-201	19.3.7.1. Loss of feed flow through the tube side of EX-201; Flow through the shell side coming in at ~94 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
	19.3.8. Manual valve open on hydrocarbon drain	19.3.8.1. Potential to send feed to Hydrocarbon Sump TK-201; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
19.4. High Pressure	19.4.1. EX-202 Exchanger Isolation	19.4.1.1. Potential thermal expansion and release with potential ignition and personnel impact	675. 8" valve on feed outlet from EX-202/203 is CSO 676. PSV-20010 (set @ 305 psig) on outlet of EX-202 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
	19.4.2. EX-203 Exchanger Isolation	19.4.2.1. Potential thermal expansion and release with potential ignition and personnel impact	675. 8" valve on feed outlet from EX-202/203 is CSO 678. PSV-20020 (set @ 305 psig) on outlet of EX-203 is sized for thermal expansion	P	4	1	2	61. No recommendations required.	
	19.4.3. Exchanger plugging - Heat Exchanger EX-201	19.4.3.1. Gradual restricted flow, see low/no flow consequences in this node							
19.5. Low Pressure	19.5.1. No additional causes identified								
19.6. High Level	19.6.1. Not applicable								
19.7. Low Level	19.7.1. Not applicable								
19.8. High Interface Level	19.8.1. Not applicable								
19.9. Low Interface Level	19.9.1. Not applicable								
19.10. High Temperature	19.10.1. Low feed flow from upstream Condensate Feed Pumps PM-111A/B/C		679. PSV-20195/20215/20225	P	4	1	2	61. No recommendations required.	

Node: 19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX- 201,202,203, 204, 205)

Drawings / References:

40-PI-9204 Redline; 40-PI-9204 ; 40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9206; 40-PI-9207

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		19.10.1.1. Loss of feed to Naphtha Stabilizer Column T-201; Potential to build up temperature in T-201; Potential overpressure of T-201 with loss of containment and potential personnel impact	(set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for blocked flow						
			544. PDAH-20235 high differential pressure alarm on Stabilizer T-201 will prompt operator response						
			680. LAL-20525 low level alarm on Stabilizer T-201 will prompt operator response						
		19.10.1.2. Loss of feed to Combi Tower T-202; Increased vapor traffic; Potential operational upset							
		19.10.1.3. Loss of feed flow through the tube side of EX-201; Flow through the shell side coming in at ~94 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.10.1.4. Loss of feed flow through the tube side of EX-202; Flow through shell side coming in at ~200 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.10.1.5. Loss of feed flow through the tube side of EX-203; Flow through shell side coming in at ~420 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.10.1.6. Loss of feed flow through the tube side of EX-204; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
		19.10.1.7. Loss of feed flow through the tube side of EX-205; Flow through shell side coming in at ~570 degF, not exceeding design temperature of tubes; Operability issue only, no hazardous consequences identified							
19.11. Low Temperature	19.11.1. Low ambient temperature	19.11.1.1. No history of issues with feed lines							
	19.11.2. Exchanger fouling	19.11.2.1. Low feed temperature into Naphtha Stabilizer Column T-201, operability issue only, no hazardous consequences identified							
19.12. Contamination/ Composition	19.12.1. No additional causes identified								
19.13. Corrosion	19.13.1. No additional causes identified								
19.14. Additional Phase	19.14.1. Not applicable								

Node: 19. Node 19: Condensate feed from Condensate feed pumps to Condensate Feed Filter/Coalescer through preheaters to Naphtha Stabilizer Column (Pumps PM-111 A/B/C, filter FL-201 A/B, Heat Exchanger EX- 201,202,203, 204, 205)

Drawings / References:

40-PI-9204 Redline; 40-PI-9204 ; 40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9206; 40-PI-9207

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
19.15. Rupture/Loss of Containment	19.15.1. Flange/gasket leaks	19.15.1.1. Release of material, potential ignition if not observed in incipient stages	3. Leak detection and repair program LDAR program 455. Preventative Maintenance (PM) for flanges/gaskets	P	3	2	2	61. No recommendations required.	
19.16. Loss of Utilities	19.16.1. Instrument air failure	19.16.1.1. Valves revert to fail-safe position							
19.17. Start-up/Shutdown hazards/ Maintenance	19.17.1. Failure to purge EX-201/2/3/4 tubes with nitrogen after a shutdown	19.17.1.1. Potential oxygen ingress with potential explosive atmosphere; Potential personnel impact	458. Start-up procedure includes nitrogen purge 459. Oxygen check after nitrogen purging	P	4	1	2	61. No recommendations required.	
19.18. General	19.18.1. No additional causes identified								

Node: 20. Node 20: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-201, Heat Exchanger EX 206, 207A/B)

Drawings / References:

40-PI-9207; 40-PI-9210; 40-PI-9210 Redline; 40-PI-9214

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
20.1. Low/No Flow	20.1.1. Level control valve LCV-20525 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	20.1.1.1. High level in Naphtha Stabilizer Column T-201; Potential carryover to Recycle Gas Compressors; Potential carryover from T-201 Condenser and downstream into Accumulator and Spheres; Potential off-spec product resulting in need to send product back and re-process; Potential roof failure and release of vapors with potential permit exceedance; Potential financial impact	763. PDI-20235 Differential Pressure Alarm will prompt operator response on high and high-high	F	4	3	4	1. Perform an engineering evaluation to provide additional safeguards for the scenario of high level in Naphtha Stabilizer Column T-101/201 resulting in potential carryover to spheres as well as loss of reboiler duty and tray damage, leading to financial impact.	
		20.1.1.2. Loss of reboiler duty; Potential tray damage in T-201 with potential financial impact and downtime	763. PDI-20235 Differential Pressure Alarm will prompt operator response on high and high-high	F	3	3	3	1. Perform an engineering evaluation to provide additional safeguards for the scenario of high level in Naphtha Stabilizer Column T-101/201 resulting in potential carryover to spheres as well as loss of reboiler duty and tray damage, leading to financial impact.	
		20.1.1.3. Loss of feed to Combi Tower T-202; Increased vapor traffic; Potential operational upset							
20.2. High Flow	20.2.1. Level control valve LCV-20525 fails open or bypass valve inadvertently opened	20.2.1.1. Loss of level of Naphtha Stabilizer Column T-201, potential blow by of vapors at 90 psig to Combi Tower T-202 rated at 50 psig; Potential overpressure between 1.5-1.8x MAWP of T-202 - potential release through gaskets; Potential ignition and personnel impact Considered a low initial likelihood due to the need to overcome a significant liquid head pressure to result in a gas blowby case	597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response	P	3	2	2	61. No recommendations required.	

Node: 20. Node 20: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-201, Heat Exchanger EX 206, 207A/B)

Drawings / References:

40-PI-9207; 40-PI-9210; 40-PI-9210 Redline; 40-PI-9214

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
20.3. Reverse/Misdirected Flow	20.3.1. Tube leak/rupture in Heat exchanger EX-206	20.3.1.1. Potential leak of hot oil into the Combi Tower feed; Loss of hot oil; Operability issue only, no hazardous consequences identified							
	20.3.2. Manual valve open on hydrocarbon drain	20.3.2.1. Potential to send feed to Hydrocarbon Sump TK-201; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
20.4. High Pressure	20.4.1. No additional causes identified								
20.5. Low Pressure	20.5.1. No additional causes identified								
20.6. High Level	20.6.1. No additional causes identified								
20.7. Low Level	20.7.1. No additional causes identified								
20.8. High Interface Level	20.8.1. Not applicable								
20.9. Low Interface Level	20.9.1. Not applicable								
20.10. High Temperature	20.10.1. TIC-20265 malfunctions	20.10.1.1. Increased hot oil flow through EX-206; Increased temperature in T-201; Potential to build up temperature in T-201; Potential increased vapor traffic; Potential overpressure of T-201; Potential to damage trays in T-201 with financial impact	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 598. TAH-20310 high temperature alarm on overheads from Stabilizer T-201 will prompt operator response	F	3	2	2	61. No recommendations required.	
	20.10.2. External fire around T-201 Naphtha Stabilizer Column	20.10.2.1. The vessel is above surface that is sloped away towards the hydrocarbon drain header and the skirt is fire-proofed, therefore accumulation of flammable liquids and external fire is not considered credible							

Node: 20. Node 20: Naphtha Stabilizer Column, Naphtha Stabilizer Reboiler, Combi Tower Feed Preheater to Combi Tower (T-201, Heat Exchanger EX 206, 207A/B)

Drawings / References:

40-PI-9207; 40-PI-9210; 40-PI-9210 Redline; 40-PI-9214

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
20.11. Low Temperature	20.11.1. TIC-20265 malfunctions	20.11.1.1. Decreased hot oil flow through EX-206; Decreased temperature in T-201; Potential off-spec product; Operability issue only, no hazardous consequences identified							
20.12. Contamination/ Composition	20.12.1. No additional causes identified								
20.13. Corrosion	20.13.1. Corrosion of T-201 Stabilizer Column Overheads	20.13.1.1. Potential loss of containment from T-201 with potential ignition and personnel impact	681. T-101/201 inspections 2x/year 463. T-101/201 AUT scan every 6 months	P	3	2	2	61. No recommendations required.	
20.14. Additional Phase	20.14.1. Not applicable								
20.15. Rupture/Loss of Containment	20.15.1. Flange/gasket leaks	20.15.1.1. Release of material, potential ignition if not observed in incipient stages	3. Leak detection and repair program LDAR program 455. Preventative Maintenance (PM) for flanges/gaskets	P	3	2	2	61. No recommendations required.	
20.16. Loss of Utilities	20.16.1. Loss of instrument air	20.16.1.1. Actuated valves will go to fail-state position							
20.17. Start-up/Shutdown hazards/ Maintenance	20.17.1. Tube leak/rupture on Heat exchanger EX-207A/B (hot oil pumps shutdown)	20.17.1.1. Potential to contaminate hot oil with product; Potential coking of Hot Oil Heater F-201 tubes over time with potential tube failure; Potential financial impact	464. Monthly hot oil sampling 682. LAH-21965 High level alarm on Hot Oil Surge Tank PV-203 will prompt operator response	F	3	2	2	61. No recommendations required.	
20.18. General	20.18.1. No additional causes identified								

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
21.1. Low/No Flow	21.1.1. Naphtha Stabilizer Overhead Pump PM-201A/B trips	21.1.1.1. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-201, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	683. FAL-20400 Low flow alarm on reflux flow to Stabilizer T-201 will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water	F	5	1	2	61. No recommendations required.	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			boot will prompt operator response 8. Standby pump, remote startup from control room 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response						
		21.1.1.2. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-201; Increased temperature in T-201; Potential to build up temperature in T-201; Potential increased vapor traffic; Potential overpressure of T-201; Potential to damage trays in T-201 with financial impact	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 598. TAH-20310 high temperature alarm on overheads from Stabilizer T-201 will prompt operator response	F	3	2	2	61. No recommendations required.	
	21.1.2. Manual valve on suction line to Naphtha Stabilizer Overhead Pump PM-201A/B inadvertently closed	21.1.2.1. Loss of suction to Naphtha Stabilizer Overhead Pump PM-201A/B; Potential to cavitate pumps with potential seal damage with potential loss of containment, potential ignition with personnel impact	686. Dual seals on pumps PM-201A/B 687. Seal failure alarm SFA-20415/20440 will shutdown pumps PM-201A/B 627. Lower Explosive Limit LEL alarms AI-10416/10441/20416/20441 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		21.1.2.2. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-201, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for	683. FAL-20400 Low flow alarm on reflux flow to Stabilizer T-201 will prompt operator response	F	5	1	2	61. No recommendations required.	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response						
		21.1.2.3. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-201; Increased temperature in T-201; Potential to build up temperature in T-201; Potential increased vapor traffic; Potential overpressure of T-201; Potential to damage trays in T-201 with financial impact	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 598. TAH-20310 high temperature alarm on overheads from Stabilizer T-201 will prompt operator response	F	3	2	2	61. No recommendations required.	
	21.1.3. Manual valve on discharge line from Naphtha Stabilizer Overhead Pump PM-201A/B inadvertently closed	21.1.3.1. Blocked discharge from pumps with potential to deadhead pumps; Pump max discharge pressure is 55 psig and downstream piping is rated for up to 285 psig, therefore no overpressure concern; Potential seal damage with potential loss of containment, potential ignition with personnel impact	686. Dual seals on pumps PM-201A/B 687. Seal failure alarm SFA-20415/20440 will shutdown pumps PM-201A/B 627. Lower Explosive Limit LEL alarms AI-10416/10441/20416/20441 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		21.1.3.2. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-201, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-209	683. FAL-20400 Low flow alarm on reflux flow to Stabilizer T-201	F	5	1	2	61. No recommendations required.	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response						
		21.1.3.3. Low/no reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-201; Increased temperature in T-201; Potential to build up temperature in T-201; Potential increased vapor traffic; Potential overpressure of T-201; Potential to damage trays in T-201 with financial impact	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 598. TAH-20310 high temperature alarm on overheads from Stabilizer T-201 will prompt operator response	F	3	2	2	61. No recommendations required.	
	21.1.4. FCV-20400 malfunctions closed or upstream/downstream valve(s) inadvertently closed	21.1.4.1. No reflux flow resulting in loss of cooling in Naphtha Stabilizer Column T-201; Increased temperature in T-201; Potential to build up temperature in T-201; Potential increased vapor traffic; Potential overpressure of T-201; Potential to damage trays in T-201 with financial impact	599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling 542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response	F	3	1	1	61. No recommendations required.	
	21.1.5. LCV-20350 malfunctions closed or upstream/downstream valve(s) inadvertently closed	21.1.5.1. Loss of flow of Y-Grade to storage, high level in Naphtha Stabilizer Accumulator pressure vessel PV-201, potential to overfill Recycle Compressor Discharge Scrubber pressure vessel PV-209	683. FAL-20400 Low flow alarm on reflux flow to Stabilizer T-201	F	5	1	2	61. No recommendations required.	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	will prompt operator response 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response						
21.2. High Flow	21.2.1. LCV-20350 on Y-Grade product line fails open or bypass inadvertently opened	21.2.1.1. Loss of level in Naphtha Stabilizer Accumulator pressure vessel PV-201, potential to run Naphtha Stabilizer Overhead Pump PM-201A/B dry, pump cavitation and damage, potential release with personnel impact	688. LALL-20365 low-low level alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 trips Naphtha Stabilizer Overhead Pumps PM-201A/B 686. Dual seals on pumps PM-201A/B 687. Seal failure alarm SFA-20415/20440 will shutdown pumps PM-201A/B	P	4	1	2	61. No recommendations required.	
	21.2.2. FCV-20400 fails open or bypass inadvertently opened on reflux line	21.2.2.1. Increased flow of reflux to Naphtha Stabilizer Column T-201; Increased liquid traffic; Potential to flood T-201, increasing deltaP, restricting vapor traffic; Potential to damage the T-201 trays with financial impact	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 544. PDAH-20235 high differential pressure alarm on Stabilizer T-201 will prompt operator response	F	2	2	1	61. No recommendations required.	
		21.2.2.2. Increased flow of reflux to Naphtha Stabilizer Column T-201; Increased liquid traffic; Potential increased pressure to the Combi Tower T-202 and increased gas volume to the RGC; Potential increased backpressure on the RGC resulting in RGC shutdown leading to flaring and potential environmental impact (unplanned flaring event)	542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 544. PDAH-20235 high differential pressure alarm on Stabilizer T-	F	1	2	1	3. Evaluate the engineering studies on flaring emissions for the Splitters.	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			201 will prompt operator response						
21.3. Reverse/Misdirected Flow	21.3.1. Inadvertent opening of Y-Grade rerun valve	21.3.1.1. Excessive Y-Grade to Feed Tank (Tk-200-1/2/3 & TK-100-14); Potential floating roof damage and release of light ends, ignition, personnel impact.	467. Lower explosive limit LEL alarm AT-9101,9102,9103,9104; AT-9301,9302,9393,9304 at Feed Tanks and operator response to cordon off area 468. Administrative procedures give direction on how to send Y-Grade to Re-Run	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	21.3.2. Manual valve open on hydrocarbon drain	21.3.2.1. Potential to send feed to Hydrocarbon Sump TK-201; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
21.4. High Pressure	21.4.1. Naphtha Stabilizer Condenser AF-201-1/2 fin Fan motor mechanical failure/ throws a blade	21.4.1.1. Loss of individual Naphtha Stabilizer Condenser AF-201-1/2 fin Fan, slow pressurization of Naphtha Stabilizer Column T-201, potential overpressure and loss of containment, ignition and personnel impact	689. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 (sized for electric power failure leading to loss of cooling) 690. XI-20315A/B Condenser Run Alarm will prompt operator response 691. VAH-20316A/B and VAH-20331A/B high vibration alarm will prompt operator response	P	4	1	2	61. No recommendations required.	
	21.4.2. PCV-20395B malfunctions open or bypass valve inadvertently open	21.4.2.1. Inadvertently sending hot vapors to Accumulator PV-201; Potential slow pressurization of Naphtha Stabilizer Column T-201,	689. PSV-20195/20215/20225 (set @ 126, 123, and	P	4	2	3	4. Provide an additional safeguard for the scenario of PCV-10395B/20395B malfunctioning open sending hot vapors	

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		potential overpressure and loss of containment, ignition and personnel impact	123 psig respectively) on Stabilizer T-201 (sized for electric power failure leading to loss of cooling)					to Accumulator PV-101/201 resulting in potential overpressure of Stabilizer Column T-101/201.	
21.5. Low Pressure	21.5.1. No additional causes identified								
21.6. High Level	21.6.1. LCV-20380 fails closed or manual valve(s) inadvertently closed	21.6.1.1. Water carryover into the reflux and Y-grade storage, loss of fractionation, operability issues							
21.7. Low Level	21.7.1. LCV-20380 malfunctions open or bypass inadvertently opened	21.7.1.1. Excessive hydrocarbon in the waste water system (Tank 5), Potential floating roof damage and release of light ends, ignition, personnel impact.	663. Lower explosive limit alarms AI-9110A/B/C at Tank TK-5 will prompt operators to cordon off the area	P	4	3	4	5. Provide additional safeguards for the scenario of the PV-101/201 Accumulator Water Boot LCV malfunctioning open leading to excessive hydrocarbons in Tank 5 and floating roof damage with release and ignition. 60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
21.8. High Interface Level	21.8.1. Not applicable								
21.9. Low Interface Level	21.9.1. Not applicable								
21.10. High Temperature	21.10.1. External fire around PV-201 Naphtha Stabilizer Accumulator	21.10.1.1. The vessel is located on a grated surface and is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
21.11. Low Temperature	21.11.1. No additional causes identified								
21.12. Contamination/ Composition	21.12.1. Elevated chlorides	21.12.1.1. Excess corrosion with potential loss of containment and potential ignition	13. Relief valves and rupture disks designed for chlorides 471. Mechanical integrity program 14. Corrosion inhibitor injection	P	4	2	3	6. Perform engineering evaluation to ensure corrosion monitoring program is adequate and properly implemented throughout the Naphtha Stabilizer overhead system to monitor for potential corrosion due to elevated chlorides.	
		21.12.1.2. Deposits causing flow restrictions, see consequences related to low/no flow in this node							
21.13. Corrosion	21.13.1. No additional causes identified								
21.14. Additional Phase	21.14.1. Not applicable								
21.15. Rupture/Loss of Containment	21.15.1. No additional causes identified								

Node: 21. Node 21: Naphtha Stabilizer Column overhead system: Naphtha Stabilizer Condenser, Accumulator, Pumps; and Y-Grade to storage (AF-201, PV-201, PM-201A/B)

Drawings / References:

40-PI-9207; 40-PI-9208; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9038

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
21.16. Loss of Utilities	21.16.1. Power loss due to lightning storm results in loss of control and off-spec product	21.16.1.1. Off-spec product - unable to divert Y-Grade and heavier material from storage to rerun during lightning storm (lowers ability to safely access valves during storm/lightning alert and heavier product needed to balance Y-Grade vapor pressure). Potential financial impact	472. Lightning arrestors 473. Lightning dissipaters	F	2	3	2	2. Evaluate the need to automate the valves on product lines to the rerun line. This will allow for faster and safer switchover to and from rerun line from control room and eliminate the need for operations to manually manipulate valves during lightning storms and severe weather events.	
		21.16.1.2. Off-spec product requires opening manual valves to divert material to rerun line with potential personnel impact during lightning in the area	473. Lightning dissipaters 472. Lightning arrestors	P	4	3	4	2. Evaluate the need to automate the valves on product lines to the rerun line. This will allow for faster and safer switchover to and from rerun line from control room and eliminate the need for operations to manually manipulate valves during lightning storms and severe weather events.	
	21.16.2. Localized loss of power	21.16.2.1. See PM-201A/B pump trip scenario in this node							
		21.16.2.2. See AF-201 fan failure scenario in this node							
	21.16.3. Loss of instrument air	21.16.3.1. Valves revert to fail-safe position							
21.17. Start-up/Shutdown hazards/ Maintenance	21.17.1. Cooling during Naphtha Stabilizer Column T-201 shutdown	21.17.1.1. Potential for vacuum as Naphtha Stabilizer Column T-201 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
	21.17.2. Naphtha Stabilizer Condenser AF-201-1/2 fin fan access for maintenance or cleaning	21.17.2.1. Currently no means for personnel to tie off for fall protection, potential fall hazard, personnel impact	448. No specific safeguards identified	P	3	3	3	8. Evaluate the need to provide means for personnel to access and tie-off for fall protection during maintenance and cleaning activities for the Naphtha Stabilizer Condenser AF-101/201-1/2 fin fans.	
21.18. General	21.18.1. No additional causes identified								

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
22.1. Low/No Flow	22.1.1. Combi Tower Bottoms Product Pump PM-204A/B trips	22.1.1.1. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	692. Second Combi Tower Bottoms Product Pump PM-204A/B available for remote startup 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
	22.1.2. Manual valve on suction to Combi Tower Bottoms Product Pump PM-204A/B inadvertently closed	22.1.2.1. Loss of suction to PM-204A/B; Potential to cavitate pump resulting in pump damage; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	22.1.3. Manual valve on discharge from Combi Tower Bottoms Product Pump PM-204A/B inadvertently closed	22.1.3.1. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			operators to cordon off the area						
		22.1.3.2. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
	22.1.4. 6" manual valve from EX-205 inadvertently closed	22.1.4.1. Product flow out from the shell side of EX-205 blocked; No overpressure concerns, as the max pressure from the upstream pumps PM-204A/B is 104 psig, and the shell side is MAWP 305 psig; High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		22.1.4.2. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			11461/11481/21461/21481 will prompt operators to cordon off the area						
	22.1.5. TCV-28320B malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	22.1.5.1. Inadvertently sending all flow through EX-212A/B; Potential for higher temperature wash water to desalter; Potential wash water steaming; Operability issue only, no hazardous consequences identified							
		22.1.5.2. Potential to back-up flow to Combi Tower; High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
	22.1.6. TCV-28320A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	22.1.6.1. No longer heating up the wash water for the Desalter using EX-212A/B; Potential to send colder temperature wash water to the Desalter; Potential cold Desalter temperature; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	22.1.7. 6" manual valve on inlet to AF-203 Bottoms Product Cooler inadvertently closed	22.1.7.1. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-	F	1	1	1	61. No recommendations required.	

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			204A/B will prompt operator response						
		22.1.7.2. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	22.1.8. 6" manual valve on outlet from AF-203 Bottoms Product Cooler inadvertently closed	22.1.8.1. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		22.1.8.2. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	22.1.9. TCV-21240A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	22.1.9.1. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		22.1.9.2. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	22.1.10. LCV-21740 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	22.1.10.1. High level in Combi Tower T-202; Potential to overfill T-202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	2	1	61. No recommendations required.	
			546. Dual seals on PM-204A/B	P	4	1	2		

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		22.1.10.2. Blocked discharge from pumps PM-204A/B; Potential to deadhead pumps; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	22.1.11. FCV-25010 fails closed or upstream/downstream manual valve(s) inadvertently closed	22.1.11.1. No consequences, this valve is normally kept closed							
22.2. High Flow	22.2.1. TCV-28320A malfunctions open or bypass inadvertently open	22.2.1.1. Potential for higher temperature wash water to desalter; Operability issue only, no hazardous consequences identified							
	22.2.2. TCV-21240B malfunctions open or bypass inadvertently open	22.2.2.1. Inadvertently bypass flow around AF-203 Bottoms Product Cooler; Potential to send higher temperature product downstream; If downstream HX-1/2 exchangers are offline, potential to send higher temperature (up to ~ 380 degF) product to the storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	448. No specific safeguards identified	F	3	2	2	61. No recommendations required.	1. Considered a low initial likelihood, as downstream HX-1/2 exchangers are rarely offline.
	22.2.3. LCV-21740 fails open or bypass inadvertently open	22.2.3.1. High flow results in low level in Combi Tower T-202; Potential loss of suction to Bottoms Pumps PM-204A/B; Potential to cavitate pump resulting in pump damage; Potential seal failure with potential loss of containment; Potential ignition and personnel impact	546. Dual seals on PM-204A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 694. LAL-21740 low level alarm on T-202 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	22.2.4. FCV-25010 malfunctions open or bypass inadvertently opened	22.2.4.1. Potential for colder temperature from EX-207; Operability issue only, no hazardous consequences identified							
22.3. Reverse/Misdirected Flow	22.3.1. 6" manual valve on bypass around EX-205 shell side inadvertently opened	22.3.1.1. Inadvertently partially bypassing product flow around EX-205; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; No damage expected to							

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Storage Tank or environmental impacts expected; Operability issue only, no hazardous consequences identified							
		22.3.1.2. Potential reduced ability to heat the feed in EX-205; Potential for low feed temperature into Naphtha Stabilizer Column T-201; Potential reduced rates; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
	22.3.2. 2" manual valve on AF-203 flush to EX-207 inadvertently opened	22.3.2.1. Potential for colder temperature from EX-207; Operability issue only, no hazardous consequences identified							
	22.3.3. 2" manual valve on diesel flush line to AF-203 inadvertently opened	22.3.3.1. Sending diesel to AF-203 inadvertently would require multiple 2" manual valves to be opened, which is not credible							
22.4. High Pressure	22.4.1. Bottoms Product Cooler AF-203-1/2 fin fan motor mechanical failure/throws a blade	22.4.1.1. Loss of individual AF-203-1/2 fin fan; Potential to send higher temperature product downstream; If downstream HX-1/2 exchangers are offline, potential to send higher temperature (up to ~ 380 degF) product to the storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	448. No specific safeguards identified	F	3	2	2	61. No recommendations required.	1. Considered a low initial likelihood, as downstream HX-1/2 exchangers are rarely offline.
22.5. Low Pressure	22.5.1. Too many fin fans in AF-203 running in winter	22.5.1.1. Potential to excessively cool the bottoms with potential to result in the lines waxing up and plugging; Potential restricted flow; See no/low flow consequences in this node							
22.6. High Level	22.6.1. No additional causes identified								
22.7. Low Level	22.7.1. No additional causes identified								
22.8. High Interface Level	22.8.1. Not applicable								
22.9. Low Interface Level	22.9.1. Not applicable								
22.10. High Temperature	22.10.1. External fire around T-202 Combi Tower	22.10.1.1. The vessel is above surface that is sloped away towards the hydrocarbon drain header and the skirt is fire-proofed, therefore accumulation of flammable liquids and external fire is not considered credible							
22.11. Low Temperature	22.11.1. Ambient conditions below freezing	22.11.1.1. No known issues identified							
22.12. Contamination/ Composition	22.12.1. No additional causes identified								
22.13. Corrosion	22.13.1. Salt build-up	22.13.1.1. Deposits causing flow restrictions, see consequences related to low/no flow in this node							
22.14. Additional Phase	22.14.1. Not applicable								
22.15. Rupture/Loss of Containment	22.15.1. No additional causes identified								

Node: 22. Node 22: Combi Tower, Combi Tower Bottoms Pump, Bottoms to Feed Preheater, Bottoms Product to Desalter Was Water Heater, Bottom Product Cooler and Combi Tower Bottoms Product to Storage (T-202, PM-204A/B, AF-203, Heat Exchanger EX-205, Heat Exchanger EX-212A/B)

Drawings / References:

40-PI-9059A; 40-PI-9206; 40-PI-9207; 40-PI-9214; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9225; 40-PI-9226

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
22.16. Loss of Utilities	22.16.1. Localized loss of power	22.16.1.1. See PM-204A/B pump trip scenario in this node							
		22.16.1.2. See AF-203 fan failure scenario in this node							
	22.16.2. Loss of instrument air	22.16.2.1. Valves revert to fail-safe position							
22.17. Start-up/Shutdown hazards/ Maintenance	22.17.1. LT-21740 maintenance	22.17.1.1. Personnel exposure to LT-21740 during maintenance activities (radiation)	476. Radiation Safety Plan	P	1	3	1	61. No recommendations required.	
	22.17.2. Cooling during Combi Tower T-202 shutdown	22.17.2.1. Potential for vacuum as Combi Tower T-203 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
22.18. General	22.18.1. No additional causes identified								

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
23.1. Low/No Flow	23.1.1. Combi Tower Overhead Pumps PM-203A/B trip	23.1.1.1. Inability to send liquids out of PV-202 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	695. Second Combi Tower Overhead Pump PM-203A/B available for remote startup 696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response	F	1	2	1	61. No recommendations required.	
		23.1.1.2. Loss of reflux flow resulting in loss of cooling in Combi Tower T-202, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi	P	4	1	2	61. No recommendations required.	

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Tower T-202 is sized for loss of cooling 695. Second Combi Tower Overhead Pump PM-203A/B available for remote startup 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response						
	23.1.2. 12" manual valve on inlet to Combi Tower Overhead Pumps PM-203A/B inadvertently closed	23.1.2.1. Loss of suction to Combi Tower Overhead Pumps PM-203A/B; Potential to cavitate pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	700. Seal failure alarm SF-21360/21385 will shutdown pumps PM-203A/B 699. Dual seals on PM-203A/B	P	4	1	2	61. No recommendations required.	
		23.1.2.2. Inability to send liquids out of PV-202 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	1	1	61. No recommendations required.	
		23.1.2.3. Loss of reflux flow resulting in loss of cooling in Combi Tower T-202, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response	P	4	1	2	61. No recommendations required.	
	23.1.3. 10" manual valve on outlet from Combi Tower Overhead Pumps PM-203A/B inadvertently closed	23.1.3.1. Blocked discharge from Combi Tower Overhead Pumps PM-203A/B; Potential to deadhead pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	700. Seal failure alarm SF-21360/21385 will shutdown pumps PM-203A/B 699. Dual seals on PM-203A/B	P	4	1	2	61. No recommendations required.	

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		23.1.3.2. Inability to send liquids out of PV-202 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	1	1	61. No recommendations required.	
		23.1.3.3. Loss of reflux flow resulting in loss of cooling in Combi Tower T-202, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response	P	4	1	2	61. No recommendations required.	
	23.1.4. FCV-21345 reflux control valve fails closed or upstream/downstream manual valve(s) inadvertently closed	23.1.4.1. Loss of reflux flow resulting in loss of cooling in Combi Tower T-202, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response	P	4	1	2	61. No recommendations required.	
	23.1.5. 6" manual valve on shell side outlet from EX-201 inadvertently closed	23.1.5.1. Potential to build up liquid level in PV-202 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	1	1	61. No recommendations required.	
		23.1.5.2. Loss of light naphtha product to storage; Potential financial impact	696. LAH-21320 high level alarm on PV-202	F	1	2	1	61. No recommendations required.	

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Accumulator will prompt operator response						
	23.1.6. LCV-21320 fails closed or upstream/downstream manual valve(s) inadvertently closed	23.1.6.1. Potential to build up liquid level in PV-202 Accumulator; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	3	1	61. No recommendations required.	
		23.1.6.2. Loss of light naphtha product to storage; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
23.2. High Flow	23.2.1. FCV-21345 reflux valve fails open or bypass valve open	23.2.1.1. Excess reflux sent to Combi Tower T-202, potential increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response	P	4	1	2	61. No recommendations required.	
	23.2.2. Level control valve LCV-21320 light naphtha to storage fails open or bypass opened	23.2.2.1. Potential loss of level in Accumulator PV-202; Loss of suction to Combi Tower Overhead Pumps PM-203A/B; Potential to cavitate pumps resulting in potential seal failure, loss of containment, potential ignition and personnel impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 700. Seal failure alarm SF-21360/21385 will shutdown pumps PM-203A/B 699. Dual seals on PM-203A/B	P	4	1	2	61. No recommendations required.	
		23.2.2.2. Potential loss of level in Accumulator PV-202; Loss of reflux to Combi Tower T-202; Loss of reflux flow resulting in loss of cooling in Combi Tower T-202, eventual increase in pressure, and potential to overpressure column, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response	P	4	1	2	61. No recommendations required.	
23.3. Reverse/Misdirected Flow	23.3.1. PCV-21315B malfunctions open or bypass inadvertently open	23.3.1.1. Inadvertently route some of the flow around the AF-202 Condensers (most of the flow would still go through the	701. PSV-21690/21710/21720	P	4	1	2	61. No recommendations required.	

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Condensers); Potential to send hot vapors to Accumulator PV-202 (would not exceed PV-202 design temperature); RCG would shutdown on high suction pressure, and the hot vapors would be unable to flow to the Hot Oil Furnace F-201; Potential to back-up pressure to the Combi Tower T-202; Potential overpressure of T-202 with loss of containment and personnel impact	(set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet						
			702. PAHH-21715 (2oo3) high-high pressure interlock on T-202 will trip heater F-201A/B Ammonia Skid and Hot Oil Pumps and will prompt operator response						
	23.3.2. 6" manual bypass around EX-201 opened	23.3.2.1. This valve is regularly open; No adverse consequences identified							
23.4. High Pressure	23.4.1. No additional causes identified								
23.5. Low Pressure	23.5.1. No additional causes identified								
23.6. High Level	23.6.1. Failure to drain water boot for PV-202 Accumulator	23.6.1.1. Potential water carryover into the reflux and Light Naphtha storage; Potential reduced reflux efficiency; Potential increased moisture in system; Potential corrosion; Potential loss of fractionation; Reduced efficiency; Operability issue only, no hazardous consequences identified							
23.7. Low Level	23.7.1. Improper drainage of water boot for PV-202 Accumulator	23.7.1.1. Potential to send light naphtha to Hydrocarbon Sump TK-201; Potential inefficient operation; Potential to overfill sump; Potential environmental impact; Potential release of hydrocarbon vapors; Potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
23.8. High Interface Level	23.8.1. Not applicable								
23.9. Low Interface Level	23.9.1. Not applicable								
23.10. High Temperature	23.10.1. AF-202 Combi Tower Condenser fin fan trip/failure	23.10.1.1. A loss of a fin fan for AF-202 would not result in a significant high temperature consequence downstream; Operability issue only, no hazardous consequences identified							

Node: 23. Node 23: Combi Tower Overhead System: Combi Tower Condenser, Accumulator, Overhead Pumps, Light Naphtha Product Cooler and Light Naphtha to Storage (AF-202-1/12, PV-202, PM-203 A/B, Heat Exchanger EX-201) to Tanks 24/25

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9215; 40-PI-9216; 40-PI-9217; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9222 ; 40-PI-9222 Redline

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	23.10.2. External fire around PV-202 Accumulator	23.10.2.1. The vessel is located on a grated surface and is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
23.11. Low Temperature	23.11.1. No additional causes identified								
23.12. Contamination/ Composition	23.12.1. No additional causes identified								
23.13. Corrosion	23.13.1. No additional causes identified								
23.14. Additional Phase	23.14.1. Not applicable								
23.15. Rupture/Loss of Containment	23.15.1. No additional causes identified								
23.16. Loss of Utilities	23.16.1. Power failure	23.16.1.1. See PM-203A/B pump trip scenario in this node							
	23.16.2. Loss of instrument air	23.16.2.1. Valves revert to fail-safe position							
23.17. Start-up/Shutdown hazards/ Maintenance	23.17.1. H2S in the LVN	23.17.1.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 109. H2S Scavenger requires good mixing to reduce the H2S present in the LVN tank 606. AI-20499/21699 (Splitter 2) Fixed H2S monitors will alert in the control room 607. AI-10499/11699 (Splitter 1) Fixed H2S monitors will alert in the control room	P	2	2	1	61. No recommendations required.	
		23.17.1.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
23.18. General	23.18.1. No additional causes identified								

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
24.1. Low/No Flow	24.1.1. Heavy Naphtha Stripper Product Pump PM-205B trips	24.1.1.1. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response	F	1	3	1	61. No recommendations required.	
		24.1.1.2. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. Confirmed that LELs AI-11461/11481/21461/21481 are located within 15 feet of T-102 and T-202.
	24.1.2. 6" manual valve on suction to Heavy Naphtha Stripper Product Pump PM-205B inadvertently closed	24.1.2.1. Blocked suction to PM-205B; Potential pump cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		24.1.2.2. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response	F	1	2	1	61. No recommendations required.	
		24.1.2.3. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	24.1.3. 6" manual valve on discharge from Heavy Naphtha Stripper Product Pump PM-205B inadvertently closed	24.1.3.1. Blocked discharge from PM-205B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		24.1.3.2. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response	F	1	2	1	61. No recommendations required.	
		24.1.3.3. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	24.1.4. 6" manual valve on EX-202 shell side outlet inadvertently closed	24.1.4.1. Blocked discharge from PM-205B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		24.1.4.2. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response	F	1	2	1	61. No recommendations required.	
		24.1.4.3. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	24.1.5. FCV-21191 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	24.1.5.1. Blocked discharge from PM-205B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area 485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B	P	4	2	3	9. Provide an additional safeguard for the scenario of FCV-11191/21191 malfunctioning closed resulting in either: 1) Pump PM-105B/205B deadhead, with seal failure and loss of containment or 2) Overfill of Naphtha Stripper T-103/203 liquids back to Combi Tower T-102/202 with possible loss of containment. (Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop). 60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		24.1.5.2. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response	P	4	2	3	9. Provide an additional safeguard for the scenario of FCV-11191/21191 malfunctioning closed resulting in either: 1) Pump PM-105B/205B deadhead, with seal failure and loss of containment or 2) Overfill of Naphtha Stripper T-103/203 liquids back to Combi Tower T-102/202 with possible loss of containment.	

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								(Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop).	
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		24.1.5.3. High level in Heavy Naphtha Stripper T-203, potential to overfill T-203 back to Combi Tower T-202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	24.1.6. 6" manual valve on product to re-run line inadvertently closed	24.1.6.1. This valve is normally closed; No adverse consequences identified							
	24.1.7. LCV-21595 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	24.1.7.1. Potential Combi Tower T-202 flooding; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			672. FAL-21191 Low Flow Alarm downstream of AF-205 Heavy Naphtha Cooler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		24.1.7.2. Potential Combi Tower T-202 flooding; leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	672. FAL-21191 Low Flow Alarm downstream of AF-205 Heavy Naphtha Cooler will prompt operator response	F	1	2	1	61. No recommendations required.	1. LAH/LAL-21595 is not tied to LCV-21595.
			673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response						
	24.1.8. T-202 Combi Tower tray pluggage	24.1.8.1. Potential Combi Tower T-202 flooding; high differential pressure; Potential to get heavy naphtha liquids in the vapor return	566. PDAH-21735 high differential pressure alarm on Combi Tower	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	T-202 will prompt operator response						
			672. FAL-21191 Low Flow Alarm downstream of AF-205 Heavy Naphtha Cooler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		24.1.8.2. Potential Combi Tower T-202 flooding; leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response	F	1	3	1	61. No recommendations required.	
24.2. High Flow	24.2.1. 6" manual valve on bypass around EX-202 shell side inadvertently opened	24.2.1.1. Reduced preheat for feed flow (at most a 10 degF difference); Operability issue only, no hazardous consequences identified							
	24.2.2. 6" manual valve on product to re-run line inadvertently open	24.2.2.1. Potential reduced flow of heavy naphtha product to storage tanks; Negligible financial loss							
	24.2.3. LCV-21595 fails open or bypass inadvertently open	24.2.3.1. Potential to draw kerosene and diesel into the heavy naphtha draw off line; Potential to dry out the tray; Potential off-spec product/quality issues; Potential financial impact	703. TAH-21617 high temperature alarm on EX-208 reboiler	F	1	3	1	61. No recommendations required.	
	24.2.4. FCV-21191 fails open or bypass valve inadvertently open	24.2.4.1. Potential to lose level in Stripper T-103; Potential pump PM-105B cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	2	3	10. Provide an additional safeguard for the scenario of FCV-11191/21191 failing open resulting in loss of level in Stripper T-103/203 and pump PM-105B/205B cavitation, with seal failure and possible loss of containment. (Note - LIC-11595/21595 on Stripper T-103/203 is tied to the FCV-11191/21191 control loop).	1. LAH/LAL-11595/21595 on Stripper T-103/203 is tied to the FIC-11191 control loop.
			485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B						
24.3. Reverse/Misdirected Flow	24.3.1. No additional causes identified								
24.4. High Pressure	24.4.1. External fire around T-203 Heavy Naphtha Stripper	24.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
24.5. Low Pressure	24.5.1. No additional causes identified								

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
24.6. High Level	24.6.1. No additional causes identified								
24.7. Low Level	24.7.1. No additional causes identified								
24.8. High Interface Level	24.8.1. Not applicable								
24.9. Low Interface Level	24.9.1. Not applicable								
24.10. High Temperature	24.10.1. Heavy Naphtha Product Cooler AF-205 fan trips	24.10.1.1. Higher temperature heavy naphtha to storage, minimal consequences at storage tanks - not exceeding storage tank design temperature; Negligible increased venting; Operability issue only, no hazardous consequences identified							
24.11. Low Temperature	24.11.1. No additional causes identified								
24.12. Contamination/ Composition	24.12.1. No additional causes identified								
24.13. Corrosion	24.13.1. No additional causes identified								
24.14. Additional Phase	24.14.1. Not applicable								
24.15. Rupture/Loss of Containment	24.15.1. No additional causes identified								
24.16. Loss of Utilities	24.16.1. Loss of power	24.16.1.1. See AF-205 fan failure scenario, this node							
		24.16.1.2. See See PM-205B pump trip scenario, this node							
	24.16.2. Loss of instrument air	24.16.2.1. Valves revert to fail-safe position							
24.17. Start-up/Shutdown hazards/ Maintenance	24.17.1. Cooling during Heavy Naphtha Stripper T-203 shutdown	24.17.1.1. Potential for vacuum as Heavy Naphtha Stripper T-203 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid	

Node: 24. Node 24: Heavy Naphtha Side Draw From T-202 to Heavy Naphtha Stripper (T-203), Stripper Reboiler, and Heavy Naphtha Stripper Bottoms Pump to Feed Preheater and Product Cooler to Storage; Overhead Vapor Return to T-202 (T-203, Heat Exchanger EX-208, PM-205A/B, AF-205)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9219; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9223

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								pulling vacuum and update procedures if necessary.	
24.18. General	24.18.1. No additional causes identified								

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
25.1. Low/No Flow	25.1.1. Level control valve LCV-21660 fails closed	25.1.1.1. Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response 591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. LAH/LAL-21660 on Kero Stripper is not tied to the LCV-21660 control loop.
		25.1.1.2. Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response	F	1	3	1	61. No recommendations required.	
	25.1.2. T-202 Combi Tower Tray Pluggage	25.1.2.1. Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		25.1.2.2. Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B	F	1	3	1	61. No recommendations required.	
	25.1.3. Kero Stripper Product Pump PM-206A/B trips or loss of power	25.1.3.1. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response 705. Standby pump available - Kero Stripper Product Pump PM-206A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		25.1.3.2. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response 705. Standby pump available - Kero Stripper Product Pump PM-206A/B	F	1	2	1	61. No recommendations required.	
	25.1.4. Manual valve closed inadvertently in Kero Stripper Product Pump PM-206A/B suction or basket strainers on suction lines plugged	25.1.4.1. Kero Stripper Product Pump PM-206A/B cavitation, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B						
			484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area						
		25.1.4.2. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		25.1.4.3. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response	F	1	2	1	61. No recommendations required.	
	25.1.5. Manual valve closed inadvertently in Kero Stripper Product Pump PM-206A/B discharge	25.1.5.1. Kero Stripper Product Pump PM-206A/B deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B						

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area						
		25.1.5.2. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response	F	1	2	1	61. No recommendations required.	
		25.1.5.3. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	25.1.6. 8" manual valve on EX-203 shell side outlet inadvertently closed	25.1.6.1. Blocked discharge from Kero Stripper Product Pump PM-206A/B; Potential pump deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		25.1.6.2. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response	F	1	2	1	61. No recommendations required.	

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		25.1.6.3. High level in Kero Stripper T-204; Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	25.1.7. 8" manual valve upstream/downstream of AF-204 Kero Product Cooler inadvertently closed	25.1.7.1. Inadvertently route all flow through either AF-204-1 or AF-204-2; Potential reduced cooling ability, sending higher temperature product to storage tanks; Not enough to exceed design temp of storage tanks (200 degF); Potential reduced flow and potential increased level in Kero Stripper T-204; Potential high level in Stripper T-204; Potential Combi Tower T-202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		25.1.7.2. Inadvertently route all flow through either AF-204-1 or AF-204-2; Potential reduced cooling ability, sending higher temperature product to storage tanks; Not enough to exceed design temp of storage tanks (200 degF); Potential reduced flow and potential increased level in Kero Stripper T-204; Potential high level in Stripper T-204; Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response	F	1	2	1	61. No recommendations required.	
	25.1.8. FCV-21221 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed or 8" manual valve on kero product to rerun inadvertently closed	25.1.8.1. Potential Combi Tower T-102 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/2	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			1481 will prompt operators to cordon off the area						
			595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response						
		25.1.8.2. Blocked discharge from Kero Stripper Product Pump PM-206A/B; Potential pump deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	491. Dual seals on PM-106A/B and PM-206A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B						
			484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area						
		25.1.8.3. Potential Combi Tower T-202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
25.2. High Flow	25.2.1. LCV-21660 fails open or bypass valve inadvertently open	25.2.1.1. Potential to draw diesel into the kero draw off line; Potential to dry out the tray; Potential off-spec product/quality issues; Potential financial impact	706. TAH-21656 high temperature alarm from EX-209 Reboiler will prompt operator response	F	1	3	1	61. No recommendations required.	
		25.2.1.2. Potential to flood the Kero Stripper T-204; Potential to send out liquids in the vapor return line back to the Combi Tower T-202; Potential damage to vapor return piping; Potential release with ignition and personnel impact	592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. LAH/LAL-21660 on Kero Stripper is not tied to the LCV-21660 control loop.
			706. TAH-21656 high temperature alarm from EX-209 Reboiler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	25.2.2. 8" manual valve on bypass around EX-203 inadvertently opened	25.2.2.1. Potential reduced ability to heat the feed in EX-203; Potential for low feed temperature into Naphtha Stabilizer Column T-201; Potential to swing T-201; Potential offspec product/quality; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		25.2.2.2. Inadvertently partially bypassing product flow around EX-203; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; No damage expected to Storage Tank or environmental impacts expected; Operability issue only, no hazardous consequences identified							
	25.2.3. FCV-21221 fails open or bypass valve inadvertently open	25.2.3.1. Loss of level in Kero Stripper T-204, potential to vapor lock the stripper, potential to cavitate Kero Stripper Product Pump PM-206A/B; Potential seal failure with loss of containment, ignition, and personnel impact	491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	25.2.4. Both Kero Product Pumps PM-206A/B running at once	25.2.4.1. Potential loss of level in Kero Stripper T-204, potential to vapor lock the stripper, potential to cavitate Kero Stripper Product Pump PM-206A/B; Potential seal failure with loss of containment, ignition, and personnel impact	704. LAL-21660 low level alarm on T-204 Kero Stripper will prompt operator response 491. Dual seals on PM-106A/B and PM-206A/B 492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
25.3. Reverse/Misdirected Flow	25.3.1. Tube leak/rupture on kero Stripper Reboiler EX-209	25.3.1.1. Potential to contaminate kero product with hot oil; Potential loss of hot oil resulting in potential need to shutdown and financial impact	707. LAL-21965 Low level alarm on Hot Oil Surge Tank pressure vessel PV 203 will	F	3	2	2	61. No recommendations required.	

Node: 25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
			708. LALL-21975 low-low level alarm on Hot Oil Surge Tank PV-203 will prompt operator response						
	25.3.2. Inadvertent opening of spec valve to rerun	25.3.2.1. Inefficient operation - sending product to rerun; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
25.4. High Pressure	25.4.1. External fire around T-204 Kero Stripper	25.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
25.5. Low Pressure	25.5.1. No additional causes identified								
25.6. High Level	25.6.1. No additional causes identified								
25.7. Low Level	25.7.1. No additional causes identified								
25.8. High Interface Level	25.8.1. Not applicable								
25.9. Low Interface Level	25.9.1. Not applicable								
25.10. High Temperature	25.10.1. Kero Product Cooler AF-204 fan trips	25.10.1.1. Potential to send higher temperature Kerosene product to downstream storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	709. XI-21215A/B and XI-21235A/B Kero Product Cooler AF-204 fan run indication will alarm and prompt operator response	F	3	2	2	61. No recommendations required.	
			710. TAH-21232A/B high temperature alarm will prompt operator response						
25.11. Low Temperature	25.11.1. No additional causes identified								
25.12. Contamination/ Composition	25.12.1. No additional causes identified								
25.13. Corrosion	25.13.1. No additional causes identified								
25.14. Additional Phase	25.14.1. Not applicable								
25.15. Rupture/Loss of Containment	25.15.1. No additional causes identified								

Node: **25. Node 25: Kero Side draw from T-202 to Kero Stripper (T-204), Stripper Reboiler, and Stripper Bottoms Pump to Heat Exchanger EX-203 Feed Preheater and to Kero Product Cooler to Storage; Vapor Return to T-202 (T-204, Heat Exchanger EX-209, PM-206A/B, AF-204)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9214; 40-PI-9220; 40-PI-9221 ; 40-PI-9221 Redline ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9224

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
25.16. Loss of Utilities	25.16.1. Loss of power	25.16.1.1. See PM-206A/B pump trip scenario, this node							
		25.16.1.2. See AF-204 fan failure scenario, this node							
	25.16.2. Loss of instrument air	25.16.2.1. Valves revert to fail-safe position							
25.17. Start-up/Shutdown hazards/ Maintenance	25.17.1. Cooling during Kero Stripper T-204 shutdown	25.17.1.1. Potential for vacuum as Kero Stripper T-204 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
25.18. General	25.18.1. No additional causes identified								

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
26.1. Low/No Flow	26.1.1. Level control valve LCV-25090 fails closed or T-202 tower tray plugs	26.1.1.1. Potential to drop diesel into the bottom of the Combi Tower T-202, potential to send out diesel with the AGO; Potential loss of level in Diesel Stripper T-205; Potential to cavitate pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B 711. LAL-25090 low level alarm on T-205 Diesel Stripper will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	1. LAH/LAL-25090 on T-205 Diesel Stripper is not tied to LCV-25090

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		26.1.1.2. Potential to drop diesel into the bottom of the Combi Tower T-202, potential to send out diesel with the AGO; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response 711. LAL-25090 low level alarm on T-205 Diesel Stripper will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. Confirmed that LELs AI-11461/11481/21461/21481 are located within 15 feet of T-102 and T-202.
	26.1.2. Diesel Stripper Product Pumps PM-207A/B trips	26.1.2.1. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			operators to cordon off the area						
	26.1.3. Manual valve closed on suction to Diesel Stripper Product Pumps PM-207A/B or basket strainers plugged on suction piping	26.1.3.1. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		26.1.3.2. Potential to cavitate pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B 596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response	P	4	1	2	61. No recommendations required.	
	26.1.4. Manual valve closed on discharge from Diesel Stripper Product Pumps PM-207A/B	26.1.4.1. Potential to deadhead pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		26.1.4.2. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	26.1.5. 6" manual valve on EX-204 shell side outlet inadvertently closed	26.1.5.1. Potential to deadhead pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B 596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response	P	4	1	2	61. No recommendations required.	
		26.1.5.2. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
			712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
	26.1.6. 6" manual valve(s) upstream/downstream AF-206 Diesel Product Cooler inadvertently closed	26.1.6.1. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response						
			712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response						
			631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		26.1.6.2. Potential to deadhead pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B	P	4	1	2	61. No recommendations required.	
			594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B						
			596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response						

Node: 26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			712. FAL-26065 low flow alarm downstream of AF-206 Diesel Product Cooler will prompt operator response						
	26.1.7. FCV-26065 malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	26.1.7.1. Potential to deadhead pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B 484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		26.1.7.2. High level in Diesel Stripper T-205; Inability to draw off diesel from Combi Tower T-202; Potential to ultimately flood level in T-202; Potential to send liquids in the vapor return lines; Potential damage to vapor return piping; Potential release with ignition and personnel impact	595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	2	3	11. Provide an additional safeguard for the scenario of FCV-16065/26065 malfunctioning closed resulting in overfill of Diesel Stripper T-105/205 and flooding of Combi Tower T-102/202 with potential loss of containment. (Note - LIC-15090/25090 on Stripper T-105/205 is tied to the FCV-16065/26065 control loop).	
26.2. High Flow	26.2.1. 6" manual bypass around EX-204 inadvertently open	26.2.1.1. Potential reduced ability to heat the feed in EX-204; Potential for low feed temperature into Naphtha Stabilizer Column T-201; Potential to swing T-201; Potential offspec product/quality; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		26.2.1.2. Inadvertently partially bypassing product flow around EX-204; Potential to send higher temperature product to Storage Tank TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	713. TAH-26055 high temperature alarm downstream AF-206 Diesel Product Cooler will prompt operator response	F	2	2	1	61. No recommendations required.	
	26.2.2. LCV-25090 fails open or bypass inadvertently open	26.2.2.1. Potential to dry out Combi Tower T-202 tray, leading to potential tray damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	26.2.3. FCV-26065 fails open or bypass inadvertently open	26.2.3.1. Potential loss of level in Diesel Stripper T-205; Potential to cavitate pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B 594. Seal failure alarm SFA-25025/25045 will	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			shutdown pumps PM-207A/B						
			484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area						
	26.2.4. Both PM-207A/B Diesel Stripper Product Pumps running at once	26.2.4.1. Potential loss of level in Diesel Stripper T-205; Potential to cavitate pumps PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	593. Dual seals on PM-207A/B	P	4	1	2	61. No recommendations required.	
			594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B						
			711. LAL-25090 low level alarm on T-205 Diesel Stripper will prompt operator response						
26.3. Reverse/Misdirected Flow	26.3.1. Tube leak/rupture on Diesel Stripper Reboiler EX-210	26.3.1.1. Potential to contaminate diesel product with hot oil; Potential loss of hot oil resulting in potential need to shutdown and financial impact	707. LAL-21965 Low level alarm on Hot Oil Surge Tank pressure vessel PV 203 will prompt operator response	F	3	2	2	61. No recommendations required.	
			708. LALL-21975 low-low level alarm on Hot Oil Surge Tank PV-203 will prompt operator response						
	26.3.2. Inadvertent opening of spec valve to rerun	26.3.2.1. Inefficient operation - sending product to rerun; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
26.4. High Pressure	26.4.1. External fire around T-205 Diesel Stripper	26.4.1.1. This vessel is located above a surface that is sloped away towards the hydrocarbon drain header, therefore accumulation of flammable liquids and external fire is not considered credible							
26.5. Low Pressure	26.5.1. No additional causes identified								
26.6. High Level	26.6.1. No additional causes identified								
26.7. Low Level	26.7.1. No additional causes identified								
26.8. High Interface Level	26.8.1. Not applicable								
26.9. Low Interface Level	26.9.1. Not applicable								

Node: **26. Node 26: Diesel Stripper (T-205, Heat Exchanger EX-210, AF-206, PM-207A/B)**

Drawings / References:

40-PI-9206; 40-PI-9214; 40-PI-9221 Redline ; 40-PI-9221 ; 40-PI-9222 ; 40-PI-9222 Redline; 40-PI-9236; 40-PI-9237

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
26.10. High Temperature	26.10.1. Diesel Product Cooler AF-206 fan trips	26.10.1.1. Potential to send higher temperature Kerosene product to downstream storage tanks TK-100-13/100-14/120-22/120-23; Max temp of tanks is 200 degF, potential to exceed tank design temperature; Potential IFR roof damage; Potential financial impact and environmental impact	714. XI-26040/26050A Diesel Product Cooler AF-206 fan run indication will alarm and prompt operator response 713. TAH-26055 high temperature alarm downstream AF-206 Diesel Product Cooler will prompt operator response	F	3	2	2	61. No recommendations required.	
26.11. Low Temperature	26.11.1. No additional causes identified								
26.12. Contamination/ Composition	26.12.1. No additional causes identified								
26.13. Corrosion	26.13.1. No additional causes identified								
26.14. Additional Phase	26.14.1. Not applicable								
26.15. Rupture/Loss of Containment	26.15.1. No additional causes identified								
26.16. Loss of Utilities	26.16.1. Loss of power	26.16.1.1. See AF-206 fan failure scenario in this node							
		26.16.1.2. See PM-207A/B pump trip scenario in this node							
26.17. Start-up/Shutdown hazards/ Maintenance	26.17.1. Cooling during Diesel Stripper T-205 shutdown	26.17.1.1. Potential for vacuum as Diesel Stripper T-205 cools during shutdown, column damage, potential loss of containment, ignition, personnel impact	448. No specific safeguards identified	P	4	3	4	7. Provide additional safeguards to prevent potential vacuum in the Naphtha Stabilizer Column T-101 due to cooling during shutdown: Possible recommendation from the PHA team: 1) Verify the shutdown procedures for the Naphtha Stabilizer Columns T-101/201, Combi Towers T-102/202, Heavy Naphtha Strippers T-103/203, Kero Strippers T-104/204, and Diesel Strippers T-105/205 include monitoring the column pressure and open up relief vents to avoid pulling vacuum and update procedures if necessary.	
26.18. General	26.18.1. No additional causes identified								

Node: **27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
27.1. Low/No Flow	27.1.1. PCV-21315A fails closed or upstream/downstream manual valve(s) inadvertently closed	27.1.1.1. Loss of suction to Recycle Gas Compressor; Potential compressor rod load damage with potential financial impact	715. PIC-23565/23611 low suction pressure shutdown will shutdown CM-201A/CM-201B compressors	F	1	3	1	61. No recommendations required.	
		27.1.1.2. Loss of vapor flow out of Combi Tower Accumulator pressure vessel PV-202, increased pressure in Combi Tower T-202, overpressure Combi Tower, loss of containment, potential ignition and personnel impact	701. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet 597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response 702. PAHH-21715 (2oo3) high-high pressure interlock on T-202 will trip heater F-201A/B Ammonia Skid and Hot Oil Pumps and will prompt operator response	P	4	1	2	61. No recommendations required.	
	27.1.2. PCV-20395A malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	27.1.2.1. Potential to overpressure PV-201 Naphtha Stabilizer Accumulator with loss of containment, potential ignition and personnel impact	716. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for blocked vapor outlet	P	4	1	2	61. No recommendations required.	
	27.1.3. Suction Scrubber pressure vessel PV-207 demister pad saturated	27.1.3.1. Slow increase in pressure on Combi Tower T-202; Potential carryover of water to RGC Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	2	1	61. No recommendations required.	
	27.1.4. 10" manual valve on suction to Recycle Compressor CM-201A/B inadvertently closed		701. PSV-21690/21710/21720	P	4	1	2	61. No recommendations required.	

Node: **27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		27.1.4.1. Potential increase in pressure in Combi Tower T-202, and potential to overpressure column, loss of containment, potential ignition and personnel impact	(set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet						
			698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response						
			597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response						
	27.1.5. XV-23576/23627 on discharge from Recycle Compressor CM-201A/B fails closed	27.1.5.1. Blocked CM-201A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	717. PSV-23574/23622 (set @ 125 psig) is sized for blocked flow	P	4	1	2	61. No recommendations required.	
			718. PAHH-23584/23626 high-high pressure alarm on CM-201A/B discharge will shutdown compressor						
	27.1.6. Recycle Compressor CM-201A or B trips	27.1.6.1. Potential increase in pressure in Combi Tower T-202, and potential to overpressure column, loss of containment, potential ignition and personnel impact	701. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet	P	4	1	2	61. No recommendations required.	
			698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response						
			597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response						
	27.1.7. Pressure control valve PCV-23685 fails closed		717. PSV-23574/23622 (set @	P	4	1	2	61. No recommendations required.	

Node: 27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		27.1.7.1. Blocked CM-201A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	125 psig) is sized for blocked flow 718. PAHH-23584/23626 high-high pressure alarm on CM-201A/B discharge will shutdown compressor						
	27.1.8. FCV-23462 fails closed or upstream/downstream manual valve(s) inadvertently closed	27.1.8.1. Blocked CM-201A/B compressor discharge; Potential overpressure with loss of containment, potential ignition and personnel impact	717. PSV-23574/23622 (set @ 125 psig) is sized for blocked flow 718. PAHH-23584/23626 high-high pressure alarm on CM-201A/B discharge will shutdown compressor	P	4	1	2	61. No recommendations required.	
27.2. High Flow	27.2.1. PCV-21315A malfunctions open or bypass inadvertently open	27.2.1.1. Reduced pressure in Combi Tower T-202, increased level in Combi Tower Accumulator pressure vessel PV-202; Potential to overfill PV-202 and carryover to Recycle Gas Compressor Suction Scrubber PV-207; Potential to overfill PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	2	1	61. No recommendations required.	
	27.2.2. PCV-20395A fails open or bypass valve open	27.2.2.1. T-201 Stabilizer pressure would drop; Potential liquid carryover to the PV-201 Accumulator; Potential overfill PV-201 and carryover to Recycle Compressor Discharge Scrubber PV-209; Potential to overfill PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will	F	5	1	2	61. No recommendations required.	

Node: 27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
	27.2.3. 10" manual valve on suction to offline CM-201A (or CM-201B) left open while the other compressor is running	27.2.3.1. Reduced flow of suction to online CM-201A/B; Operability issue only, no hazardous consequences identified							
	27.2.4. Pressure control valve PCV-23685 fails open (compressor discharge)	27.2.4.1. Reduced residence time in PV-208 Lube Oil Separator Vessel; Potential inability to separate out off-gas; Potential to damage the CM-201A/B compressor casing, slide valve, and bearings; Potential to send contaminated lube oil to Heater; Potential asset damage; Potential environmental impact	719. LSLL-23668 low-low level switch on PV-208 Lube Oil Separator will shutdown CM-201A/B compressor	F	2	3	2	61. No recommendations required.	
	27.2.5. FCV-23462 malfunctions open or bypass valve inadvertently open	27.2.5.1. Potential low backpressure on the Discharge Scrubber PV-209; Potential to carryover liquids to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	720. LAL-23540 low level alarm on PV-209 will prompt operator response 721. PAL-23663 low pressure alarm on PV-209 outlet will prompt operator response	F	5	2	4	14. Provide an additional safeguard for the scenario of FCV-13462/23462 malfunctioning open on the outlet of PV-109/PV-209 Discharge Scrubber causing low backpressure on PV-109/PV-209 resulting in a potential liquid carryover to the Hot Oil Furnace F-101/F-201 fuel inlet causing potential explosion with personnel impace and significant downtime.	
	27.2.6. Pressure control valve PCV-23460 fails open to flare	27.2.6.1. Unintended flaring event; Potential environmental impact	448. No specific safeguards identified	E	1	4	2	61. No recommendations required.	
27.3. Reverse/Misdirected Flow	27.3.1. BDV-23575/23619 malfunctions open when not needed	27.3.1.1. Inadvertently sending compressor suction gas to vent header; Potential unplanned flaring event with potential environmental impact	448. No specific safeguards identified	E	1	4	2	61. No recommendations required.	
27.4. High Pressure	27.4.1. No additional causes identified								
27.5. Low Pressure	27.5.1. No additional causes identified								
27.6. High Level	27.6.1. Suction scrubber pump PM-213A/B failure	27.6.1.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-207; Potential to overfill PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	2	1	61. No recommendations required.	
	27.6.2. Manual valve(s) on suction to Suction scrubber pump PM-213A/B inadvertently closed	27.6.2.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-207; Potential to overfill PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response	F	1	2	1	61. No recommendations required.	

Node: 27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B						
		27.6.2.2. Loss of suction to pumps PM-213A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	722. LAL-23507 low level alarm on Suction Scrubber P-207 will prompt operator response 723. LALL-23507 low-low alarm on Suction Scrubber P-207 will shutdown pumps PM-213A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	27.6.3. Manual valve(s) on discharge from Suction scrubber pump PM-213A/B inadvertently closed	27.6.3.1. Potential high level in Recycle Gas Compressor Suction Scrubber PV-207; Potential to overfill PV-207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	696. LAH-21320 high level alarm on PV-202 Accumulator will prompt operator response 693. LAHH-23505 high-high level alarm on Suction Scrubber PV-207 will shutdown compressors CM-201A/B	F	1	2	1	61. No recommendations required.	
		27.6.3.2. Loss of discharge from pumps PM-213A/B with potential deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	722. LAL-23507 low level alarm on Suction Scrubber P-207 will prompt operator response 723. LALL-23507 low-low alarm on Suction Scrubber P-207 will shutdown pumps PM-213A/B 512. Lower Explosive Limit LEL alarm AI-11361/21361 will	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operators to cordon off the area						
	27.6.4. Discharge scrubber pump PM-217A/B failure	27.6.4.1. Potential high level in PV-209 Discharge Scrubber; Potential to overfill PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 724. Standby pump PM-217A/B available for remote start-up	F	5	1	2	61. No recommendations required.	
	27.6.5. Manual valve on suction to Discharge Scrubber Pump PM-217A/B inadvertently closed	27.6.5.1. Potential to overfill PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response	F	5	1	2	61. No recommendations required.	
		27.6.5.2. Loss of suction to pumps PM-217A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response						
	27.6.6. Manual valve on discharge from Discharge Scrubber Pump PM-217A/B inadvertently closed	27.6.6.1. Blocked discharge from pumps PM-217A/B with potential deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area 684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		27.6.6.2. Potential to overfill PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response	F	5	1	2	61. No recommendations required.	
27.7. Low Level	27.7.1. Suction scrubber pump PM-213A/B fails to stop	27.7.1.1. Potential to pump out all level in Suction Scrubber PV-207; Potential loss of suction to pumps PM-213A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	722. LAL-23507 low level alarm on Suction Scrubber P-207 will prompt operator response 723. LALL-23507 low-low alarm on Suction	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Scrubber P-207 will shutdown pumps PM-213A/B						
			512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area						
	27.7.2. Discharge scrubber pump PM-117A/B fails to stop	27.7.2.1. Potential to pump out all level in Discharge Scrubber PV-109; Potential loss of suction to pumps PM-117A/B with potential cavitation; Potential seal failure with loss of containment, potential ignition and personnel impact	518. LAL-13540 low level alarm on PV-109 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			519. LALL-13540 low-low level alarm on PV-109 will shut off PM-117A/B						
			512. Lower Explosive Limit LEL alarm AI-11361/21361 will prompt operators to cordon off the area						
27.8. High Interface Level	27.8.1. Not applicable								
27.9. Low Interface Level	27.9.1. Not applicable								
27.10. High Temperature	27.10.1. Electric Gas Preheater HTR-201 stuck in "on" position	27.10.1.1. High suction temperature, increased load on Recycle Compressor CM-202A/B, high discharge temperature from compressor, potential overheating of compressor lube oil and compressor damage (bearings, etc.)	725. TAH-23510 High Temperature Alarm on Heater H-201 will prompt operator response	F	1	1	1	61. No recommendations required.	
			726. TAHH-23510 High-High Temperature Alarm on Electric Gas Preheater HTR-201 will prompt operator response						
			727. TAH-23625/23583 high temperature alarm on CM-201A/B Compressor discharge will prompt operator response						
			728. TAHH-23625/23583 high						

Node: 27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			temperature alarm on CM-201A/B Compressor discharge will prompt operator response						
	27.10.2. TCV-23515 receives incorrect signal (opens inadvertently)	27.10.2.1. Inadvertently sending compressor discharge flow around Gas-to-Gas Exchanger EX-211; Inability to cool discharge gas with EX-211; Potential to send higher temperature discharge gas to downstream AF-208 (design temp 350 degF); Not expected to exceed design temp of AF-208; Operability issue only, no hazardous consequences identified							
		27.10.2.2. Inadvertently sending compressor discharge flow around Gas-to-Gas Exchanger EX-211; Inability to heat suction gas with EX-211; Potential to send lower temperature gas to compressor; Potential liquids in compressor resulting in potential compressor damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	27.10.3. AF-208 Gas Aftercooler Fan Failure	27.10.3.1. Reduced ability to cool CM-201A/B compressor discharge gas; Potential higher BTUs; Potential increased offgas to Heater; Potential increased emissions with potential environmental impact	729. CM-201A/B run permissive requires at least one AF-208 fan running to remain online	F	1	3	1	61. No recommendations required.	
27.11. Low Temperature	27.11.1. Electric Gas Preheater HTR-201 trips when needed (winter)	27.11.1.1. Lower Recycle Compressor CM-202A/B suction temperature, potential for liquids to reach compressor, compressor damage (no loss of containment)	730. Gas to Gas Heat Exchanger EX-211 downstream of Electric Gas Preheater HTR-201 731. TAL-23564/23610 low temperature alarm on suction to Recycle Compressor CM-201A/B will prompt operator response	F	1	2	1	61. No recommendations required.	
	27.11.2. TCV-23515 receives incorrect signal (closes inadvertently)	27.11.2.1. Inadvertently sending compressor discharge flow through Gas-to-Gas Exchanger EX-211, further cooling the discharge gas when not needed; Potential liquid carryover PV-209 Discharge Scrubber; Potential to overfill PV-209 and send liquid to Hot Oil Furnace F-201 fuel inlet, potential for damage of burners, heater firebox, potential explosion and personnel impact and damage with downtime greater than 90 days	684. LAH-23540 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response 685. LAH-23538 High level alarm on Recycle Compressor Discharge Scrubber pressure vessel PV-209 will prompt operator response	F	5	2	4	12. Provide an additional safeguard for the scenario of inadvertently sending CM-101A/B / CM-201A/B compressor discharge gas through the EX-111/211 Gas-to-Gas Exchanger, further cooling the gas and leading to potential PV-109/209 Discharge Scrubber liquid carryover to the Hot Oil Furnace F-101/201 fuel inlet causing potential explosion with personnel impace and significant downtime.	

Node: **27. Node 27: From PV-202 Combi Tower Accumulator to Recycle Gas Compressor, Recycle Compressor Suction Knockout Drum, Recycle Cooler, Recycle Gas Knockout Drum, Recycle Gas Knockout Pump to Fuel Gas System, (PV-207, AF-208, PV-209, PM-217, PM-213) (Recycle Compressor Vendor Package evaluated in Node 11)**

Drawings / References:

40-PI-9205 ; 40-PI-9205 Redline; 40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9213; 40-PI-9218 Redline; 40-PI-9218 ; 40-PI-9228 Redline; 40-PI-9228 ; E000353-01-101 SHT 1 ; E000353-01-101 SHT 2; E000353-01-101 SHT 3; E000353-01-101 SHT 4; E000353-01-101 SHT 5; E000353-02-102 SHT 1; E000353-02-102 SHT 2

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
27.12. Contamination/ Composition	27.12.1. Product infiltration into compressor oil	27.12.1.1. Contamination of Recycle Compressor CM-202A/B oil; Potential to damage the CM-201A/B compressor casing, slide valve, and bearings; Potential to send contaminated lube oil to Heater; Potential asset damage; Potential environmental impact	25. Coalescing Filters in PV-108/208	F	2	3	2	61. No recommendations required.	
27.13. Corrosion	27.13.1. No specific causes identified								
27.14. Additional Phase	27.14.1. Not applicable								
27.15. Rupture/Loss of Containment	27.15.1. Tube leak/rupture on EX-211 Gas-to-Gas Exchanger	27.15.1.1. Potential to overpressure and blow compressor seals; Potential to contaminate compressor lube oil; Potential compressor damage and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	27.15.2. Recycle Compressor CM-201A/B mechanical seal leaks	27.15.2.1. Loss of barrier fluid; Potential loss of containment of barrier fluid; Potential release of lube oil with entrained gas; Potential environmental/financial impact	732. LSLL-23668 low-low level switch on PV-208 Lube Oil Separator will shutdown CM-201A/B compressor	F	1	3	1	61. No recommendations required.	
27.16. Loss of Utilities	27.16.1. Loss of power	27.16.1.1. Compressor CM-201A/B shuts down upon loss of power; Potential flaring event with potential environmental impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	27.16.2. Loss of instrument air	27.16.2.1. Valves revert to fail-safe position							
27.17. Start-up/Shutdown hazards/ Maintenance	27.17.1. Recycle Compressor CM-201A/B accessibility issues (congested area)	27.17.1.1. Potential personnel impacts while performing maintenance activities	733. Platforms built with access to CM-201A/B valves	P	1	4	2	61. No recommendations required.	
27.18. General	27.18.1. Compressor CM-201A/B designed for clean-gas service	27.18.1.1. Potential compressor wear and tear due to dirty gas service, requiring excessive maintenance activities; Potential for environmental impacts and financial impacts due to increased flaring events	448. No specific safeguards identified	F	2	5	4	13. Perform an engineering study to determine if a different type of Recycle Gas Compressor should replace the existing CM-101A/B / CM-201A/B compressors, as existing compressors are designed for clean-gas service only. Financial risk only.	

Node: **28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air**

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
28.1. Low/No Flow	28.1.1. One hot oil Hot Oil Circulation Pump PM-202A/B/C/D trips	28.1.1.1. Reduced flow of hot oil to the heat exchangers; Decreased heat transfer, operability issue only, no hazardous consequences identified							
	28.1.2. Manual valve closed in Hot Oil Circulation Pump PM-202A/B/C/D suction	28.1.2.1. Loss of suction to Hot Oil Circulation Pump PM-202A/B/C/D; Potential pump cavitation with seal failure and loss of containment, potential ignition and personnel impact	734. VAH-20485/20505/20525/20545 high vibration alarm on Hot Oil Circulation Pump PM-202A/B/C/D will	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response 735. Dual seals on PM-202A/B/C/D						
	28.1.3. Manual valve closed in Hot Oil Circulation Pump PM-202A/B/C/D discharge	28.1.3.1. Blocked discharge from Hot Oil Circulation Pump PM-202A/B/C/D; Potential pump deadhead with seal failure and loss of containment, potential ignition and personnel impact	734. VAH-20485/20505/20525/20545 high vibration alarm on Hot Oil Circulation Pump PM-202A/B/C/D will prompt operator response 735. Dual seals on PM-202A/B/C/D	P	4	1	2	61. No recommendations required.	
	28.1.4. Multiple Hot Oil Circulation Pumps PM-202A/B/C/D trip	28.1.4.1. Reduced hot oil flow, potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	736. All Hot Oil Circulation Pumps PM-202A/B/C/D down - trips Hot Oil Heater F-201 737. PALL-20625 low-low pressure alarm on Hot Oil Heater F-201 discharge will trip F-201 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D	P	4	1	2	61. No recommendations required.	
28.2. High Flow	28.2.1. All Hot Oil Circulation Pumps PM-202A/B/C/D online at once	28.2.1.1. No adverse consequences identified							
28.3. Reverse/Misdirected Flow	28.3.1. No specific causes identified								
28.4. High Pressure	28.4.1. Pressure control valve PCV-21940 set too high	28.4.1.1. Potential to pop the rupture disk, resulting in PSV-21945 release and unintended flaring event, potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.4.2. Hot oil blocked in at Hot Oil Heater F-201	28.4.2.1. Potential to overpressure hot oil piping, loss of containment, release, personnel impact	739. Pressure safety valve PSV-20630 set @ 300 psig, sized for hydraulic expansion	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D						
28.5. Low Pressure	28.5.1. Pressure control valve PCV-21940 set too low	28.5.1.1. Insufficient suction head for Hot Oil Circulation Pump PM-202A/B/C/D, potential for insufficient pump suction head, pump cavitation and potential pump damage	734. VAH-20485/20505/20525/20545 high vibration alarm on Hot Oil Circulation Pump PM-202A/B/C/D will prompt operator response 735. Dual seals on PM-202A/B/C/D	P	4	1	2	61. No recommendations required.	
28.6. High Level	28.6.1. Thermal expansion of heat transfer fluid on startup	28.6.1.1. Potential to overfill Hot Oil Surge Tank pressure vessel PV-203, requires manual removal of excess heat transfer fluid after cooling it down and removing via vac truck; Potential downtime required to cool down the heat transfer fluid prior to removal (~1 to 5 days of downtime)	31. Startup procedures to account for thermal expansion	F	2	3	2	61. No recommendations required.	
28.7. Low Level	28.7.1. Thermal expansion of heat transfer fluid on startup requires low level in Hot Oil Surge Tank pressure vessel PV-203 at startup	28.7.1.1. Hot Oil Circulation Pump PM-202A/B/C/D cavitation at startup/shutdown due to insufficient suction head, potential seal failure and loss of containment, potential ignition and personnel impact	31. Startup procedures to account for thermal expansion 734. VAH-20485/20505/20525/20545 high vibration alarm on Hot Oil Circulation Pump PM-202A/B/C/D will prompt operator response 735. Dual seals on PM-202A/B/C/D	P	4	1	2	61. No recommendations required.	
28.8. High Interface Level	28.8.1. Not applicable								
28.9. Low Interface Level	28.9.1. Not applicable								
28.10. High Temperature	28.10.1. Reduced heat transfer fluid flow	28.10.1.1. Reduced hot oil flow, potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	737. PALL-20625 low-low pressure alarm on Hot Oil Heater F-201 discharge will trip F-201 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D						
	28.10.2. FCV-21681 on hot oil from Kero Stripper Reboiler EX-209 fails open or bypass inadvertently open	28.10.2.1. Potential reduced hot oil flow to other heat exchangers; Potential off-spec product with financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.10.3. FCV-20521 fails open or bypass inadvertently open	28.10.3.1. Increased temperature in Naphtha Stabilizer Column T-201, push heavier components overhead and contaminate Y-Grade product, product quality issues, potential to exceed Maximum Allowable Working Temperature (MAWT) of T-201 of 375F, mechanical integrity issues, potential release, potential ignition and personnel impact	598. TAH-20310 high temperature alarm on overheads from Stabilizer T-201 will prompt operator response 542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling	P	4	1	2	61. No recommendations required.	
	28.10.4. FCV-20475 fails open or bypass inadvertently open	28.10.4.1. Increase in Combi Tower T-202 temperature and pressure, potential column overpressure, loss of containment, potential ignition and personnel impact	697. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for loss of cooling 597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response 702. PAHH-21715 (2oo3) high-high pressure interlock on T-202 will trip heater F-201A/B Ammonia Skid and Hot Oil Pumps and will prompt operator response	P	4	1	2	61. No recommendations required.	
	28.10.5. FCV-20522 fails open or bypass inadvertently open	28.10.5.1. Inadvertent bypass of hot oil flow around EX-207 Combi Tower Feed Preheater; Potential reduced heat to Combi Tower T-202 feed; Potential off-spec product; Potential trip to Heater;	738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-	F	2	3	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Potential flaring; Potential downtime; Potential environmental and environmental impact	24104A/B/C/D and TI-24124A/B/C/D						
	28.10.6. FCV-26030 on hot oil from Diesel Stripper Reboiler EX-110 fails open or bypass inadvertently open	28.10.6.1. Potential reduced hot oil flow to other heat exchangers; Potential off-spec product with financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
28.11. Low Temperature	28.11.1. FCV-21681 malfunctions closed or manual valve(s) inadvertently closed	28.11.1.1. Loss of hot oil circulation through EX-209 Kero Stripper Reboiler; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.11.2. FCV-20521 fails closed or manual valve(s) inadvertently closed	28.11.2.1. Loss of hot oil circulation through Naphtha Stabilizer Reboiler EX-206; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.11.3. FCV-20475 fails closed or manual valve(s) inadvertently closed	28.11.3.1. Loss of heat to Combi Tower T-202 feed; Potential off-spec product; Potential trip to Heater; Potential flaring; Potential downtime; Potential environmental and environmental impact	738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D	F	2	3	2	61. No recommendations required.	
	28.11.4. FCV-26030 fails closed or manual valve(s) inadvertently closed	28.11.4.1. Loss of hot oil circulation through Diesel Stripper Reboiler EX-210; Potential off-spec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
28.12. Contamination/ Composition	28.12.1. No additional causes identified								
28.13. Corrosion	28.13.1. No specific causes identified								
28.14. Additional Phase	28.14.1. Not applicable								
28.15. Rupture/Loss of Containment	28.15.1. No additional causes identified								
28.16. Loss of Utilities	28.16.1. Loss of instrument air	28.16.1.1. Valves revert to fail-safe position							
	28.16.2. Loss of power	28.16.2.1. See Hot Oil Heater F-201 trip scenarios, this node							
28.17. Start-up/Shutdown hazards/ Maintenance	28.17.1. Hot Oil Circulation Pump PM-202A/B/C/D aging and hot oil quality	28.17.1.1. High frequency of Hot Oil Circulation Pump PM-202A/B/C/D seal failures (currently weekly). Pump temperatures require significant time for cooling before maintenance, requiring downtime	448. No specific safeguards identified	F	2	5	4	15. Implement the following hot oil system improvements to minimize the frequency of hot oil system maintenance: 1) Additional spare hot oil pumps; 2) Hot oil filtration; 3) Hot Oil Day Tank; 4) Nitrogen blanket on Hot Oil Day Tank and Hot Oil Surge Drums; 5) Hot Oil Seal Coolers; 6) Upgrade heat transfer fluid.	
	28.17.2. H2S in the CEMS building	28.17.2.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S	P	2	2	1	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			610. AT-10581/20581 located inside CEMS 1/2 buildings will alert in the control room						
		28.17.2.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
28.18. General	28.18.1. **What if vessel temperature control loop fails open or closed?**	28.18.1.1. Fails open - increased heating of hot oil; potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 737. PALL-20625 low-low pressure alarm on Hot Oil Heater F-201 discharge will trip F-201 698. FALL-21350 low-low flow alarm on Hot Oil Heater F-201 will prompt operator response 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D	P	4	1	2	61. No recommendations required.	
		28.18.1.2. Fails closed - Hot Oil Heater F-201 shuts down on low pressure; Unit would go to re-run mode; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.18.2. **What if fuel supply is cut off?**	28.18.2.1. Loss of firing; Hot Oil Heater F-201 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.18.3. **What if flame fails?**	28.18.3.1. Potential for unburned fuel in Hot Oil Heater F-201 firebox, ignition and heat damage, personnel impact	33. 20 minute purge cycle prior to startup 32. Hot Oil Heater F-101/201 cells must be balanced, loss of a burner in one cell requires cutting burner on the other (BMS)	P	4	1	2	61. No recommendations required.	
	28.18.4. **What if air damper fails closed?**	28.18.4.1. Mechanical failure (all dampers in manual) - pressure swings on ID fan, air imbalance, potential for high Hot Oil Heater F-301 duct temperatures; Potential heater damage and downtime	741. Pressure alarm high-high PAH-24137 at common duct alarm will prompt operator response 742. Pressure alarm high-high PAHH-24137 at common duct	F	3	1	1	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			trips Hot Oil Heater F-201 743. Pressure alarm high-high PAHH-24102/24122 on individual cells trips Hot Oil Heater F-201						
	28.18.5. **What if air damper fails open?**	28.18.5.1. Positive pressure and Hot Oil Heater F-201 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.18.6. **What if blower or motor fails?**	28.18.6.1. Potential for over temperature of Hot Oil Heater F-201/ducts, heater damage	744. FD Fan BM-203 or ID Fan BM-204 shutdown trips Hot Oil Heater F-201 745. Temperature alarm high TAH-24105/24125 - high duct temperature alarm will prompt operator response 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D	F	3	1	1	61. No recommendations required.	
	28.18.7. **What if fuel supply pressure increases?**	28.18.7.1. Increased heating of hot oil; potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201	P	4	1	2	61. No recommendations required.	
	28.18.8. **What if water is entrained in fuel supply?**	28.18.8.1. Loss of flame at one or more burners; Potential for unburned fuel in Hot Oil Heater F-201 firebox, ignition and heat damage, personnel impact	33. 20 minute purge cycle prior to startup 32. Hot Oil Heater F-101/201 cells must be balanced, loss of a	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			burner in one cell requires cutting burner on the other (BMS)						
	28.18.9. **What if fuel supply regulator fails open?**	28.18.9.1. Increased heating of hot oil; potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	747. FCV-24100/24120 fuel supply control valve will further control the fuel gas supply pressure 740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201	P	4	1	2	61. No recommendations required.	
	28.18.10. **What if fuel supply regulator fails closed?**	28.18.10.1. Loss of firing; Hot Oil Heater F-201 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.18.11. **What if fuel contains excessive solids?**	28.18.11.1. Potential for burner pluggage, loss of flame at one or more burners; Hot Oil Heater F-201 shuts down; Potential off-spec product with potential environmental and financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	28.18.12. **What if burner tube skin temperature increases? (Caused by flow restriction in tube)**	28.18.12.1. Potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201	P	4	1	2	61. No recommendations required.	
	28.18.13. **What if burner tube skin temperature decreases?**	28.18.13.1. Not considered a stand-alone credible scenario							
	28.18.14. **What if burner tube ruptures?**	28.18.14.1. Potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201						
	28.18.15. **What if solids or coke builds up on tube external surfaces?**	28.18.15.1. Potential to damage heater tubes; Potential reduced heating efficiency; Operability issue only, no hazardous consequences identified							
	28.18.16. **What if the BTU value of the gas changes?**	28.18.16.1. Potential increased gas usage on the low end; Potential reduced heating efficiency; Operability issue only, no hazardous consequences identified							
	28.18.17. **What if solids build up on tube internal surfaces?**	28.18.17.1. Flow restriction in tubes; Potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D 746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201	P	4	1	2	61. No recommendations required.	
	28.18.18. **What if the purge is inadequate? (BMS Sequence Failure)**	28.18.18.1. Potential for burner explosion on restart if un reacted fuel is present	39. Timer failure results in burner management system (BMS) trip of the Hot Oil Furnace F-101/201 748. AIC-24103/24123 low oxygen analyzer alarm will prompt operator response	P	4	1	2	61. No recommendations required.	
	28.18.19. **What if hot oil flow through the tubes stops?**	28.18.19.1. Flow restriction in tubes; Potential for overheating of Hot Oil Heater F-201 tubes, tube failure, loss of containment, ignition, heater damage, potential personnel impact and environmental impact	740. PAHH-24009/24029 high-high pressure alarm will trip Hot Oil Heater F-201 738. High flue gas duct temperature - trips Hot Oil Heater F-201 TI-24104A/B/C/D and TI-24124A/B/C/D	P	4	1	2	61. No recommendations required.	

Node: 28. Node 28: Hot Oil Heater, F-201; hot oil system: Hot oil Surge Tank, Hot Oil Circulation Pumps, Hot oil heat exchanger (PV-203, PM-202), including natural gas and combustion air

Drawings / References:

40-PI-9210; 40-PI-9210 Redline; 40-PI-9211 Redline 2; 40-PI-9211 Redline; 40-PI-9211 ; 40-PI-9212; 40-PI-9219; 40-PI-9220; 40-PI-9236

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			746. TAHH-20615 high high temperature alarm will trip Hot Oil Heater F-201						

Node: 29. Node 29: Aqueous Ammonia Heater System

Drawings / References:

40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
29.1. Low/No Flow (Ammonia)	29.1.1. Ammonia Forwarding Pump PM-120A/B trips	29.1.1.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	749. Redundant Ammonia Forwarding Pumps PM-120A/B 750. PIT-24815 low pump discharge pressure alarm will prompt operator response 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
	29.1.2. Manual valve(s) on suction to Ammonia Forwarding Pump PM-120A/B inadvertently closed or upstream strainer plugged	29.1.2.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	750. PIT-24815 low pump discharge pressure alarm will prompt operator response 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	
		29.1.2.2. NH3 Forwarding Pump PM-120A/B cavitation, potential seal damage, release of aqueous ammonia, personnel impact	750. PIT-24815 low pump discharge pressure alarm will prompt operator response 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	29.1.3. Manual valve(s) on discharge from Ammonia Forwarding Pump PM-120A/B inadvertently closed	29.1.3.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	750. PIT-24815 low pump discharge pressure alarm will prompt operator response 751. Analyzer indicator AI-20575A/C NOx	E	1	2	1	61. No recommendations required.	

Node: **29. Node 29: Aqueous Ammonia Heater System**

Drawings / References:

40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			analyzers will prompt operator response						
		29.1.3.2. NH3 Forwarding Pump PM-120A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	753. PSV-24812A/B (set @ 95 psig) is sized for blocked discharge 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	29.1.4. Closure of vapor return line on tank filling	29.1.4.1. Potential overpressure of Ammonia Storage Tank PV-106 with release of aqueous ammonia to surrounding area and potential personnel impact	448. No specific safeguards identified	P	4	3	4	18. Perform engineering evaluation to verify that PSV-14806 on Ammonia Storage Tank PV-106 is sized for loss of vapor return during truck filling resulting in potential overpressure of PV-106. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
	29.1.5. Failure of expansion joint EJ1500	29.1.5.1. Release of aqueous ammonia to surrounding area and potential personnel impact	751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	P	3	2	2	61. No recommendations required.	
	29.1.6. Failure of expansion joint EJ-1610	29.1.6.1. Release of aqueous ammonia to surrounding area and potential personnel impact	751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	P	3	2	2	61. No recommendations required.	
	29.1.7. FCV-24821 fails closed or upstream/downstream manual valve(s) inadvertently closed or upstream filters FIL-2500A/B plugged	29.1.7.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	750. PIT-24815 low pump discharge pressure alarm will prompt operator response 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	2	1	61. No recommendations required.	
		29.1.7.2. NH3 Forwarding Pump PM-120A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	753. PSV-24812A/B (set @ 95 psig) is sized for blocked discharge 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
	29.1.8. NH3 injection nozzles plugged at furnace SCR	29.1.8.1. Potential inadequate ammonia injection; Potential inability to control air emissions with potential environmental/financial impact	751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	3	1	61. No recommendations required.	

Node: **29. Node 29: Aqueous Ammonia Heater System**

Drawings / References:

40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		29.1.8.2. NH3 Forwarding Pump PM-120A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	753. PSV-24812A/B (set @ 95 psig) is sized for blocked discharge 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	
29.2. Low/No Flow (Air)	29.2.1. Air blower failure BM-205A/B	29.2.1.1. Inability to send dilution air, inability to disperse ammonia properly; Potential reduced ability to control air emissions with potential environmental/financial impact	751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response 752. FIT-24820 low flow alarm on pump discharge will prompt operator response 754. Spare BM-205A/B available for start-up	E	1	1	1	61. No recommendations required.	
29.3. High Flow	29.3.1. NOx analyzer out of calibration, flow control valve FCV-24821 fails open or flow element (FE) reading low, or flow control valve FCV bypass open or leaking through	29.3.1.1. High level of ammonia injection; Potential high NH3 concentration out of the stack with potential environmental/financial impact	51. NOx analyzers calibrated daily	E	1	4	2	61. No recommendations required.	
	29.3.2. Ammonia Forwarding Pump PM-120A/B fail to turn off	29.3.2.1. Potential to pull vacuum on PV-106 Ammonia Storage Tank; Potential release of aqueous ammonia with potential personnel impact	559. LIT-14805 low level interlock on PV-106 will trip pumps PM-119A/B and PM-120A/B	P	3	3	3	19. Perform engineering evaluation to verify that PSV-14807 on PV-106 Ammonia Storage Tank is sized for vacuum. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
29.4. Reverse/Misdirected Flow	29.4.1. No specific causes identified								
29.5. High Pressure	29.5.1. PCV-24813 pressure regulator fails fully open	29.5.1.1. Inability to reduce PM-120A/B pump discharge pressure to 80 psig; Potential to send up to 95 psig ammonia back to PV-106 Ammonia Storage Tank; Not expected to result in an overpressure scenario; No adverse consequences identified							
29.6. Low Pressure	29.6.1. PCV-24813 pressure regulator fails fully closed	29.6.1.1. NH3 Forwarding Pump PM-120A/B deadhead, potential seal damage, release of aqueous ammonia, personnel impact	753. PSV-24812A/B (set @ 95 psig) is sized for blocked discharge 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	P	3	1	1	61. No recommendations required.	

Node: 29. Node 29: Aqueous Ammonia Heater System
Drawings / References:
40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
29.7. High Level	29.7.1. Overfilling Ammonia Storage Tank PV-106	29.7.1.1. Potential overpressure of Ammonia Storage Tank PV-106 with release of aqueous ammonia to surrounding area and potential personnel impact	560. Filling PV-106 Ammonia Storage Tank is a continuously monitored operation 561. LIT-14805 high level alarm will prompt operator response	P	4	1	2	61. No recommendations required.	
29.8. Low Level	29.8.1. Failure to re-fill Ammonia Storage Tank PV-106	29.8.1.1. Inability to send ammonia to heater system; Potential inability to control air emissions with potential environmental/financial impact	750. PIT-24815 low pump discharge pressure alarm will prompt operator response 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response 752. FIT-24820 low flow alarm on pump discharge will prompt operator response	E	1	1	1	61. No recommendations required.	
29.9. High Interface Level	29.9.1. Not applicable								
29.10. Low Interface Level	29.10.1. Not applicable								
29.11. High Temperature	29.11.1. Air Heater HTR-1640 control failure	29.11.1.1. Excess hot air supply, potential catalyst damage with potential environmental/financial impact	751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response 562. Temperature indicator TI-14827 high NH3/air temperature indication will prompt operator response	E	1	2	1	61. No recommendations required.	
	29.11.2. External fire around PV-106 Ammonia Storage Tank	29.11.2.1. Potential thermal expansion and release with potential personnel impact	448. No specific safeguards identified	P	3	3	3	20. Perform engineering evaluation to determine if Ammonia Storage Tank PV-106 requires overpressure protection for external fire scenario. If so, determine if PSV-14806 on PV-106 is sized appropriately and update the PSV sizing documentation. If evaluation determines PSV-14806 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
29.12. Low Temperature	29.12.1. Heater unit failure HTR-1640		751. Analyzer indicator AI-20575A/C NOx	E	1	2	1	61. No recommendations required.	

Node: 29. Node 29: Aqueous Ammonia Heater System

Drawings / References:

40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		29.12.1.1. Reduced vaporization of NH3 prior to heater injection, potential impaired distribution of NH3 into Hot Oil Furnace H-101/201; Potential environmental/financial impact	analyzers will prompt operator response 52. Temperature indicator TI-14827 low NH3/air temperature indication will prompt operator response						
29.13. Contamination/ Composition	29.13.1. Out of spec ammonia - high or low aqueous NH3 concentration	29.13.1.1. Higher ammonia with potential environmental/financial impact	563. Ammonia spec is checked upon arrival 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
		29.13.1.2. Higher NOX with potential environmental/financial impact	563. Ammonia spec is checked upon arrival 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
		29.13.1.3. Potential mineralization on vaporizer; Potential pluggage of vaporizer with potential reduced/loss of flow of ammonia; Potential environmental/financial impact	563. Ammonia spec is checked upon arrival 751. Analyzer indicator AI-20575A/C NOx analyzers will prompt operator response	E	1	1	1	61. No recommendations required.	
29.14. Corrosion	29.14.1. No specific causes identified								
29.15. Additional Phase	29.15.1. Not applicable								
29.16. Rupture/Loss of Containment	29.16.1. No additional causes identified								
29.17. Loss of Utilities	29.17.1. Loss of instrument air	29.17.1.1. Loss of programmable logic controller (PLC) cabinet purge; Potential to overheat the PLC with potential environmental/impact	755. Pressure switch low PSL-24824 on instrument air line to system (including cabinet purge) will prompt operator response	E	1	3	1	61. No recommendations required.	
	29.17.2. Loss of power	29.17.2.1. See pump trip scenario, this node							
		29.17.2.2. See blower trip scenario, this node							
29.18. Start-up/Shutdown hazards/ Maintenance	29.18.1. Driver pulls away while loading or hose rupture	29.18.1.1. Potential release from hose/piping, personnel impact	564. Wheel chocks required during loading	P	3	2	2	61. No recommendations required.	

Node: 29. Node 29: Aqueous Ammonia Heater System

Drawings / References:

40-PI-9212; 40-PI-9213; 13J0010_100101; 13J0039_100101

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
29.19. General	29.19.1. No additional causes identified								

Node: 30. Node 30: Natural Gas/Fuel Gas Header

Drawings / References:

40-PI-9228 ; 40-PI-9228 Redline; 40-PI-9231; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
30.1. Low/No Flow	30.1.1. PCV-20355 fails closed or upstream/downstream manual valve(s) inadvertently closed	30.1.1.1. Loss of make-up natural gas to PV-201 Naphtha Stabilizer Accumulator; Inability to send naphtha to Combi Tower T-202; Potential higher temperature in T-202; Potential higher temperature to the heater F-201; Potential increased vapor traffic from T-201; Potential off-spec product with potential financial impact	544. PDAH-20235 high differential pressure alarm on Stabilizer T-201 will prompt operator response	F	1	3	1	61. No recommendations required.	
	30.1.2. FCV-21760 fails closed or upstream/downstream manual valve(s) inadvertently closed	30.1.2.1. Loss of stripping gas to T-202 Combi Tower; Potential off-spec AGO product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	30.1.3. FCV-23480 fails closed or upstream/downstream manual valve(s) inadvertently closed	30.1.3.1. Loss of stripping gas to T-202 Combi Tower; Potential off-spec AGO product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	30.1.4. FCV-25070 fails closed or upstream/downstream manual valve(s) inadvertently closed	30.1.4.1. Loss of sweep gas to flare header; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	30.1.5. PCV-25080 fails fully closed	30.1.5.1. Loss of sweep gas to flare header; Potential incomplete combustion; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	30.1.6. Manual valve closed in main header or supply line to users	30.1.6.1. Unit/plant shutdown - see individual gas users for specific scenarios							
30.2. High Flow	30.2.1. PCV-20355 malfunctions fully open or bypass valve inadvertently open	30.2.1.1. Potential to overpressure T-201 Stabilizer with potential loss of containment, ignition and personnel impact	716. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for blocked vapor outlet 756. PCV-20395C on vapor overheads from PV-201 will open to flare on high overhead pressure	P	4	1	2	61. No recommendations required.	
	30.2.2. FCV-21760 malfunctions open or bypass inadvertently opened	30.2.2.1. Excess stripping gas at 65 psig, increase in Combi Tower T-202 pressure; Potential to overpressure T-202 with loss of containment, potential ignition and personnel impact	701. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet	P	4	1	2	61. No recommendations required.	

Node: **30. Node 30: Natural Gas/Fuel Gas Header**

Drawings / References:

40-PI-9228 ; 40-PI-9228 Redline; 40-PI-9231; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response						
			757. PAH-21715A/B/C high pressure alarm will prompt operator response						
	30.2.3. FCV-23480 malfunctions open or bypass inadvertently opened	30.2.3.1. Excess stripping gas, increase in Combi Tower T-202 pressure; Potential to overpressure T-202 with loss of containment, potential ignition and personnel impact	701. PSV-21690/21710/21720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) on Combi Tower T-202 is sized for blocked vapor outlet	P	4	1	2	61. No recommendations required.	
			757. PAH-21715A/B/C high pressure alarm will prompt operator response						
			597. PAH-21271 High Pressure Alarm on Combi Tower T-202 will prompt operator response						
	30.2.4. FCV-25070 malfunctions open or bypass inadvertently opened	30.2.4.1. Potential for excess gas flow to users; No high pressure or environmental concerns identified; Operability issue only, no hazardous consequences identified							
30.3. Reverse/Misdirected Flow	30.3.1. No specific causes identified								
30.4. High Pressure	30.4.1. PCV-25080 fails fully open	30.4.1.1. Inability to reduce natural gas to 40 psig; Potential larger flame at flare; Potential heater shutdown; Potential environmental/financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
30.5. Low Pressure	30.5.1. No additional causes identified								
30.6. High Level	30.6.1. No additional causes identified								
30.7. Low Level	30.7.1. No additional causes identified								
30.8. High Interface Level	30.8.1. Not applicable								
30.9. Low Interface Level	30.9.1. Not applicable								

Node: **30. Node 30: Natural Gas/Fuel Gas Header**
Drawings / References:
40-PI-9228 ; 40-PI-9228 Redline; 40-PI-9231; 40-PI-9235; 88-P1-9244

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
30.10. High Temperature	30.10.1. No specific causes identified								
30.11. Low Temperature	30.11.1. No specific causes identified								
30.12. Contamination/ Composition	30.12.1. Contaminants in Natural Gas - pipe scale	30.12.1.1. Potential for blockage of heater burners, control valves, regulator; Potential financial/environmental impact	54. Filter FL-208 natural gas filter/separator on gas supply 55. Strainers on gas line to heaters	F	1	3	1	61. No recommendations required.	
30.13. Corrosion	30.13.1. Corrosion on natural gas piping	30.13.1.1. Potential long-term mechanical integrity issues, no acute hazards identified							
30.14. Additional Phase	30.14.1. Not applicable								
30.15. Rupture/Loss of Containment	30.15.1. No additional causes identified								
30.16. Loss of Utilities	30.16.1. Loss of instrument air	30.16.1.1. Valves revert to fail-safe position							
30.17. Start-up/Shutdown hazards/ Maintenance	30.17.1. No additional causes identified								
30.18. General	30.18.1. No additional causes identified								

Node: **31. Node 31: Hydrocarbon Drain**
Drawings / References:
40-PI-9232

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
31.1. Low/No Flow	31.1.1. Hydrocarbon Sump Pump PM-218A/B trips	31.1.1.1. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 601. LAH-22501 high level in TK-201 Sump will kick on other pump PM-218A/B 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **31. Node 31: Hydrocarbon Drain**

Drawings / References:

40-PI-9232

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response to cordon off the area						
	31.1.2. Manual valve closed in Hydrocarbon Sump Pump PM-218A/B discharge	31.1.2.1. Hydrocarbon Sump Pump PM-218A/B deadheaded, potential pump damage; Potential seal failure with potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		31.1.2.2. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
31.2. High Flow	31.2.1. Drain valve left open	31.2.1.1. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	31.2.2. High rate of flow from vacuum truck to sump	31.2.2.1. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 31. Node 31: Hydrocarbon Drain

Drawings / References:

40-PI-9232

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
31.3. Reverse/Misdirected Flow	31.3.1. Reverse flow from Feed Tanks to sump	31.3.1.1. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	F	1	2	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	31.3.2. Send Y-Grade to sump (drain valve left open from pumps PM-201A/B)	31.3.2.1. Potential release through Hydrocarbon Sump TK-201 vent; Potential ignition and personnel impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area 600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
31.4. High Pressure	31.4.1. No additional causes identified								
31.5. Low Pressure	31.5.1. No additional causes identified								
31.6. High Level	31.6.1. No additional causes identified								
31.7. Low Level	31.7.1. No additional causes identified								
31.8. High Interface Level	31.8.1. Not applicable								
31.9. Low Interface Level	31.9.1. Not applicable								

Node: 31. Node 31: Hydrocarbon Drain

Drawings / References:

40-PI-9232

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
31.10. High Temperature	31.10.1. Drain valve left open sending hot material to sump	31.10.1.1. Potential release through Hydrocarbon Sump TK-201 vent; Potential ignition and personnel impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area 600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
31.11. Low Temperature	31.11.1. No specific causes identified								
31.12. Contamination/ Composition	31.12.1. Trash in sump (for example from vacuum trucks) can plug pump screen or damage pump	31.12.1.1. Hydrocarbon Sump Pump PM-218A/B deadheaded, potential pump damage; Potential seal failure with potential ignition and personnel impact	600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		31.12.1.2. Potential to overfill the Hydrocarbon Sump TK-201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area 600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
31.13. Corrosion	31.13.1. No specific causes identified								
31.14. Additional Phase	31.14.1. Not applicable								
31.15. Rupture/Loss of Containment	31.15.1. No additional causes identified								

Node: 31. Node 31: Hydrocarbon Drain

Drawings / References:

40-PI-9232

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
31.16. Loss of Utilities	31.16.1. Loss of power	31.16.1.1. See pump trip scenario, this node							
31.17. Start-up/Shutdown hazards/ Maintenance	31.17.1. Maintenance activities at sump	31.17.1.1. Potential for vapor release during maintenance activities, ignition of vapors from maintenance activity, personnel impact	57. Hot work permit required for work in Class I, Div. 1 area with continuous monitoring 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area	P	3	2	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
31.18. General	31.18.1. No additional causes identified								

Node: 32. Node 32: Wastewater Drain

Drawings / References:

40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9232; 40-PI-9233

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
32.1. Low/No Flow	32.1.1. Contact Water Sump Pump PM-215A/B trip	32.1.1.1. Potential to overfill Contact Water Sump TK-202; Potential release through TK-202 atmospheric vent; Potential to flood unit with potential environmental impact	758. LAH-22504 high level alarm on TK-202 will prompt operator response	E	2	3	2	61. No recommendations required.	
	32.1.2. Manual valve on discharge from Contact Water Sump Pump PM-215A/B inadvertently closed	32.1.2.1. Potential to overfill Contact Water Sump TK-202; Potential release through TK-202 atmospheric vent; Potential to flood unit with potential environmental impact	758. LAH-22504 high level alarm on TK-202 will prompt operator response	E	2	2	1	61. No recommendations required.	
		32.1.2.2. Potential to deadhead Contact Water Sump Pump PM-215A/B with potential pump damage and financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
32.2. High Flow	32.2.1. Drain valve left open or opened inadvertently	32.2.1.1. Potential to send hydrocarbons to Contact Water Sump TK-202, with vent to atmosphere, send to Gulf Coast Water Authority; Potential environmental/financial impact	569. LI-18500A high interface level alarm on TK 5-0 will prompt operator response	F	1	2	1	61. No recommendations required.	
32.3. Reverse/Misdirected Flow	32.3.1. Check valve failure	32.3.1.1. Flow of contact water into TK-202 from Tank 5; Potential to overfill TK-202; Potential release through TK-202 atmospheric vent; Potential to flood unit with potential environmental impact	759. LAH-22504 high level alarm on TK-202 will prompt operator response and will turn on both pumps PM-215A/B	E	2	3	2	61. No recommendations required.	
32.4. High Pressure	32.4.1. No additional causes identified								
32.5. Low Pressure	32.5.1. No additional causes identified								

Node: **32. Node 32: Wastewater Drain**

Drawings / References:

40-PI-9209 Redline; 40-PI-9209 ; 40-PI-9232; 40-PI-9233

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
32.6. High Level	32.6.1. No additional causes identified								
32.7. Low Level	32.7.1. No specific causes identified								
32.8. High Interface Level	32.8.1. Not applicable								
32.9. Low Interface Level	32.9.1. Not applicable								
32.10. High Temperature	32.10.1. No specific causes identified								
32.11. Low Temperature	32.11.1. No specific causes identified								
32.12. Contamination/ Composition	32.12.1. Trash in sump (for example from vacuum trucks) can plug pump screen or damage pump	32.12.1.1. Potential loss of pumping capacity; Potential to overfill TK-202; Potential release through TK-202 atmospheric vent; Potential to flood unit with potential environmental impact	759. LAH-22504 high level alarm on TK-202 will prompt operator response and will turn on both pumps PM-215A/B	E	2	3	2	61. No recommendations required.	
32.13. Corrosion	32.13.1. No specific causes identified								
32.14. Additional Phase	32.14.1. Not applicable								
32.15. Rupture/Loss of Containment	32.15.1. No additional causes identified								
32.16. Loss of Utilities	32.16.1. Power interruption	32.16.1.1. See pump trip scenario, this node							
32.17. Start-up/Shutdown hazards/ Maintenance	32.17.1. Access to outfall #1 and #2 for sampling	32.17.1.1. Requires personnel to go through standing water; Potential snakes in area with potential personnel impact	448. No specific safeguards identified	P	2	3	2	61. No recommendations required.	
32.18. General	32.18.1. No additional causes identified								

Node: **33. Node 33: Air Compressor System**

Drawings / References:

1318317146 SHT 1; 1318317146 SHT 2; 1318317146 SHT 3; 1318317146 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
33.1. Low/No Flow	33.1.1. CM-102A/B/CM-202A/B Air Compressors trip	33.1.1.1. Cabinets using air purge for electrical classification unprotected; Potential flammable atmosphere, potential ignition and personnel impact	760. Standby Air Compressors CM-202A/B automatic	P	3	2	2	61. No recommendations required.	

Node: **33. Node 33: Air Compressor System**

Drawings / References:

1318317146 SHT 1; 1318317146 SHT 2; 1318317146 SHT 3; 1318317146 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			startup - lead/lag system 59. Pressure alarm low PAL-40001 unit air header low air pressure alarm will prompt operator response						
		33.1.1.2. Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	760. Standby Air Compressors CM-202A/B automatic startup - lead/lag system	F	1	3	1	61. No recommendations required.	
	33.1.2. Air filter plugged	33.1.2.1. Cabinets using air purge for electrical classification unprotected; Potential flammable atmosphere, potential ignition and personnel impact	59. Pressure alarm low PAL-40001 unit air header low air pressure alarm will prompt operator response 571. Air filters are on a scheduled preventative maintenance schedule	P	3	2	2	61. No recommendations required.	
		33.1.2.2. Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
33.2. High Flow	33.2.1. CM-102A/B/CM-202A/B Air Compressors both online	33.2.1.1. No adverse consequences identified							
33.3. Reverse/Misdirected Flow	33.3.1. No specific causes identified								
33.4. High Pressure	33.4.1. High air pressure from CM-102A/B/CM-202A/B Air Compressors due to line regulator failure.	33.4.1.1. Maximum Air Compressors CM-202A/B supply is lower than pressure rating of downstream systems (listed at 175 psig); No adverse consequences identified							
33.5. Low Pressure	33.5.1. No additional causes identified								
33.6. High Level	33.6.1. No additional causes identified								
33.7. Low Level	33.7.1. No specific causes identified								
33.8. High Interface Level	33.8.1. Not applicable								
33.9. Low Interface Level	33.9.1. Not applicable								
33.10. High Temperature	33.10.1. External fire around PV-211 Air Receiver	33.10.1.1. Potential thermal expansion and release with potential ignition and personnel impact	448. No specific safeguards identified	P	3	3	3	21. Perform engineering evaluation to determine if PV-111/211 Air Receiver	

Node: **33. Node 33: Air Compressor System**
Drawings / References:
1318317146 SHT 1; 1318317146 SHT 2; 1318317146 SHT 3; 1318317146 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								requires overpressure protection for external fire scenario. If so, determine if PSV-13470 on PV-111 and PSV-23470 on PV-211 are sized appropriately and update the PSV sizing documentation. If evaluation determines PSV-13470 on PV-111 and PSV-23470 on PV-211 are not sized for this scenario, develop a plan to add adequate overpressure protection.	
33.11. Low Temperature	33.11.1. Low ambient temperature	33.11.1.1. Potential freezing air dryers; Potential loss of instrument air to downstream users; Plant shutdown, valves should revert to their fail-safe positions; Potential financial impact/downtime	448. No specific safeguards identified	F	2	5	4	22. Complete the existing project to replace the air compressor system with more reliable equipment, including heat tracing on equipment and building enclosure.	
33.12. Contamination/ Composition	33.12.1. Water in line - desiccant dryer not removing moisture from air or poor desiccant maintenance	33.12.1.1. Potential to accumulate water in pneumatic valves or lines that can freeze during winter conditions damaging valve actuator or loss of valve actuation, respectively; Potential financial impact	60. Lines with potential to see water are heat traced	F	1	3	1	61. No recommendations required.	
	33.12.2. Accumulation of water in Air Receiver pressure vessel PV-211, resulting in corrosion (receiver located upstream of instrument air dryer)	33.12.2.1. Failure of shell integrity on the Air Receiver pressure vessel PV-211; Potential loss of containment with potential financial impact	761. Air Receiver pressure vessel PV-211 drain trap (and manual valve cracked open)	F	1	3	1	61. No recommendations required.	
		33.12.2.2. Potential for rust particles in air headers; Potential build-up in instrument air equipment; Potential to block off instrument air tubing to control valve users; See valve fail-state scenarios in this PHA							
33.13. Corrosion	33.13.1. Potential corrosion due to wet air	33.13.1.1. Potential increased corrosion in air compressor system, ultimate potential loss of containment and financial impact	448. No specific safeguards identified	F	2	4	3	22. Complete the existing project to replace the air compressor system with more reliable equipment, including heat tracing on equipment and building enclosure.	
33.14. Additional Phase	33.14.1. Not applicable								
33.15. Rupture/Loss of Containment	33.15.1. No additional causes identified								
33.16. Loss of Utilities	33.16.1. Loss of power to facility	33.16.1.1. Loss of pressure causing operation valves fail to safe position							
	33.16.2. Power surge	33.16.2.1. Circuit breakers trip, fuses blow; potential to damage various powered equipment; Valves will go to fail-safe position							
33.17. Start-up/Shutdown hazards/ Maintenance	33.17.1. No additional causes identified								
33.18. General	33.18.1. No additional causes identified								

Node: **33. Node 33: Air Compressor System**

Drawings / References:

1318317146 SHT 1; 1318317146 SHT 2; 1318317146 SHT 3; 1318317146 SHT 4

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark

Node: **34. Node 34: Condensate Feed through Desalter**

Drawings / References:

40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
34.1. Low/No Flow	34.1.1. Condensate Booster Pump PM-121A/B/C trip	34.1.1.1. Inability to feed condensate to the pump's respective Desalter; Potential inefficient operation of the Desalter; Potential inability to send feed back to the unit; Potential downtime (less than 5 days) and potential financial impact	573. Spare Condensate Booster Pump PM-121C	F	2	3	2	61. No recommendations required.	
	34.1.2. Valve(s) upstream of booster pumps (PM-121A/B/C) inadvertently closed.	34.1.2.1. Loss of suction to pump PM-121A/B/C; Potential pump cavitation with seal failure, potential loss of containment of condensate, potential ignition and personnel impact	574. Seal failure alarm SFA-18026/18066/18046 will shutdown pumps PM-121A/B/C 351. Fire deluge in the condensate booster pump area	P	4	1	2	61. No recommendations required.	
		34.1.2.2. Inability to feed condensate to the pump's respective Desalter; Potential inefficient operation of the Desalter; Potential inability to send feed back to the unit; Potential downtime (less than 5 days) and potential financial impact	573. Spare Condensate Booster Pump PM-121C	F	2	3	2	61. No recommendations required.	
	34.1.3. Valve(s) downstream of booster pumps closed (or plugged).	34.1.3.1. Blocked discharge from pump PM-121A/B/C; Potential pump deadhead with seal failure, potential loss of containment of condensate, potential ignition and personnel impact	351. Fire deluge in the condensate booster pump area 574. Seal failure alarm SFA-18026/18066/18046 will shutdown pumps PM-121A/B/C	P	4	1	2	61. No recommendations required.	
		34.1.3.2. Inability to feed condensate to the pump's respective Desalter; Potential inefficient operation of the Desalter; Potential inability to send feed back to the unit; Potential downtime (less than 5 days) and potential financial impact	573. Spare Condensate Booster Pump PM-121C	F	2	3	2	61. No recommendations required.	
	34.1.4. PDCV-28100 malfunctions closed	34.1.4.1. Blocked discharge from pump PM-121A/B/C; Potential pump deadhead with seal failure, potential loss of containment of condensate, potential ignition and personnel impact	351. Fire deluge in the condensate booster pump area 574. Seal failure alarm SFA-18026/18066/18046 will shutdown pumps PM-121A/B/C	P	4	1	2	61. No recommendations required.	
		34.1.4.2. Inability to feed condensate to the pump's respective Desalter; Potential inefficient operation of the Desalter; Potential	573. Spare Condensate Booster Pump PM-121C	F	2	3	2	61. No recommendations required.	

Node: 34. Node 34: Condensate Feed through Desalter

Drawings / References:
40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		inability to send feed back to the unit; Potential downtime (less than 5 days) and potential financial impact							
	34.1.5. Manual valve(s) on suction to PM-126A/B Demulsifier Metering Pumps inadvertently closed	34.1.5.1. Inability to send additional demulsifier to desalter pumps; Potential emulsification with potential inefficient separation in the Desalter D-101/201; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		34.1.5.2. Potential loss of suction to PM-126A/B Demulsifier Metering Pump; Potential seal failure and loss of containment to surrounding area; Potential environmental impact	448. No specific safeguards identified	E	1	3	1	61. No recommendations required.	
	34.1.6. Manual valve(s) on discharge from PM-126A/B Demulsifier Metering Pumps inadvertently closed	34.1.6.1. Inability to send additional demulsifier to desalter pumps; Potential emulsification with potential inefficient separation in the Desalter D-101/201; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		34.1.6.2. Blocked discharge from PM-126A/B Demulsifier Metering Pump; Potential pump deadhead; Potential seal failure and loss of containment to surrounding area; Potential environmental impact	582. PSV-18009/18010 (set @ 305 psig) is sized for blocked outlet	E	1	1	1	61. No recommendations required.	
	34.1.7. PM-126A/B Demulsifier Metering Pump trip	34.1.7.1. Inability to send additional demulsifier to desalter pumps; Potential emulsification with potential inefficient separation in the Desalter D-101/201; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
34.2. High Flow	34.2.1. Large slug of water enters system in condensate	34.2.1.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 34. Node 34: Condensate Feed through Desalter

Drawings / References:
40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	34.2.2. PDCV-28100 fails open	34.2.2.1. Potential inability to send out flow from the Desalter, liquid full; Potential overpressure of D-101/201 with potential loss of containment, ignition, and personnel impact	577. PAH-18191/28191 high pressure alarm on Desalter D-101/201 will prompt operator response 352. PSV-18190/28190 (set @ 305 psig) on Desalter D-101/201 are sized for blocked outlet	P	4	1	2	61. No recommendations required.	
34.3. Reverse/Misdirected Flow	34.3.1. Reverse flow of condensate into the demulsifier injection system (Condensate Booster Pump PM-121A/B/C trip)	34.3.1.1. Potential to overfill demulsifier tank TK-108; Potential release of condensate and demulsifier into area; Potential ignition and personnel impact	575. LAH-18008 high level alarm on TK-108 will prompt operator response 350. Lower Explosive Limit alarms in area AI-18007 and AI-18004 will prompt operators to cordon off the area 576. FALL-18022/18062/18042 low-low flow alarm on discharge of pumps PM-121A/B/C will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	34.3.2. Valve misalignment on pump switch (while C pump is operating).	34.3.2.1. Potential loss of feed return to the unit; Potential downtime (less than 5 days) and potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
		34.3.2.2. Potential to direct flow from both Splitters to one desalter; Potential inability to send out flow from the Desalter, liquid full; Potential overpressure of D-101/201 with potential loss of containment, ignition, and personnel impact	577. PAH-18191/28191 high pressure alarm on Desalter D-101/201 will prompt operator response 352. PSV-18190/28190 (set @ 305 psig) on Desalter D-101/201 are sized for blocked outlet	P	4	1	2	61. No recommendations required.	
	34.3.3. 2" manual valve on condensate flush to EX-112A/B or to EX-212A/B inadvertently open	34.3.3.1. Potential to contaminate AGO product with feed; Potential off-spec product with financial impact (contamination of the tank)	448. No specific safeguards identified	F	3	3	3	23. Provide a safeguard for the scenario of inadvertently opening the 2" manual valve on the condensate flush line to EX-112A/B or EX-212A/B resulting in a potential contamination of AGO product with feed.	

Node: 34. Node 34: Condensate Feed through Desalter

Drawings / References:
40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
34.4. High Pressure	34.4.1. PCV-18165/28165 on overhead from D-101/201 Desalter malfunctions closed or upstream/downstream manual valve(s) inadvertently closed	34.4.1.1. Blocked overhead outlet from Desalter D-101/201; Potential overpressure of D-101/201 with potential loss of containment, ignition, and personnel impact	352. PSV-18190/28190 (set @ 305 psig) on Desalter D-101/201 are sized for blocked outlet 577. PAH-18191/28191 high pressure alarm on Desalter D-101/201 will prompt operator response	P	4	1	2	61. No recommendations required.	
		34.4.1.2. Blocked discharge from pump PM-121A/B/C; Potential pump deadhead with seal failure, potential loss of containment of condensate, potential ignition and personnel impact	351. Fire deluge in the condensate booster pump area 574. Seal failure alarm SFA-18026/18066/18046 will shutdown pumps PM-121A/B/C	P	4	1	2	61. No recommendations required.	
		34.4.1.3. Inability to feed condensate to the pump's respective Desalter; Potential inefficient operation of the Desalter; Potential inability to send feed back to the unit; Potential downtime (less than 5 days) and potential financial impact	577. PAH-18191/28191 high pressure alarm on Desalter D-101/201 will prompt operator response	F	2	3	2	61. No recommendations required.	
	34.4.2. External fire around D-101/201 Desalter	34.4.2.1. Potential thermal expansion and release with potential ignition and personnel impact	580. PSV-18190/28190 (set @ 305 psig) on Desalter D-101/201 are sized for external fire	P	4	1	2	61. No recommendations required.	
34.5. Low Pressure	34.5.1. PCV-18165/28165 (back pressure control valve) fails open or bypass inadvertently open	34.5.1.1. Potential loss of pressure in Desalter D-101/201; Potential increased level in T-101/201, potential loss of heat in T-101/201; Potential product quality issues; Potential to pressure up the Combi Tower T-102/202; Potential to overpressure T-102/202 with loss of containment, potential ignition and personnel impact	480. PSV-11690/11710/11720 (set @ 52.5 psig, 50 psig, and 51.25 psig respectively) is sized for loss of cooling 578. LAH-10525 high level alarm on Stabilizer T-101 will prompt operator response 652. LAH-20525 high level alarm on Stabilizer T-201 will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 34. Node 34: Condensate Feed through Desalter

Drawings / References:
40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		34.5.1.2. Potential loss of pressure in Desalter D-101/201; Potential to expose the grid DT-101A/B/C with potential to damage the grid from exposure; Potential financial impact and downtime	579. LSL-18145 low level switch will shut off power for the grids DT-101A/B/C 670. LSL-28145 low level switch will shut off power for grids DT-201A/B/C	F	2	3	2	61. No recommendations required.	
34.6. High Level	34.6.1. Overfilling TK-108 Demulsifier Chemical Storage Tank	34.6.1.1. Potential release of demulsifier to surrounding area with potential environmental impact	575. LAH-18008 high level alarm on TK-108 will prompt operator response	E	1	3	1	61. No recommendations required.	
34.7. Low Level	34.7.1. Failure to refill TK-108 Demulsifier Chemical Storage Tank	34.7.1.1. Inability to send additional demulsifier to desalter pumps; Potential emulsification with potential inefficient separation in the Desalter D-101/201; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	581. LAL-18008 low level alarm on TK-108 will prompt operator response	F	1	3	1	61. No recommendations required.	
		34.7.1.2. Potential loss of suction to PM-126A/B Demulsifier Metering Pump; Potential seal failure and loss of containment to surrounding area; Potential environmental impact	581. LAL-18008 low level alarm on TK-108 will prompt operator response	E	1	3	1	61. No recommendations required.	
34.8. High Interface Level	34.8.1. High water level in Desalter D-101/201	34.8.1.1. Potential to damage the Desalter grids DT-101A/B/C or DT-201A/B/C; Potential financial impact	583. LAH-18140 high level alarm on D-101 will prompt operator response 671. LAH-28140 high level alarm on D-201 will prompt operator response	F	2	3	2	61. No recommendations required.	
34.9. Low Interface Level	34.9.1. Low water level in Desalter D-101/201	34.9.1.1. Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
34.10. High Temperature	34.10.1. No specific causes identified								
34.11. Low Temperature	34.11.1. Cold Desalter temperatures	34.11.1.1. Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
34.12. Contamination/ Composition	34.12.1. No additional causes identified								
34.13. Corrosion	34.13.1. No additional causes identified								
34.14. Additional Phase	34.14.1. Not applicable								

Node: 34. Node 34: Condensate Feed through Desalter

Drawings / References:
40-PI-9006; 40-PI-9057; 40-PI-9058; 40-PI-9059A; 40-PI-9064; 40-PI-9066; 40-PI-9206; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
34.15. Rupture/Loss of Containment	34.15.1. No additional causes identified								
34.16. Loss of Utilities	34.16.1. Loss of power	34.16.1.1. Pneumatic and solenoid valves fail in designed position. Loss of desalter operability; See consequence 34.1.1.1							
		34.16.1.2. See pump trip scenarios, this node							
	34.16.2. Loss of instrument air	34.16.2.1. Pneumatic and solenoid valves fail in designed position. Loss of desalter operability; See consequence 34.1.1.1							
34.17. Start-up/Shutdown hazards/ Maintenance	34.17.1. Failure to shut off water after starting up and before shutting down	34.17.1.1. Potential to short out the Desalter grids DT-101A/B/C; Potential to damage the grids with potential financial impact	584. Desalter start-up and shutdown procedures	F	2	2	1	61. No recommendations required.	
34.18. General	34.18.1. No additional causes identified								

Node: 35. Node 35: Wash Water System

Drawings / References:
40-PI-9058; 40-PI-9059; 40-PI-9059A; 40-PI-9060; 40-PI-9061; 40-PI-9063; 40-PI-9064; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
35.1. Low/No Flow	35.1.1. Valve(s) upstream of wash water pumps PM-122A/B inadvertently closed or strainer plugged	35.1.1.1. Inability to send wash water to the desalter system; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	585. FAL-18393/18387 low flow alarm on PM-122A/B discharge will prompt operator response and will shutdown the pumps	F	1	2	1	61. No recommendations required.	
		35.1.1.2. Inability to send wash water to EX-112A/B and EX-212A/B exchangers; Potential tube rupture; Potential to contaminate wash water with AGO; Potential water vaporization in exchangers, potential overpressure and potential personnel impact	353. FAL-18305/18310 low flow alarm on discharge of pumps will prompt operator response 354. PSV-18315/28315 (set @ 305 psig) on EX-112A/B and EX-212A/B is sized for thermal expansion of liquid and vapor	P	4	1	2	61. No recommendations required.	
		35.1.1.3. Potential loss of suction to pumps PM-122A/B; Potential pump cavitation, potential seal failure with loss of containment of wash water to surrounding area; No environmental impact; Operability issue only, no hazardous consequences identified							
	35.1.2. Valve downstream of wash water pumps closed (or plugged) or plugged flow element on pump discharge	35.1.2.1. Inability to send wash water to the desalter system; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	353. FAL-18305/18310 low flow alarm on discharge of pumps	F	1	2	1	61. No recommendations required.	

Node: 35. Node 35: Wash Water System

Drawings / References:

40-PI-9058; 40-PI-9059; 40-PI-9059A; 40-PI-9060; 40-PI-9061; 40-PI-9063; 40-PI-9064; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			will prompt operator response						
		35.1.2.2. Inability to send wash water to EX-112A/B and EX-212A/B exchangers; Potential tube rupture; Potential to contaminate wash water with AGO; Potential water vaporization in exchangers, potential overpressure and potential personnel impact	353. FAL-18305/18310 low flow alarm on discharge of pumps will prompt operator response 354. PSV-18315/28315 (set @ 305 psig) on EX-112A/B and EX-212A/B is sized for thermal expansion of liquid and vapor	P	4	1	2	61. No recommendations required.	
		35.1.2.3. Blocked discharge from pumps PM-122A/B; Potential pump deadhead, potential seal failure with loss of containment of wash water to surrounding area; No environmental impact; Operability issue only, no hazardous consequences identified							
	35.1.3. FCV-18305/18310 fails closed or upstream/downstream manual valve(s) inadvertently closed	35.1.3.1. Inability to send wash water to the desalter system; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
		35.1.3.2. Inability to send wash water to EX-112A/B and EX-212A/B exchangers; Potential tube rupture; Potential to contaminate wash water with AGO; Potential water vaporization in exchangers, potential overpressure and potential personnel impact	354. PSV-18315/28315 (set @ 305 psig) on EX-112A/B and EX-212A/B is sized for thermal expansion of liquid and vapor 586. TAH-18320/28320 high temperature alarm on EX-112A/B and EX-212A/B will prompt operator response	P	4	1	2	61. No recommendations required.	
		35.1.3.3. Blocked discharge from pumps PM-122A/B; Potential pump deadhead, potential seal failure with loss of containment of wash water to surrounding area; No environmental impact; Operability issue only, no hazardous consequences identified							
	35.1.4. XV-18000 on wash water line before the desalter vessel mix valve fails closed	35.1.4.1. Inability to send wash water to the desalter system; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
		35.1.4.2. Blocked water outlet from EX-112A/B and EX-212A/B exchangers; Potential tube rupture; Potential to contaminate wash water with AGO; Potential water vaporization in exchangers, potential overpressure and potential personnel impact	354. PSV-18315/28315 (set @ 305 psig) on EX-112A/B and EX-212A/B is sized for	P	4	1	2	61. No recommendations required.	

Node: **35. Node 35: Wash Water System**

Drawings / References:

40-PI-9058; 40-PI-9059; 40-PI-9059A; 40-PI-9060; 40-PI-9061; 40-PI-9063; 40-PI-9064; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			thermal expansion of liquid and vapor 586. TAH-18320/28320 high temperature alarm on EX-112A/B and EX-212A/B will prompt operator response						
		35.1.4.3. Blocked discharge from pumps PM-122A/B; Potential pump deadhead, potential seal failure with loss of containment of wash water to surrounding area; No environmental impact; Operability issue only, no hazardous consequences identified							
35.2. High Flow	35.2.1. FCV-18305/18310 malfunctions open.	35.2.1.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response	P	4	1	2	61. No recommendations required.	
	35.2.2. Both Wash Water Pumps PM-122A/B on at the same time	35.2.2.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer	P	4	1	2	61. No recommendations required.	

Node: **35. Node 35: Wash Water System**

Drawings / References:

40-PI-9058; 40-PI-9059; 40-PI-9059A; 40-PI-9060; 40-PI-9061; 40-PI-9063; 40-PI-9064; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Accumulator pressure vessel PV-201 water boot will prompt operator response						
35.3. Reverse/Misdirected Flow	35.3.1. High pressure in condensate inlet of desalter.	35.3.1.1. The maximum discharge pressure of the PM-121A/B pumps is 187 psig, and the wash water pumps discharge at 220 psig, therefore reverse flow of condensate into the wash water is not a credible scenario							
	35.3.2. Manual wash water bypass valve around EX-112A/B or EX-212A/B inadvertently opened	35.3.2.1. Inadvertently bypassing wash water flow around EX-112A/B and EX-212A/B exchangers; Potential tube rupture; Potential to contaminate wash water with AGO; Potential water vaporization in exchangers, potential overpressure and potential personnel impact	354. PSV-18315/28315 (set @ 305 psig) on EX-112A/B and EX-212A/B is sized for thermal expansion of liquid and vapor	P	4	1	2	61. No recommendations required.	
35.4. High Pressure	35.4.1. Blocked in liquids in EX-113A/B/C exchangers	35.4.1.1. Potential corrosion, potential thermal expansion with release and personnel impact	615. PSV-18351/18355/18335 (set @ 305 psig) is sized for thermal expansion and vapor generation	P	3	2	2	61. No recommendations required.	
35.5. Low Pressure	35.5.1. No additional causes identified								
35.6. High Level	35.6.1. No additional causes identified								
35.7. Low Level	35.7.1. No additional causes identified								
35.8. High Interface Level	35.8.1. No additional causes, see Node 36								
35.9. Low Interface Level	35.9.1. No additional causes, see Node 36								
35.10. High Temperature	35.10.1. High flow on the shell side (AGO side) of EX-112A/B or EX-212A/B	35.10.1.1. Potential for higher temperature wash water to desalter. Operational issue, no hazardous consequences identified							
35.11. Low Temperature	35.11.1. Low water supply temperature.	35.11.1.1. Reduced temperature at mixing. Potential cold Desalter temperature; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
35.12. Contamination/ Composition	35.12.1. High amount of sodium bi sulfide in water	35.12.1.1. Potential to corrode piping and vessels, potential release of condensate to atmosphere; Potential ignition and personnel impact	589. Preventative Maintenance (PM) inspections 590. Injection of catalyzed oxygen is monitored as part of operator rounds	P	3	2	2	61. No recommendations required.	
35.13. Corrosion	35.13.1. No additional causes identified								

Node: 35. Node 35: Wash Water System

Drawings / References:

40-PI-9058; 40-PI-9059; 40-PI-9059A; 40-PI-9060; 40-PI-9061; 40-PI-9063; 40-PI-9064; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
35.14. Additional Phase	35.14.1. Not applicable								
35.15. Rupture/Loss of Containment	35.15.1. Tube leak/rupture in EX-113A/B/C	35.15.1.1. Wash water flows to waste water; Reduced flow of wash water to desalters; Potential increased waste water flow to Tank 5, not expected to result in a tank overfill; Potential offspec product with potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
	35.15.2. Tube leak/rupture in E-112A/B Bottoms Product to Desalter Wash Water Heater	35.15.2.1. Water contaminates the AGO stream. Product quality issues (offspec AGO product)	587. TAL-18320/28320 low temperature alarm on E-112A/B and E-212A/B will prompt operator response 588. TAL-11263A/B and TAL-21263A/B low temperature alarm from AF-103/203 will prompt operator response	F	3	2	2	61. No recommendations required.	
35.16. Loss of Utilities	35.16.1. Loss of power	35.16.1.1. See pump trip scenarios, this node							
	35.16.2. Loss of instrument air	35.16.2.1. Valves revert to fail-safe position							
35.17. Start-up/Shutdown hazards/ Maintenance	35.17.1. Reverse flow of condensate into the wash water system during shutdown	35.17.1.1. Potential contamination of wash water with condensate; Potential inefficient chloride removal; Potential inability to remove chlorides from feed; Potential quality issues; Potential financial impact	448. No specific safeguards identified	F	1	4	2	24. Implement a backflow preventer on the wash water injection point to the Desalter system to prevent reverse flow of condensate into the wash water during shutdown.	
35.18. General	35.18.1. No additional causes identified								

Node: 36. Node 36: Waste Water

Drawings / References:

40-PI-9045; 40-PI-9058; 40-PI-9059; 40-PI-9061; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
36.1. Low/No Flow	36.1.1. LCV-18130/28130 fails closed or manual valve in water outlet line inadvertently closed.	36.1.1.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling	P	4	1	2	61. No recommendations required.	

Node: 36. Node 36: Waste Water

Drawings / References:

40-PI-9045; 40-PI-9058; 40-PI-9059; 40-PI-9061; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response						
			543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response						
		36.1.1.2. Potential waste water on the Desalter grids DT-101A/B/C or DT-201A/B/C, potential to damage	583. LAH-18140 high level alarm on D-101 will prompt operator response	F	2	3	2	61. No recommendations required.	
			671. LAH-28140 high level alarm on D-201 will prompt operator response						
	36.1.2. Plugged exchanger EX-113A/B/C or manual valve(s) on inlet/outlet of exchanger inadvertently closed	36.1.2.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling	P	4	1	2	61. No recommendations required.	
			599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling						
			7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response						
			543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response						

Node: 36. Node 36: Waste Water

Drawings / References:

40-PI-9045; 40-PI-9058; 40-PI-9059; 40-PI-9061; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		36.1.2.2. Potential waste water on the Desalter grids DT-101A/B/C or DT-201A/B/C, potential to damage	583. LAH-18140 high level alarm on D-101 will prompt operator response 671. LAH-28140 high level alarm on D-201 will prompt operator response	F	2	3	2	61. No recommendations required.	
	36.1.3. Plugged waste water cooler (AF-109, AF-209).	36.1.3.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response 543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response	P	4	1	2	61. No recommendations required.	
		36.1.3.2. Potential waste water on the Desalter grids DT-101A/B/C or DT-201A/B/C, potential to damage	583. LAH-18140 high level alarm on D-101 will prompt operator response 671. LAH-28140 high level alarm on D-201 will prompt operator response	F	2	3	2	61. No recommendations required.	
		36.1.3.3. Potential reduced ability to cool waste water in AF-109/209; Potential to send higher temperature waste water to downstream tank T-5; The maximum temperature we can send to Gulf Coast is 140 degF; Potential to send waste water exceeding 140 degF to Gulf Coast; Potential financial impact - Gulf Coast fining or requiring a shutdown	612. TAH-18353/28353 high temperature alarm on AF-109/209 outlet will prompt operator response	F	1	3	1	61. No recommendations required.	

Node: 36. Node 36: Waste Water

Drawings / References:

40-PI-9045; 40-PI-9058; 40-PI-9059; 40-PI-9061; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
36.2. High Flow	36.2.1. LCV-18130/28130 malfunctions open or bypass valve inadvertently opened	36.2.1.1. High flow of waste water to waste water tank T-5, however not enough city water pressure to result in an overfill of T-5; Potential to send grease out with the waste water from T-5; Potential environmental/financial impact	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response	F	1	2	1	61. No recommendations required.	
		36.2.1.2. Potential to drain the Desalter D-101/201, potential to send oil to the waste water system; Potential for Gulf Coast to require a shutdown	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response	F	1	2	1	61. No recommendations required.	
36.3. Reverse/Misdirected Flow	36.3.1. AF-109/209 cooler bypass valve inadvertently opened.	36.3.1.1. Inability to cool waste water in AF-109/209; Potential to send higher temperature waste water to downstream tank T-5; The maximum temperature we can send to Gulf Coast is 140 degF; Potential to send waste water exceeding 140 degF to Gulf Coast; Potential financial impact - Gulf Coast fining or requiring a shutdown	612. TAH-18353/28353 high temperature alarm on AF-109/209 outlet will prompt operator response	F	1	3	1	61. No recommendations required.	
36.4. High Pressure	36.4.1. Blocked in liquids in EX-113A/B/C exchangers (spare)	36.4.1.1. Potential corrosion, potential thermal expansion with release and personnel impact	448. No specific safeguards identified	P	3	3	3	25. Perform engineering evaluation to determine the need for overpressure protection from thermal expansion on the tube side of exchangers EX-113A/B/C and develop a plan to install thermal relief protection if deemed necessary.	
36.5. Low Pressure	36.5.1. Operational upset or condensation causes vacuum in exchangers EX-113A/B/C while blocked in for maintenance (hot water cooled down too quickly)	36.5.1.1. Collapse of vessel with release of water or condensate. Potential ignition, personnel impact.	448. No specific safeguards identified	P	4	1	2	61. No recommendations required.	
36.6. High Level	36.6.1. No additional causes identified								
36.7. Low Level	36.7.1. No additional causes identified								
36.8. High Interface Level	36.8.1. No specific causes identified								
36.9. Low Interface Level	36.9.1. No specific causes identified								
36.10. High Temperature	36.10.1. Fan failure on AF-109/209	36.10.1.1. Inability to cool waste water in AF-109/209; Potential to send higher temperature waste water to downstream tank T-5; The maximum temperature we can send to Gulf Coast is 140 degF; Potential to send waste water exceeding 140 degF to Gulf Coast; Potential financial impact - Gulf Coast fining or requiring a shutdown	612. TAH-18353/28353 high temperature alarm on AF-109/209 outlet will prompt operator response	F	1	3	1	61. No recommendations required.	

Node: 36. Node 36: Waste Water

Drawings / References:

40-PI-9045; 40-PI-9058; 40-PI-9059; 40-PI-9061; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	36.10.2. Waste water AF-109/209 cooler louver failure.	36.10.2.1. Potential reduced ability to cool waste water in AF-109/209; Potential to send higher temperature waste water to downstream tank T-5; The maximum temperature we can send to Gulf Coast is 140 degF; Potential to send waste water exceeding 140 degF to Gulf Coast; Potential financial impact - Gulf Coast fining or requiring a shutdown	612. TAH-18353/28353 high temperature alarm on AF-109/209 outlet will prompt operator response	F	1	3	1	61. No recommendations required.	
36.11. Low Temperature	36.11.1. Low ambient temperature.	36.11.1.1. Potential freezing of valves/piping. Potential release of water to atmosphere with potential ignition, potential personnel impact and financial impact	355. Most desalter water piping is heat traced and insulated 613. TAL-18353/28353 low temperature alarm on AF-109/209 outlet will prompt operator response	P	3	4	4	26. Perform engineering evaluation to determine additional freeze protection requirements in the Desalter units, in order to mitigate the likelihood of low ambient temperatures causing water pipe/valves freezing resulting in a loss of containment.	
36.12. Contamination/ Composition	36.12.1. No additional causes identified								
36.13. Corrosion	36.13.1. No additional causes identified								
36.14. Additional Phase	36.14.1. Not applicable								
36.15. Rupture/Loss of Containment	36.15.1. No additional causes identified								
36.16. Loss of Utilities	36.16.1. Loss of power	36.16.1.1. See AF-109/209 fan failure scenario, this node							
	36.16.2. Loss of instrument air	36.16.2.1. Valves revert to fail-safe position							
36.17. Start-up/Shutdown hazards/ Maintenance	36.17.1. Reverse flow from Tank T-5 back to the Desalter D-101/201	36.17.1.1. Operability issue only, no hazardous consequences identified							
36.18. General	36.18.1. Spare heat exchanger EX-113B isolated	36.18.1.1. Potential thermal growth or bio-growth and pitting. Potential corrosion leading to failure. Potential release of water and condensate, potential ignition and personnel exposure	614. EX-113A/B/C Exchangers are included in the facility Mechanical Integrity (MI) program	P	3	3	3	27. Perform engineering evaluation to determine the need to rotate the spare exchanger EX-113A/B/C (in and out of service) to prevent long-term isolation of the exchanger resulting in thermal/bio-growth and pitting, corrosion, and failure.	

Node: 37. Node 37: Mud Wash

Drawings / References:

40-PI-9058; 40-PI-9058A; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
37.1. Low/No Flow	37.1.1. Mud Water Pump PM-123B/C trip	37.1.1.1. Loss of ability to provide mud wash to the Desalter D-101/201. Sediment build up in the desalter vessels; Potential grid damage DT-101B/C; Potential product quality issues; Potential financial impact	360. Desalter sampling will indicate sediment build up (samples taken daily)	F	2	3	2	61. No recommendations required.	

Node: 37. Node 37: Mud Wash

Drawings / References:

40-PI-9058; 40-PI-9058A; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	37.1.2. Valve(s) upstream of mud water pumps (PM-123B/C) inadvertently closed	37.1.2.1. Loss of suction to pumps PM-123B/C; Potential pump damage (cavitation). Potential small release of mud wash water and condensate to atmosphere and personnel exposure.	616. Procedure GPS-1315 includes Mud Wash Pump PM-123B/C valve alignment.	P	3	2	2	61. No recommendations required.	
	37.1.3. Valve(s) downstream of mud water pumps PM-123B/C closed (or plugged)	37.1.3.1. Blocked discharge from pump PM-123B/C; Potential pump damage (deadheading). Potential release of mud wash water and condensate to atmosphere and personnel exposure.	616. Procedure GPS-1315 includes Mud Wash Pump PM-123B/C valve alignment.	P	3	2	2	61. No recommendations required.	
	37.1.4. Plugging of internal mud wash header.	37.1.4.1. Potential reduced ability to provide mud wash to the Desalter D-101/201. Sediment build up in the desalter vessels; Potential grid damage DT-101B/C; Potential product quality issues; Potential financial impact	360. Desalter sampling will indicate sediment build up (samples taken daily)	F	2	3	2	61. No recommendations required.	
		37.1.4.2. Potential loss of suction to pumps PM-123B/C; Potential pump damage (cavitation). Potential small release of mud wash water and condensate to atmosphere and personnel exposure.	616. Procedure GPS-1315 includes Mud Wash Pump PM-123B/C valve alignment.	P	3	2	2	61. No recommendations required.	
	37.1.5. FCV-18260 fails closed or upstream/downstream manual valve(s) inadvertently closed	37.1.5.1. Loss of ability to provide mud wash to the Desalter D-101/201. Sediment build up in the desalter vessels; Potential grid damage DT-101B/C; Potential product quality issues; Potential financial impact	360. Desalter sampling will indicate sediment build up (samples taken daily)	F	2	3	2	61. No recommendations required.	
		37.1.5.2. Blocked discharge from pump PM-123B/C; Potential pump damage (deadheading). Potential release of mud wash water and condensate to atmosphere and personnel exposure.	448. No specific safeguards identified	P	3	3	3	28. Provide a safeguard for the scenario of FCV-18260 failing closed, causing blocked discharge from pump PM-123B/C with potential loss of containment. Possible recommendation from the PHA team: 1) Consider adding a low flow switch downstream of pump PM-123B/C that will shutdown pumps.	
37.2. High Flow	37.2.1. FCV-18260 malfunctions open or bypass valve inadvertently opened	37.2.1.1. Potential build up of level in Desalter; Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling 7. LAH-10360 High Level Alarm on	P	4	1	2	61. No recommendations required.	

Node: 37. Node 37: Mud Wash

Drawings / References:

40-PI-9058; 40-PI-9058A; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response						
			543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response						
		37.2.1.2. Potential waste water on the Desalter grids DT-101A/B/C or DT-201A/B/C, potential to damage	583. LAH-18140 high level alarm on D-101 will prompt operator response	F	2	3	2	61. No recommendations required.	
			671. LAH-28140 high level alarm on D-201 will prompt operator response						
37.3. Reverse/Misdirected Flow	37.3.1. Reverse flow into the firewater system (Mud Wash Pump PM-123A/B/C - seal failure/check valve)	37.3.1.1. Reverse flow of water/condensate to the fire water system through the seal flush tubing; Potential release from fire water monitor with potential environmental/financial impact	448. No specific safeguards identified	F	2	5	4	29. Complete the incident investigation (Incident #3190149) for the incident that occurred on 3/31/2024 involving the reverse flow of water/condensate to the fire water system, and implement corrective actions as identified.	1. Incident that occurred on 3/31/2024 - still under investigation.
	37.3.2. 2" manual valve on waste water (downstream of Mud Wash Pumps PM-123B/C) inadvertently opened	37.3.2.1. Inadvertently route mud wash to the outlet of the Desalter D-101/201, ultimately sending flow to the Stabilizer Towers T-101/201; Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling	P	4	1	2	61. No recommendations required.	
			599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling						
			7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response						
			543. LAH-20360 High Level Alarm on Naphtha Stabilizer						

Node: 37. Node 37: Mud Wash

Drawings / References:

40-PI-9058; 40-PI-9058A; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Accumulator pressure vessel PV-201 water boot will prompt operator response						
	37.3.3. Inadvertently opening 1" manual valve on pump PM-123A/B/C suction header to flare	37.3.3.1. Inadvertently send mud wash water to flare; Potential long-term corrosion issues on flare piping with potential release, potential ignition and personnel impact	448. No specific safeguards identified	P	3	3	3	30. Set the 1" manual valve on suction header to Mud Wash pumps PM-123A/B/C to flare as CSO to prevent inadvertently routing water to flare.	
37.4. High Pressure	37.4.1. Blocked in liquids in piping upstream/downstream Mud Wash Pumps PM-123A/B/C	37.4.1.1. Potential thermal expansion and release with potential personnel impact	448. No specific safeguards identified	P	3	3	3	31. Perform engineering evaluation to ensure the piping upstream/downstream of Mud Wash Pumps PM-123A/B/C has adequate thermal expansion relief valves. Develop plan to install additional relief valve if required and provide adequate access for maintenance.	
37.5. Low Pressure	37.5.1. No additional causes identified								
37.6. High Level	37.6.1. No additional causes identified								
37.7. Low Level	37.7.1. No specific causes identified								
37.8. High Interface Level	37.8.1. Not applicable								
37.9. Low Interface Level	37.9.1. Not applicable								
37.10. High Temperature	37.10.1. No additional causes identified								
37.11. Low Temperature	37.11.1. Low ambient temperature.	37.11.1.1. Potential freezing of valves/piping. Potential release of water/condensate to atmosphere with potential ignition, potential personnel impact and financial impact	617. Piping to/from Mud Wash Pumps PM-123B/C is heat traced and insulated	P	3	4	4	26. Perform engineering evaluation to determine additional freeze protection requirements in the Desalter units, in order to mitigate the likelihood of low ambient temperatures causing water pipe/valves freezing resulting in a loss of containment.	
37.12. Contamination/ Composition	37.12.1. Spare pump blocked in and shut off	37.12.1.1. Potential corrosion of lines. Potential release of condensate and water with potential personnel impact	448. No specific safeguards identified	P	3	3	3	32. Perform engineering evaluation to determine the need to rotate the spare pump PM-123A/B/C to prevent long-term isolation of a single pump resulting in corrosion and release, and add to rotation schedule if deemed necessary.	
37.13. Corrosion	37.13.1. No additional causes identified								
37.14. Additional Phase	37.14.1. Not applicable								

Node: **37. Node 37: Mud Wash**

Drawings / References:

40-PI-9058; 40-PI-9058A; 40-PI-9241

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
37.15. Rupture/Loss of Containment	37.15.1. No additional causes identified								
37.16. Loss of Utilities	37.16.1. Loss of power	37.16.1.1. See pump PM-123A/B/C scenario, this node							
	37.16.2. Loss of instrument air	37.16.2.1. Valves revert to fail-safe position							
37.17. Start-up/Shutdown hazards/ Maintenance	37.17.1. Increased frequency of Desalter mud washing	37.17.1.1. Increased amount of solid build-up in tank T-5, requiring more frequent tank clean-out; Potential increased wear on the tank; Potential damage to the floating roof; Potential release of solid build-up to Gulf Coast; Potential environmental/financial impact	618. Regular maintenance on tank T-5 to remove solid build-up (during scheduled outages)	F	3	3	3	33. Provide an additional safeguard to mitigate the likelihood of solid build-up in waste water tank T-5 resulting in potential damage to the floating roof and potential release of solid build-up to Gulf Coast.	
37.18. General	37.18.1. No additional causes identified								

Node: **38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25**

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
38.1. Low/No Flow	38.1.1. Manual valve closed downstream of rundown chiller EX-114 to tank TK-120-24/25	38.1.1.1. Loss of Light Naphtha flow to storage, deadhead of overhead pump not expected as reflux continue to flow, Combi Tower Accumulator pressure vessel PV-102/ 202 fills, potential to overfill accumulator to Recycle Compressor Suction Scrubber pressure vessel PV-107/ 207 compressor suction drum and to compressor, compressor damage	619. LAH-11320/21320 High Level Alarm on PV-102/202 Accumulator will prompt operator response 620. LAHH-13505/23505 High-High Level Alarm on PV-107/207 Suction Scrubber will shutdown Splitter 1/2 RGCs	F	1	1	1	35. Perform safeguard verification for the Tank Farm Nodes (38-45) to ensure that each safeguard listed performs the correct interlock or alarm as stated in the PHA report. Update the P&IDs to show interlocks and/or alarms.	
		38.1.1.2. Loss of Light Naphtha flow to storage from chiller, loss of heat load on glycol circuit, low delta temperature on chiller units, potential chiller compressor damage, release of glycol/ refrigerant and minor environmental impact, potential equipment damage	361. TT-5002/5004 supply water temperature will shutdown chiller compressor units on low heat load on the glycol system 621. FSL-5001/5002 low flow switch will shutdown chiller compressor units	E	1	1	1	61. No recommendations required.	
		38.1.1.3. Loss of Light Naphtha flow to storage from chiller, potential to overpressure chiller exchanger naphtha side, potential loss of containment of naphtha with potential personnel impact	448. No specific safeguards identified	P	4	3	4	34. Perform engineering evaluation to verify that PSV-9675/9676 on rundown/recirculation chillers EX-114/115 is sized for blocked exchanger	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								outlet. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9675/9676 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
	38.1.2. Manual valve closed on suction of Naphtha Recirculation Pump P-6 or strainer plugs	38.1.2.1. Loss of flow to Light Naphtha Recirculation Pump P-6, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		38.1.2.2. Loss of Light Naphtha flow to storage from chiller, loss of heat load on glycol circuit, low delta temperature on chiller units, potential chiller compressor damage, release of glycol/ refrigerant and minor environmental impact, potential equipment damage	361. TT-5002/5004 supply water temperature will shutdown chiller compressor units on low heat load on the glycol system 621. FSL-5001/5002 low flow switch will shutdown chiller compressor units	E	1	1	1	61. No recommendations required.	
		38.1.2.3. Loss of flow to Light Naphtha Recirculation Pump P-6, and recirculation chiller, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will	F	1	1	1	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response 362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number						
		38.1.2.4. Loss of Light Naphtha flow to storage from chiller, potential to overpressure chiller exchanger naphtha side, potential loss of containment of naphtha with potential personnel impact	448. No specific safeguards identified	P	4	3	4	34. Perform engineering evaluation to verify that PSV-9675/9676 on rundown/recirculation chillers EX-114/115 is sized for blocked exchanger outlet. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9675/9676 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
	38.1.3. Recirculation Pump P-6 stops	38.1.3.1. Loss of Light Naphtha flow to storage from chiller, potential to overpressure chiller exchanger naphtha side, potential loss of containment of naphtha with potential personnel impact	448. No specific safeguards identified	P	4	3	4	34. Perform engineering evaluation to verify that PSV-9675/9676 on rundown/recirculation chillers EX-114/115 is sized for blocked exchanger outlet. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9675/9676 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
		38.1.3.2. Loss of Light Naphtha flow to storage from chiller, loss of heat load on glycol circuit, low delta temperature on chiller units, potential chiller compressor damage, release of glycol/ refrigerant and minor environmental impact, potential equipment damage	361. TT-5002/5004 supply water temperature will shutdown chiller compressor units on low heat load on the glycol system 621. FSL-5001/5002 low flow switch will shutdown chiller compressor units	E	1	2	1	61. No recommendations required.	
		38.1.3.3. Loss of flow to Light Naphtha Recirculation Pump P-6, and recirculation chiller, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19	F	1	1	1	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			on rundown/ recirculation chiller outlet temperature will prompt operator response 362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number						
	38.1.4. 10" valve closed on discharge of Recirculation Pump P6 or MOV-9439/9449 closed on inlet to TK-120-24 or MOV-9431/9441 closed on inlet to TK-120-25	38.1.4.1. Loss of flow from Light Naphtha Recirculation Pump P-6, potential deadhead and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		38.1.4.2. Loss of Light Naphtha flow to storage from chiller, loss of heat load on glycol circuit, low delta temperature on chiller units, potential chiller compressor damage, release of glycol/ refrigerant and minor environmental impact, potential equipment damage	361. TT-5002/5004 supply water temperature will shutdown chiller compressor units on low heat load on the glycol system 621. FSL-5001/5002 low flow switch will shutdown chiller compressor units	E	1	1	1	61. No recommendations required.	
		38.1.4.3. Loss of Light Naphtha flow to storage from chiller, potential to overpressure chiller exchanger naphtha side, potential loss of containment of naphtha with potential personnel impact	448. No specific safeguards identified	P	4	3	4	34. Perform engineering evaluation to verify that PSV-9675/9676 on rundown/recirculation chillers EX-114/115 is sized for blocked exchanger outlet. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9675/9676 is not sized for this scenario, develop a	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								plan to add adequate overpressure protection.	
	38.1.5. MOV-9449/9441 eductor closed and MOV-9439/ 31 diffuser open on tanks TK-120-24/25	38.1.5.1. Potential reduced tank mixing, poor H2S scavenger efficiency; Potential need for additional scavenger to the tank to prevent increased H2S content; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
		38.1.5.2. Increased flow to diffuser, not an issue under normal circumstances, potential issue when sending light naphtha to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	622. TK-120-24/25 Initial Fill Procedure (GPS 348) includes guidelines for maximum flowrates 623. TK-120-24/25 and TK-100-15 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9449/9441/9442 and opening diffuser MOV-9439/9431/9432 626. TK-120-24/25 and TK-100-15 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	38.1.6. MOV-9421/ 9429/ 9410/ 9412 closed on outlet of tank TK-120-24/25 to Recirculation Pump P-6 when tank is in recirculation	38.1.6.1. Loss of flow to Light Naphtha Recirculation Pump P-6, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		38.1.6.2. Loss of Light Naphtha flow to storage from chiller, loss of heat load on glycol circuit, low delta temperature on chiller units, potential chiller compressor damage, release of glycol/ refrigerant and minor environmental impact, potential equipment damage	361. TT-5002/5004 supply water temperature will shutdown chiller compressor units on low heat load on the glycol system	E	1	1	1	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			621. FSL-5001/5002 low flow switch will shutdown chiller compressor units						
		38.1.6.3. Loss of flow to Light Naphtha Recirculation Pump P-6, and recirculation chiller, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response 362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number	F	1	1	1	61. No recommendations required.	
		38.1.6.4. Loss of Light Naphtha flow to storage from chiller, potential to overpressure chiller exchanger naphtha side, potential loss of containment of naphtha with potential personnel impact	448. No specific safeguards identified	P	4	3	4	34. Perform engineering evaluation to verify that PSV-9675/9676 on rundown/recirculation chillers EX-114/115 is sized for blocked exchanger outlet. If PSV is sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9675/9676 is not sized for this scenario, develop a plan to add adequate overpressure protection.	
	38.1.7. MOV-9421/ 9429/ 9410/ 9412 closed on outlet of tank TK-120-24/25 to Transfer Pump P-2 when tank is in transfer to pipeline	38.1.7.1. Loss of flow to Light Naphtha Recirculation Pump P-6, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			voting and alarm to prompt operator response						
	38.1.8. MOV-9414/ 9415 closed or MOV-9412/9410 closed on suction to Transfer Pump P-2A/B or strainer plugs	38.1.8.1. Decreased flow of naphtha from naphtha tank TK-120-24/25 to Transfer Pump P-2A/B, loss of flow to Light Naphtha P-2A/B, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed 376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	38.1.9. Blocked outlet of Transfer Pump P-2A/B or 16" valve closed on product manifold out of facility	38.1.9.1. Decreased flow of naphtha from naphtha tank TK-120-24/25 to Transfer Pump P-2A/B, not expected to overpressure downstream piping, loss of flow to Light Naphtha P-2A/B, potential deadhead and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed 376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
	38.1.10. P-20A/B/C glycol pump trip	38.1.10.1. Loss of glycol flow to Light Naphtha Rundown/Recirculation Chillers EX-114/115, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance	362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number	F	1	1	1	61. No recommendations required.	
			363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response						
			365. Temperature alarm high TAH-15/ 19 on naphtha side of rundown/ recirculation chiller outlet temperature						
			366. Flow switch FS-1/2 on low flow shutdown on glycol after chiller compressors with alarm						
		38.1.10.2. Loss of glycol flow to Compressor Chiller exchangers CH-1A/1B, potential to damage the chiller compressor, increased temperature of tanks TK-120-24/25; increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number	F	1	1	1	61. No recommendations required.	
			363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response						

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response						
38.2. High Flow	38.2.1. High glycol flow (2 or 3 pumps P-20A/B/C online when not needed)	38.2.1.1. High glycol flow through Compressor Chiller exchangers CH-1A/1B, no equipment issues from increased glycol flow through the chiller exchangers; Operability issue only, no hazardous consequences identified							
		38.2.1.2. Increased flow of glycol through Naphtha Rundown/ Recirculation chillers EX-114/115; decreased naphtha recirculation/ rundown temperature, operability issue, no hazardous consequences identified							
	38.2.2. MOV-9413 open on pump P-2A/B bypass	38.2.2.1. Reduced suction to pumps P-2A/B, reduced pump discharge pressure; Potential reduced efficiency due to low suction pressure and potential reduced flow of naphtha to pipeline; Potential financial impact	377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response	F	1	2	1	61. No recommendations required.	
	38.2.3. Both pumps P-2A/B running at once	38.2.3.1. Flow is maximized under normal circumstances. Flow is regulated downstream by receiver, operability issue, no hazardous consequences identified							
38.3. Reverse/Misdirected Flow	38.3.1. 8" Rundown chiller bypass open	38.3.1.1. Bypass flow around the Light Naphtha Rundown Chiller EX-114, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response	F	1	1	1	61. No recommendations required.	
			364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response						
			362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number						
		38.3.1.2. Loss of Light Naphtha flow to Rundown chiller; Potential inefficient chiller operation; No adverse consequences identified							
	38.3.2. 8" Rundown line open to second tank when not intended	38.3.2.1. Routing flow to a tank that is not in operation; No stand-alone consequences identified							

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	38.3.3. Manual Recirculation chiller bypass open	38.3.3.1. Loss of flow to Naphtha Recirculation Chiller EX-115, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance	363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response 362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number	F	1	1	1	61. No recommendations required.	
		38.3.3.2. Loss of Light Naphtha flow to Rundown chiller; Potential inefficient chiller operation; No adverse consequences identified							
	38.3.4. 10" recirculation line open to second tank when not intended	38.3.4.1. Routing flow to a tank that is not in operation; No stand-alone consequences identified							
	38.3.5. 8" valve V-9106 from Y-grade spheres open	38.3.5.1. Potential to flow Y-grade to Heavy Naphtha Tank TK-100-15 or Light Naphtha Tank TK-120-24/25, increased light naphtha true vapor pressure (TVP), potential major release from Light Naphtha Tank TK-120-24/25, potential vapor cloud and vapor cloud explosion, personnel injury, environmental impact, asset damage	448. No specific safeguards identified	P	4	3	4	36. Set the 8" valve (V-9106) from Spheres as CSC to prevent inadvertently routing Y-grade to Naphtha Tanks (TK-200-24/25, TK-100-15), or remove this spool piece entirely. See P&ID 88-PI-9226	
	38.3.6. Tube leak on rundown chiller EX-114	38.3.6.1. Potential for naphtha to flow into the glycol system, potential to overpressure glycol system exchangers and equipment; Potential release of glycol and naphtha to grade through the PSV-9675; Potential pool fire with personnel impact	625. EX-114/115 Rundown and Recirculation Chillers are located inside of a berm in a remote location of the tank farm unit	P	4	1	2	61. No recommendations required.	1. Low likelihood of ignition.
	38.3.7. Tube leak on recirculation chiller EX-115	38.3.7.1. Potential to flow glycol into the naphtha tanks TK-120-24/25; Potential to shutdown chiller on loss of flow; increased temperature of tanks TK-120-24/25; increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance; Potential environmental/financial impact	362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number 363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response	F	1	1	1	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response						
	38.3.8. MOV-9439/ 9449 open on naphtha inlet to tank TK-120-24 when not intended (on initial start-up)	38.3.8.1. Potential issue when sending light naphtha to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	623. TK-120-24/25 and TK-100-15 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9449/9441/9442 and opening diffuser MOV-9439/9431/9432 626. TK-120-24/25 and TK-100-15 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	38.3.9. 6" manual valve X-9202 open on low suction line from tank TK-120-24 when not intended	38.3.9.1. Potential to draw water in naphtha product, off-spec product, operability issue, no hazardous consequences identified							
	38.3.10. 6" manual valve X-9206 open on low suction line from tank TK-120-25 when not intended	38.3.10.1. Potential to draw water in naphtha product, off-spec product, operability issue, no hazardous consequences identified							
	38.3.11. 12" valve V-9120/9122 open and tank outlet MOV open from the light virgin naphtha LVN tank not in recirculation service to the other LVN tank (both in LVN service)	38.3.11.1. Inadvertently route flow to the closed off recirculation tank; No adverse consequences identified							
	38.3.12. Valve misalignment on naphtha transfer header - Tank TK-120-24 and 120-25 in light; tank TK-100-15 heavy	38.3.12.1. (Flow Tank TK-120-24 or TK-120-25 to TK-100-15) Potential major flow of light naphtha to heavy naphtha, potential issue with inventory tracking, potential off-spec product, operability issue, potential deviation on environmental permitting - incorrect product in tank, environmental impact	375. SAAB level transmitter LT-9409 on light naphtha tank TK-120-24 will alarm on unintended flow direction (intentional flow direction is manually set) 374. SAAB level transmitter LT-9401 on light naphtha tank TK-120-25 will alarm on unintended flow direction (intentional flow direction is manually set) 380. SAAB level transmitter LT-9402 on heavy naphtha tank	F	1	2	1	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			TK-100-15 will alarm on unintended flow direction (intentional flow direction is manually set)						
		38.3.12.2. (Flow Tank TK-100-15 to TK-120-24/25) Potential major flow of heavy naphtha to light naphtha, potential issue with inventory tracking, potential off-spec product; potential financial impact	375. SAAB level transmitter LT-9409 on light naphtha tank TK-120-24 will alarm on unintended flow direction (intentional flow direction is manually set) 374. SAAB level transmitter LT-9401 on light naphtha tank TK-120-25 will alarm on unintended flow direction (intentional flow direction is manually set) 380. SAAB level transmitter LT-9402 on heavy naphtha tank TK-100-15 will alarm on unintended flow direction (intentional flow direction is manually set)	F	1	2	1	61. No recommendations required.	
38.4. High Pressure	38.4.1. External fire on EX-114/115	38.4.1.1. Increased pressure and overpressure of Rundown Chiller, potential vessel rupture and personnel impact	448. No specific safeguards identified	P	4	3	4	37. Perform engineering evaluation to determine if EX-114/115 Rundown/Recirculation Exchangers require overpressure protection for external fire scenario. If so, develop a plan to add adequate overpressure protection.	
	38.4.2. External fire on Light Naphtha Tanks TK-120-24/25	38.4.2.1. Increased pressure and overpressure of Light Naphtha Tanks TK-120-24/25, potential vessel rupture and personnel impact	382. Foam chamber for tank seals 383. Several open vents on floating roof tank for tank TK-120-24/25 (design for 10,000BPH inflow/ 15,000 BPH outflow) are sized for external fire	P	4	1	2	61. No recommendations required.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	38.4.3. Blocked in naphtha on inlet/ outlet and thermal expansion	38.4.3.1. Potential to overpressure naphtha tanks by thermal expansion, release of naphtha, personnel injury, environmental impact	384. Thermal Pressure Safety Valves across naphtha piping circuit	P	4	1	2	61. No recommendations required.	
38.5. Low Pressure	38.5.1. Chiller compressor trips	38.5.1.1. Low refrigerant pressure, increased glycol temperature to Light Naphtha Rundown/Recirculation Chiller, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance	362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number 363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response	F	1	1	1	61. No recommendations required.	
38.6. High Level	38.6.1. Improper lineup - lining up rundown to a full tank	38.6.1.1. Potential overfill of Light Naphtha Tank to grade, potential release of light naphtha, pool fire, potential personnel injury	385. Level alarm high LAH-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response 386. Level alarm hihi LAHH-9401/ 9409 on Naphtha Tank TK-120-24/25 alarm with operator response 387. level switch high LSH-9401/ 9409 mechanical switch will close inlet motor operated valve MOV to tanks TK-120-24/25 stopping naphtha flow to storage	P	4	1	2	61. No recommendations required.	
38.7. Low Level	38.7.1. More recirculation outflow than inflow (operator error)	38.7.1.1. Loss of flow to Light Naphtha Recirculation Pump P-6, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
		38.7.1.2. Increased flow of naphtha out of naphtha tank TK-120-24/25 to Transfer Pump P-2A/B, low level and loss of flow to Light Naphtha pump P-2A/B, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			376. Normal operating procedure for pump P-2A/B includes verifying pump line-up						
			377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response						
			369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
		38.7.1.3. Increased flow of naphtha out of naphtha tank TK-120-24/25, decreased level in TK-120-24/25, potential to land roof, potential air emission, permit deviation, potential environmental impact	388. Level alarm low LAL-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response and failure indication	F	1	2	1	61. No recommendations required.	
		38.7.1.4. Increased flow of naphtha out of naphtha tank TK-120-24/25, decreased level in TK-120-24/25, potential to land roof,	389. Tanks are in remote location - low personnel presence	P	4	1	2	61. No recommendations required.	1. May not draw in air in the event of landing

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		potential explosive atmosphere in vapor space, potential explosion, personnel injury, environmental impact, asset damage	388. Level alarm low LAL-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response and failure indication						roof/ Tank vapor space may not reach lower explosive limit in the event of landing roof
	38.7.2. High naphtha flow from Transfer Pumps P-2A/B (3100 vs 400 gpm design vs required) out to product terminal	38.7.2.1. Increased flow of naphtha out of naphtha tank TK-120-24/25, decreased level in TK-120-24/25, potential to land roof, potential explosive atmosphere in vapor space, potential explosion, personnel injury, environmental impact, asset damage	389. Tanks are in remote location - low personnel presence 388. Level alarm low LAL-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response and failure indication	P	4	1	2	61. No recommendations required.	1. May not draw in air in the event of landing roof/ Tank vapor space may not reach lower explosive limit in the event of landing roof
		38.7.2.2. Loss of flow to Light Naphtha Recirculation Pump P-6, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	367. Flow switch FS-9402 low flow shutdown on pump P-6 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		38.7.2.3. Increased flow of naphtha out of naphtha tank TK-120-24/25 to Transfer Pump P-2A/B, low level and loss of flow to Light Naphtha pump P-2A/B, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed 376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
			369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
		38.7.2.4. Increased flow of naphtha out of naphtha tank TK-120-24/25, decreased level in TK-120-24/25, potential to land roof, potential air emission, permit deviation, potential environmental impact	388. Level alarm low LAL-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response and failure indication	F	1	2	1	61. No recommendations required.	
38.8. High Interface Level	38.8.1. Not applicable								
38.9. Low Interface Level	38.9.1. Not applicable								
38.10. High Temperature	38.10.1. No additional causes identified								
38.11. Low Temperature	38.11.1. Low ambient temperature	38.11.1.1. No consequences of note.							
38.12. Contamination/ Composition	38.12.1. Mixer failure	38.12.1.1. Potential reduced mixing efficiency; No adverse consequences identified, operability issue only							
	38.12.2. Loss of hydrogen sulfide scavenger flow	38.12.2.1. Decreased flow of hydrogen sulfide scavenger, increased hydrogen sulfide content, potential off-spec H2S, inability to send product until it is treated; Potential financial impact	448. No specific safeguards identified	F	1	4	2	61. No recommendations required.	
38.13. Corrosion	38.13.1. No additional causes identified								
38.14. Additional Phase	38.14.1. Not applicable								
38.15. Rupture/Loss of Containment	38.15.1. Mechanical Roof Failure	38.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential vapor release and ignition, potential flash fire/ explosion, personnel injury, environmental impact, asset damage	390. Lower explosive limit LEL alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will identify seal damage on roof	P	4	2	3	38. Evaluate the need to implement a monthly third-party inspection of the roofs for Naphtha Tanks TK-120-24/25, TK-100-15 to mitigate the likelihood of roof damage leading to vapor release.	

Node: 38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	38.15.2. Refrigerant leak on Chiller compressors	38.15.2.1. Potential refrigerant R-134A leak to atmosphere; refrigerant is VOC exempt; No adverse consequences identified							
		38.15.2.2. Increased glycol temperature to Light Naphtha Rundown/Recirculation Chiller, increased tank temperature, increased light naphtha true vapor pressure (TVP), potential release from Light Naphtha Tank TK-120-24/25, potential off-spec product, environmental exceedance	362. Daily lab sample on rundown header to verify Reid vapor pressure (RVP) number 363. TAH-9401/ 9409 High Temperature Alarm on Tank TK-120-24/25 temperature will prompt operator response 364. Temperature alarm high TAH-15/ 19 on rundown/ recirculation chiller outlet temperature will prompt operator response	F	1	1	1	61. No recommendations required.	
	38.15.3. Sight-glass failure (flow indicator FI-9409)	38.15.3.1. Potential major naphtha spill to grade releasing contents of tank, potential pool fire, personnel injury, environmental impact, asset damage	391. Water draw procedure states to open the valve downstream of the sight-glass before opening the valve upstream of the sight-glass to prevent tank head crushing the sight-glass 392. Secondary containment around tank	P	3	2	2	61. No recommendations required.	
38.16. Loss of Utilities	38.16.1. Loss of power	38.16.1.1. See pump trip scenarios, this node							
38.17. Start-up/Shutdown hazards/ Maintenance	38.17.1. H2S in the feed to the plant	38.17.1.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S	P	2	2	1	61. No recommendations required.	

Node: **38. Node 38: Naphtha Rundown/ Recirculation Chiller and Light Virgin Naphtha Tanks TK-120-24/25**

Drawings / References:

88-P1-9217; 88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9236; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			605. AT-9410A/B Fixed H2S monitors located on the Naphtha manifold and will alert in the control room						
			109. H2S Scavenger requires good mixing to reduce the H2S present in the LVN tank						
		38.17.1.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
	38.17.2. 4" valve (X-9203) on TK-120-24 left closed when TK-120-25 is taken out of service for maintenance (or vice versa)	38.17.2.1. Blocked in PSV relief path for other tanks connected to the naphtha header (TK-120-24/25, TK-100-15) with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	39. Update the Tank Line-Up operating procedure for TK-120-24/25 and TK-100-15 to ensure that, when one tank is taken out of service, the 4" valve (X-9203, X-9207, and X-9155) is set as CSO to ensure an open relief path for the PSVs on the connecting tanks (TK-120-24/25 and TK-100-15).	
38.18. General	38.18.1. No additional causes identified								

Node: **39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25**

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
39.1. Low/No Flow	39.1.1. 8" valve closed on heavy naphtha rundown header from one or both splitter towers	39.1.1.1. Blocked discharge from Heavy Naphtha Stripper Product Pumps PM-105B/205B; Potential pump deadhead; Potential seal failure with loss of containment, potential ignition and personnel impact	484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			485. Seal failure alarm SFA-11480/21480 will shutdown pump PM-105B/205B						
		39.1.1.2. High level in Heavy Naphtha Stripper T-103/203, potential to overfill T-103/203 back to Combi Tower T-102/202, leading to products Kero, Diesel, and AGO going off-spec, operability/product quality issues; Potential financial impact	18. FAL-11191 Low Flow Alarm downstream of AF-105 Heavy Naphtha Cooler will prompt operator response	F	1	1	1	61. No recommendations required.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			672. FAL-21191 Low Flow Alarm downstream of AF-205 Heavy Naphtha Cooler will prompt operator response						
			19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response						
			673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response						
		39.1.1.3. High level in Heavy Naphtha Stripper T-103/203, potential to overfill T-103/203 back to Combi Tower T-102/202; high differential pressure; Potential to get heavy naphtha liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
			566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response						
			18. FAL-11191 Low Flow Alarm downstream of AF-105 Heavy Naphtha Cooler will prompt operator response						
			672. FAL-21191 Low Flow Alarm downstream of AF-205 Heavy Naphtha Cooler will prompt operator response						
			19. LAH-11595 Level Alarm High on Heavy Naphtha Stripper T-103 will prompt operator response						
			673. LAH-21595 Level Alarm High on Heavy Naphtha Stripper T-203 will prompt operator response						

Node: **39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25**

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	39.1.2. MOV-9432/ 9442 closed on naphtha inlet to tank TK-100-15 or MOV-9431/ 9441 closed on naphtha inlet to tank TK-120-25	39.1.2.1. See consequences associated with cause 39.1.1 above, no additional consequences identified							
	39.1.3. MOV-9421/ 9422/ 9411/ 9412 closed on outlet of tank TK-120-25/TK-100-15	39.1.3.1. Decreased flow of naphtha from naphtha tank TK-120-25/TK-100-15; Loss of flow to Light Naphtha P-2A/B, potential cavitation and seal leak, release of naphtha to grade, potential ignition and personnel impact	377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	39.1.4. MOV-9414/ 9415 Blocked inlet valve to Transfer Pump P-2A/B or strainer plugs	39.1.4.1. Decreased flow of naphtha from naphtha tank TK-120-24/25 to Transfer Pump P-2A/B, loss of flow to Light Naphtha P-2A/B, potential cavitation and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed 376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6,	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: **39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25**

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			P-2A/B with 1oo2 voting and alarm to prompt operator response						
		39.1.4.2. Decreased flow of naphtha from naphtha tank TK-100-15 to Transfer Pump P-2A/B when only one heavy naphtha tank in service, potential overfill of tank, spilling naphtha to grade, potential pool fire, personnel injury, environmental impact	393. Level alarm high and high high LAH/ LAHH-9402 on Naphtha Tank TK-100-15 with operator response 380. SAAB level transmitter LT-9402 on heavy naphtha tank TK-100-15 will alarm on unintended flow direction (intentional flow direction is manually set) 394. level switch high LSH-9402 mechanical switch will close inlet motor operated valve MOV to tanks TK-100-15 stopping heavy naphtha flow to storage	P	4	1	2	61. No recommendations required.	
		39.1.4.3. Inability to flow product to pipeline; Potential reduced throughput; Potential financial impact	376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response	F	2	2	1	61. No recommendations required.	
	39.1.5. Transfer Pump P-2A/B stops	39.1.5.1. Decreased flow of naphtha from naphtha tank TK-100-15 to Transfer Pump P-2A/B when only one heavy naphtha tank in service, potential overfill of tank, spilling naphtha to grade, potential pool fire, personnel injury, environmental impact	393. Level alarm high and high high LAH/ LAHH-9402 on Naphtha Tank TK-100-15 with operator response 380. SAAB level transmitter LT-9402 on heavy naphtha tank TK-100-15 will alarm on unintended flow direction (intentional	P	4	1	2	61. No recommendations required.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			flow direction is manually set)						
			394. level switch high LSH-9402 mechanical switch will close inlet motor operated valve MOV to tanks TK-100-15 stopping heavy naphtha flow to storage						
		39.1.5.2. Inability to flow product to pipeline; Potential reduced throughput; Potential financial impact	376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response	F	2	2	1	61. No recommendations required.	
	39.1.6. Blocked outlet of Transfer Pump P-2A/B or 16" valve closed on product manifold out of facility	39.1.6.1. Decreased flow of naphtha from naphtha tank TK-120-25 and TK-100-15 to Transfer Pump P-2A/B, loss of flow to Heavy Naphtha P-2A/B, potential deadhead and seal leak, release of naphtha to atmosphere, potential ignition and personnel impact	624. Pump P-2A/B permissives prevent the pumps from starting if MOV-9414/9415 is closed 376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
		39.1.6.2. Decreased flow of naphtha from naphtha tank TK-100-15 to Transfer Pump P-2A/B when only one heavy naphtha tank in service, potential overfill of tank, spilling naphtha to grade, potential pool fire, personnel injury, environmental impact	393. Level alarm high and high high LAH/ LAHH-9402 on Naphtha Tank TK-100-15 with operator response 380. SAAB level transmitter LT-9402 on heavy naphtha tank TK-100-15 will alarm on unintended flow direction (intentional flow direction is manually set) 394. level switch high LSH-9402 mechanical switch will close inlet motor operated valve MOV to tanks TK-100-15 stopping heavy naphtha flow to storage	P	4	1	2	61. No recommendations required.	
		39.1.6.3. Inability to flow product to pipeline; Potential reduced throughput; Potential financial impact	376. Normal operating procedure for pump P-2A/B includes verifying pump line-up 377. Flow switch FS-9407/ 09 low flow shutdown on pump P-2A/B and will alarm to prompt operator response	F	2	2	1	61. No recommendations required.	
39.2. High Flow	39.2.1. Water draw valves left open (TK-100-15)	39.2.1.1. Increased flow of water and eventually hydrocarbon liquid to sump and to Water Tank 5, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	1	1	61. No recommendations required.	
39.3. Reverse/Misdirected Flow	39.3.1. Valve misalignment on naphtha transfer header - Tank TK-120-24 in light; tank TK-100-15 or TK-120-25 in heavy	39.3.1.1. (Flow Tank TK-120-24 to TK-120-25) Potential major flow of light naphtha to heavy naphtha, potential issue with inventory tracking, potential off-spec product, operability issue, potential	375. SAAB level transmitter LT-9409 on light naphtha tank TK-	F	1	2	1	61. No recommendations required.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		deviation on environmental permitting - incorrect product in tank, environmental impact	120-24 will alarm on unintended flow direction (intentional flow direction is manually set) 374. SAAB level transmitter LT-9401 on light naphtha tank TK-120-25 will alarm on unintended flow direction (intentional flow direction is manually set)						
		39.3.1.2. (Flow Tank TK-120-25 to TK-120-24) Potential major flow of heavy naphtha to light naphtha, potential issue with inventory tracking, potential off-spec product; potential financial impact	375. SAAB level transmitter LT-9409 on light naphtha tank TK-120-24 will alarm on unintended flow direction (intentional flow direction is manually set) 374. SAAB level transmitter LT-9401 on light naphtha tank TK-120-25 will alarm on unintended flow direction (intentional flow direction is manually set)	F	1	2	1	61. No recommendations required.	
	39.3.2. MOV-9432/ 9442 open on naphtha inlet to tank TK-100-15 when not intended (on initial start-up)	39.3.2.1. Potential issue when sending heavy naphtha to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	623. TK-120-24/25 and TK-100-15 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9449/9441/9442 and opening diffuser MOV-9439/9431/9432 626. TK-120-24/25 and TK-100-15 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	39.3.3. motor operated valve MOV-9431/ 9441 open on naphtha inlet to tank TK-120-25 when not intended (on initial start-up)	39.3.3.1. Potential issue when sending heavy naphtha to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	623. TK-120-24/25 and TK-100-15 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9449/9441/9442 and opening diffuser MOV-9439/9431/9432	P	4	1	2	61. No recommendations required.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			626. TK-120-24/25 and TK-100-15 tanks are grounded and bonded for start-up						
	39.3.4. 6" manual valve X-9206 open on low suction line from tank TK-120-25 when not intended	39.3.4.1. Potential to draw water in naphtha product, off-spec product, operability issue, no hazardous consequences identified							
	39.3.5. 6" manual valve X-9154 open on low suction line from tank TK-100-15 when not intended	39.3.5.1. Potential to draw water in naphtha product, off-spec product, operability issue, no hazardous consequences identified							
	39.3.6. 12" valve V-9121 open and tank outlet MOV open from the heavy virgin naphtha HVN tank to recirculation header	39.3.6.1. (Flow Tank TK-100-15 to TK-120-24/25) Potential major flow of heavy naphtha to light naphtha, potential issue with inventory tracking, potential off-spec product; potential financial impact	628. Quality checks on the LVN tanks (twice per shift)	F	2	2	1	61. No recommendations required.	
	39.3.7. 16" manual jumper valve X-9239 inadvertently open on naphtha product manifold to feed tanks and outbound motor operated valve MOV open on Feed Tanks	39.3.7.1. Potential to send heavy naphtha to feed tank, potential excess rerun and economic loss, potential increased Feed tank emissions, potential environmental impact	381. SAAB level transmitter LT-9101/ 9102/ 9103 on feed tank TK-200-1/2/3 will alarm on unintended flow direction (intentional flow direction is manually set)	F	1	2	1	61. No recommendations required.	
39.4. High Pressure	39.4.1. External fire on Heavy Naphtha Tank TK-100-15	39.4.1.1. Increased pressure and overpressure of Heavy Naphtha Tank TK-100-15, potential vessel rupture and personnel impact	382. Foam chamber for tank seals 395. Several open vents on floating roof tank for tank TK-100-15 (design for 10,000BPH inflow/ 15,000 BPH outflow) sized for external fire	P	4	1	2	61. No recommendations required.	
	39.4.2. External fire on Heavy Naphtha Tanks TK-120-25	39.4.2.1. Increased pressure and overpressure of Heavy Naphtha Tanks TK-120-25, potential vessel rupture and personnel impact	382. Foam chamber for tank seals 383. Several open vents on floating roof tank for tank TK-120-24/25 (design for 10,000BPH inflow/ 15,000 BPH outflow) are sized for external fire	P	4	1	2	61. No recommendations required.	
	39.4.3. Gravity feeding to Heavy Naphtha between Tanks TK-120-25 and TK-100-15 (high inflow/ outflow)	39.4.3.1. Potential to exceed tank design inflow/ outflow rate (if high level in one tank and low level in other); Potential damage to IFR on both tanks; Potential increased environmental emissions; Potential environmental and financial impact	643. Tank-to-Tank Transfer Procedure (0305) updated to prevent flow rates exceeding the floating roof maximum flow rate limit of 2500 BPH.	F	4	3	4	40. Update the P&IDs and the tank drawings for the Kero, Diesel, AGO, Naphtha, and Feed Tanks to reflect the internal vacuum breakers.	

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	39.4.4. Blocked in naphtha and thermal expansion	39.4.4.1. Potential to overpressure naphtha tanks by thermal expansion, release of naphtha, personnel injury, environmental impact	384. Thermal Pressure Safety Valves across naphtha piping circuit	P	4	1	2	61. No recommendations required.	
39.5. Low Pressure	39.5.1. No additional causes identified								
39.6. High Level	39.6.1. Improper lineup - lining up rundown to a full tank	39.6.1.1. Potential overfill of Heavy Naphtha Tank to grade, potential release of heavy naphtha, pool fire, potential personnel injury	393. Level alarm high and high high LAH/ LAHH-9402 on Naphtha Tank TK-100-15 with operator response 385. Level alarm high LAH-9401/ 9409 on Naphtha Tank TK-120-24/25 with operator response 394. level switch high LSH-9402 mechanical switch will close inlet motor operated valve MOV to tanks TK-100-15 stopping heavy naphtha flow to storage 387. level switch high LSH-9401/ 9409 mechanical switch will close inlet motor operated valve MOV to tanks TK-120-24/25 stopping naphtha flow to storage	P	4	1	2	61. No recommendations required.	
39.7. Low Level	39.7.1. Both pumps P-2A/B running at once	39.7.1.1. Flow is maximized under normal circumstances. Flow is regulated downstream by receiver, operability issue, no hazardous consequences identified							
39.8. High Interface Level	39.8.1. Not applicable								
39.9. Low Interface Level	39.9.1. Not applicable								
39.10. High Temperature	39.10.1. No additional causes identified								
39.11. Low Temperature	39.11.1. Low ambient temperature	39.11.1.1. No consequences of note.							
39.12. Contamination/ Composition	39.12.1. Mixer failure	39.12.1.1. Not expected to result in an impact; mixer is not currently in service, no hazardous consequences identified							

Node: 39. Node 39: Heavy Virgin Naphtha Tanks TK-100-15 and TK-120-25

Drawings / References:

88-P1-9226; 88-P1-9228; 88-P1-9231; 88-P1-9237; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
39.13. Corrosion	39.13.1. No specific causes identified								
39.14. Additional Phase	39.14.1. Not applicable								
39.15. Rupture/Loss of Containment	39.15.1. Mechanical Roof Failure	39.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential vapor release and ignition, potential flash fire/ explosion, personnel injury, environmental impact, asset damage	390. Lower explosive limit LEL alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will identify seal damage on roof 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response	P	4	2	3	38. Evaluate the need to implement a monthly third-party inspection of the roofs for Naphtha Tanks TK-120-24/25, TK-100-15 to mitigate the likelihood of roof damage leading to vapor release. 60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
39.16. Loss of Utilities	39.16.1. No additional causes identified								
39.17. Start-up/Shutdown hazards/ Maintenance	39.17.1. 4" valve (X-9155) on TK-100-15 left closed when TK-120-25 is taken out of service for maintenance (or vice versa)	39.17.1.1. Blocked in PSV relief path for other tanks connected to the naphtha header (TK-120-25, TK-100-15) with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	39. Update the Tank Line-Up operating procedure for TK-120-24/25 and TK-100-15 to ensure that, when one tank is taken out of service, the 4" valve (X-9203, X-9207, and X-9155) is set as CSO to ensure an open relief path for the PSVs on the connecting tanks (TK-120-24/25 and TK-100-15).	
39.18. General	39.18.1. No additional causes identified								

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
40.1. Low/No Flow	40.1.1. Plugged Kero Filters	40.1.1.1. High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 566. PDAH-21735 high differential pressure alarm on Combi Tower	P	4	1	2	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			T-202 will prompt operator response						
			493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
		40.1.1.2. High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
		40.1.1.3. Kero Stripper Product Pump PM-106A/B or PM-206A/B deadhead, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	492. Seal failure alarm SFA-11500/11520/21500/21520 will shutdown pumps PM-106A/B and PM-206A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			484. Lower explosive limit LEL-11481/11461/21481/21461 will alarm and prompt operators to cordon off area 491. Dual seals on PM-106A/B and PM-206A/B						
	40.1.2. Manual valve closed on kero rundown header from one or both splitter towers	40.1.2.1. See consequences associated with cause 40.1.1							
	40.1.3. Manual valve(s) closed on inlet of Filters FLT-11/ 12A/B/C/ 13A/B or 21/ 22A/B/C/ 23A/B	40.1.3.1. See consequences associated with cause 40.1.1							
	40.1.4. Manual valve(s) closed on outlet of Filters FLT-11/ 23	40.1.4.1. See consequences associated with cause 40.1.1							
		40.1.4.2. Decreased flow of kero to storage, Blocked outlet of filters; Potential to overpressure filters with damage to filters, loss of containment of Kero, potential ignition and personnel impact	448. No specific safeguards identified	P	4	3	4	41. Perform engineering evaluation to verify that PSV-9601 thru PSV-9612 on the Filters FL-11 thru FL-23 are sized for blocked filter outlet. If PSVs are sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9601 thru PSV-9612 on the Filters FL-11 thru FL-23 are not sized for this scenario, develop a plan to add adequate overpressure protection.	
	40.1.5. MOV-9231/41; MOV-9034/44; MOV-9232/42 closed on kero inlet to tank TK-100-11/20 and TK-150-10 when in rundown	40.1.5.1. See consequences associated with cause 40.1.1							
		40.1.5.2. Decreased flow of kero to storage, Blocked outlet of filters; Potential to overpressure filters with damage to filters, loss of containment of Kero, potential ignition and personnel impact	448. No specific safeguards identified	P	4	3	4	41. Perform engineering evaluation to verify that PSV-9601 thru PSV-9612 on the Filters FL-11 thru FL-23 are sized for blocked filter outlet. If PSVs are sized appropriately, update PSV sizing documentation. If evaluation determines PSV-9601 thru PSV-9612 on the Filters FL-11 thru FL-23 are not sized for this scenario, develop a plan to add adequate overpressure protection.	
		40.1.5.3. Potential reduced throughput; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will	F	2	2	1	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
	40.1.6. 12" or 10" valve closed on suction to Kero Transfer Pump P-4 or strainer plugs	40.1.6.1. Loss of flow to Kero to Transfer Pump P-4, potential cavitation and seal leak, release of kero to atmosphere, potential seal fire, personnel injury, environmental impact	629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area 396. Flow switch FS-9212 low flow shutdown on pump P-4 with alarm	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		40.1.6.2. Inability to transfer kero between tanks; Potential to overfill tank (TK-100-20/11, TK-150-10); Potential release of kero to surrounding area, potential ignition, personnel impact, potential environmental impact	407. Level alarm high LAH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 with operator response 408. Level alarm hihi LAHH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 alarm with operator response 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	40.1.7. Transfer Pump P-4 stops	40.1.7.1. Inability to transfer kero between tanks; Potential to overfill tank (TK-100-20/11, TK-150-10); Potential release of kero to surrounding area, potential ignition, personnel impact, potential environmental impact	407. Level alarm high LAH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 with operator response 408. Level alarm hihi LAHH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 alarm with operator response 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			operator to cordon off the area						
	40.1.8. 10" valve closed on discharge of Transfer Pump P-4	40.1.8.1. Loss of flow from Kero to Transfer Pump P-4, potential deadhead and seal leak, release of kero to atmosphere, potential seal fire, personnel injury, environmental impact	629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area 396. Flow switch FS-9212 low flow shutdown on pump P-4 with alarm	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		40.1.8.2. Inability to transfer kero between tanks; Potential to overfill tank (TK-100-20/11, TK-150-10); Potential release of kero to surrounding area, potential ignition, personnel impact, potential environmental impact	407. Level alarm high LAH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 with operator response 408. Level alarm hihi LAHH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 alarm with operator response 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	40.1.9. MOV-9221/ 22/ 24 closed on outlet of tank TK-100-11/20 and TK-150-10 to Transfer Pump P-1A/B when tank is in transfer to pipeline	40.1.9.1. No flow of kero from kero tanks TK-100-11/20 and 150-10 to Transfer Pump P-1A/B, loss of flow to kero P-1A/B, potential cavitation and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting	P	4	1	2	61. No recommendations required.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		40.1.9.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response 490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B 630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)	P	4	1	2	61. No recommendations required.	
		40.1.9.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will	F	1	2	1	61. No recommendations required.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.10. MOV-9205/06/07 closed on outlet of tank TK-100-11/20 and TK-150-10 to Transfer Pump P-1A/B when tank is in transfer to pipeline	40.1.10.1. No flow of kero from kero tanks TK-100-11/20 and 150-10 to Transfer Pump P-1A/B, loss of flow to kero P-1A/B, potential cavitation and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 638. Transfer Pump P-1A/B permissive, pump will not run if valves MOV-9205/6/7/8/9 are closed 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
		40.1.10.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	638. Transfer Pump P-1A/B permissive, pump will not run if valves MOV-9205/6/7/8/9 are closed	P	4	1	2	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response						
			566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response						
			493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.10.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	638. Transfer Pump P-1A/B permissive, pump will not run if valves MOV-9205/6/7/8/9 are closed	F	1	2	1	61. No recommendations required.	
			493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.11. 12" manual valve blocked on inlet valve to Transfer Pump P-1A/B or strainer plugs	40.1.11.1. No flow of kero from kero tanks TK-100-11/20 and 150-10 to Transfer Pump P-1A/B, loss of flow to kero P-1A/B, potential cavitation and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm	P	4	1	2	61. No recommendations required.	
			368. Pump is in remote location and inside tank berm and separate containment						
			397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting						
			398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		40.1.11.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
			566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response						
			493. LAH-11660 high level alarm on Kero						

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.11.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.12. Transfer Pump P-1A/B stops	40.1.12.1. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero	486. PDAH-11735 high differential pressure alarm on Combi Tower	P	4	1	2	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	T-102 will prompt operator response 566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response 490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B 630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.12.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response 490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 591. FAL-21221 low flow alarm downstream of AF-204 will	F	1	2	1	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			shutdown pumps PM-206A/B 630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.13. Blocked 8" manual outlet valve of Transfer Pump P-1A/B	40.1.13.1. Loss of flow from kero P-1A/B, potential deadhead and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
		40.1.13.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response 566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response 493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.13.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.14. MOV-9211 fails closed when not performing diesel flush on outlet of pump P-1A/B	40.1.14.1. Loss of flow from kero P-1A/B, potential deadhead and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm	P	4	1	2	61. No recommendations required.	
			368. Pump is in remote location and inside tank berm and separate containment						
			397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector						

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting						
			398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		40.1.14.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	
			566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response						
			493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.14.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-	493. LAH-11660 high level alarm on Kero Stripper T-104 will	F	1	2	1	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	prompt operator response 592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response 490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B 591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B 630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.15. MOV-9212 fails closed or 16" valve closed on product manifold out of facility	40.1.15.1. Loss of flow from kero P-1A/B, potential deadhead and seal leak, release of kero to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	42. Field verify the location and tag numbers of existing product motor operated valve MOVs to pipeline and update P&IDs accordingly. The product MOVs are not shown on the P&IDs. An example of this includes MOV-9212 on kero to pipeline.	
		40.1.15.2. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; high differential pressure; Potential to get kero liquids in the vapor return line; Potential damage to vapor return piping; Potential release with ignition and personnel impact	486. PDAH-11735 high differential pressure alarm on Combi Tower T-102 will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			566. PDAH-21735 high differential pressure alarm on Combi Tower T-202 will prompt operator response						
			493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response						
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
		40.1.15.3. Potential Kero tank overfill (TK-100-20/11, TK-150-10); High level in Kero Stripper T-104/204; Potential Combi Tower T-102/202 flooding; leading to products Diesel and AGO going off-spec, operability/product quality issues; Potential financial impact	493. LAH-11660 high level alarm on Kero Stripper T-104 will prompt operator response	F	1	2	1	61. No recommendations required.	
			592. LAH-21660 high level alarm on Kero Stripper T-204 will prompt operator response						
			490. FAL-11221 low flow alarm downstream of AF-104 will shutdown pumps PM-106A/B						
			591. FAL-21221 low flow alarm downstream of AF-204 will shutdown pumps PM-206A/B						

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			630. Multiple Kero Tanks available (TK-100-20/11, TK-150-10)						
	40.1.16. 4" gate valve or 2" ball closed on relief header to tanks TK-100-11/20; TK-150-10	40.1.16.1. Potential blocked outlet of thermal PSVs, release of kero to grade, potential personnel injury and environmental impact (The P&ID for TK-100-11 has the 4" gate valve shown as CSC; The P&ID for TK-150-10 has an upstream 2" ball valve that is not listed as CSO; The P&ID for TK-100-20 does not have the 4" gate valve listed as CSO)	448. No specific safeguards identified	P	4	3	4	43. Identify manual valves that need to be car sealed open to ensure there is an open flow path for thermal relief valves on Kero, Diesel, Naphtha, AGO, and Feed Tanks, as well as preventing cross contamination of the product. Car seal these valves open and update P&IDs accordingly.	
	40.1.17. Failure to perform water draw when needed	40.1.17.1. Increased water level in kero tank TK-100-11/20 and 150-10, potential water in kero product, off-spec product, potential financial impact	632. Procedure (GPS 0314) states to perform water draw off on Kero Tanks prior to certifying the tanks	F	2	3	2	61. No recommendations required.	
40.2. High Flow	40.2.1. Both pumps P-1A/B running at once	40.2.1.1. Flow is maximized under normal circumstances. Flow is regulated downstream by terminal operations, No hazardous consequences identified							
	40.2.2. MOV-9210 fails open on pump P-1A/B bypass	40.2.2.1. Increased flow of kero recycling around Transfer Pump P-1A/B, decreased flow to terminal, delay in transfer, logistical impact to terminals; Potential to increase temperature on pumps P-1A/B with potential pump damage, potential seal failure, release, ignition and personnel impact	368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
	40.2.3. Water draw valves left open	40.2.3.1. Increased flow of water and eventually hydrocarbon liquid to sump and to Water Tank 5, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the	F	3	1	1	61. No recommendations required.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			water draw from the tank						
40.3. Reverse/Misdirected Flow	40.3.1. 10" valve open on recirculation from filters	40.3.1.1. No adverse consequences identified; Unintended additional cleaning to Kero Tanks							
	40.3.2. 10" recirculation line open to second tank when not intended	40.3.2.1. Potential to flow kero to tank that is not lined up; No adverse consequences identified							
	40.3.3. 8" or 10" Rundown line open from filter rundown line to second tank when not intended	40.3.3.1. Potential to flow kero to tank that is not lined up; No adverse consequences identified							
	40.3.4. MOV-9241/ 42/ 44 open on kero eductor inlet to tank TK-100-11/20 and TK-150-10 when not intended (on initial start-up)	40.3.4.1. Potential issue when sending kero to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	633. TK-100-11/20 and TK-150-10 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9244/9242/9241 and opening diffuser MOV-9234/9232/9231 634. TK-100-11/20 and TK-150-10 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	40.3.5. 6" manual valve open on low suction line from tank TK-100-11/20 and TK-150-10 when not intended	40.3.5.1. Potential to draw water in kero product, off-spec product, potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	40.3.6. 12" valve V-9129/30/31 open and tank outlet MOV open on two different kero tanks	40.3.6.1. In this scenario (sending flow out of Kero tank), pump P-4 would not be running, therefore flow would not be sent back to the Kero tanks; Two MOV outlet valves open on different kero tanks would result in potentially sending uncertified kero flow to pipeline; Potential financial impact from sending uncertified kero to pipeline	635. Procedure (GPS 0304) states proper tank line-up 636. Customer provides instructions in daily operational "night notes" on tank rundown	F	4	1	2	61. No recommendations required.	
	40.3.7. Valve misalignment on kero/ diesel transfer header - Tank TK-100-11/20 and TK-150-10	40.3.7.1. (Flow Tank any kero tank to another) Potential to flow kero from one tank to the other tank, tank level will balance, operability issue, no hazardous consequences identified							
		40.3.7.2. (Flow of diesel to kero tanks or kero to diesel tanks) Potential major flow of kero to diesel or vice versa, potential issue with inventory tracking, potential off-spec product; No environmental impact (distillate products able to be stored in same tanks); Potential financial impact due to off-spec product	635. Procedure (GPS 0304) states proper tank line-up 402. SAAB level transmitter LT-9202 on kero tank TK-100-21 will alarm on unintended flow direction (intentional flow direction is manually set) 400. SAAB level transmitter LT-9201 on	F	2	1	1	61. No recommendations required.	

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			kero tank TK-150-10 will alarm on unintended flow direction (intentional flow direction is manually set)						
			401. SAAB level transmitter LT-9204 on kero tank TK-100-11 will alarm on unintended flow direction (intentional flow direction is manually set)						
			404. SAAB level transmitter LT-9203 on diesel tank TK-100-21 will alarm on unintended flow direction (intentional flow direction is manually set)						
			405. SAAB level transmitter LT-9306 on diesel tank TK-100-12 will alarm on unintended flow direction (intentional flow direction is manually set)						
	40.3.8. 16" manual valve V-9237 inadvertently open on kero product manifold to feed tanks and outbound MOV open on Feed Tanks	40.3.8.1. Potential to send kero to feed tank, potential excess rerun and economic loss; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
40.4. High Pressure	40.4.1. External fire on Kero Tanks TK-100-11/20 and TK-150-10	40.4.1.1. Increased pressure and overpressure of Kero Tanks TK-100-11/20 and TK-150-10, potential vessel rupture, financial impact	382. Foam chamber for tank seals	P	4	1	2	61. No recommendations required.	
			403. Several open vents on floating roof tank for tanks TK-100-11/20 and TK-150-10 are sized for external fire						
	40.4.2. Gravity feeding between Kero Tanks TK-100-11/20 and TK-150-10	40.4.2.1. Potential to exceed tank design inflow/ outflow rate (if high level in one tank and low level in other); Potential damage to IFR on both tanks; Potential increased environmental emissions; Potential environmental and financial impact	643. Tank-to-Tank Transfer Procedure (0305) updated to prevent flow rates exceeding the floating roof maximum flow rate limit of 2500 BPH.	F	4	3	4	40. Update the P&IDs and the tank drawings for the Kero, Diesel, AGO, Naphtha, and Feed Tanks to reflect the internal vacuum breakers.	

Node: 40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	40.4.3. Blocked in kero and thermal expansion	40.4.3.1. Potential to overpressure piping by thermal expansion, release of kero, personnel injury, environmental impact	406. Thermal Pressure Safety Valves across kero piping circuit	P	4	1	2	44. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-9239 are car sealed open and update P&ID to show valves CSO. Also update P&ID to include field verified PSV tag numbers.	
40.5. Low Pressure	40.5.1. No additional causes identified								
40.6. High Level	40.6.1. Improper lineup - lining up rundown to a full tank	40.6.1.1. Potential overfill of Kero Tank to grade, potential release of kero, pool fire, potential personnel injury	407. Level alarm high LAH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 with operator response 408. Level alarm hihi LAHH-9201/02/04 on Kero Tank TK-100-11/20 and TK-150-10 alarm with operator response 409. level switch high LSH-9201/02/04 mechanical switch will close inlet motor operated valve MOV to tanks Kero Tank TK-100-11/20 and TK-150-10 stopping kero flow to storage 635. Procedure (GPS 0304) states proper tank line-up	P	4	1	2	61. No recommendations required.	
40.7. Low Level	40.7.1. No additional causes identified								
40.8. High Interface Level	40.8.1. Not applicable								
40.9. Low Interface Level	40.9.1. Not applicable								
40.10. High Temperature	40.10.1. No additional causes identified								
40.11. Low Temperature	40.11.1. Low ambient temperature	40.11.1.1. No consequences of note.							
40.12. Contamination/ Composition	40.12.1. Mixer failure	40.12.1.1. Potential reduced tank mixing, no off-spec product issues; No adverse consequences identified							
40.13. Corrosion	40.13.1. No specific causes identified								

Node: **40. Node 40: Kero Tanks 100-20/ 100-11/ 150-10 and Recirc Filters/ Pumps**

Drawings / References:

88-P1-9220; 88-P1-9221; 88-P1-9222; 88-P1-9229; 88-P1-9232; 88-P1-9234; 88-P1-9239 ; 88-P1-9235; 88-P1-9217; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9230; 88-P1-9231

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
40.14. Additional Phase	40.14.1. Not applicable								
40.15. Rupture/Loss of Containment	40.15.1. Mechanical Roof Failure	40.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential vapor release and ignition, potential flash fire/ explosion, personnel injury, environmental impact, asset damage	390. Lower explosive limit LEL alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will identify seal damage on roof 637. Monthly through hatch inspection on Kero Tanks TK-100-11/20, TK-150-11	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
40.16. Loss of Utilities	40.16.1. Loss of power	40.16.1.1. See pump trip scenarios, this node							
40.17. Start-up/Shutdown hazards/ Maintenance	40.17.1. 4" valve (V-9159) on TK-100-20 left closed when need (or 4" valves V-9070 for TK-100-11 or V-9066 for TK-150-10)	40.17.1.1. Blocked in PSV relief path for other tanks connected to the Kero header with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	45. Update the Tank Line-Up operating procedure for TK-100-20/11 and TK-150-10 to ensure that, when one tank is taken out of service, one of the 4" valves on the other two in-service tanks is set to CSO to ensure an open relief path for the PSVs on the connecting tanks.	
40.18. General	40.18.1. No additional causes identified								

Node: **41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps**

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
41.1. Low/No Flow	41.1.1. 8" valve closed on diesel rundown header from one or both splitter towers	41.1.1.1. Potential to deadhead Diesel Stripper Product Pumps PM-107A/B or PM-207A/B; Potential seal failure; Potential loss of containment, ignition and personnel impact	498. Dual seals on PM-107A/B 593. Dual seals on PM-207A/B 499. Seal failure alarm SFA-15025/15045 will shutdown pumps PM-107A/B 594. Seal failure alarm SFA-25025/25045 will shutdown pumps PM-207A/B	P	4	1	2	61. No recommendations required.	
		41.1.1.2. High level in Diesel Stripper T-105/205; Inability to draw off diesel from Combi Tower T-102/202; Potential to ultimately flood level in T-102/202; Potential to send liquids in the vapor return lines;	502. LAH-15090 high level alarm on Diesel Stripper T-105 will	P	4	1	2	61. No recommendations required.	

Node: **41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps**

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Potential damage to vapor return piping; Potential release with ignition and personnel impact	prompt operator response 596. LAH-25090 high level alarm on Diesel Stripper T-205 will prompt operator response 16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response						
		41.1.1.3. No flow of diesel to storage, not expected to impact tanks; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
		41.1.1.4. Loss of diesel flow to exchanger EX-104/204; Loss of pre-heat; Potential reduced ability to heat the feed in EX-104/204; Potential for low feed temperature into Naphtha Stabilizer Column T-101/201; Potential to swing T-101/201; Potential offspec product/quality; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
	41.1.2. MOV-9336/46 9233/43 closed on diesel inlet to tank TK-100-12/21 when in rundown	41.1.2.1. See consequences associated with cause 41.1.1							
	41.1.3. 12" or 10" valve closed on suction to Transfer Pump P-4 or strainer plugs	41.1.3.1. Loss of flow to Diesel to Transfer Pump P-4, potential cavitation and seal leak, release of diesel to atmosphere, potential seal fire from seal, personnel injury, environmental impact	396. Flow switch FS-9212 low flow shutdown on pump P-4 with alarm 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
	41.1.4. Transfer Pump P-4 stops	41.1.4.1. Decreased flow of diesel between tanks, delay in transfer, operability issue, no hazardous consequences identified							
	41.1.5. 10" valve closed on discharge of Transfer Pump P-4	41.1.5.1. Loss of flow from Transfer Pump P-4, potential deadhead and seal leak, release of diesel to atmosphere, potential seal fire from seal, personnel injury, environmental impact	396. Flow switch FS-9212 low flow shutdown on pump P-4 with alarm 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	41.1.6. 10" valve closed on discharge of Transfer Pump P-4 to other tank (when performing tank transfer)	41.1.6.1. Loss of flow from Transfer Pump P-4, potential deadhead and seal leak, release of diesel to atmosphere, potential seal fire from seal, personnel injury, environmental impact	396. Flow switch FS-9212 low flow shutdown on pump P-4 with alarm 629. Lower Explosive Limit LEL alarm AI-9201/2/3/4 around kero/diesel manifold that will prompt operator to cordon off the area 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting						
			398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		41.1.6.2. Decreased flow of diesel between tanks, delay in transfer, operability issue, no hazardous consequences identified							
	41.1.7. MOV-9336/46 9233/43 closed on diesel inlet to tank TK-100-12/21 when transferring tank to tank	41.1.7.1. See consequences associated with cause 41.1.6							
	41.1.8. MOV-9223 /9326 closed on outlet of tank TK-100-12/21 to Transfer Pump P-1A/B when tank is in transfer to pipeline	41.1.8.1. Decreased flow of diesel from Diesel Tanks TK-100-12/21 to Transfer Pump P-1A/B, loss of flow to P-1A/B, potential cavitation and seal leak, release of diesel to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm	P	4	1	2	61. No recommendations required.	
			368. Pump is in remote location and inside tank berm and separate containment						
			397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting						
			398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		41.1.8.2. Decreased flow of diesel from diesel tank TK-100-12/21 to Transfer Pump P-1A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	41.1.9. MOV-9208/ 09 closed on outlet of tank TK-100-12/21 to Transfer Pump P-1A/B when tank is in transfer to pipeline	41.1.9.1. Decreased flow of diesel from Diesel Tanks TK-100-12/21 to Transfer Pump P-1A/B, loss of flow to P-1A/B, potential cavitation and seal leak, release of diesel to grade, potential seal fire, personnel injury, environmental impact	638. Transfer Pump P-1A/B permissive, pump will not run if valves MOV-9205/6/7/8/9 are closed	P	4	1	2	61. No recommendations required.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting						
		41.1.9.2. Decreased flow of diesel from diesel tank TK-100-12/21 to Transfer Pump P-1A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	41.1.10. 12" manual valve blocked on inlet valve to Transfer Pump P-1A/B or strainer plugs	41.1.10.1. Decreased flow of diesel from Diesel Tanks TK-100-12/21 to Transfer Pump P-1A/B, loss of flow to P-1A/B, potential cavitation and seal leak, release of diesel to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-	P	4	1	2	61. No recommendations required.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			4, and P-1A/B with 1oo2 voting						
		41.1.10.2. Decreased flow of diesel from diesel tank TK-100-12/21 to Transfer Pump P-1A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	41.1.11. Transfer Pump P-1A/B stops	41.1.11.1. Decreased flow of diesel from diesel tank TK-100-12/21 to Transfer Pump P-1A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	41.1.12. Blocked 8" manual outlet valve of Transfer Pump P-1A/B	41.1.12.1. Blocked flow out of Transfer Pump P-1A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
		41.1.12.2. Blocked flow out of Transfer Pump P-1A/B, potential deadhead and seal leak, release of diesel to grade, potential seal fire, personnel injury, environmental impact	399. Flow switch FS-9202/04 low flow shutdown on pump P-1A/B with alarm 368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
	41.1.13. MOV-9211 fails closed when not performing diesel flush on outlet of pump P-1A/B	41.1.13.1. See consequences associated with cause 41.1.12							
	41.1.14. MOV-9212 fails closed or 16" valve closed on product manifold out of facility	41.1.14.1. See consequences associated with cause 41.1.12							
	41.1.15. Failure to perform water draw when needed	41.1.15.1. Increased water level in diesel tank TK-100-12/21, potential water in diesel product, off-spec product, potential financial impact	639. Procedure (GPS 0314) states to perform water draw off on Diesel Tanks prior to certifying the tanks	F	2	3	2	61. No recommendations required.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	41.1.16. 4" gate valve or 2" ball closed on relief header to tank TK-100-12/21	41.1.16.1. Potential blocked outlet of thermal PSVs, release of diesel to grade, potential personnel injury and environmental impact	448. No specific safeguards identified	P	4	3	4	43. Identify manual valves that need to be car sealed open to ensure there is an open flow path for thermal relief valves on Kero, Diesel, Naphtha, AGO, and Feed Tanks, as well as preventing cross contamination of the product. Car seal these valves open and update P&IDs accordingly.	
	41.1.17. MOV-9211 closed to product on diesel flush when flushing	41.1.17.1. This valve is a normally closed during diesel flushing operation, no hazardous consequences identified							
41.2. High Flow	41.2.1. Both pumps P-1A/B running at once	41.2.1.1. Flow is maximized under normal circumstances. Flow is regulated downstream by terminal operations, operability issue, no hazardous consequences identified							
	41.2.2. MOV-9210 fails open on pump P-1A/B bypass	41.2.2.1. Increased flow of diesel recycling around Transfer Pump P-1A/B, decreased flow to terminal, delay in transfer, logistical impact to terminals; Potential to increase temperature on pumps P-1A/B with potential pump damage, potential seal failure, release, ignition and personnel impact	368. Pump is in remote location and inside tank berm and separate containment 397. analyzer transmitter AT-9201/ 02/ 03/ 04 gas detector around the kero/diesel manifold with automatic shutdown of pumps P-4, and P-1A/B with 1oo4 voting 398. flame transmitter XT-9201/ 02 flame detector directed at pumps with automatic shutdown of pumps P-4, and P-1A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
	41.2.3. Water draw valves left open	41.2.3.1. Increased flow of water and eventually hydrocarbon liquid to sump and to Water Tank 5, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	1	1	61. No recommendations required.	
	41.2.4. 8" valve open on diesel flush when NOT flushing	41.2.4.1. Several other valves will be closed downstream, no stand-alone consequences identified							
	41.2.5. MOV-9211 open to product on diesel flush when flushing	41.2.5.1. Several other valves will be closed downstream, no stand-alone consequences identified							

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
41.3. Reverse/Misdirected Flow	41.3.1. 10" recirculation line open to second tank when not intended	41.3.1.1. Potential to flow diesel to tank that is not lined up; No adverse consequences identified							
	41.3.2. MOV open on diesel eductor inlet to tank TK-100-12/21 when not intended (on initial start-up)	41.3.2.1. Potential issue when sending kero to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	640. Tk-100-12/21 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9346/9243 and opening diffuser MOV-9336/9233 641. Tk-100-12/21 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	41.3.3. 6" manual valve open on low suction line from tank TK-100-12/21 when not intended	41.3.3.1. Potential to draw water in diesel product, off-spec product, potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	41.3.4. 12" valve V-9132/33 open and tank outlet MOV open on both diesel tanks	41.3.4.1. In this scenario (sending flow out of Diesel tank), pump P-4 would not be running, therefore flow would not be sent back to the Diesel tanks; Two MOV outlet valves open on different Diesel tanks would result in potentially sending uncertified diesel flow to pipeline; Potential financial impact from sending uncertified diesel to pipeline	635. Procedure (GPS 0304) states proper tank line-up 636. Customer provides instructions in daily operational "night notes" on tank rundown	F	3	1	1	61. No recommendations required.	
	41.3.5. Valve misalignment on kero/ diesel transfer header - Tank TK-100-12/21	41.3.5.1. (Flow Tank TK-100-12/21 to other) Potential to flow diesel from one tank to the other tank, tank level will balance, operability issue, no hazardous consequences identified							
		41.3.5.2. (Flow of diesel to kero tanks or kero to diesel tanks) Potential major flow of kero to diesel or vice versa, potential issue with inventory tracking, potential off-spec product; No environmental impact (distillate products able to be stored in same tanks); Potential financial impact due to off-spec product	635. Procedure (GPS 0304) states proper tank line-up 402. SAAB level transmitter LT-9202 on kero tank TK-100-21 will alarm on unintended flow direction (intentional flow direction is manually set) 400. SAAB level transmitter LT-9201 on kero tank TK-150-10 will alarm on unintended flow direction (intentional flow direction is manually set)	F	2	1	1	61. No recommendations required.	

Node: **41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps**

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			401. SAAB level transmitter LT-9204 on kero tank TK-100-11 will alarm on unintended flow direction (intentional flow direction is manually set) 404. SAAB level transmitter LT-9203 on diesel tank TK-100-21 will alarm on unintended flow direction (intentional flow direction is manually set) 405. SAAB level transmitter LT-9306 on diesel tank TK-100-12 will alarm on unintended flow direction (intentional flow direction is manually set)						
	41.3.6. 8" diesel flush valve open to any AGO tank when not intended	41.3.6.1. Several other valves will be closed downstream, no stand-alone consequences identified							
	41.3.7. 8" or 12" valve open on diesel flush to feed pumps	41.3.7.1. Potential to send diesel to splitters, potential excess rerun and economic loss, potential major unit upset; Potential financial impact	642. Shutdown Procedure (GPS-1100) includes a step for diesel flush	F	2	2	1	61. No recommendations required.	
	41.3.8. 8" valve open on diesel flush to exchanger pumps P-HX	41.3.8.1. Several other valves will be closed downstream, no stand-alone consequences identified							
	41.3.9. Diesel flush open to AGO exchanger HX1/2 when in AGO service	41.3.9.1. Potential to send diesel to AGO tank, potential issue with inventory tracking; No environmental impact (distillate product); No off-spec issues, additional diesel in AGO, potential lower flash; No adverse consequences identified							
	41.3.10. 8" manual valve open on diesel flush to AGO manifold on 8"-9319 on 88-PI-9230	41.3.10.1. Potential to send diesel to AGO tank, potential issue with inventory tracking; No environmental impact (distillate product); No off-spec issues, additional diesel in AGO, potential lower flash; No adverse consequences identified							
41.4. High Pressure	41.4.1. External fire on Diesel Tanks TK-100-12/21	41.4.1.1. Increased pressure and overpressure of Diesel Tanks TK-100-12/21, potential vessel rupture, financial impact	382. Foam chamber for tank seals 410. Several open vents on floating roof tank for tank Diesel	P	4	1	2	61. No recommendations required.	

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			Tanks TK-100-12/21 are sized for external fire						
	41.4.2. Gravity feeding between Diesel Tanks TK-100-12/21	41.4.2.1. Potential to exceed tank design inflow/ outflow rate (if high level in one tank and low level in other); Potential damage to IFR on both tanks; Potential financial impact	643. Tank-to-Tank Transfer Procedure (0305) updated to prevent flow rates exceeding the floating roof maximum flow rate limit of 2500 BPH.	F	4	3	4	40. Update the P&IDs and the tank drawings for the Kero, Diesel, AGO, Naphtha, and Feed Tanks to reflect the internal vacuum breakers.	
	41.4.3. Blocked in Diesel and thermal expansion	41.4.3.1. Potential to overpressure piping by thermal expansion, release of kero, personnel injury, environmental impact	406. Thermal Pressure Safety Valves across kero piping circuit	P	4	1	2	44. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-9239 are car sealed open and update P&ID to show valves CSO. Also update P&ID to include field verified PSV tag numbers.	
41.5. Low Pressure	41.5.1. No additional causes identified								
41.6. High Level	41.6.1. Improper lineup - lining up rundown to a full tank	41.6.1.1. Potential overfill of Diesel Tank to grade, potential release of Diesel, pool fire, potential personnel injury	411. Level alarm high LAH-9203/ 9306 on Diesel Tank TK-100-12/21 with operator response 412. Level alarm hihi LAH-9203/ 9306 on Diesel Tank TK-100-12/21 alarm with operator response 413. level switch high LSH-9203/ 9306 on Diesel Tank TK-100-12/21 mechanical switch will close inlet motor operated valve MOV to tanks Diesel Tanks TK-100-12/21 stopping diesel flow to storage	P	4	1	2	61. No recommendations required.	
41.7. Low Level	41.7.1. No additional causes identified								
41.8. High Interface Level	41.8.1. Not applicable								
41.9. Low Interface Level	41.9.1. Not applicable								
41.10. High Temperature	41.10.1. No additional causes identified								

Node: 41. Node 41: Diesel Tanks 100-12/ 100-21 and Pumps

Drawings / References:

88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9221; 88-P1-9222; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9229; 88-P1-9230; 88-P1-9231; 88-P1-9233; 88-P1-9234; 88-P1-9239 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
41.11. Low Temperature	41.11.1. Low ambient temperature	41.11.1.1. No consequences of note.							
41.12. Contamination/ Composition	41.12.1. Mixer failure	41.12.1.1. Potential reduced tank mixing; Potential stratification in tank; No off-spec product issues; Operability issue only, no hazardous consequences identified							
41.13. Corrosion	41.13.1. No specific causes identified								
41.14. Additional Phase	41.14.1. Not applicable								
41.15. Rupture/Loss of Containment	41.15.1. Mechanical Roof Failure	41.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential diesel release and ignition, potential pool fire requiring tank replacement, personnel injury, environmental impact, asset damage	390. Lower explosive limit LEL alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will identify seal damage on roof 644. Monthly through hatch inspection Diesel Tanks TK-100-12/21	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
41.16. Loss of Utilities	41.16.1. Loss of power	41.16.1.1. See pump trip scenarios, this node							
41.17. Start-up/Shutdown hazards/ Maintenance	41.17.1. 4" valve (V-9074) on TK-100-12 left closed when TK-100-21 is taken out of service for maintenance (or vice versa)	41.17.1.1. Blocked in PSV relief path for other tanks connected to the diesel header with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	46. Update the Tank Line-Up operating procedure for TK-100-12/21 to ensure that, when one tank is taken out of service, the 4" valve (V-9074 for TK-100-12, X-9163 for TK-100-21) is set as CSO to ensure an open relief path for the PSVs on the connecting tanks.	
41.18. General	41.18.1. No additional causes identified								

Node: 42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
42.1. Low/No Flow	42.1.1. Manual valve closed on AGO rundown header from one or both splitter towers	42.1.1.1. High level in Combi Tower T-102/202; Potential to overfill T-102/202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107/207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response	F	1	1	1	61. No recommendations required.	

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response						
		42.1.1.2. Combi Tower Bottoms Product Pumps PM-104A/B or PM-204A/B deadheaded, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	474. Dual seals on PM-104A/B 546. Dual seals on PM-204A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B 547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		42.1.1.3. Inability to heat the feed in EX-105/205; Potential for low feed temperature into Naphtha Stabilizer Column T-101/201; Potential reduced rates; Potential financial impact	448. No specific safeguards identified	F	1	3	1	61. No recommendations required.	
	42.1.2. MOV-9335/45; 9237/47; 9338/48 closed on AGO inlet to tank TK-120-22/23 and TK-100-13	42.1.2.1. See consequences associated with cause 42.1.1							
	42.1.3. 12" valve closed on suction to Transfer Pump P-5 or strainer plugs	42.1.3.1. Loss of flow to AGO to Transfer Pump P-5, potential cavitation and seal leak, release of AGO to atmosphere, potential ignition and personnel impact	414. Flow switch FS-9630 low flow shutdown on pump P-5 with alarm 415. analyzer transmitter AT-9301/02/ 03/ 04 gas detector	P	4	1	2	61. No recommendations required.	

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting						
	42.1.4. Transfer Pump P-5 stops	42.1.4.1. Decreased flow of AGO tank to tank transfer rate, delay in transfer; No adverse consequences identified							
	42.1.5. 10" valve closed on discharge of Transfer Pump P-5	42.1.5.1. Loss of flow to AGO from Transfer Pump P-5, potential deadhead and seal leak, release of AGO to atmosphere, potential seal fire from seal, personnel injury, environmental impact	414. Flow switch FS-9630 low flow shutdown on pump P-5 with alarm 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
		42.1.5.2. Decreased flow of AGO tank to tank transfer rate, delay in transfer; No adverse consequences identified							
	42.1.6. MOV-9335/45; 9237/47; 9338/48 closed on AGO inlet to tank TK-120-22/23 and TK-100-13 when transferring tank to tank	42.1.6.1. See consequences associated with 42.1.5							
	42.1.7. MOV-9325 /9327/ 9328 closed on outlet of tank TK-120-22/23 and TK-100-13 to Transfer Pump P-3A/B when tank is in transfer to pipeline	42.1.7.1. Decreased flow of AGO from AGO Tanks TK-120-22/23 and TK-100-13 to Transfer Pump P-3A/B, loss of flow to AGO P-3A/B, potential cavitation and seal leak, release of AGO to grade, potential ignition and personnel impact	417. Flow switch FS-9302/04 low flow shutdown on pump P-3A/B with alarm 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of	P	4	1	2	61. No recommendations required.	

Node: 42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting 647. Pump P-3A/B permissive prevents pump from starting unless MOV-9325 /9327/ 9328 are open						
		42.1.7.2. Decreased flow of AGO from AGO tank to Transfer Pump P-3A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	42.1.8. MOV-9305/ 06/ 07 closed on outlet of tank TK-120-22/23 and TK-100-13 to Transfer Pump P-3A/B when tank is in transfer to pipeline	42.1.8.1. Decreased flow of AGO from AGO Tanks TK-120-22/23 and TK-100-13 to Transfer Pump P-3A/B, loss of flow to AGO P-3A/B, potential cavitation and seal leak, release of AGO to grade, potential ignition and personnel impact	417. Flow switch FS-9302/04 low flow shutdown on pump P-3A/B with alarm 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting 648. Pump P-3A/B permissive prevents pump from starting unless MOV-9305/6/7 are open	P	4	1	2	61. No recommendations required.	
		42.1.8.2. Decreased flow of AGO from AGO tank to Transfer Pump P-3A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	42.1.9. MOV-9311/12 closed on inlet valve to Transfer Pump P-3A/B or strainer plugs	42.1.9.1. Decreased flow of AGO from AGO Tanks TK-120-22/23 and TK-100-13 to Transfer Pump P-3A/B, loss of flow to AGO P-	417. Flow switch FS-9302/04 low flow	P	4	1	2	61. No recommendations required.	

Node: 42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		3A/B, potential cavitation and seal leak, release of AGO to grade, potential ignition and personnel impact	shutdown on pump P-3A/B with alarm 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting						
	42.1.10. Transfer Pump P-3A/B stops	42.1.10.1. Decreased flow of AGO from AGO tank to Transfer Pump P-3A/B, not expected to result in overfill of tank as movement tank is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	42.1.11. Blocked 14" manual outlet valve of Transfer Pump P-3A/B	42.1.11.1. Blocked flow from Transfer Pump P-3A/B, potential deadhead and seal leak, release of AGO to grade, potential ignition and personnel impact	417. Flow switch FS-9302/04 low flow shutdown on pump P-3A/B with alarm 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting 648. Pump P-3A/B permissive prevents pump from starting unless MOV-9305/6/7 are open	P	4	1	2	61. No recommendations required.	
		42.1.11.2. Decreased flow of AGO from AGO tank to Transfer Pump P-3A/B, not expected to result in overfill of tank as movement tank	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		is not receiving rundown, transfer shutdown, logistical impact to terminals; Potential financial impact							
	42.1.12. Product Manifold MOV fails closed or 16" valve closed on product manifold out of facility	42.1.12.1. See consequences associated with cause 42.1.11							
	42.1.13. 4" gate valve or 2" ball closed on relief header to AGO Tanks TK-120-22/23 and TK-100-13	42.1.13.1. Potential blocked outlet of thermal PSVs, release of AGO to grade, potential personnel injury and environmental impact	448. No specific safeguards identified	P	4	3	4	43. Identify manual valves that need to be car sealed open to ensure there is an open flow path for thermal relief valves on Kero, Diesel, Naphtha, AGO, and Feed Tanks, as well as preventing cross contamination of the product. Car seal these valves open and update P&IDs accordingly.	
42.2. High Flow	42.2.1. Both pumps P-3A/B running at once	42.2.1.1. Flow is maximized under normal circumstances. Flow is regulated downstream by terminal operations, operability issue, no hazardous consequences identified							
	42.2.2. MOV-9308 fails open on pump P-3A/B bypass	42.2.2.1. Increased flow of AGO recycling around Transfer Pump P-3A/B, decreased flow to terminal, delay in transfer, logistical impact to terminals; Potential to increase temperature on pumps P-3A/B with potential pump damage, potential seal failure, release, ignition and personnel impact	415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting	P	4	1	2	61. No recommendations required.	
42.3. Reverse/Misdirected Flow	42.3.1. 10" recirculation line open to other tank when not intended	42.3.1.1. Potential to flow AGO to tank that is not lined up; No adverse consequences identified							
	42.3.2. MOV to eductor open and MOV to diffuser closed on tanks TK-120-22/23 and TK-100-13 when not intended	42.3.2.1. Potential issue when sending AGO to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	645. Tk-100-13, Tk-120-22/23 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9348/9345/9347 and opening diffuser MOV-9338/9335/9337 646. Tk-100-13, Tk-120-22/23 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	42.3.3. 12" valve V-9089/90/91 open and tank outlet MOV open on two different AGO tanks	42.3.3.1. In this scenario (sending flow out of AGO tank), pump P-5 would not be running, therefore flow would not be sent back to the AGO tanks;	635. Procedure (GPS 0304) states proper tank line-up	F	3	1	1	61. No recommendations required.	

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		Two MOV outlet valves open on different AGO tanks would result in potentially sending uncertified AGO flow to pipeline; Potential financial impact from sending uncertified AGO to pipeline	636. Customer provides instructions in daily operational "night notes" on tank rundown						
	42.3.4. Valve misalignment on AGO transfer header - Tank TK-100-13; 120-22/23	42.3.4.1. (Flow Tank TK-100-13; 120-22/23 to other) Potential to flow AGO from one tank to the other tank, tank level will balance, off-spec of low quality and high quality AGO (chlorides); Potential financial impact	636. Customer provides instructions in daily operational "night notes" on tank rundown	F	1	3	1	61. No recommendations required.	
	42.3.5. 16" manual valve X-9241 inadvertently open on AGO product manifold to feed tanks	42.3.5.1. Potential to send AGO to feed tank, potential excess rerun and economic loss; Potential financial impact	448. No specific safeguards identified	F	2	3	2	61. No recommendations required.	
	42.3.6. AGO valve open from AGO manifold to AGO exchanger HX1/2	42.3.6.1. AGO is currently flowing through exchangers HX1/2; No consequences identified, normal operation							
	42.3.7. Feed valve open from AGO manifold to AGO exchanger HX1/2	42.3.7.1. Feed is currently flowing through exchangers HX1/2; No consequences identified, normal operation							
	42.3.8. Feed and AGO lined up to AGO exchanger HX1/2	42.3.8.1. Feed and AGO are currently flowing through exchangers HX1/2; No consequences identified, normal operation							
	42.3.9. Tube leak on AGO Exchanger HX-1/2	42.3.9.1. Potential to send Feed into AGO Product, off-spec AGO; potential environmental permitting issue, environmental/financial impact (potential off-spec of a full AGO tank)	448. No specific safeguards identified	F	3	4	4	47. Ensure that the AGO Exchangers HX-1/2 are included in the mechanical integrity program to mitigate the likelihood of tube leaks. Perform evaluation for the need for additional safeguards if deemed necessary. Financial risk only.	
	42.3.10. 16" valve open to AGO Pig launcher	42.3.10.1. There is a locking mechanism on the hatch of the AGO PIG Launcher that is difficult to open; Inadvertently opening the 16" valve to the PIG launcher would be an intentional action and is therefore not considered to be credible							
42.4. High Pressure	42.4.1. External fire on AGO Tanks TK-100-13; 120-22/23	42.4.1.1. Increased pressure and overpressure of AGO Tanks TK-100-13; 120-22/23, potential vessel rupture, financial impact	382. Foam chamber for tank seals 418. Several open vents on floating roof tank for tank AGO Tanks TK-100-13; 120-22/23 are sized for external fire	P	4	1	2	61. No recommendations required.	
	42.4.2. External fire on AGO exchanger HX1/2 Shell side	42.4.2.1. Increased pressure and overpressure of AGO exchanger HX1/2 Shell side potential vessel rupture, financial impact	419. Pressure Safety Valve PSV-9631/9632 on feed shell side of exchanger HX1/2 set at 125 psig routed to a feed tank and sized for external fire	P	4	1	2	48. Perform evaluation of PSV-9631/9632 on the shell-side of AGO Exchanger HX1/2 to determine if the discharge location needs to be re-routed in order to provide protection against external fire (currently routed to the Feed Tanks). Update P&ID (88-PI-9218) to include the tag numbers of the shell-side PSVs.	

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	42.4.3. External fire on AGO exchanger HX1/2 Tube side	42.4.3.1. Increased pressure and overpressure of AGO exchanger HX1/2 Shell side, potential vessel rupture, financial impact	420. Pressure Safety Valve PSV-9633/34 on AGO tube side of exchanger HX1/2 set at 125 psig routed to a feed tank and sized for external fire	P	4	1	2	49. Perform evaluation of PSV-9633/9634 on the tube-side of AGO Exchanger HX1/2 to determine if the discharge location needs to be re-routed in order to provide protection against external fire (currently routed to the Feed Tanks).	
	42.4.4. Gravity feeding between AGO Tanks TK-100-13; 120-22/23	42.4.4.1. Potential to exceed tank design inflow/ outflow rate (if high level in one tank and low level in other); Potential damage to IFR on both tanks; Potential financial impact	643. Tank-to-Tank Transfer Procedure (0305) updated to prevent flow rates exceeding the floating roof maximum flow rate limit of 2500 BPH.	F	4	3	4	40. Update the P&IDs and the tank drawings for the Kero, Diesel, AGO, Naphtha, and Feed Tanks to reflect the internal vacuum breakers.	
	42.4.5. Blocked in AGO and thermal expansion	42.4.5.1. Potential to overpressure piping by thermal expansion, release of AGO, personnel injury, environmental impact	649. Thermal Pressure Safety Valves across AGO piping circuit	P	4	1	2	44. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-9239 are car sealed open and update P&ID to show valves CSO. Also update P&ID to include field verified PSV tag numbers.	
42.5. Low Pressure	42.5.1. No additional causes identified								
42.6. High Level	42.6.1. Improper lineup - lining up rundown to a full tank	42.6.1.1. Potential overfill of AGO Tank to grade, potential release of AGO, pool fire, potential personnel injury	421. Level alarm high LAH-9305/ 07/ 08 on AGO Tank TK-100-13; 120-22/23 with operator response 422. Level alarm hihi LAH-9305/ 07/ 08 on AGO Tank TK-100-13; 120-22/23 alarm with operator response 423. level switch high LSH-9305/ 07/ 08 on AGO Tank TK-100-13; 120-22/23 mechanical switch will close inlet motor operated valve MOV to tanks AGO Tanks stopping AGO flow to storage	P	4	1	2	61. No recommendations required.	
42.7. Low Level	42.7.1. No additional causes identified								
42.8. High Interface Level	42.8.1. Not applicable								
42.9. Low Interface Level	42.9.1. Not applicable								

Node: **42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps**

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
42.10. High Temperature	42.10.1. Tank Heater turns on (currently locked out)	42.10.1.1. The tank heaters are locked out and disconnected, therefore this is not a credible scenario.							
42.11. Low Temperature	42.11.1. Low ambient temperature	42.11.1.1. No consequences of note.							
42.12. Contamination/ Composition	42.12.1. No additional causes identified								
42.13. Corrosion	42.13.1. H2S and/or water in AGO tanks	42.13.1.1. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
42.14. Additional Phase	42.14.1. Not applicable								
42.15. Rupture/Loss of Containment	42.15.1. Mechanical Roof Failure	42.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential AGO release and ignition, potential pool fire requiring tank replacement, personnel injury, environmental impact, asset damage	650. Lower explosive limit LEL alarms AT-9301/9302/9303/9304 around the AGO Manifold will identify seal damage on roof 653. Monthly through hatch inspection on AGO Tanks Tk-100-13, TK-120-22/23	P	4	1	2	61. No recommendations required.	
	42.15.2. Sight-glass failure (flow indicator FI-9305/07/08)	42.15.2.1. Potential major AGO spill to grade releasing contents of tank, potential pool fire, personnel injury, environmental impact, asset damage	391. Water draw procedure states to open the valve downstream of the sight-glass before opening the valve upstream of the sight-glass to prevent tank head crushing the sight-glass 392. Secondary containment around tank	P	3	2	2	61. No recommendations required.	
42.16. Loss of Utilities	42.16.1. Loss of power	42.16.1.1. See pump trip scenarios, this node							
42.17. Start-up/Shutdown hazards/ Maintenance	42.17.1. H2S in the AGO	42.17.1.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S 608. AT-9301 Fixed H2S monitors located by the transfer pumps in the AGO manifold	P	2	2	1	61. No recommendations required.	

Node: 42. Node 42: AGO Tanks 100-13/ 120-22/ 120-23 and Pumps

Drawings / References:

88-P1-9217; 88-P1-9218; 88-P1-9219; 88-P1-9223; 88-P1-9224; 88-P1-9225; 88-P1-9226; 88-P1-9227 ; 88-P1-9227 Redline; 88-P1-9230; 88-P1-9231; 88-P1-9239

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			will alert in the control room						
		42.17.1.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
	42.17.2. 4" valve (V-9083) on TK-100-13 left closed when TK-120-22 or TK-120-23 is taken out of service for maintenance (or 4" valves V-9062 for TK-120-22, V-9078 for TK-120-23)	42.17.2.1. Blocked in PSV relief path for other tanks connected to the AGO header with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	50. Update the Tank Line-Up operating procedure for TK-100-13, TK-120-22/23 to ensure that, when one tank is taken out of service, the 4" valve (V-9083 for TK-100-13, V-9062 for TK-120-22, V-9078 for TK-120-23) is set as CSO to ensure an open relief path for the PSVs on the connecting tanks.	
42.18. General	42.18.1. No additional causes identified								

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
43.1. Low/No Flow	43.1.1. Closed valves: manual valve closed inadvertently on suction side on condensate feed pump PM-111A/B/C	43.1.1.1. Potential mechanical damage to condensate feed pump PM-111A/B/C (cavitation); Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C 446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		43.1.1.2. Loss of feed to the unit, resulting in shutdown and financial impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C	F	2	2	1	61. No recommendations required.	
	43.1.2. Loss of feed from tank farm	43.1.2.1. See consequences associated with cause 43.1.1							
	43.1.3. Closed valves: Manual valve closed inadvertently on discharge side of condensate		1. Flow alarm low FAL-12475 low flow alarm	P	4	1	2		1. The location of the

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	feed pump PM-111A/B/C including exchanger valves	43.1.3.1. Potential mechanical damage to condensate feed pump PM-111A/B/C (deadhead); Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	and operator response to trip condensate feed pumps PM-111A/B/C 446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C					60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		43.1.3.2. Loss of feed to the unit, resulting in shutdown and financial impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C	F	2	2	1	61. No recommendations required.	
	43.1.4. MOV-9180/9181/9170/9171/9160/9161 in condensate feed pump PM-111A/B/C discharge fails closed on discharge side of pump	43.1.4.1. See consequences associated with cause 43.1.3							
	43.1.5. Condensate feed pump PM-111A/B/C mechanical failure or loss of power	43.1.5.1. Loss of feed to the unit, resulting in shutdown and financial impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C	F	2	2	1	61. No recommendations required.	
	43.1.6. 12" Manual valve closed, MOV-60010 or pressure control valve PCV-60020 closed on main Magellan 12" line to Feed Tanks	43.1.6.1. Magellan Line has been removed from service (air gapped on Magellan side), therefore no consequences of concern for Magellan line							
	43.1.7. 20" Manual valve closed or MOV closed on main KMCC line to Feed Tank Inlets	43.1.7.1. Decreased flow of Feed to Feed Tanks TK-200-1/2/3, Feed will not be transferred to unit from the same tank that is being filled from pipeline, not expected to result in low level, no impact tanks, delay in feed transfer; Potential financial impact (potential need to reduce throughput)	651. Pipeline Feed Receipt Procedure (GPS 0303) includes correct valve line-up	F	2	2	1	61. No recommendations required.	
	43.1.8. MOV-9121/22/23 closed on outlet of tank TK-200/1/2/3 to Feed Pump PM-111A/B/C when tank is feeding to Splitter Units	43.1.8.1. Decreased flow of Feed from Feed Tanks TK-200-1/2/3 to Feed Pumps PM-111A/B/C, loss of flow to Feed PM-111A/B/C, potential cavitation; Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C 446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area						truck would drive into the dike area.
			2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C						
		43.1.8.2. Loss of flow to Feed Pumps PM-111A/B/C, not expected to result in overfill of tank as movement tank is not receiving rundown, loss of feed to Splitter and unit shutdown, financial impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C	F	2	2	1	61. No recommendations required.	
	43.1.9. 12" manual valve closed on feed to Feed Pumps PM-111A/B/C or strainer plugs	43.1.9.1. See consequences associated with cause 43.1.8							
	43.1.10. 4" Manual valve(s) closed on hydrocarbon header from one or both splitter towers	43.1.10.1. Hydrocarbon Sump Pump PM-118A/B or PM-218A/B deadheaded, potential pump damage; Potential seal failure with potential ignition and personnel impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response						
			567. LAH-12501 high level in TK-101 Sump will kick on other pump PM-118A/B						
			601. LAH-22501 high level in TK-201 Sump will kick on other pump PM-218A/B						
			451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
		43.1.10.2. Potential to overfill the Hydrocarbon Sump TK-101/201, release through atmospheric vent; Potential runoff to stormwater drain (contained in Tank 5 Wastewater Tank); Potential environmental/financial impact	56. LAH-12501/12502 redundant high level alarms on TK-101 Hydrocarbon Sump will prompt operator response	F	1	1	1	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			600. LAH-22501/22502 redundant high level alarms on Sump TK-201 will prompt operator response 451. Lower explosive limit LEL alarms AI-10416/20416 near Sumps TK-101/201 will prompt operator response to cordon off the area						
	43.1.11. Manual valve(s) closed on hydrocarbon header from Flare knockout drum	43.1.11.1. Potential for liquid carryover to flare, environmental impact, potential burning liquid fallout from flare, personnel impact	549. LAHH-30040 high-high level alarm on PV-305 will initiate unit shutdown 548. LAHH-30045 high-high level alarm on PV-305 will auto start pumps PM-312A/B	P	4	1	2	61. No recommendations required.	
		43.1.11.2. Potential for liquid carryover to flare, resulting in shutdown and financial impact	549. LAHH-30040 high-high level alarm on PV-305 will initiate unit shutdown 548. LAHH-30045 high-high level alarm on PV-305 will auto start pumps PM-312A/B	F	2	1	1	61. No recommendations required.	
43.2. High Flow	43.2.1. All pumps PM-111A/B/C online	43.2.1.1. Increased feed to Naphtha Stabilizer Column T-101/201, reduced temperature in T-101/201, potential for off-spec product, potential financial impact	578. LAH-10525 high level alarm on Stabilizer T-101 will prompt operator response 652. LAH-20525 high level alarm on Stabilizer T-201 will prompt operator response 429. Flow control valve FCV-12425/ 22425 will control flow to each Splitter tower and will alarm at high flow and prompt operator response	F	2	2	1	61. No recommendations required.	

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		43.2.1.2. Increased flow of Feed out of Feed Tanks TK-200-1/2/3, decreased level in Feed tanks, potential to land roof, potential explosive atmosphere in vapor space, potential explosion, personnel injury, environmental impact, asset damage	430. Level alarm low LAL-9101/02/03 on Feed Tank TK-200-1/2/3 with operator response and failure indication 429. Flow control valve FCV-12425/ 22425 will control flow to each Splitter tower and will alarm at high flow and prompt operator response 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		43.2.1.3. Increased flow of Feed out of Feed Tanks TK-200-1/2/3, decreased level in Feed Tanks TK-200-1/2/3, loss of flow to Feed Pumps PM-111A/B/C; Potential cavitation, seal failure and release of hydrocarbons; Potential ignition and personnel impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C 446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
43.3. Reverse/Misdirected Flow	43.3.1. Any 12" manual valve open between two feed tanks on the feed manifold (suction of Feed Pump PM-111A/B/C) when not intended	43.3.1.1. Potential to flow Feed to unintended Feed tank, not expected to result in an overfill as tanks level will balance by gravity, operability issue, no hazardous consequences identified							
	43.3.2. 6" manual valve open on low suction line from tank TK-200-1/2/3 when not intended	43.3.2.1. Potential to draw water in Feed Splitter 1/2, potential high water content in feed, potential unit upset on Splitter 1/2, operability issue, no hazardous consequences identified							
	43.3.3. Valve misaligned between two feed tanks on the feed manifold (suction of Feed Pump PM-111A/B/C)	43.3.3.1. Potential to flow Feed to unintended Feed tank, not expected to result in an overfill as tanks level will balance by gravity, operability issue, no hazardous consequences identified							

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	43.3.4. 10" manual valve opened on jumper from Tank TK-200-1 rerun inlet to TK-200-1 outlet (9215)	43.3.4.1. Potential to send rerun to feed, not expected to result in any operability issue, no hazardous consequences identified							
	43.3.5. 10" manual valve opened on Tank TK-200-2 rerun inlet to TK-200-2 outlet (9216)	43.3.5.1. Potential to send rerun to feed, not expected to result in any operability issue, no hazardous consequences identified							
	43.3.6. 20" motor operated valve MOV open on KMCC feed to Tank TK-100-14 and Tank inlet motor operated valve MOV-9450 open when not intended	43.3.6.1. Potential overfill of Rerun Tank TK-100-14 to grade, potential release of condensate/rerun, potential pool fire or vapor cloud and explosion or flash fire, potential personnel injury, environmental impact, asset damage	432. Level alarm high LAH-9400 on Rerun Tank TK-100-14 with operator response 433. Level alarm hihi LAHH-9400 on Rerun Tank TK-100-14 alarm with operator response 434. level switch high LSH-9400 on Rerun Tank TK-100-14 mechanical switch will close all inlet motor operated valve MOVs to Rerun Tank 651. Pipeline Feed Receipt Procedure (GPS 0303) includes correct valve line-up	P	4	1	2	61. No recommendations required.	
43.4. High Pressure	43.4.1. External fire on Feed Tanks TK-200-1/2/3	43.4.1.1. Increased pressure and overpressure of Feed Tanks TK-200-1/2/3, potential vessel rupture, financial impact	382. Foam chamber for tank seals 431. Several open vents on floating roof tank for Feed Tanks TK-200-1/2/3 are sized for external fire	P	4	1	2	61. No recommendations required.	
	43.4.2. Gravity feeding between tank TK-200-1/2/3	43.4.2.1. Potential to exceed tank design inflow/ outflow rate (if high level in one tank and low level in other); Potential damage to IFR on both tanks; Potential financial impact	643. Tank-to-Tank Transfer Procedure (0305) updated to prevent flow rates exceeding the floating roof maximum flow rate limit of 2500 BPH.	F	4	3	4	40. Update the P&IDs and the tank drawings for the Kero, Diesel, AGO, Naphtha, and Feed Tanks to reflect the internal vacuum breakers.	
	43.4.3. Blocked in Feed and thermal expansion	43.4.3.1. Potential to overpressure piping by thermal expansion, release of Feed, personnel injury, environmental impact	435. Thermal Pressure Safety Valves across Feed piping circuit	P	4	1	2	44. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-9239 are car sealed open and update P&ID to show valves CSO. Also update P&ID to include field verified PSV tag numbers. 51. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-	

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
								9246 are car sealed open and update P&ID to show valves as CSO.	
43.5. Low Pressure	43.5.1. No additional causes identified								
43.6. High Level	43.6.1. Improper lineup - lining up receiving feed into a full tank	43.6.1.1. Potential overfill of Feed Tank TK-200-1/2/3 to grade, potential release of condensate, potential pool fire or vapor cloud and explosion or flash fire, potential personnel injury, environmental impact, asset damage	426. Level alarm high LAH-9101/ 02/ 03 on Feed Tank TK-200-1/2/3 with operator response 427. Level alarm hihi LAHH-9101/ 02/ 03 on Feed Tank TK-200-1/2/3 alarm with operator response 428. level switch high LSH-9101/ 02/ 03 on Feed Tank TK-200-1/2/3 mechanical switch will close inlet motor operated valve MOV to Feed Tanks stopping Feed flow to storage and eventually shutting the pipeline flow on high pressure 651. Pipeline Feed Receipt Procedure (GPS 0303) includes correct valve line-up	P	4	1	2	61. No recommendations required.	
43.7. Low Level	43.7.1. No additional causes identified								
43.8. High Interface Level	43.8.1. Not applicable								
43.9. Low Interface Level	43.9.1. Not applicable								
43.10. High Temperature	43.10.1. High ambient temperature	43.10.1.1. No consequences of note.							
43.11. Low Temperature	43.11.1. Low ambient temperature	43.11.1.1. No consequences of note.							
43.12. Contamination/ Composition	43.12.1. Mixer failure	43.12.1.1. Potential reduced tank mixing, potential build-up of solids in Feed Pump PM-111A/B/C strainers; Potential mechanical damage to condensate feed pump PM-111A/B/C (cavitation); Potential seal failure and release of hydrocarbons; Potential ignition and personnel impact	1. Flow alarm low FAL-12475 low flow alarm and operator response to trip condensate feed pumps PM-111A/B/C	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	1. The location of the Condensate Feed Pumps PM-111A/B/C

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			446. PM-111A/B/C located in feed tank dike/containment 447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 2. Tandem seal - level switch on seal pot trips condensate feed pump PM-111A/B/C						does not have close by ignition sources - only sources would be if a truck would drive into the dike area.
		43.12.1.2. Potential reduced tank mixing; Potential stratification in tank; No off-spec product issues; Potential process upset; Potential shutdown with potential throughput interruption and financial impact	460. PAH-10395 high pressure alarm on Accumulator PV-101 overheads will prompt operator response 542. PAH-20395 high pressure alarm on Accumulator PV-201 overheads will prompt operator response	F	2	3	2	61. No recommendations required.	
43.13. Corrosion	43.13.1. Excess water in the feed	43.13.1.1. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							
43.14. Additional Phase	43.14.1. Not applicable								
43.15. Rupture/Loss of Containment	43.15.1. Mechanical Roof Failure	43.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential Feed release and ignition, potential pool fire or vapor explosion, requiring tank replacement, personnel injury, environmental impact, asset damage	447. Area lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area 654. Monthly through hatch inspection on Feed Tanks Tk-200-1/2/3	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
43.16. Loss of Utilities	43.16.1. Loss of power	43.16.1.1. See pump trip scenarios, this node							
43.17. Start-up/Shutdown hazards/ Maintenance	43.17.1. H2S in the feed to the plant	43.17.1.1. Potential for personnel exposure to H2S in the event of a leak or release	604. Personal protection monitors will alert at 10 ppm H2S	P	2	3	2	61. No recommendations required.	
		43.17.1.2. Potential increased corrosion to piping and equipment due to H2S; Long-term asset damage, no acute consequences identified							

Node: 43. Node 43: Feed Tank and Feed Pumps PM-111A/B/C

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9239 ; 88-P1-9246; 88-P1-9251

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	43.17.2. Condensate Feed Pumps PM-111A/B/C rated for 1472 gpm, unit flows up to 1600 gpm	43.17.2.1. Potential excessive wear on Condensate Feed Pumps, potential financial impact	448. No specific safeguards identified	F	1	4	2	52. Review the design pump curve for Condensate Feed Pumps PM-111A/B/C to determine the maximum flowrate. The P&ID lists the design flowrate as 1472 gpm, but the current flowrate of the pump is 1600 gpm. If design review determines the pump is operating outside of design limits, evaluate the need to re-rate the pump for a higher maximum flowrate.	
	43.17.3. Gauging or sampling Feed Tanks TK-200-1/2/3 from top	43.17.3.1. Potential personnel exposure to toxic gas when gauging the tank, sampling the tank or doing a movement from or to the tank	371. Four gas monitors will alert operator of hydrogen sulfide 372. Third party inspectors includes wearing breathing air for sampling 373. Hatch has a vapor plug which will prevent immediate exposure to hydrogen sulfide - will allow time for response to gas monitor	P	2	2	1	53. Add signage to Rerun Tank TK-100-14, Feed Tanks TK-200-1/2/3 and Light Naphtha Tanks TK-120-24/25 stating that hydrogen sulfide H2S may be present to warn inspectors gauging the tank.	
	43.17.4. 4" valve (X-9001) on TK-200-1 left closed when TK-200-2 or TK-200-3 is taken out of service for maintenance (or 4" valves X-9016 for TK-200-2, X-9029 for TK-200-3)	43.17.4.1. Blocked in PSV relief path for other tanks connected to the feed header with potential overpressure of tank, loss of containment, personnel impact	448. No specific safeguards identified	P	4	3	4	54. Update the Tank Line-Up operating procedure for TK-200-1/2/3 to ensure that, when one tank is taken out of service, the 4" valve (X-9001 for TK-200-1, X-9016 for TK-200-2, X-9029 for TK-2003) is set as CSO to ensure an open relief path for the PSVs on the connecting tanks.	
43.18. General	43.18.1. No additional causes identified								

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
44.1. Low/No Flow	44.1.1. 8" Manual valve closed on Rerun header from one or both splitter towers	44.1.1.1. Combi Tower Bottoms Product Pumps PM-104A/B or PM-204A/B deadheaded, potential seal failure, release of hydrocarbons and ignition, potential personnel impact	474. Dual seals on PM-104A/B 546. Dual seals on PM-204A/B 475. Seal Failure Alarm SF-11410/11430 will shutdown pumps PM-104A/B	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			547. Seal Failure Alarm SF-21410/21430 will shutdown pumps PM-204A/B 631. Lower Explosive Limit LEL alarms AI-11461/11481/21461/21481 will prompt operators to cordon off the area						
		44.1.1.2. High level in Combi Tower T-102/202; Potential to overfill T-102/202; Potential to overfill the Recycle Gas Compressor Suction Scrubber PV-107/207 and send liquids to the RGC; Potential to damage compressor leading to financial impact	16. LAH-11740 High Level Alarm on Combi Tower T-102 will prompt operator response 595. LAH-21740 High Level Alarm on Combi Tower T-202 will prompt operator response 17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-204A/B will prompt operator response	F	1	1	1	61. No recommendations required.	
		44.1.1.3. Decreased flow of product to storage, not expected to impact Rerun Tank TK-100-14; Potential financial impact	17. FAL-11250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-104A/B will prompt operator response 545. FAL-21250 low flow alarm on discharge of Combi Tower Bottoms Product Pumps PM-	F	2	2	1	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			204A/B will prompt operator response						
	44.1.2. motor operated valve MOV-9430/40 closed on Rerun inlet to Rerun tank TK-100-14 rerun rundown to TK-100-14 only	44.1.2.1. See consequences associated with cause 44.1.1							
	44.1.3. 12" valve closed on suction to Pump P-7 or strainer plugs	44.1.3.1. Loss of Rerun flow to Pump P-7, potential cavitation and seal leak, release of rerun to atmosphere, potential seal fire from seal, personnel injury, environmental impact, asset damage	436. Flow switch FS-9404 low flow shutdown on pump P-7 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		44.1.3.2. Loss of Rerun flow to Pumps P-7 and P-HX, potential cavitation and seal leak of P-HX, release of rerun/ condensate to grade, potential seal fire or explosion, personnel injury, environmental impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm 424. analyzer transmitter AT-9101/ 02/ 03/ 04 gas detector around the Feed manifold with automatic shutdown of	P	4	1	2	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pump PM-111A/B/C and P-HX with 2oo4 voting						
			425. flame transmitter XT-9101/ 02 flame detector directed at pumps with automatic shutdown of pumps PM-111A/B/C and P-HX with 1oo2 voting						
		44.1.3.3. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	
	44.1.4. Pump P-7 stops	44.1.4.1. Loss of Rerun flow to Pumps P-7 and P-HX, potential cavitation and seal leak of P-HX, release of rerun/ condensate to grade, potential seal fire or explosion, personnel injury, environmental impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	P	4	1	2	61. No recommendations required.	
			424. analyzer transmitter AT-9101/ 02/ 03/ 04 gas detector around the Feed manifold with automatic shutdown of pump PM-111A/B/C and P-HX with 2oo4 voting						
			425. flame transmitter XT-9101/ 02 flame detector directed at pumps with automatic shutdown of pumps PM-111A/B/C and P-HX with 1oo2 voting						
		44.1.4.2. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	
	44.1.5. 8" valve closed on discharge of Pump P-7	44.1.5.1. Blocked flow from Pump P-7, potential deadhead and seal leak, release of rerun to atmosphere, potential seal fire from seal, personnel injury, environmental impact, asset damage	436. Flow switch FS-9404 low flow shutdown on pump P-7 with alarm	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
			369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown						

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
			415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting						
			416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting						
		44.1.5.2. Loss of Rerun flow to Pumps P-7 and P-HX, potential cavitation and seal leak of P-HX, release of rerun/ condensate to grade, potential seal fire or explosion, personnel injury, environmental impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	P	4	1	2	61. No recommendations required.	
			424. analyzer transmitter AT-9101/ 02/ 03/ 04 gas detector around the Feed manifold with automatic shutdown of pump PM-111A/B/C and P-HX with 2oo4 voting						
			425. flame transmitter XT-9101/ 02 flame detector directed at pumps with automatic shutdown of pumps PM-111A/B/C and P-HX with 1oo2 voting						
		44.1.5.3. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	44.1.6. 12" valve closed on suction to Pump P-HX or strainer plugs	44.1.6.1. See consequences associated with cause 44.1.5							
	44.1.7. Pump P-HX stops	44.1.7.1. Blocked flow from Pump P-7, potential deadhead and seal leak, release of rerun to atmosphere, potential seal fire from seal, personnel injury, environmental impact, asset damage	436. Flow switch FS-9404 low flow shutdown on pump P-7 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown pumps P-6, P-2A/B with 1oo4 voting 370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response 415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting 416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
		44.1.7.2. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	
	44.1.8. 10" valve closed on discharge of Pump P-HX or 8" valve closed on inlet of AGO Exchangers HX-1/2	44.1.8.1. Blocked flow from Pump P-7, potential deadhead and seal leak, release of rerun to atmosphere, potential seal fire from seal, personnel injury, environmental impact, asset damage	436. Flow switch FS-9404 low flow shutdown on pump P-7 with alarm 369. Lower Explosive Limit alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will shutdown	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			pumps P-6, P-2A/B with 1oo4 voting						
			370. XT-9401/9402 Flame Detectors will shutdown pumps P-6, P-2A/B with 1oo2 voting and alarm to prompt operator response						
			415. analyzer transmitter AT-9301/ 02/ 03/ 04 gas detector around the AGO manifold with automatic shutdown of pumps P-5, and P-3A/B with 1oo4 voting						
			416. flame transmitter XT-9301/ 02 flame detector directed at pumps with automatic shutdown of pumps P-5, and P-3A/B with 1oo2 voting						
		44.1.8.2. Blocked flow from pump P-HX; Potential pump deadhead and seal leak of P-HX, release of rerun/ condensate to grade, potential seal fire or explosion, personnel injury, environmental impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	P	4	1	2	61. No recommendations required.	
			424. analyzer transmitter AT-9101/ 02/ 03/ 04 gas detector around the Feed manifold with automatic shutdown of pump PM-111A/B/C and P-HX with 2oo4 voting						
			425. flame transmitter XT-9101/ 02 flame detector directed at pumps with automatic shutdown of pumps PM-111A/B/C and P-HX with 1oo2 voting						
		44.1.8.3. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
	44.1.9. 8" valve closed on outlet of Feed Exchangers HX-1/2	44.1.9.1. See consequences 44.1.8.1 and 44.1.8.2 above							
		44.1.9.2. No flow of Rerun tank to tank transfer rate, delay in transfer, not expected to overpressure Exchanger HX1/2, operability issue, no hazardous consequences identified							
	44.1.10. Inlet motor operated valve MOV closed to all Feed tanks TK-200-1/2/3 when transferring TK-100-14 through pumps P-7 and P-HX	44.1.10.1. See consequences 44.1.8.1 and 44.1.8.2 above							
		44.1.10.2. No flow of Rerun tank to tank transfer rate, delay in transfer, not expected to overpressure Exchanger HX1/2, operability issue, no hazardous consequences identified							
	44.1.11. All 8" manual valves closed on feed manifold outlet of tank TK-200/1/2/3 to Transfer Pump P-HX when transferring from tank to tank	44.1.11.1. Loss of Rerun flow to Pumps P-7 and P-HX, potential cavitation and seal leak of P-HX, release of rerun/ condensate to grade, potential seal fire or explosion, personnel injury, environmental impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm 424. analyzer transmitter AT-9101/ 02/ 03/ 04 gas detector around the Feed manifold with automatic shutdown of pump PM-111A/B/C and P-HX with 2oo4 voting 425. flame transmitter XT-9101/ 02 flame detector directed at pumps with automatic shutdown of pumps PM-111A/B/C and P-HX with 1oo2 voting	P	4	1	2	61. No recommendations required.	
		44.1.11.2. Decreased flow of Rerun tank to tank transfer rate, delay in transfer; Potential financial impact	437. Flow switch FS-9125 low flow shutdown on pump P-HX with alarm	F	1	2	1	61. No recommendations required.	
	44.1.12. 4" gate valve or 2" ball closed on relief header to Rerun Tank Tk-100-14	44.1.12.1. Potential blocked outlet of thermal PSVs, release of Feed to grade, potential personnel injury and environmental impact	448. No specific safeguards identified	P	4	3	4	43. Identify manual valves that need to be car sealed open to ensure there is an open flow path for thermal relief valves on Kero, Diesel, Naphtha, AGO, and Feed Tanks, as well as preventing cross contamination of the product. Car seal these valves open and update P&IDs accordingly.	
44.2. High Flow	44.2.1. No additional causes identified								
44.3. Reverse/Misdirected Flow	44.3.1. 8" rerun to Pump P-HX suction header open and 12" open to other Feed tank when not intended	44.3.1.1. Potential overfill of Feed Tank TK-200-1/2/3 to grade, potential release of condensate, potential pool fire or vapor cloud	426. Level alarm high LAH-9101/ 02/ 03 on Feed Tank TK-200-	P	4	1	2	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
		and explosion or flash fire, potential personnel injury, environmental impact, asset damage	1/2/3 with operator response 427. Level alarm hihi LAHH-9101/ 02/ 03 on Feed Tank TK-200-1/2/3 alarm with operator response 428. level switch high LSH-9101/ 02/ 03 on Feed Tank TK-200-1/2/3 mechanical switch will close inlet motor operated valve MOV to Feed Tanks stopping Feed flow to storage and eventually shutting the pipeline flow on high pressure						
	44.3.2. motor operated valve MOV-9141/42/43 to eductor open and motor operated valve MOV-9131/32/33 to diffuser closed on tanks TK-200-1/2/3 when sending rerun to Feed Tanks on initial fill	44.3.2.1. Potential issue when sending rerun to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	655. Tk-200-1/2/3 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9141/9142/9143 and opening diffuser MOV-9131/9132/9133 656. Tk-200-1/2/3 tanks are grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	44.3.3. motor operated valve MOV-9430 to eductor open and motor operated valve MOV-9440 to diffuser closed on tanks TK-100-14 initial fill	44.3.3.1. Potential issue when sending feed to eductors on initial fill (start-up), potential high flow causing roof to go off axis and roof damage, potential static charge build up and tank fire/ explosion, personnel injury, environmental impact and asset damage	657. Tk-100-14 Initial Fill Procedure (GPS 348) includes guidelines for closing the eductor MOV-9430 and opening diffuser MOV-9440 658. Tk-100-14 tank is grounded and bonded for start-up	P	4	1	2	61. No recommendations required.	
	44.3.4. 6" manual valve open on low suction line from tank TK-100-14 when not intended	44.3.4.1. Potential to draw water in Feed Splitter 1/2, potential high water content in feed, potential unit upset on Splitter 1/2, operability issue, no hazardous consequences identified							
44.4. High Pressure	44.4.1. External fire on Rerun Tank TK-100-14	44.4.1.1. Increased pressure and overpressure of Rerun Tank TK-100-14, potential vessel rupture, financial impact	382. Foam chamber for tank seals 438. Several open vents on floating roof tank for Rerun Tank	P	4	1	2	61. No recommendations required.	

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			TK-100-14 are rated for external fire						
	44.4.2. Blocked in Rerun/ condensate and thermal expansion	44.4.2.1. Potential to overpressure piping by thermal expansion, release of condensate/ rerun, personnel injury, environmental impact	439. Thermal Pressure Safety Valves across condensate/ rerun piping circuit	P	4	1	2	44. Verify manual valves on inlet and outlet piping of PSVs on drawing 88-PI-9239 are car sealed open and update P&ID to show valves CSO. Also update P&ID to include field verified PSV tag numbers. 55. 1) Verify that the operating procedures include information that the hydrocarbon drain header is routed Feed Tank TK-200-1/2/3 from Rerun Tank TK-100-14 to prevent overfill in the event that a PSV is leaking by or relieving. 2) Update the operating procedure to ensure that the hydrocarbon header is car sealed open to at least one Feed Tank or Rerun Tank TK-100-14 to allow an open flow path for thermal PSVs.	
44.5. Low Pressure	44.5.1. No additional causes identified								
44.6. High Level	44.6.1. Improper lineup - lining up rundown to a full tank	44.6.1.1. Potential overfill of Rerun Tank TK-100-14 to grade, potential release of condensate/rerun, potential pool fire or vapor cloud and explosion or flash fire, potential personnel injury, environmental impact, asset damage	432. Level alarm high LAH-9400 on Rerun Tank TK-100-14 with operator response 433. Level alarm hihi LAHH-9400 on Rerun Tank TK-100-14 alarm with operator response 434. level switch high LSH-9400 on Rerun Tank TK-100-14 mechanical switch will close all inlet motor operated valve MOVs to Rerun Tank	P	4	1	2	61. No recommendations required.	
44.7. Low Level	44.7.1. No additional causes identified								
44.8. High Interface Level	44.8.1. Not applicable								
44.9. Low Interface Level	44.9.1. Not applicable								
44.10. High Temperature	44.10.1. High ambient temperature	44.10.1.1. No consequences of note.							
44.11. Low Temperature	44.11.1. Low ambient temperature	44.11.1.1. No consequences of note.							

Node: 44. Node 44: Rerun Tank 100-14 and Rerun Pump P-HX

Drawings / References:

88-P1-9215; 88-P1-9216; 88-P1-9217; 88-P1-9218; 88-P1-9224; 88-P1-9227 Redline; 88-P1-9227 ; 88-P1-9246

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
44.12. Contamination/ Composition	44.12.1. No additional causes identified								
44.13. Corrosion	44.13.1. No additional causes identified								
44.14. Additional Phase	44.14.1. Not applicable								
44.15. Rupture/Loss of Containment	44.15.1. Mechanical Roof Failure	44.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential condensate release and ignition, potential pool fire requiring tank replacement, personnel injury, environmental impact, asset damage	390. Lower explosive limit LEL alarms AT-9401/9402/9403/9404 around the Naphtha Manifold will identify seal damage on roof 659. Monthly through hatch inspection Rerun Tank Tk-100-14	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
44.16. Loss of Utilities	44.16.1. Loss of power	44.16.1.1. See pump trip scenarios, this node							
44.17. Start-up/Shutdown hazards/ Maintenance	44.17.1. Gauging or sampling Rerun Tank TK-100-14 from top	44.17.1.1. Potential personnel exposure to toxic gas when gauging the tank, sampling the tank or doing a movement from or to the tank	371. Four gas monitors will alert operator of hydrogen sulfide 372. Third party inspectors includes wearing breathing air for sampling 373. Hatch has a vapor plug which will prevent immediate exposure to hydrogen sulfide - will allow time for response to gas monitor	P	2	2	1	53. Add signage to Rerun Tank TK-100-14, Feed Tanks TK-200-1/2/3 and Light Naphtha Tanks TK-120-24/25 stating that hydrogen sulfide H2S may be present to warn inspectors gauging the tank.	
44.18. General	44.18.1. No additional causes identified								

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
45.1. Low/No Flow	45.1.1. P-17 Discharge pump or PM-18A/B Transfer Pump fails.	45.1.1.1. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds.	P	4	1	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump						
	45.1.2. Manual valve on tank outlet inadvertently closed.	45.1.2.1. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump	P	4	1	2	61. No recommendations required.	
		45.1.2.2. Potential to cavitation pump P-17, PM-18A/B; Potential seal failure; Release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response 661. Flow switch FSL-18530/18550 low flow shutdown on pumps PM-18A/B with alarm and operator response	P	4	1	2	61. No recommendations required.	
	45.1.3. LCV-18500 fails closed	45.1.3.1. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-	P	4	1	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			5 will prompt operator response 662. P-17 Discharge Pump available						
		45.1.3.2. Potential to deadhead pump PM-18A/B; Potential seal failure; Release of HCs and water to ground; Potential ignition and personnel impact	661. Flow switch FSL-18530/18550 low flow shutdown on pumps PM-18A/B with alarm and operator response 663. Lower explosive limit alarms AI-9110A/B/C at Tank TK-5 will prompt operators to cordon off the area 357. High level alarm on LT-9624 on tank T-5 will prompt operator response	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	45.1.4. P-19 Skimmer pump fails.	45.1.4.1. Potential overfill of oil, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	2	2	61. No recommendations required.	
	45.1.5. Tank inlet line inadvertently closed.	45.1.5.1. Potential to discharge water from desalter to the splitter process. Potential vaporization of water in downstream naphtha stabilizer and combi tower. Potential overpressure of T-101/201, loss of containment, ignition and personnel impact	527. PSV-10195/10215/10225 (set @ 126/123/123 respectively) on Stabilizer T-101 is sized for loss of cooling 599. PSV-20195/20215/20225 (set @ 126, 123, and 123 psig respectively) on Stabilizer T-201 is sized for loss of cooling 7. LAH-10360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-101 water boot will prompt operator response	P	4	1	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			543. LAH-20360 High Level Alarm on Naphtha Stabilizer Accumulator pressure vessel PV-201 water boot will prompt operator response						
	45.1.6. Manual valve closed on water draw from Y-Grade Spheres	45.1.6.1. This valve is a normally closed and water draw from spheres routed to flare knockout drum, no hazardous consequences identified							
	45.1.7. Manual valve closed or plugging on water from containment areas	45.1.7.1. Potential to flood containment area, spilling oily water to ground, potential environmental/financial impact	441. Level switch high LSH-9629/9632 will turn Sump Pumps P-13/14 and P-15/16 online and will alarm and prompt operator response	F	2	2	1	61. No recommendations required.	
	45.1.8. Pump P-13/14/15/16 strainer plugs	45.1.8.1. Potential to overfill Lift station #1/2, spilling oily water to ground, potential environmental impact	440. Spare pump available for Pumps P-13/14 and P-15/16 441. Level switch high LSH-9629/9632 will turn Sump Pumps P-13/14 and P-15/16 online and will alarm and prompt operator response	F	2	2	1	61. No recommendations required.	
		45.1.8.2. Potential sump pump P-13/14/15/16 cavitation and seal leak, release inside of containment sump, asset damage only	664. Level switch high LSH-9629/9632 will alarm and prompt operator response	F	2	3	2	61. No recommendations required.	
	45.1.9. Pump P-13/14/15/16 stops on lift station #1/2	45.1.9.1. Potential to overfill Lift station #1/2, spilling oily water to ground, potential environmental impact	440. Spare pump available for Pumps P-13/14 and P-15/16 441. Level switch high LSH-9629/9632 will turn Sump Pumps P-13/14 and P-15/16 online and will alarm and prompt operator response	F	2	2	1	61. No recommendations required.	
	45.1.10. 4" manual valve closed on suction of Pump P-19 or strainer plugs	45.1.10.1. No flow to Skimmer Pump P-19, potential cavitation and seal leak, release of hydrocarbon to grade, potential seal or pool fire, potential personnel injury, environmental impact, asset damage	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 442. Flow switch low FSL-18580 low flow	P	4	1	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			shutdown on pump P-19 with alarm						
		45.1.10.2. Potential overfill of oil, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	2	2	61. No recommendations required.	
	45.1.11. Pump P-19 stops	45.1.11.1. Potential overfill of oil, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	2	2	61. No recommendations required.	
	45.1.12. Manual valve closed on Discharge of Pump P-19	45.1.12.1. Blocked flow from Skimmer Pump P-19, potential deadhead and seal leak, release of hydrocarbon to grade, potential seal or pool fire, potential personnel injury, environmental impact, asset damage	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 442. Flow switch low FSL-18580 low flow shutdown on pump P-19 with alarm	P	4	1	2	61. No recommendations required.	
		45.1.12.2. Potential overfill of oil, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	2	2	61. No recommendations required.	
	45.1.13. Manual valve closed on suction of Pump P-17 or strainer plugs	45.1.13.1. Cavitation of pump P-17; Potential seal failure; Release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds.	P	4	1	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			357. High level alarm on LT-9624 on tank T-5 will prompt operator response 443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response						
		45.1.13.2. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response 356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump	P	4	1	2	61. No recommendations required.	
	45.1.14. Manual valve closed on Discharge of Pump P-17	45.1.14.1. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response 356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump	P	4	1	2	61. No recommendations required.	
		45.1.14.2. Deadhead of pump P-17; Potential seal failure; Release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624	P	4	1	2	61. No recommendations required.	

Node: **45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19**

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response						
	45.1.15. Manual valve closed on suction of Pump PM-18A/B or strainer plugs	45.1.15.1. Potential to cavitation pump PM-18A/B; Potential seal failure; Release of HCs and water to ground; Potential ignition and personnel impact	356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 661. Flow switch FSL-18530/18550 low flow shutdown on pumps PM-18A/B with alarm and operator response	P	4	1	2	61. No recommendations required.	
		45.1.15.2. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response 356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump	P	4	1	2	61. No recommendations required.	
	45.1.16. Pump PM-18A/B stops	45.1.16.1. Potential overfill of tank 5 with release of HCs and water to ground; Potential ignition and personnel impact	443. Flow switch FS-9626 low flow shutdown on pump P-17 with alarm and operator response	P	4	1	2	61. No recommendations required.	

Node: **45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19**

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			356. Field level indication on LT-9624 checked on hourly operator rounds. 357. High level alarm on LT-9624 on tank T-5 will prompt operator response 660. Three pumps available for tank 5: PM-18A/B Transfer Pumps and P-17 Discharge Pump						
45.2. High Flow	45.2.1. Manual valve left open on water draw from Y-Grade Spheres	45.2.1.1. Potential to send Y Grade to Lift Station #1. Potential release to atmosphere from lift station #1, potential vapor cloud and ignition, fire explosion, personnel impact environmental impact, asset damage	665. Lower explosive limit LEL alarms AT-9301/9302/9303/9304 around the AGO Manifold will prompt operators to cordon off the area 666. PV-410 (GPS 0315) Water Draw Procedure 667. PV-411 (GPS 0316) Water Draw Procedure	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
	45.2.2. Pump P-19 left online	45.2.2.1. Potential to send water to Rerun Tank TK-100-14, not expected to overflow TK-100-14, operability issue, no hazardous consequences identified							
	45.2.3. Pump P-17 online while pump PM-18A/B online	45.2.3.1. Increased flow of water from Pump P-17/ PM-18A/B, decreased water level in Tank TK-5, potential to draw down level in tank and send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank 444. Water pumps PM-18A/B automatically shut off at a pre-set level in Tank TK-5	F	3	2	2	61. No recommendations required.	
45.3. Reverse/Misdirected Flow	45.3.1. Tank 5 bypass line inadvertently opened.	45.3.1.1. Potential to bypass tank 5. Oil is sent to tank T-100-14 Operational issue. No hazardous consequence identified.							
	45.3.2. Containment water motor operated valve MOVs left in wrong position directed to outfall as opposed to lift station #1/2	45.3.2.1. These MOVs are locked out, therefore this scenario is not credible							
	45.3.3. Rerun inlet valve open on tank TK-200-1 when skimming oil from TK-5	45.3.3.1. Potential to send hydrocarbon skim to Feed Tank TK-200-1, operability issue, no hazardous consequences identified							

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
45.4. High Pressure	45.4.1. External fire on Water Tank TK-5	45.4.1.1. Increased pressure and overpressure of Water Tank TK-5, potential vessel rupture, financial impact	382. Foam chamber for tank seals 445. Several open vents on floating roof tank for Water Tank TK-5 rated for external fire	P	4	1	2	61. No recommendations required.	
45.5. Low Pressure	45.5.1. No additional causes identified								
45.6. High Level	45.6.1. No additional causes identified								
45.7. Low Level	45.7.1. Level Control Valve LCV-18500 fails open or manual bypass valve open	45.7.1.1. Increased flow of water from Pump PM-18A/B, decreased water level in Tank TK-5, potential to draw down level in tank and send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank 444. Water pumps PM-18A/B automatically shut off at a pre-set level in Tank TK-5	F	3	2	2	61. No recommendations required.	
45.8. High Interface Level	45.8.1. Skimmer pump fails.	45.8.1.1. Potential overfill of oil, potential to send hydrocarbon to gulf coast water authority; Potential to impact biologics at water treatment; Potential financial impact	379. LT-18500B with High Level Alarm on tank T-5 will prompt operator response 378. Water draw procedure states to monitor the water sump level and periodically sample the water draw from the tank	F	3	2	2	61. No recommendations required.	
45.9. Low Interface Level	45.9.1. Low oil interface level in tank 5	45.9.1.1. Potential increased corrosion in skimmer line, potential long-term mechanical integrity issues, no acute hazardous consequences identified							
45.10. High Temperature	45.10.1. No additional causes identified								
45.11. Low Temperature	45.11.1. Low ambient temperature	45.11.1.1. No consequences of note							
45.12. Contamination/ Composition	45.12.1. Development of emulsion layer in Tank-5.	45.12.1.1. Potential to discharge out of spec waste water; Potential to impact biologics at water treatment; Potential financial impact	378. Water draw procedure states to monitor the water sump level and periodically sample the	F	3	2	2	61. No recommendations required.	

Node: 45. Node 45: Waste Water TK-5 and Pumps P-18A/B P-17 and P-19

Drawings / References:

88-P1-9215; 88-P1-9238 ; 88-P1-9238 Redline; 88-P1-9239 ; 88-P1-9250

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			water draw from the tank 358. Chemical treatment of emulsion layer 359. Ability to pump to splitter feed tank.						
45.13. Corrosion	45.13.1. Corrosion of metal in water service	45.13.1.1. Potential increased corrosion of piping/equipment; potential long-term mechanical integrity issues, no acute hazardous consequences identified							
45.14. Additional Phase	45.14.1. Not applicable								
45.15. Rupture/Loss of Containment	45.15.1. Mechanical Roof Failure	45.15.1.1. Potential to damage roof, tipping roof, liquid hydrocarbon on top of internal floating roof IFR, potential increased emission, potential HC release and ignition, potential pool fire requiring tank replacement, personnel injury, environmental impact, asset damage	663. Lower explosive limit alarms AI-9110A/B/C at Tank TK-5 will prompt operators to cordon off the area 668. Monthly through hatch inspection Tank 5 completed by operations. Visual inspection of tank roof and seal.	P	4	3	4	56. Evaluate the need to implement a regular third-party inspection of the roofs for Waste Water Tank TK-5 to mitigate the likelihood of roof damage leading to vapor release. 60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
45.16. Loss of Utilities	45.16.1. Loss of power	45.16.1.1. See pump trip scenarios, this node							
45.17. Start-up/Shutdown hazards/ Maintenance	45.17.1. Solid buildup in Waste Water Tank TK-5	45.17.1.1. Potential to impact biologics at water treatment; Potential financial impact; Potential to contribute to roof damage	669. Tk-5 Solid Removal Procedure (GPS 0358) 663. Lower explosive limit alarms AI-9110A/B/C at Tank TK-5 will prompt operators to cordon off the area 668. Monthly through hatch inspection Tank 5 completed by operations. Visual inspection of tank roof and seal.	P	4	1	2	60. Ensure that lower explosive limit LEL alarms AT-9101/9102/9103/9104 will prompt operators to cordon off the area.	
45.18. General	45.18.1. No additional causes identified								

Node: 46. Global

Drawings / References:

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
46.1. External events	46.1.1. 8.2. External events - Tornado	46.1.1.1. Potential equipment damage	62. Inclement weather incidents addressed in Emergency Response Plan (AP 0926)						
	46.1.2. External events - Hurricane	46.1.2.1. Potential equipment damage	63. Hurricane Procedure - Plan (AP 0912)	F	3	3	3	57. (Global Node) Update the KM Hurricane Plan to be site-specific and address: - timing of plant shutdown - provisions for personnel riding out the storm on site to return home prior to storm to make home preparations - details/checklist to verify tanks will not lift off foundations	
	46.1.3. 8.3. External events - Flooding	46.1.3.1. This site has adequate drainage; No outstanding concerns identified by the PHA team							
46.2. Leak/Rupture	46.2.1. Leak from any piping flange/ pump seal	46.2.1.1. Potential release of gas/ liquid to atmosphere, potential fire/explosion, personnel injury, environmental impact	64. Combustible gas detectors located throughout facility with action to emergency shutdown ESD 65. Mechanical integrity program for piping inspection 3. Leak detection and repair program LDAR program						
	46.2.2. Leak from any manual drain valve	46.2.2.1. Potential release of gas/ liquid to atmosphere, potential fire/explosion, personnel injury, environmental impact	64. Combustible gas detectors located throughout facility with action to emergency shutdown ESD 66. Lock out tag out LOTO/ maintenance procedures to replace plugs 3. Leak detection and repair program LDAR program						
46.3. Maintenance Hazards	46.3.1. Inaccessible valve/ equipment	46.3.1.1. See HF/Flow switch FL checklists							
	46.3.2. Impacts to equipment from external sources	46.3.2.1. Potential release of gas/ liquid to atmosphere, potential fire/explosion, personnel injury, environmental impact	64. Combustible gas detectors located throughout facility with action to emergency shutdown ESD 66. Lock out tag out LOTO/ maintenance						1. Bollard installation project in progress

Node: 46. Global
Drawings / References:

Deviation	Causes	Consequences	Safeguards	CAT	S	L	RR	Recommendations	Remark
			procedures to replace plugs						
46.4. Operating Procedures	46.4.1. Inadequate operating procedures	46.4.1.1. See HF checklists							
46.5. Human Factors	46.5.1. Equipment with same name	46.5.1.1. See HF checklists							
46.6. Utility Failure	46.6.1. Loss of air	46.6.1.1. Emergency shutdowns ESDs will close, control valves will go to fail-safe positions, plant will safely shutdown, decreased production, operability issue							
	46.6.2. Loss of power	46.6.2.1. All pumps, compressors, programmable logic controllers PLCs, trip							
46.7. Regulatory Boundary Review	46.7.1. Existing EPA RMP and OSHA PSM Boundaries fail to include potential equipment in the tank farm that could be considered as covered under each regulation.	46.7.1.1. There is an open MOC to reassess the PSM boundaries for the facility. In addition, this PHA included an analysis of the tank farm to assess risks.		P	1	5	2	58. (Global Node) Review the PSM boundaries in the Tank Farm to determine if all applicable equipment is included in the covered process.	
46.8. P&ID Updates	46.8.1. Inaccurate piping and instrumentation diagrams	46.8.1.1. Potential missed equipment, inaccurate process safety information. Potential regulatory concern.		F	3	4	4	59. (Global Node) Update the Piping and Instrumentation Diagrams (P&IDs) for the Galena Park facility per the redlines during the PHA.	